

NEBIANS

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Physics

Grade 12 · Science Stream

Complete study notes for the NEB / CDC secondary education curriculum. Every unit of the official syllabus, with derivations, worked examples, diagrams, tables, summaries, practice questions and full model question papers.

About this book

These notes cover the complete official curriculum for Physics, Grade 12, as published by the Curriculum Development Centre (CDC), Sanathimi, and examined by the National Examination Board (NEB).

How this book is organised

Each chapter maps to one unit of the CDC curriculum and opens with the exact syllabus points that unit must cover, so you can tick off the syllabus as you study. Inside a chapter you will find:

- **Concept sections** following the order of the official curriculum.
- **Derivations** set out step by step, the way a full-mark answer should be written.
- **Worked examples** with the complete solution method, not just the answer.
- **Figures, tables and graphs** drawn to be redrawn by hand in the exam hall.
- **Key points, exam tips and common mistakes** pulled out of the running text.
- **A chapter summary and practice questions with answers**, in NEB question style.

Syllabus coverage at a glance

Content area	Chapters	Teaching hours
Mechanics	3	22
Heat and Thermodynamics	2	12
Waves and Optics	8	26
Electricity and Magnetism	6	35
Modern Physics	6	33

Examination pattern

- 3hr, FM 75 written (+25 internal) | Group A MCQ 11x1=11 | Group B short 8x5=40 | Group C long 3x8=24

PLEASE READ THIS

This is a community study aid, not an official CDC or NEB publication, and it does not replace your prescribed textbook. It was prepared from the official CDC curriculum documents and past NEB question papers. Edition 1.0 is a first release: if you spot an error, report it in the NEBians app so the next edition can fix it for everybody.

Edition 1.0 · 2082 BS (2025–26 AD) · Prepared for the NEBians open learning community · Free to share and print for study purposes.

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UNIT 1

Rotational dynamics

Mechanics • 7 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Equation of angular motion; Relation between linear and angular kinematics
- Kinetic energy of rotation of rigid body
- Moment of inertia; Radius of gyration
- Moment of inertia of a uniform rod
- Torque and angular acceleration for a rigid body
- Work and power in rotational motion
- Angular momentum; conservation of angular momentum

A spinning flywheel, a turning grinding stone at a mill in Bhaktapur, the rotor of a Kulekhani turbine — none of these move from place to place, yet they clearly have motion and energy. To describe them we need a second set of mechanics, built on angle instead of distance. Every idea you learned in Grade 11 has a rotational twin: displacement becomes angle, mass becomes moment of inertia, force becomes torque. Learn the dictionary and most of this unit writes itself.

THE TRANSLATION TABLE IS THE WHOLE UNIT

Every rotational formula in this chapter is a linear formula with the symbols swapped: $s \rightarrow \theta$, $v \rightarrow \omega$, $a \rightarrow \alpha$, $m \rightarrow I$, $F \rightarrow \tau$, $p \rightarrow L$. If you can recall the linear result, you can rebuild the rotational one.

1.1 Equation of angular motion; Relation between linear and angular kinematics

When a rigid body rotates about a fixed axis, every particle of it moves in a circle centred on that axis. Different particles travel different distances in the same time, but all of them sweep the **same angle**. That is why angle, not distance, is the natural variable.

The **angular displacement** θ is measured in radians, defined by

$$\theta = \frac{s}{r}$$

where s is the arc length and r the radius. The **angular velocity** and **angular acceleration** are then

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

with units rad s^{-1} and rad s^{-2} .

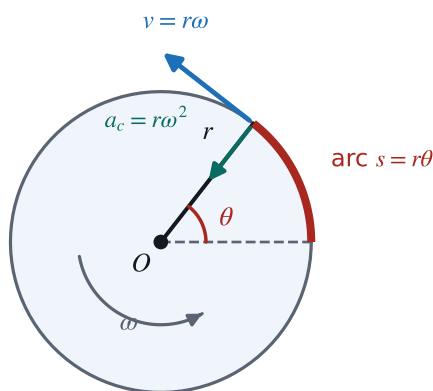


Figure 1. A rigid body rotating about an axis through O perpendicular to the page. Every point sweeps the same angle θ , but a point at radius r moves with speed $v = r\omega$ along the tangent.

If α is **constant**, integrating twice gives the equations of angular motion, exactly parallel to the linear ones:

$$\omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2, \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

and the angle turned in the n^{th} second is $\theta_n = \omega_0 + \frac{\alpha}{2}(2n - 1)$.

Because $s = r\theta$ and r is fixed for a given particle, differentiating gives the **link between the linear and angular pictures**:

$$v = r\omega, \quad a_t = r\alpha, \quad a_c = \frac{v^2}{r} = r\omega^2$$

Here a_t is the **tangential** acceleration (it changes the speed) and a_c is the **centripetal** acceleration (it changes the direction). The resultant linear acceleration has magnitude $a = \sqrt{a_t^2 + a_c^2}$.

Linear quantity	Rotational quantity	Connection
Displacement s (m)	Angular displacement θ (rad)	$s = r\theta$
Velocity v (m s ⁻¹)	Angular velocity ω (rad s ⁻¹)	$v = r\omega$
Acceleration a (m s ⁻²)	Angular acceleration α (rad s ⁻²)	$a_t = r\alpha$
Mass m (kg)	Moment of inertia I (kg m ²)	$I = \sum mr^2$
Force F (N)	Torque τ (N m)	$\tau = rF\sin\phi$
Momentum $p = mv$	Angular momentum $L = I\omega$	$L = rps\sin\phi$

RADIANS, NEVER DEGREES OR RPM

$v = r\omega$, $s = r\theta$ and $\tau = I\alpha$ are true **only** when angles are in radians. Convert first: 1 rev = 2π rad, and N rev min⁻¹ means $\omega = 2\pi N/60$ rad s⁻¹.

WORKED EXAMPLE 1.1

Problem. A grinding wheel starts from rest and reaches 300 rev min^{-1} in 10 s with uniform angular acceleration. Find (a) the angular acceleration, (b) the number of revolutions made, and (c) the linear speed of a point 0.40 m from the axis at the end.

Solution. (a) $\omega = \frac{2\pi \times 300}{60} = 10\pi = 31.4 \text{ rad s}^{-1}$, so

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{10\pi - 0}{10} = \pi = 3.14 \text{ rad s}^{-2}$$

(b) $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(\pi)(10)^2 = 50\pi \text{ rad}$. Number of revolutions $= 50\pi/2\pi = 25$ revolutions.

(c) $v = r\omega = 0.40 \times 31.4 = 12.6 \text{ m s}^{-1}$.

1.2 Kinetic energy of rotation of rigid body

Think of the rigid body as a collection of particles of masses $m_1, m_2, \dots$ at perpendicular distances $r_1, r_2, \dots$ from the axis. All share the same ω , so particle i has speed $v_i = r_i\omega$.

ROTATIONAL KINETIC ENERGY

The total kinetic energy is the sum of the kinetic energies of the particles:

$$E_k = \sum_i \frac{1}{2} m_i v_i^2 = \sum_i \frac{1}{2} m_i (r_i \omega)^2 = \frac{1}{2} \left(\sum_i m_i r_i^2 \right) \omega^2$$

The bracket contains only the masses and their distances from the axis — it is a property of the body and the axis, not of the motion. Calling it the **moment of inertia** I ,

$$E_k = \frac{1}{2} I \omega^2$$

Compare $\frac{1}{2}mv^2$: the moment of inertia plays exactly the role that mass plays in translation. A body that is also moving forward (a rolling wheel) has total kinetic energy $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$.

1.3 Moment of inertia; Radius of gyration

MOMENT OF INERTIA

The moment of inertia of a body about a given axis is the sum of the products of the mass of each particle and the square of its perpendicular distance from that axis: $I = \sum m_i r_i^2$, or $I = \int r^2 dm$ for a continuous body. Its SI unit is kg m^2 and its dimensional formula is $[ML^2]$.

Moment of inertia measures **rotational inertia** — the opposition a body offers to a change in its angular velocity. Unlike mass it is not a single number for a body: it depends on (i) the total mass, (ii) how that mass is distributed, and (iii) **which axis** is chosen. Mass placed far from the axis counts much more, because the distance is squared.

Two theorems shorten most calculations:

- **Parallel axis theorem.** $I = I_{cm} + Md^2$, where I_{cm} is about a parallel axis through the centre of mass and d is the separation.
- **Perpendicular axis theorem** (plane laminae only). $I_z = I_x + I_y$, the z -axis being perpendicular to the lamina.

Body (mass M)	Axis	Moment of inertia	Radius of gyration k
Thin rod, length L	$\perp$ through centre	$ML^2/12$	$L/\sqrt{12}$
Thin rod, length L	$\perp$ through one end	$ML^2/3$	$L/\sqrt{3}$
Ring, radius R	central, $\perp$ to plane	MR^2	R
Disc, radius R	central, $\perp$ to plane	$MR^2/2$	$R/\sqrt{2}$
Solid cylinder, radius R	own axis	$MR^2/2$	$R/\sqrt{2}$
Solid sphere, radius R	diameter	$2MR^2/5$	$R\sqrt{2/5}$
Hollow sphere, radius R	diameter	$2MR^2/3$	$R\sqrt{2/3}$

The **radius of gyration** k is the distance from the axis at which the whole mass could be concentrated as a single point without changing the moment of inertia:

$$I = Mk^2 \quad \Rightarrow \quad k = \sqrt{\frac{I}{M}}$$

WORKED EXAMPLE 1.2

Problem. A solid disc flywheel of mass 20 kg and radius 0.50 m spins at 120 rev min^{-1} . Find its moment of inertia, radius of gyration and rotational kinetic energy.

Solution.

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(20)(0.50)^2 = 2.5 \text{ kg m}^2$$

$$k = \sqrt{I/M} = \sqrt{2.5/20} = \sqrt{0.125} = 0.354 \text{ m}$$

which is indeed $R/\sqrt{2}$. With $\omega = 2\pi(120)/60 = 4\pi = 12.57 \text{ rad s}^{-1}$,

$$E_k = \frac{1}{2}I\omega^2 = \frac{1}{2}(2.5)(12.57)^2 = 197 \text{ J}$$

1.4 Moment of inertia of a uniform rod

UNIFORM THIN ROD, AXIS THROUGH THE CENTRE

Take a rod of mass M and length L , of uniform linear mass density $\lambda = M/L$. Let the axis pass through the centre C , perpendicular to the rod. Consider a small element of length dx at a distance x from C ; its mass is $dm = \lambda dx = (M/L) dx$ and its moment of inertia is $x^2 dm$.

Summing over the whole rod, x runs from $-L/2$ to $+L/2$:

$$I_C = \int_{-L/2}^{+L/2} x^2 \frac{M}{L} dx = \frac{M}{L} \left[\frac{x^3}{3} \right]_{-L/2}^{+L/2} = \frac{M}{3L} \left(\frac{L^3}{8} + \frac{L^3}{8} \right)$$

$$I_C = \frac{ML^2}{12}$$

SAME ROD, AXIS THROUGH ONE END

Now put the axis at the end A , so x runs from 0 to L :

$$I_A = \int_0^L x^2 \frac{M}{L} dx = \frac{M}{L} \left[\frac{x^3}{3} \right]_0^L = \frac{ML^2}{3}$$

Check with the parallel axis theorem: the end is a distance $d = L/2$ from the centre of mass, so

$$I_A = I_C + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{12} + \frac{ML^2}{4} = \frac{ML^2}{3} \quad \checkmark$$

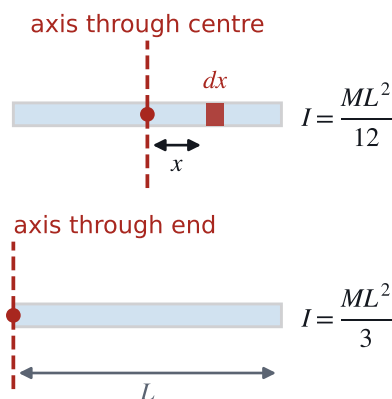


Figure 2. Moment of inertia of a uniform rod. The shaded element of mass $dm = (M/L)dx$ sits a distance x from the axis. Shifting the axis from the centre to the end raises I from $ML^2/12$ to $ML^2/3$.

1.5 Torque and angular acceleration for a rigid body

A force applied to a body that is free to rotate produces a turning effect called **torque** (or moment of force). If the force $\vec{F}$ acts at a point whose position vector from the axis is $\vec{r}$, and ϕ is the angle between them,

$$\tau = rF\sin\phi = F (r\sin\phi) = F \times (\text{perpendicular distance from the axis})$$

The quantity $d = r\sin\phi$ is the **moment arm**. Torque is largest when the force is applied perpendicular to $\vec{r}$ and as far from the axis as possible — which is why a door handle is fitted at the edge, not next to the hinge. The SI unit is N m; torque is a vector, $\vec{\tau} = \vec{r} \times \vec{F}$.

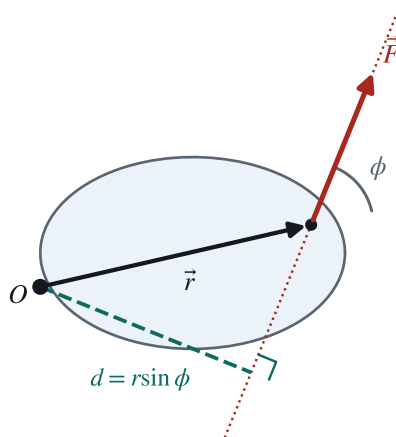


Figure 3. Torque about the axis at O . Only the component $F\sin\phi$ perpendicular to $\vec{r}$ turns the body; the moment arm is $d = r\sin\phi$.

TORQUE EQUALS $I\alpha$

Take one particle of mass m_i at distance r_i . The tangential force on it is, by Newton's second law, $F_i = m_i a_{ti} = m_i r_i \alpha$, because every particle of a rigid body shares the same α . The torque of this force about the axis is

$$\tau_i = F_i r_i = m_i r_i^2 \alpha$$

Adding over all particles, the internal forces cancel in pairs (Newton's third law), so the resultant external torque is

$$\tau = \sum_i m_i r_i^2 \alpha = \left(\sum_i m_i r_i^2 \right) \alpha$$

$$\tau = I\alpha$$

This is the rotational form of $F = ma$.

WORKED EXAMPLE 1.3

Problem. A uniform rod of mass 2 kg and length 1 m is hinged at one end and held horizontal. It is released from rest. Find its initial angular acceleration and the initial linear acceleration of the free end. Take $g = 9.8 \text{ m s}^{-2}$.

Solution. The weight acts at the centre of mass, 0.5 m from the hinge, so the torque about the hinge is

$$\tau = Mg\frac{L}{2} = 2 \times 9.8 \times 0.5 = 9.8 \text{ N m}$$

About the end, $I = ML^2/3 = 2(1)^2/3 = 0.667 \text{ kg m}^2$. Hence

$$\alpha = \frac{\tau}{I} = \frac{9.8}{0.667} = 14.7 \text{ rad s}^{-2}$$

The free end has $a_t = L\alpha = 1 \times 14.7 = 14.7 \text{ m s}^{-2}$, which is **greater than** g . A coin resting on the far end of the rod is therefore left behind when the rod is released.

1.6 Work and power in rotational motion

When a torque τ turns a body through a small angle $d\theta$, the force $F = \tau/r$ acts through an arc $ds = r d\theta$, so the work done is

$$dW = F ds = \frac{\tau}{r} (r d\theta) = \tau d\theta$$

For a constant torque, $W = \tau\theta$. The **power** is the rate of doing this work:

$$P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau\omega$$

the exact analogue of $P = Fv$. And since $\tau = I\alpha = I\omega \frac{d\omega}{d\theta}$,

$$W = \int \tau d\theta = \int_{\omega_1}^{\omega_2} I\omega d\omega = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

which is the **work-energy theorem for rotation**: the work done by the resultant torque equals the change in rotational kinetic energy.

WORKED EXAMPLE 1.4

Problem. An electric motor at a rice mill in Jhapa delivers a constant torque of 25 N m while running at $1500 \text{ rev min}^{-1}$. Find the power output, and the work done in one minute.

Solution. $\omega = \frac{2\pi \times 1500}{60} = 50\pi = 157.1 \text{ rad s}^{-1}$.

$$P = \tau\omega = 25 \times 157.1 = 3927 \text{ W} \approx 3.93 \text{ kW}$$

In 60 s , $W = Pt = 3927 \times 60 = 2.36 \times 10^5 \text{ J}$.

1.7 Angular momentum; conservation of angular momentum

ANGULAR MOMENTUM

The angular momentum of a particle about a point is the moment of its linear momentum: $L = rps \sin \phi = mvr \sin \phi$. For a rigid body rotating about a fixed axis, $L = I\omega$. Its SI unit is $\text{kg m}^2 \text{ s}^{-1}$ (or J s).

Differentiating $L = I\omega$ for a rigid body (I constant about a fixed axis),

$$\frac{dL}{dt} = I \frac{d\omega}{dt} = I\alpha = \tau$$

So **torque is the rate of change of angular momentum**, just as force is the rate of change of linear momentum. Written this way the result is more general: it holds even when I changes.

PRINCIPLE OF CONSERVATION OF ANGULAR MOMENTUM

If the resultant external torque on a system is zero, then $dL/dt = 0$, so $L = I\omega$ is constant:

$$I_1\omega_1 = I_2\omega_2$$

A body that reduces its moment of inertia must spin faster, and vice versa.

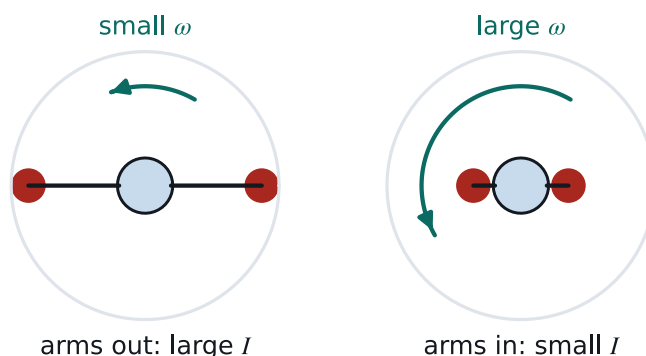


Figure 4. Conservation of angular momentum, seen from above. Pulling the masses in reduces I , so ω rises to keep $L = I\omega$ fixed.

Everyday consequences, all examined regularly:

- An **ice skater or a dancer** spins faster on folding the arms in.
- A **diver** tucks to somersault quickly, then stretches out to slow the spin before entering the water.
- A **planet** moves faster when nearer the Sun; since gravity is a central force its torque about the Sun is zero. This is Kepler's second law.
- A spinning **top or a bicycle wheel** resists being tipped over, because tipping it would change the direction of $\vec{L}$.

ANGULAR MOMENTUM IS CONSERVED, KINETIC ENERGY IS NOT

When a skater pulls the arms in, L stays the same but $E_k = L^2/2I$ **increases** because I falls. The extra energy comes from the muscular work done in pulling the arms inwards against the outward (centrifugal) effect. Conversely, when two rotating discs are coupled together, energy is lost to friction while L is conserved.

WORKED EXAMPLE 1.5

Problem. A student sits on a frictionless rotating stool with arms outstretched, holding weights. The moment of inertia of the system is 6 kg m^2 and it rotates at 2 rad s^{-1} . He pulls his arms in, reducing the moment of inertia to 2 kg m^2 . Find the new angular velocity and the change in kinetic energy.

Solution. No external torque acts about the vertical axis, so angular momentum is conserved:

$$I_1\omega_1 = I_2\omega_2 \Rightarrow 6 \times 2 = 2 \times \omega_2 \Rightarrow \omega_2 = 6 \text{ rad s}^{-1}$$

$$E_1 = \frac{1}{2}(6)(2)^2 = 12 \text{ J}, \quad E_2 = \frac{1}{2}(2)(6)^2 = 36 \text{ J}$$

The kinetic energy increases by 24 J, supplied by the work the student does in pulling the weights inwards.

Chapter summary

- For constant α : $\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\theta$; and $s = r\theta$, $v = r\omega$, $a_t = r\alpha$, $a_c = r\omega^2$.
- Rotational kinetic energy $E_k = \frac{1}{2}I\omega^2$, where $I = \sum m_i r_i^2 = \int r^2 dm$ is the moment of inertia.
- Radius of gyration: $I = Mk^2$, so $k = \sqrt{I/M}$. Parallel axis: $I = I_{cm} + Md^2$; perpendicular axis (laminae): $I_z = I_x + I_y$.
- Uniform rod of mass M , length L : $I = ML^2/12$ about the centre and $ML^2/3$ about one end.
- Torque $\tau = rF\sin\phi$ and $\tau = I\alpha$, the rotational form of $F = ma$.
- Work $W = \tau\theta$, power $P = \tau\omega$, and $W = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$.
- Angular momentum $L = I\omega$ with $\tau = dL/dt$. If $\tau = 0$ then $I_1\omega_1 = I_2\omega_2$: angular momentum is conserved, but kinetic energy need not be.

Practice questions

Group A — Multiple choice (1 mark each)

1. The moment of inertia of a uniform rod of mass M and length L about an axis through one end and perpendicular to it is [1] (a) $ML^2/12$ (b) $ML^2/3$ (c) $ML^2/2$ (d) ML^2
2. The rotational analogue of mass is [1] (a) torque (b) angular velocity (c) moment of inertia (d) angular momentum
3. A body of moment of inertia 2 kg m^2 rotates at 3 rad s^{-1} . Its rotational kinetic energy is [1] (a) 3 J (b) 6 J (c) 9 J (d) 18 J
4. A skater spinning with arms outstretched folds her arms in. Her angular velocity increases because [1] (a) her angular momentum increases (b) an external torque acts on her (c) her moment of inertia decreases while L is constant (d) her kinetic energy is conserved
5. The power delivered by a torque τ at angular velocity ω is [1] (a) $\tau\theta$ (b) $\tau\omega$ (c) $I\omega$ (d) τ/ω

ANSWERS TO GROUP A

1. (b) — integrating $x^2 dm$ from 0 to L gives $ML^2/3$. **2.** (c) — I appears in $\tau = I\alpha$ exactly where m appears in $F = ma$. **3.** (c) — $E_k = \frac{1}{2}(2)(3)^2 = 9 \text{ J}$. **4.** (c) — no external torque, so L is fixed and ω must rise. **5.** (b) — $P = dW/dt = \tau d\theta/dt = \tau\omega$.

Group B — Short answer (5 marks each)

1. Define moment of inertia and radius of gyration. On what factors does the moment of inertia of a body depend? [5]
2. Derive the relation $\tau = I\alpha$ for a rigid body rotating about a fixed axis. [5]

- A flywheel of mass 50 kg in the form of a uniform disc of radius 0.40 m rotates at 300 rev min^{-1} . Calculate its kinetic energy and the constant torque needed to stop it in 10 s. [5]
- A disc of moment of inertia 2 kg m^2 rotating at 10 rad s^{-1} is suddenly coupled to a stationary coaxial disc of moment of inertia 3 kg m^2 . Find the common angular velocity and the loss of kinetic energy. [5]

ANSWERS TO GROUP B

1. Outline: state $I = \sum m_i r_i^2$ (unit kg m^2 , dimensions $[ML^2]$) and $k = \sqrt{I/M}$. I depends on the total mass, on the distribution of that mass about the axis, and on the position and direction of the axis itself.

2. Outline: for particle i , $F_i = m_i r_i \alpha$, so $\tau_i = m_i r_i^2 \alpha$; sum over all particles (internal torques cancel in pairs) to get $\tau = (\sum m_i r_i^2) \alpha = I \alpha$.

3. $I = \frac{1}{2}MR^2 = \frac{1}{2}(50)(0.40)^2 = 4 \text{ kg m}^2$; $\omega = 2\pi(300)/60 = 31.42 \text{ rad s}^{-1}$. $E_k = \frac{1}{2}(4)(31.42)^2 = 1974 \text{ J}$. Retardation $\alpha = 31.42/10 = 3.142 \text{ rad s}^{-2}$, so $\tau = I\alpha = 4 \times 3.142 = 12.6 \text{ N m}$.

4. Angular momentum is conserved: $2 \times 10 = (2 + 3)\omega$, so $\omega = 4 \text{ rad s}^{-1}$. $E_1 = \frac{1}{2}(2)(10)^2 = 100 \text{ J}$, $E_2 = \frac{1}{2}(5)(4)^2 = 40 \text{ J}$; loss = 60 J, dissipated as heat by friction between the coupled discs.

Group C — Long answer (8 marks each)

- (a) Define moment of inertia and radius of gyration. [3] (b) Derive an expression for the moment of inertia of a uniform rod of mass M and length L about an axis through its centre and perpendicular to its length, and hence, using the parallel axis theorem, about an axis through one end. [5]
- (a) Show that the kinetic energy of a rigid body rotating about a fixed axis is $\frac{1}{2}I\omega^2$. [3] (b) A flywheel of mass 100 kg and radius of gyration 0.50 m rotates at 600 rev min^{-1} . A constant retarding torque brings it to rest in 40 s. Find its moment of inertia, its initial kinetic energy, the retarding torque, and the number of revolutions it makes before stopping. [5]

ANSWER TO GROUP C QUESTION 2(B)

$$I = Mk^2 = 100 \times (0.50)^2 = 25 \text{ kg m}^2.$$

$$\omega_0 = 2\pi(600)/60 = 20\pi = 62.83 \text{ rad s}^{-1}, \text{ so } E_k = \frac{1}{2}(25)(62.83)^2 = 4.93 \times 10^4 \text{ J}.$$

$$\alpha = (0 - 62.83)/40 = -1.571 \text{ rad s}^{-2}, \text{ so the retarding torque is } |\tau| = I|\alpha| = 25 \times 1.571 = 39.3 \text{ N m}.$$

$$\theta = \frac{1}{2}(\omega_0 + 0)t = \frac{1}{2}(62.83)(40) = 1257 \text{ rad}, \text{ so the number of revolutions} = 1257/2\pi = 200.$$

UNIT 2

Periodic motion

Mechanics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Equation of simple harmonic motion (SHM)
- Energy in SHM
- Application of SHM: vertical oscillation of mass suspended from coiled spring
- Angular SHM; simple pendulum
- Oscillatory motion: Damped oscillation, Forced oscillation and resonance

A motion that repeats itself in equal intervals of time is **periodic**. If, in addition, the body moves to and fro about a fixed point, the motion is **oscillatory**. The swinging of a temple bell, the trembling of the ground during an earthquake, the vibration of a sitar string and the current in a radio circuit are all oscillations, and all of them are built out of one simple pattern — simple harmonic motion. This unit sets up that pattern, its energy, two standard applications, and what happens when friction or an external driver is added.

WHAT MAKES A MOTION SIMPLE HARMONIC

A motion is simple harmonic if the acceleration is **proportional to the displacement from the mean position and always directed towards it**. Every result in this chapter follows from that one sentence, so learn it exactly.

2.1 Equation of simple harmonic motion (SHM)

SIMPLE HARMONIC MOTION

Simple harmonic motion is the oscillation of a body in which the restoring force (and hence the acceleration) is directly proportional to the displacement from the mean position and is directed opposite to that displacement:

$$F = -kx \quad \text{or} \quad a = -\omega^2 x, \quad \omega^2 = \frac{k}{m}$$

The constant ω is the **angular frequency** (rad s^{-1}). The minus sign is the physics: it says the force always pushes the body back towards $x = 0$.

FROM $a = -\omega^2 x$ **TO** $x = A \sin(\omega t + \phi)$

Write the acceleration as $a = v \, dv/dx$ and separate the variables:

$$v \frac{dv}{dx} = -\omega^2 x \implies \int v \, dv = -\omega^2 \int x \, dx$$

$$\frac{v^2}{2} = -\frac{\omega^2 x^2}{2} + C$$

The body is momentarily at rest at the extreme position $x = A$ (the **amplitude**), so $C = \omega^2 A^2/2$ and

$$v = \omega \sqrt{A^2 - x^2}$$

Writing $v = dx/dt$ and separating again,

$$\int \frac{dx}{\sqrt{A^2 - x^2}} = \int \omega \, dt \implies \sin^{-1} \frac{x}{A} = \omega t + \phi$$

$$x = A \sin(\omega t + \phi)$$

where the constant of integration ϕ is the **initial phase** or epoch.

Differentiating the displacement gives the velocity and acceleration:

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi), \quad a = \frac{dv}{dt} = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 x$$

The **period** is the time for one complete oscillation and the **frequency** is its reciprocal:

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}, \quad f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Position	Displacement	Speed	Acceleration
Mean position ($x = 0$)	0	maximum, $A\omega$	0
Extreme position ($x = \pm A$)	$\pm A$	0	maximum, $\omega^2 A$

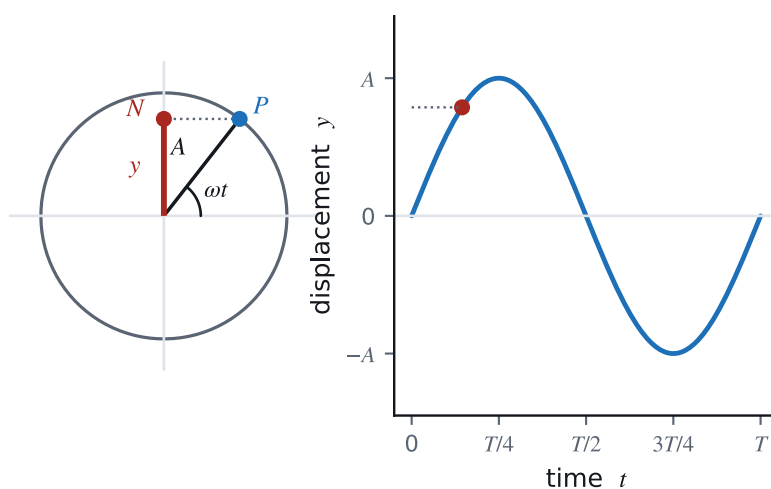


Figure 5. SHM is the projection of uniform circular motion on a diameter. As P moves round the circle of radius A with angular velocity ω , its foot N executes $y = A \sin \omega t$.

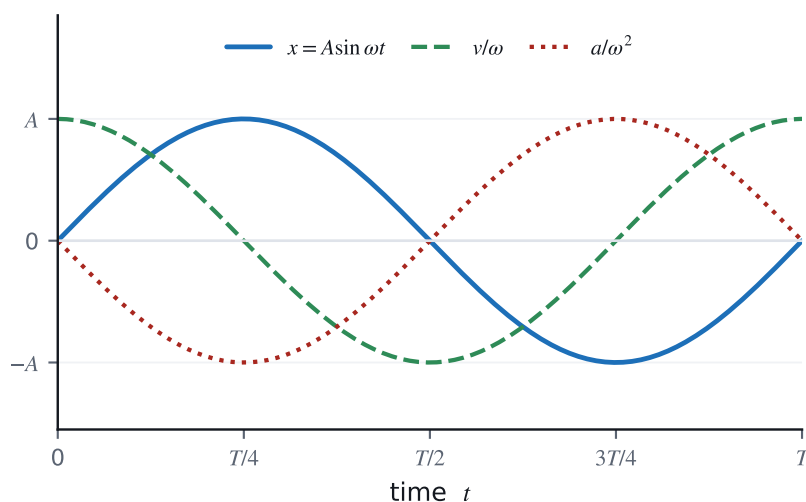


Figure 6. Displacement, velocity and acceleration in SHM. Velocity leads displacement by $\pi/2$; acceleration is exactly out of phase with displacement (π).

WORKED EXAMPLE 2.1

Problem. A particle executes SHM of amplitude 5 cm and period 2 s. Find (a) its maximum velocity, (b) its maximum acceleration, and (c) its velocity and acceleration when the displacement is 3 cm.

Solution. $\omega = 2\pi/T = 2\pi/2 = \pi = 3.14 \text{ rad s}^{-1}$ and $A = 0.05 \text{ m}$.

(a) $v_{\max} = A\omega = 0.05 \times 3.14 = 0.157 \text{ m s}^{-1}$.

(b) $a_{\max} = A\omega^2 = 0.05 \times (3.14)^2 = 0.493 \text{ m s}^{-2}$.

(c) $v = \omega \sqrt{A^2 - x^2} = 3.14 \sqrt{0.05^2 - 0.03^2} = 3.14 \times 0.04 = 0.126 \text{ m s}^{-1}$, and $a = \omega^2 x = 9.87 \times 0.03 = 0.296 \text{ m s}^{-2}$, directed towards the mean position.

2.2 Energy in SHM

The restoring force is $F = -kx$ with $k = m\omega^2$. The work done against it in moving from 0 to x is stored as **potential energy**:

$$U = \int_0^x kx \, dx = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2$$

The **kinetic energy** follows from $v = \omega \sqrt{A^2 - x^2}$:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$$

Adding them, the x^2 terms cancel:

$$E = K + U = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2 = \text{constant}$$

TOTAL ENERGY IN SHM

The total energy is constant, proportional to the **square of the amplitude** and to the **square of the frequency**. Doubling the amplitude quadruples the energy. Energy is continuously exchanged between K and U : all kinetic at the mean position, all potential at the extremes.

Averaged over a full cycle, $\langle K \rangle = \langle U \rangle = \frac{1}{4}m\omega^2 A^2$, i.e. each is half the total.

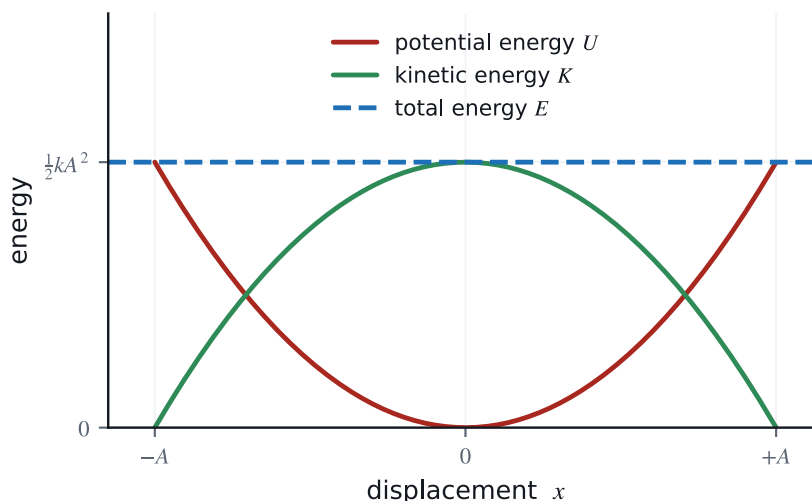


Figure 7. Kinetic and potential energy against displacement. The two parabolas always add to the constant total $E = \frac{1}{2}kA^2$.

WORKED EXAMPLE 2.2

Problem. A body of mass 0.5 kg attached to a spring of force constant 200 N m^{-1} oscillates with amplitude 0.10 m. Find the period, the total energy, and the kinetic and potential energies when the displacement is 0.05 m.

Solution. $\omega = \sqrt{k/m} = \sqrt{200/0.5} = 20 \text{ rad s}^{-1}$, so

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{20} = 0.314 \text{ s}$$

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(200)(0.10)^2 = 1.0 \text{ J}$$

At $x = 0.05 \text{ m}$; $U = \frac{1}{2}(200)(0.05)^2 = 0.25 \text{ J}$, so $K = E - U = 0.75 \text{ J}$. The speed there is $v = \sqrt{2K/m} = \sqrt{2(0.75)/0.5} = 1.73 \text{ m s}^{-1}$.

2.3 Application of SHM: vertical oscillation of mass suspended from coiled spring

Hang a mass m from a light spiral spring of force constant k . The spring stretches by an amount e until the elastic force balances the weight:

$$mg = ke \quad \Rightarrow \quad k = \frac{mg}{e}$$

This stretched position is the new mean position.

PERIOD OF VERTICAL OSCILLATION

Pull the mass a further distance x down and release it. The spring now pulls up with force $k(e + x)$ while the weight mg acts down, so the resultant (restoring) force is

$$F = mg - k(e + x) = ke - ke - kx = -kx$$

The weight cancels exactly — it only shifts the mean position, it does not affect the motion about it. Hence $ma = -kx$, i.e.

$$a = -\frac{k}{m}x$$

which is SHM with $\omega = \sqrt{k/m}$, so

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{e}{g}}$$

The second form is useful in the laboratory: measure the static extension e produced by the load and you can predict the period, or measure T and e and obtain $g = 4\pi^2 e/T^2$.

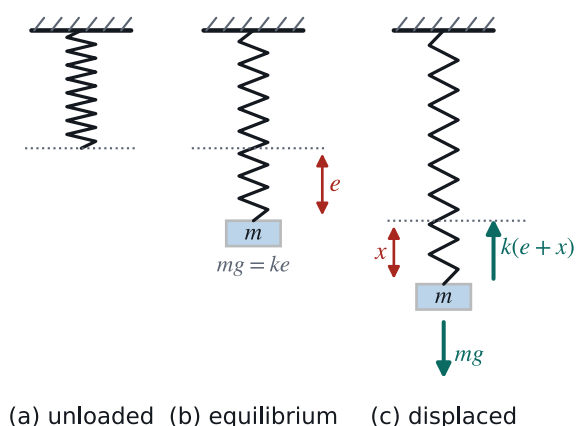


Figure 8. Vertical oscillation of a mass on a spring. The weight is cancelled by the static extension e , leaving a restoring force $-kx$ about the new mean position.

WORKED EXAMPLE 2.3

Problem. A mass of 0.5 kg hung from a light spiral spring stretches it by 10 cm. Find the force constant of the spring and the period of vertical oscillation. Take $g = 9.8 \text{ m s}^{-2}$.

Solution.

$$k = \frac{mg}{e} = \frac{0.5 \times 9.8}{0.10} = 49 \text{ N m}^{-1}$$

$$T = 2\pi\sqrt{\frac{e}{g}} = 2\pi\sqrt{\frac{0.10}{9.8}} = 2\pi(0.101) = 0.635 \text{ s}$$

The frequency is $f = 1/T = 1.58 \text{ Hz}$. Note that the answer needs only e and g — the mass cancels, because a heavier mass also stretches the spring more.

2.4 Angular SHM; simple pendulum

In **angular SHM** the body is displaced through an angle θ and the restoring **torque** is proportional to that angle:

$$\tau = -c\theta \implies I \frac{d^2\theta}{dt^2} = -c\theta \implies T = 2\pi\sqrt{\frac{I}{c}}$$

where c is the torsional constant and I the moment of inertia about the axis. A torsion pendulum and the coil of a moving-coil galvanometer behave this way.

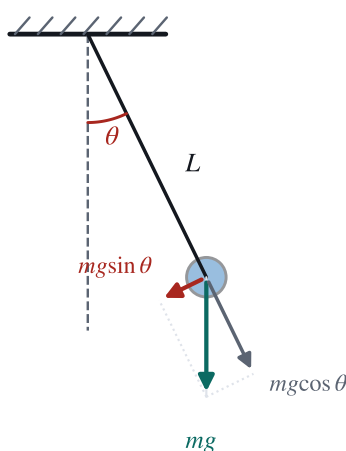


Figure 9. Simple pendulum. The component $mg\sin\theta$ is the restoring force; $mg\cos\theta$ merely balances the tension.

PERIOD OF A SIMPLE PENDULUM

A simple pendulum is a point mass (bob) of mass m hung by a light inextensible string of length L from a rigid support. When the string makes an angle θ with the vertical, the weight mg is resolved into $mg\cos\theta$ along the string (balanced by the tension) and $mg\sin\theta$ perpendicular to it. The second component is the restoring force:

$$F = -mg\sin\theta$$

For **small** angles, $\sin\theta \approx \theta$ in radians, and the displacement along the arc is $x = L\theta$, so

$$F = -mg\theta = -\frac{mg}{L}x \implies a = -\frac{g}{L}x$$

This is SHM with $\omega^2 = g/L$, hence

$$T = 2\pi\sqrt{\frac{L}{g}}$$

The four **laws of the simple pendulum** follow directly: $T \propto \sqrt{L}$ (law of length); $T \propto 1/\sqrt{g}$ (law of gravity); T is independent of the **mass** of the bob; and T is independent of the **amplitude** so long as it is small (law of isochronism). A **seconds pendulum** has $T = 2$ s, so it takes one second for each swing.

THE SMALL-ANGLE CONDITION IS NOT OPTIONAL

$T = 2\pi\sqrt{L/g}$ comes from replacing $\sin\theta$ by θ , which is accurate only for amplitudes of a few degrees (up to about 4°). For larger swings the real period is longer, and the motion is periodic but no longer simple harmonic.

WORKED EXAMPLE 2.4

Problem. Find the length of a seconds pendulum at a place where $g = 9.8 \text{ m s}^{-2}$. If this pendulum clock is carried to a place where $g = 9.78 \text{ m s}^{-2}$, how much time will it lose in one day?

Solution. From $T = 2\pi\sqrt{L/g}$,

$$L = \frac{gT^2}{4\pi^2} = \frac{9.8 \times 2^2}{4\pi^2} = 0.993 \text{ m}$$

At the new place the same pendulum has period

$$T' = 2\pi\sqrt{\frac{0.993}{9.78}} = 2.002 \text{ s}$$

The clock still counts every swing as 2 s. In one real day it makes $86400/2.002 = 43156$ oscillations and therefore records only $43156 \times 2 = 86312$ s.

Time lost per day = $86400 - 86312 = 88$ s, i.e. about 1 minute 28 s slow.

2.5 Oscillatory motion: Damped oscillation, Forced oscillation and resonance

Real oscillators lose energy to friction and air resistance, so the amplitude falls with time. This is **damped oscillation**. For small speeds the resistive force is proportional to the velocity, $F_d = -bv$, and Newton's second law gives

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

whose light-damping solution is

$$x = A_0 e^{-bt/2m} \sin(\omega' t + \phi), \quad \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

The amplitude decays exponentially as $A = A_0 e^{-bt/2m}$, and because $E \propto A^2$ the energy decays twice as fast. Damping also makes the oscillation slightly **slower** ($\omega' < \omega_0$).

Case	Condition	Behaviour	Example
Under-damped	b small	oscillates with decaying amplitude	pendulum in air
Critically damped	$b^2 = 4mk$	returns to rest in the shortest time , no oscillation	dead-beat galvanometer
Over-damped	b large	returns to rest slowly, no oscillation	pendulum in thick oil

If an external periodic force $F_0 \sin \omega_d t$ is applied, the oscillator settles into a **forced oscillation** at the **driver's** frequency ω_d , not its own. The steady amplitude is

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (b\omega_d/m)^2}}$$

RESONANCE

Resonance is the large increase in the amplitude of a forced oscillation that occurs when the frequency of the driving force becomes equal to the natural frequency of the oscillator ($\omega_d = \omega_0$). At resonance the driver feeds energy to the oscillator in step with its motion, and the amplitude is limited only by damping.

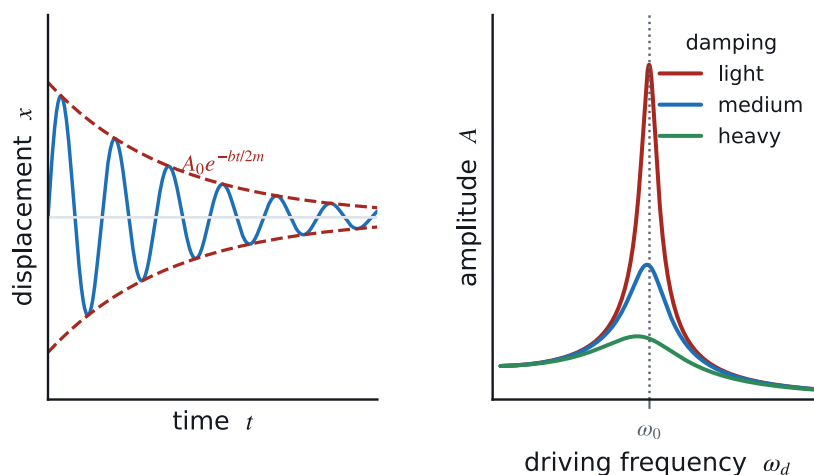


Figure 10. Left: amplitude of a damped oscillation decays as $A_0 e^{-bt/2m}$. Right: resonance curves — lighter damping gives a taller, sharper peak at $\omega_d = \omega_0$.

Resonance is useful and dangerous in equal measure: it lets a radio pick out one station, makes a sitar's sympathetic strings sing, and allows a small push at the right moment to swing a child high. But it also destroyed the Tacoma Narrows bridge, is why soldiers break step on a bridge, and is why buildings whose natural frequency matches the shaking of an earthquake — as some did in the Gorkha earthquake of 2015 — suffer the worst damage.

WORKED EXAMPLE 2.5

Problem. A damped oscillator has mass 0.2 kg and damping constant $b = 0.04 \text{ kg s}^{-1}$. After what time does (a) the amplitude fall to $1/e$ of its initial value, and (b) the energy fall to $1/e$ of its initial value?

Solution. (a) $A = A_0 e^{-bt/2m}$, so $A = A_0/e$ when $bt/2m = 1$:

$$t = \frac{2m}{b} = \frac{2 \times 0.2}{0.04} = 10 \text{ s}$$

(b) $E \propto A^2$, so $E = E_0 e^{-bt/m}$, and $E = E_0/e$ when

$$t = \frac{m}{b} = \frac{0.2}{0.04} = 5 \text{ s}$$

The energy decays in half the time the amplitude does.

Chapter summary

- SHM: $a = -\omega^2 x$, with $x = A \sin(\omega t + \phi)$, $v = A \omega \cos(\omega t + \phi) = \omega \sqrt{A^2 - x^2}$ and $T = 2\pi/\omega = 2\pi\sqrt{m/k}$.
- Speed is maximum ($A\omega$) at the mean position; acceleration is maximum ($\omega^2 A$) at the extremes. SHM is the projection of uniform circular motion on a diameter.
- $U = \frac{1}{2}m\omega^2 x^2$, $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, and the total $E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2$ is constant.
- Mass on a vertical spring: $mg = ke$ and $T = 2\pi\sqrt{m/k} = 2\pi\sqrt{e/g}$; the weight only shifts the mean position.
- Angular SHM: $\tau = -c\theta$ gives $T = 2\pi\sqrt{I/c}$. Simple pendulum: $T = 2\pi\sqrt{L/g}$ for small amplitude, independent of the mass of the bob.
- Damped oscillation: $A = A_0 e^{-bt/2m}$, $\omega' = \sqrt{k/m - b^2/4m^2}$; critical damping returns the system to rest fastest.
- A forced oscillator vibrates at the driver's frequency; at resonance ($\omega_d = \omega_0$) the amplitude is maximum and is limited only by damping.

Practice questions

Group A — Multiple choice (1 mark each)

1. In simple harmonic motion the acceleration is [1] (a) constant (b) proportional to displacement and in the same direction (c) proportional to displacement and oppositely directed (d) inversely proportional to displacement
2. The total energy of a particle executing SHM is proportional to [1] (a) A (b) A^2 (c) $1/A$ (d) $\sqrt{A}$
3. The period of a simple pendulum does **not** depend on [1] (a) its length (b) the acceleration due to gravity (c) the mass of the bob (d) any of these
4. A mass m hanging from a spring of force constant k oscillates with period [1] (a) $2\pi\sqrt{k/m}$ (b) $2\pi\sqrt{m/k}$ (c) $2\pi\sqrt{mk}$ (d) $\frac{1}{2\pi}\sqrt{m/k}$
5. At resonance the amplitude of a forced oscillation is [1] (a) zero (b) minimum (c) maximum (d) independent of damping

ANSWERS TO GROUP A

1. (c) — this is the definition, $a = -\omega^2 x$. **2.** (b) — $E = \frac{1}{2}m\omega^2 A^2$. **3.** (c) — mass cancels between the restoring force and the inertia. **4.** (b) — $\omega = \sqrt{k/m}$, so $T = 2\pi/\omega = 2\pi\sqrt{m/k}$. **5.** (c) — the driving frequency matches the natural frequency, so energy is fed in step with the motion.

Group B — Short answer (5 marks each)

1. Define simple harmonic motion. Show that the motion of a mass attached to a spring obeying Hooke's law is simple harmonic, and obtain its period. [5]
2. Derive expressions for the kinetic and potential energies of a particle in SHM and show that the total energy is constant. [5]
3. A particle of mass 0.2 kg executes SHM of amplitude 0.10 m with angular frequency 10 rad s^{-1} . Find its total energy, and its kinetic and potential energies at a displacement of 0.05 m. [5]
4. A load of 0.5 kg stretches a light spiral spring by 5 cm. Find the force constant of the spring and the period of vertical oscillation of the load. ($g = 9.8 \text{ m s}^{-2}$) [5]

ANSWERS TO GROUP B

1. Outline: state the definition; for a spring $F = -kx$, so $a = -(k/m)x = -\omega^2 x$ with $\omega = \sqrt{k/m}$, which is SHM; hence $T = 2\pi\sqrt{m/k}$.
2. Outline: $U = \int_0^x kx \, dx = \frac{1}{2}m\omega^2 x^2$; $K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$; adding, $E = \frac{1}{2}m\omega^2 A^2$, independent of x .
3. $E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}(0.2)(10)^2(0.10)^2 = 0.1 \text{ J}$. At $x = 0.05 \text{ m}$: $U = \frac{1}{2}(0.2)(100)(0.05)^2 = 0.025 \text{ J}$ and $K = 0.1 - 0.025 = 0.075 \text{ J}$.
4. $k = mg/e = (0.5 \times 9.8)/0.05 = 98 \text{ N m}^{-1}$; $T = 2\pi\sqrt{e/g} = 2\pi\sqrt{0.05/9.8} = 0.449 \text{ s}$.

Group C — Long answer (8 marks each)

1. (a) Define simple harmonic motion and, starting from $a = -\omega^2 x$, derive the expression $x = A\sin(\omega t + \phi)$ and hence $v = \omega\sqrt{A^2 - x^2}$. [5] (b) Sketch how the kinetic and potential energies vary with displacement, and state where each is maximum. [3]
2. (a) Derive an expression for the period of a simple pendulum, stating clearly the assumptions made. [5] (b) A seconds pendulum is set up in Kathmandu, where $g = 9.79 \text{ m s}^{-2}$. Find its length. If it is then taken to a place where $g = 9.81 \text{ m s}^{-2}$, how much time will it gain in a day? [3]

ANSWER TO GROUP C QUESTION 2(B)

$$L = \frac{gT^2}{4\pi^2} = \frac{9.79 \times 4}{4\pi^2} = 0.992 \text{ m}.$$

$$\text{At the new place, } T' = 2\pi\sqrt{0.992/9.81} = 1.998 \text{ s}.$$

In one real day the pendulum makes $86400/1.998 = 43244$ oscillations, and the clock records $43244 \times 2 = 86488 \text{ s}$.

$$\text{Gain per day} = 86488 - 86400 \approx 88 \text{ s}.$$

UNIT 3

Fluid statics

Mechanics • 9 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Fluid statics: Pressure in a fluid; Buoyancy
- Surface tension: Theory of surface tension; Surface energy
- Angle of contact, capillarity and its applications
- Fluid Dynamics: Newton's formula for viscosity in a liquid; Coefficient of viscosity
- Poiseuille's formula and its application
- Stokes law and its applications
- Equation of continuity and its applications
- Bernoulli's equation and its applications

A fluid is anything that flows. Because a fluid at rest cannot support a shearing force, its mechanics is written in terms of **pressure** rather than force. This unit runs from fluids at rest (pressure, upthrust), through the forces in a liquid **surface** (surface tension, capillarity), to fluids in motion (viscosity, Poiseuille, Stokes, continuity, Bernoulli).

WHERE THE MARKS ARE

Four derivations recur in almost every paper: $P = h\rho g$, the capillary rise $h = 2T\cos\theta/r\rho g$, terminal velocity from Stokes' law, and Bernoulli's equation. Learn these four and the numericals become routine.

3.1 Fluid statics: Pressure in a fluid; Buoyancy

Pressure is the normal force per unit area, $P = F/A$, in pascals ($1 \text{ Pa} = 1 \text{ N m}^{-2}$). It is a **scalar**: at a point in a fluid at rest it is the same in every direction.

PRESSURE DUE TO A LIQUID COLUMN

Take a horizontal area A at depth h below the free surface of a liquid of density ρ . The column standing on it has volume Ah , mass ρAh and weight ρAhg , all carried by that area:

$$P = \frac{\rho Ahg}{A} = h\rho g$$

Pressure therefore depends only on **depth**, density and g — never on the shape of the vessel. The **absolute** pressure is $P = P_0 + h\rho g$; $h\rho g$ alone is the **gauge** pressure. By **Pascal's law** pressure applied to an enclosed fluid is transmitted undiminished, so a small force on a narrow piston balances a large load on a wide one — the hydraulic press and brake.

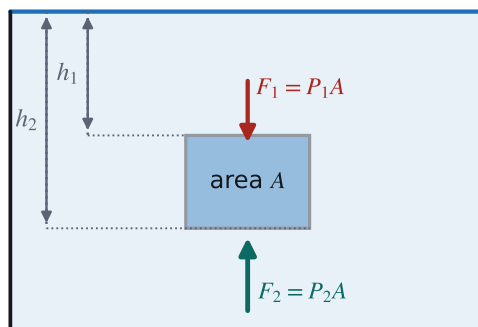


Figure 11. Origin of upthrust. The bottom face lies deeper, so $P_2 > P_1$ and the net vertical force $U = (P_2 - P_1)A = \rho g V$ acts upwards.

ARCHIMEDES' PRINCIPLE FROM PRESSURE

A cylinder of cross-section A is immersed with its top face at depth h_1 and its bottom face at h_2 . Sideways forces cancel in pairs; vertically $F_1 = h_1 \rho g A$ acts down and $F_2 = h_2 \rho g A$ up. As $h_2 > h_1$ the resultant — the **upthrust** — acts upwards:

$$U = F_2 - F_1 = (h_2 - h_1)A\rho g = V\rho g$$

Since $V\rho$ is the mass of liquid displaced, the upthrust equals the **weight of fluid displaced**.

A body floats when the upthrust equals its weight, so a body of density ρ_b and volume V floating with volume V_s submerged in liquid of density ρ_l has $V\rho_b g = V_s\rho_l g$, giving

$$\frac{V_s}{V} = \frac{\rho_b}{\rho_l}$$

Ice (917 kg m^{-3}) in sea water (1025 kg m^{-3}) therefore shows only about 11% of itself above the surface.

WORKED EXAMPLE 3.1

Problem. (a) Find the absolute pressure 10 m below the surface of Phewa Lake. (b) A block weighs 4.9 N in air and 3.92 N in water. Find its volume, density and relative density. ($P_0 = 1.013 \times 10^5 \text{ Pa}$, $\rho_w = 1000 \text{ kg m}^{-3}$, $g = 9.8 \text{ m s}^{-2}$)

Solution. (a) $P = P_0 + h\rho g = 1.013 \times 10^5 + (10)(1000)(9.8) = 1.99 \times 10^5 \text{ Pa}$.

(b) Upthrust $U = 4.9 - 3.92 = 0.98 \text{ N} = V\rho_w g$, so

$$V = \frac{0.98}{1000 \times 9.8} = 1.0 \times 10^{-4} \text{ m}^3$$

Mass = $4.9/9.8 = 0.5 \text{ kg}$, so $\rho = 0.5/10^{-4} = 5000 \text{ kg m}^{-3}$ and the relative density is 5.

3.2 Surface tension: Theory of surface tension; Surface energy

A needle floats on water and rain gathers into spherical drops: a liquid surface behaves like a stretched elastic membrane.

SURFACE TENSION

Surface tension is the force per unit length acting on an imaginary line drawn on a liquid surface, perpendicular to the line and tangential to the surface: $T = F/L$. Its unit is N m^{-1} and its dimensions are $[MT^{-2}]$.

Theory. Molecules attract one another (**cohesion**) within a sphere of influence of radius about 10^{-9} m . A molecule deep inside is pulled equally in all directions, so the resultant on it is zero; one in the surface layer has liquid only below it and feels a **net inward pull**. Work must be done to bring a molecule up to the surface, so the liquid keeps its area as **small as possible** — which is why free drops are spherical.

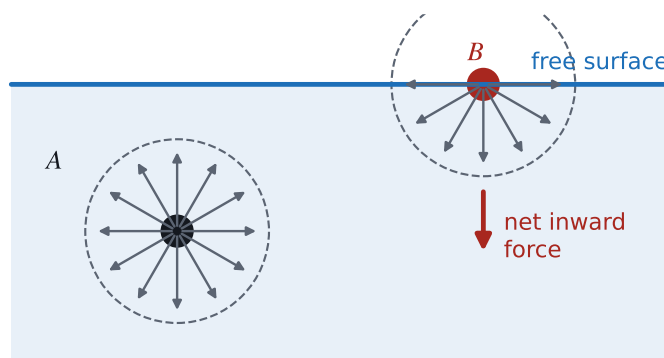


Figure 12. Molecular explanation of surface tension. Molecule A inside the liquid is pulled equally in all directions; molecule B in the surface has liquid only below it and feels a net inward force.

Surface energy. A soap film on a frame of width l pulls its light sliding wire inwards with force $F = 2Tl$ (the 2 because a film has **two** surfaces). Pulling the wire out a distance x does work

$$W = 2Tl \times x = T (2lx) = T \Delta A$$

since $2lx$ is the increase in surface area, so $T = W/\Delta A$.

SURFACE TENSION EQUALS SURFACE ENERGY

Surface energy is the work done in increasing the area by unity (J m^{-2}); surface tension is force per unit length (N m^{-1}). As $1 \text{ J m}^{-2} = 1 \text{ N m}^{-1}$, the two are numerically equal.

The excess pressure inside a curved surface is $2T/r$ for a drop or an air bubble in a liquid (one surface) and $4T/r$ for a soap bubble (two). Surface tension falls as temperature rises and is lowered by detergents.

WORKED EXAMPLE 3.2

Problem. Find the work done in blowing a soap bubble of radius 3 cm and the excess pressure inside it, given $T = 0.03 \text{ N m}^{-1}$.

Solution. A soap bubble has two surfaces, so the area created is

$$\Delta A = 2 \times 4\pi r^2 = 8\pi(0.03)^2 = 2.26 \times 10^{-2} \text{ m}^2$$

$$W = T \Delta A = 0.03 \times 2.26 \times 10^{-2} = 6.79 \times 10^{-4} \text{ J}$$

$$\text{Excess pressure} = \frac{4T}{r} = \frac{4 \times 0.03}{0.03} = 4 \text{ Pa.}$$

3.3 Angle of contact, capillarity and its applications

ANGLE OF CONTACT

The angle of contact is the angle between the tangent to the liquid surface at the point of contact and the solid surface, measured through the liquid.

Whether it is acute or obtuse depends on the competition between **adhesion** (liquid–solid attraction) and **cohesion** (liquid–liquid). The rise or fall of a liquid in a tube of fine bore is called **capillarity**.

	Water on glass	Mercury on glass
Stronger force	adhesion	cohesion
Meniscus	concave	convex
Angle of contact	acute (about 8° , taken as 0°)	obtuse (about 140°)
Behaviour	wets glass, rises	does not wet, is depressed

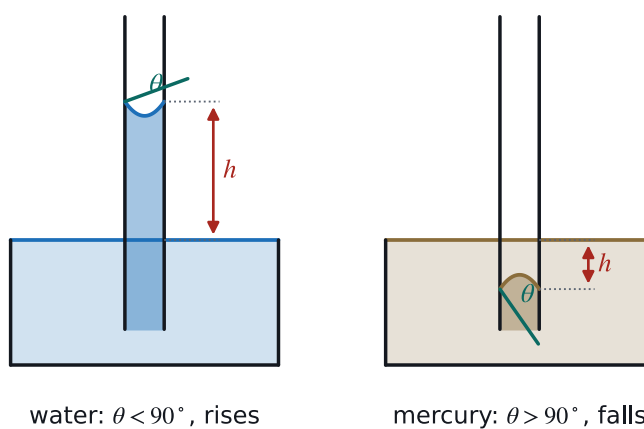


Figure 13. Capillarity. Water ($\theta < 90^\circ$) has a concave meniscus and rises; mercury ($\theta > 90^\circ$) has a convex meniscus and is depressed.

HEIGHT OF LIQUID IN A CAPILLARY TUBE

A tube of internal radius r stands in a liquid of density ρ , surface tension T and angle of contact θ . The liquid touches it along a circle of length $2\pi r$, and the vertical component of the tension per unit length is $T\cos\theta$, so the upward force is

$$F = 2\pi r T \cos \theta$$

This supports the weight of the raised column of height h , $W = (\pi r^2 h) \rho g$. Equating $F = W$,

$$2\pi r T \cos \theta = \pi r^2 h \rho g \quad \Rightarrow \quad h = \frac{2T \cos \theta}{r \rho g}$$

So $h \propto 1/r$ — the narrower the tube, the higher the rise (**Jurin's law**). If $\theta > 90^\circ$, $\cos \theta$ is negative and h is a depression, as with mercury.

Applications: oil rises up a lamp wick; sap reaches the leaves of a tree; ink soaks into blotting paper; and farmers plough their fields to **break** the soil capillaries, so water is not drawn up and lost by evaporation.

INTERNAL RADIUS, AND THE SIGN OF $\cos \theta$

In $h = 2T \cos \theta / r \rho g$, r is the **internal** radius. For water on clean glass $\cos \theta = 1$, but for mercury $\cos 140^\circ = -0.766$: get that sign wrong and a depression becomes a rise.

WORKED EXAMPLE 3.3

Problem. Find the rise of water in a capillary tube of internal diameter 0.4 mm. ($T = 0.072 \text{ N m}^{-1}$, $\theta = 0^\circ$, $\rho = 1000 \text{ kg m}^{-3}$, $g = 9.8 \text{ m s}^{-2}$)

Solution. Here $r = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$ and $\cos 0^\circ = 1$, so

$$h = \frac{2T \cos \theta}{r \rho g} = \frac{2 \times 0.072 \times 1}{(2 \times 10^{-4})(1000)(9.8)} = \frac{0.144}{1.96} = 0.0735 \text{ m}$$

The water rises 7.35 cm.

3.4 Fluid Dynamics: Newton's formula for viscosity in a liquid; Coefficient of viscosity

In **streamline** (laminar) flow a liquid moves in layers that slide over one another, each dragging on its neighbours; this internal friction is **viscosity**. If layers a distance dx apart differ in velocity by dv , then dv/dx is the **velocity gradient**, and Newton found the viscous force on a contact area A to be

$$F = -\eta A \frac{dv}{dx}$$

the minus sign showing that it opposes the relative motion.

COEFFICIENT OF VISCOSITY

The coefficient of viscosity is the tangential viscous force per unit area per unit velocity gradient, $\eta = F/(A \, dv/dx)$. Its SI unit is N s m^{-2} (Pa s), its dimensions are $[ML^{-1}T^{-1}]$, and $1 \text{ Pa s} = 10 \text{ poise}$.

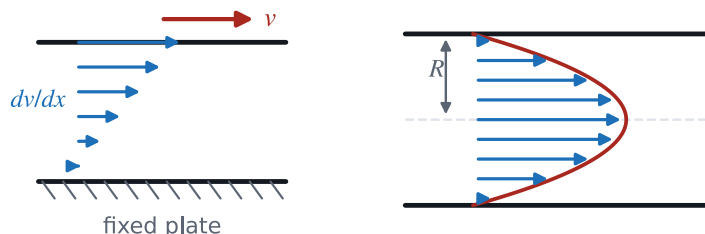


Figure 14. Left: velocity gradient dv/dx between layers of a liquid dragged by a moving plate. Right: the parabolic velocity profile of laminar flow in a tube, used in Poiseuille's formula.

Viscosity is to fluids what friction is to solids, except that the force depends on **speed**. Heating makes a liquid **less** viscous (warm honey pours easily) but a gas **more** viscous.

WORKED EXAMPLE 3.4

Problem. A plate of area 0.1 m^2 is dragged at 0.5 m s^{-1} over a 1 mm layer of oil of viscosity 0.8 Pa s on a fixed table. What force is needed?

Solution. The velocity falls uniformly to zero across the layer, so $dv/dx = 0.5/10^{-3} = 500 \text{ s}^{-1}$ and

$$F = \eta A \frac{dv}{dx} = 0.8 \times 0.1 \times 500 = 40 \text{ N}$$

3.5 Poiseuille's formula and its application

For steady laminar flow through a horizontal capillary of radius r and length l under a pressure difference P , the volume flowing per second is

$$Q = \frac{\pi P r^4}{8 \eta l}$$

POISEUILLE'S FORMULA BY THE METHOD OF DIMENSIONS

Let Q depend on the pressure gradient P/l , the radius r and η :

$$Q = K \left(\frac{P}{l} \right)^a r^b \eta^c$$

With $[Q] = [L^3 T^{-1}]$, $[P/l] = [ML^{-2} T^{-2}]$, $[r] = [L]$ and $[\eta] = [ML^{-1} T^{-1}]$,

$$L^3 T^{-1} = (ML^{-2} T^{-2})^a (L)^b (ML^{-1} T^{-1})^c$$

Comparing powers of M : $a + c = 0$; of T : $-2a - c = -1$; of L : $-2a + b - c = 3$. These give $a = 1$, $c = -1$, $b = 4$:

$$Q = K \frac{Pr^4}{\eta l}, \quad K = \frac{\pi}{8} \text{ from the full theory.}$$

THE FOURTH POWER OF THE RADIUS

$Q \propto r^4$: halving the radius cuts the flow to **one sixteenth**, which is why a slight narrowing of an artery makes the heart work so much harder.

It is used to measure η , to study blood flow, and to fix the bore of hypodermic needles and drip sets.

WORKED EXAMPLE 3.5

Problem. Water ($\eta = 1.0 \times 10^{-3} \text{ Pa s}$) flows through a horizontal tube of length 0.5 m and radius 1 mm under a pressure difference of 1000 Pa. Find Q , and Q if r is halved.

Solution.

$$Q = \frac{\pi Pr^4}{8\eta l} = \frac{\pi(1000)(10^{-3})^4}{8(1.0 \times 10^{-3})(0.5)} = \frac{\pi \times 10^{-9}}{4 \times 10^{-3}} = 7.85 \times 10^{-7} \text{ m}^3 \text{ s}^{-1}$$

i.e. $0.785 \text{ cm}^3 \text{ s}^{-1}$. Halving r divides Q by 16, giving $4.91 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}$.

3.6 Stokes law and its applications

A small sphere of radius r moving with velocity v through a fluid of viscosity η meets a retarding force

$$F = 6\pi\eta rv$$

This is **Stokes' law**, valid for slow, streamline motion; dimensional analysis of $F \propto \eta^a r^b v^c$ gives $a = b = c = 1$.

TERMINAL VELOCITY

A sphere of radius r and density ρ falling through a liquid of density σ feels its weight down, and upthrust and drag up. The drag grows with speed until the three balance and the sphere falls at a constant **terminal velocity** v_t :

$$\frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi\eta r v_t$$

$$6\pi\eta r v_t = \frac{4}{3}\pi r^3 (\rho - \sigma)g \quad \implies \quad v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

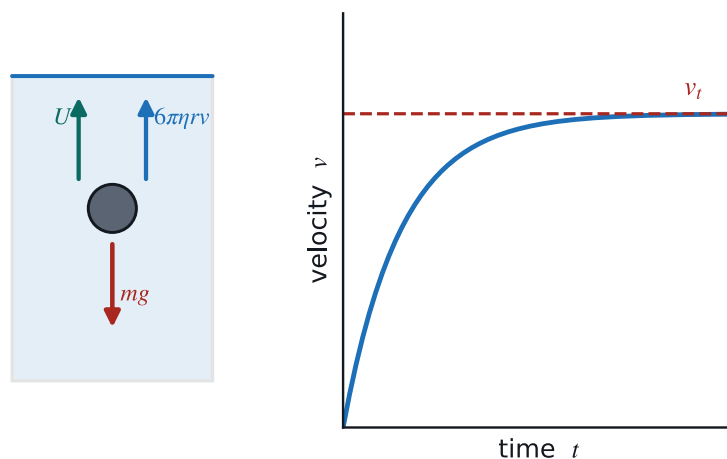


Figure 15. A sphere falling through a viscous liquid. When weight is balanced by upthrust plus drag the velocity becomes constant at the terminal value v_t .

Applications: **Millikan's oil-drop experiment** finds the drop radius and hence the electronic charge; since $v_t \propto r^2$ tiny droplets float as cloud while large drops fall fast; and η is found by timing a ball-bearing falling through the liquid.

WORKED EXAMPLE 3.6

Problem. A steel ball of radius 1 mm and density 7800 kg m^{-3} falls through oil of density 900 kg m^{-3} and viscosity 1.0 Pa s . Find v_t .

Solution.

$$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta} = \frac{2(10^{-3})^2(7800 - 900)(9.8)}{9 \times 1.0} = 1.50 \times 10^{-2} \text{ m s}^{-1}$$

i.e. about 1.5 cm s^{-1} .

3.7 Equation of continuity and its applications

Take steady, incompressible flow through a tube whose cross-section changes from A_1 to A_2 . In time Δt the liquid entering occupies a length $v_1 \Delta t$, so the mass entering is $\rho A_1 v_1 \Delta t$ and that leaving is $\rho A_2 v_2 \Delta t$. No liquid is created or lost, so

$$\rho A_1 v_1 = \rho A_2 v_2 \quad \Rightarrow \quad A_1 v_1 = A_2 v_2 = \text{constant}$$

This is the **equation of continuity** — the conservation of mass. Since $v \propto 1/A$, **a fluid speeds up where the pipe narrows**: a thumb over a hosepipe makes water shoot further, and a river runs fast through a gorge but slow over a plain.

WORKED EXAMPLE 3.7

Problem. Water flows at 2 m s^{-1} through a pipe of internal diameter 6 cm that narrows to 3 cm. Find the speed there and the volume flow rate.

Solution. Since $A \propto d^2$,

$$v_2 = v_1 \left(\frac{d_1}{d_2} \right)^2 = 2 \left(\frac{6}{3} \right)^2 = 8 \text{ m s}^{-1}$$

$$Q = A_1 v_1 = \pi (0.03)^2 (2) = 5.65 \times 10^{-3} \text{ m}^3 \text{ s}^{-1} = 5.65 \text{ litre s}^{-1}$$

3.8 Bernoulli's equation and its applications

BERNOULLI'S PRINCIPLE

For the steady streamline flow of an ideal (non-viscous, incompressible) fluid, the sum of pressure, kinetic and potential energy per unit volume is constant along a streamline:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

BERNOULLI'S EQUATION FROM THE WORK-ENERGY THEOREM

Take a tube of flow with sections A_1 (height h_1 , pressure P_1 , speed v_1) and A_2 (height h_2 , pressure P_2 , speed v_2). In time Δt a mass $m = \rho V$ enters one end and an equal mass leaves the other, where $V = A_1 v_1 \Delta t = A_2 v_2 \Delta t$. The entry pressure does work $P_1 V$ and work $P_2 V$ is done against the exit pressure, so $W = (P_1 - P_2)V$. By the work-energy theorem this is the gain in kinetic plus potential energy:

$$(P_1 - P_2)V = \left(\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \right) + (mgh_2 - mgh_1)$$

Putting $m = \rho V$ and dividing by V ,

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

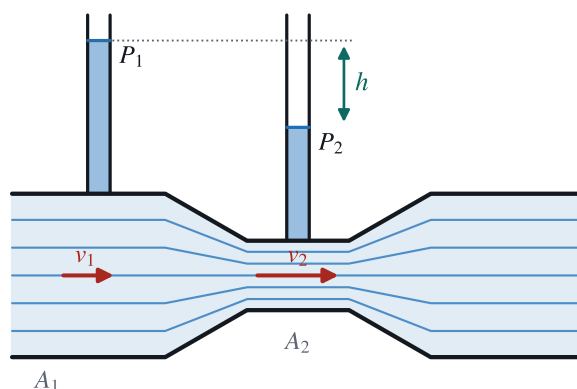


Figure 16. Venturi meter. By continuity the liquid speeds up in the throat; by Bernoulli's equation its pressure there falls, so the manometer column is shorter.

Applications.

1. **Venturi meter.** In a horizontal pipe $h_1 = h_2$, so $P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$; with $A_1v_1 = A_2v_2$ the measured pressure difference gives the rate of flow.
2. **Torricelli's theorem.** For a hole a depth h below the free surface of an open tank the pressure is atmospheric at both points and the surface speed is negligible, so $\frac{1}{2}\rho v^2 = \rho gh$, giving $v = \sqrt{2gh}$ — the speed of free fall through the same height.
3. **Aerofoil lift.** Air travels faster over the curved upper surface of a wing, so the pressure above is lower and there is a net upward force — the same effect lifts a tin roof off in a storm.
4. **Atomiser and Bunsen burner.** Fast air across the top of a tube lowers the pressure there, and atmospheric pressure drives the liquid up.

BERNOULLI'S EQUATION HAS CONDITIONS

It holds only for **steady, streamline, non-viscous, incompressible** flow and only **along one streamline**. Do not use it for turbulent flow, or for a real viscous liquid in a long pipe where energy is lost as heat.

Chapter summary

- $P = h\rho g$ (gauge), $P = P_0 + h\rho g$ (absolute); pressure depends on depth only, and Pascal's law transmits it undiminished.
- Upthrust $U = V\rho g$ = weight of fluid displaced; a floating body has $V_s/V = \rho_b/\rho_l$.
- $T = F/L$ (N m^{-1}) equals the surface energy $W/\Delta A$ (J m^{-2}). Excess pressure: $2T/r$ in a drop, $4T/r$ in a soap bubble.
- Capillary rise $h = 2T\cos\theta/r\rho g$, so $h \propto 1/r$.
- Viscosity $F = \eta A \, dv/dx$, η in Pa s, $[ML^{-1}T^{-1}]$.
- Poiseuille $Q = \pi Pr^4/8\eta l$ — the **fourth power** of the radius.
- Stokes $F = 6\pi\eta r v$, giving $v_t = 2r^2(\rho - \sigma)g/9\eta$.
- Continuity $A_1v_1 = A_2v_2$ (mass) and Bernoulli $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ (energy), giving $v = \sqrt{2gh}$ for efflux.

Practice questions

Group A — Multiple choice (1 mark each)

1. The SI unit of the coefficient of viscosity is [1] (a) poise (b) Pa s (c) N s^{-1} (d) $\text{m}^2 \text{s}^{-1}$
2. A body floats in water with three-quarters of its volume submerged. Its relative density is [1] (a) 0.25 (b) 0.50 (c) 0.75 (d) 1.33
3. Water rises to a height h in a capillary tube of radius r . In a tube of radius $r/2$ it rises to [1] (a) $h/2$ (b) h (c) $2h$ (d) $4h$
4. In Poiseuille's formula Q is proportional to [1] (a) r (b) r^2 (c) r^3 (d) r^4
5. Bernoulli's equation expresses the conservation of [1] (a) mass (b) momentum (c) energy (d) angular momentum

ANSWERS TO GROUP A

1. (b) — from $\eta = F/(A \, dv/dx)$, the unit is $\text{N s m}^{-2} = \text{Pa s}$. 2. (c) — the fraction submerged equals the ratio of densities. 3. (c) — $h \propto 1/r$. 4. (d) — $Q = \pi Pr^4/8\eta l$. 5. (c) — it is the work-energy theorem per unit volume of fluid.

Group B — Short answer (5 marks each)

1. Define surface tension and surface energy and show that they are numerically equal. [5]
2. A capillary tube of internal radius 0.5 mm is dipped vertically in water of surface tension 0.072 N m^{-1} . Calculate the rise. ($\theta = 0^\circ$, $\rho = 1000 \text{ kg m}^{-3}$) [5]
3. State Stokes' law and derive the terminal velocity of a sphere in a viscous liquid. Hence find the terminal velocity of an air bubble of radius 1 mm rising through a liquid of density 1200 kg m^{-3} and viscosity 0.5 Pa s . [5]
4. Water flows at 3 m s^{-1} along a horizontal pipe of cross-section 10 cm^2 , which narrows to 5 cm^2 . Find the speed there and the pressure difference between the two sections. [5]

ANSWERS TO GROUP B

1. Outline: $T = F/L$; surface energy $= W/\Delta A$. For a film on a frame of width l pulled out a distance x , $W = 2Tl$, $x = T(2lx) = T\Delta A$, so $1 \text{ N m}^{-1} = 1 \text{ J m}^{-2}$.
2. $h = \frac{2T \cos \theta}{r \rho g} = \frac{2(0.072)}{(5 \times 10^{-4})(1000)(9.8)} = \frac{0.144}{4.9} = 0.0294 \text{ m} = 2.94 \text{ cm}$.
3. For a rising bubble the upthrust exceeds the weight; taking the air density as negligible, $v_t = \frac{2r^2 \sigma g}{9\eta} = \frac{2(10^{-3})^2(1200)(9.8)}{9(0.5)} = 5.23 \times 10^{-3} \text{ m s}^{-1}$, about 5.2 mm s^{-1} upwards.
4. $v_2 = A_1 v_1 / A_2 = (10 \times 3) / 5 = 6 \text{ m s}^{-1}$; for a horizontal pipe $P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2}(1000)(36 - 9) = 1.35 \times 10^4 \text{ Pa}$.

Group C — Long answer (8 marks each)

1. (a) Define angle of contact and explain why water wets glass but mercury does not. [3] (b) Derive an expression for the rise of a liquid in a capillary tube and calculate the rise of water in a tube of internal radius 0.1 mm. ($T = 0.072 \text{ N m}^{-1}$, $\theta = 0^\circ$, $\rho = 1000 \text{ kg m}^{-3}$) [5]
2. (a) State Bernoulli's principle and derive Bernoulli's equation for the streamline flow of an ideal liquid. [5] (b) A tank on the ground holds water to a depth of 3 m. A small hole is made in its side 1.2 m below the surface. Find the speed of efflux and how far from the foot of the tank the jet strikes the ground. [3]

ANSWERS TO GROUP C

1. (b) $h = \frac{2T \cos \theta}{r \rho g} = \frac{2(0.072)}{(10^{-4})(1000)(9.8)} = \frac{0.144}{0.98} = 0.147 \text{ m} = 14.7 \text{ cm}$.
2. (b) $v = \sqrt{2gh} = \sqrt{2(9.8)(1.2)} = 4.85 \text{ m s}^{-1}$. The hole is 1.8 m above the ground and the jet leaves horizontally, so $t = \sqrt{2(1.8)/9.8} = 0.606 \text{ s}$ and $x = vt = 2.94 \text{ m}$.

UNIT 4

First Law of Thermodynamics

Heat and Thermodynamics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Thermodynamic systems
- Work done during volume change
- Heat and work; Internal energy and First law of thermodynamics
- Thermodynamic processes: Adiabatic, isochoric, isothermal and isobaric
- Heat capacities of an ideal gas at constant pressure and volume and relation between them
- Isothermal and Adiabatic processes for an ideal gas

Thermodynamics is the book-keeping of energy. A gas trapped in a cylinder can be heated, compressed, allowed to expand, or left alone — and in every case the energy that goes in must equal the energy that comes out plus whatever stays behind inside the gas. That single statement, written carefully, is the first law. This unit teaches you to write it correctly, to compute the work from a pressure–volume graph, and to apply it to the four processes that every NEB paper asks about.

WHAT THE EXAM WANTS FROM THIS UNIT

Three things repeat every year: (i) state the first law with correct sign conventions, (ii) derive $C_p - C_v = R$, and (iii) derive $PV^\gamma = \text{constant}$ and the work done in an isothermal or adiabatic change. Everything else is numerical substitution into those results.

4.1 Thermodynamic systems

A **system** is the definite quantity of matter we choose to study — usually a fixed mass of gas in a cylinder. Everything outside it that can exchange energy with it is the **surroundings**. The real or imagined surface separating the two is the **boundary**.

Type of system	Exchanges matter?	Exchanges energy?	Example
Open	yes	yes	water boiling in an open pan
Closed	no	yes	gas sealed in a metal cylinder
Isolated	no	no	hot tea in a perfect thermos flask

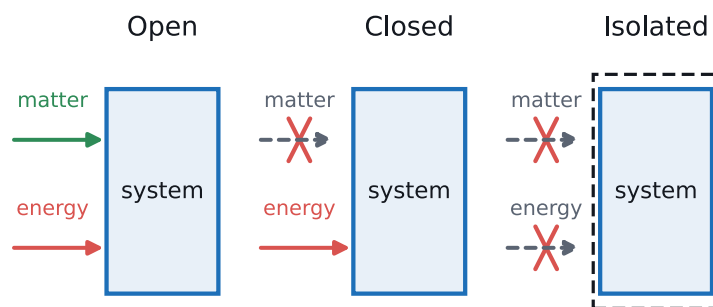


Figure 17. The three kinds of thermodynamic system, classified by what can cross the boundary.

The condition of a system is fixed by its **state variables** P , V , T and the amount of gas n . For an ideal gas these are tied together by the equation of state

$$PV = nRT, \quad R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

so only two of them can be chosen freely. That is why a state can be shown as a single point on a P - V graph, and a process as a curve. Such a graph is called an **indicator diagram**.

A system is in **thermodynamic equilibrium** when it is simultaneously in mechanical equilibrium (no unbalanced pressure), thermal equilibrium (one uniform temperature) and chemical equilibrium (no reaction in progress). Only then do P and T have single definite values, and only then can the process be drawn as a smooth curve. Such an idealised, infinitely slow process is called **quasi-static**.

STATE FUNCTION AND PATH FUNCTION

A **state function** depends only on the present state, not on how the system got there — pressure, volume, temperature and internal energy are state functions. A **path function** depends on the route taken between two states — heat and work are path functions. This is why we write ΔU but never " ΔQ ".

4.2 Work done during volume change

Consider a gas at pressure P inside a cylinder closed by a frictionless piston of area A . The gas pushes the piston with force $F = PA$. If the piston moves out through a small distance dx , so slowly that the pressure stays essentially constant during the move, the work done **by the gas** is

$$dW = F dx = PA dx = P dV$$

since $A dx = dV$ is the increase in volume. For a finite change from V_1 to V_2 we add up all such slices:

$$W = \int_{V_1}^{V_2} P dV$$

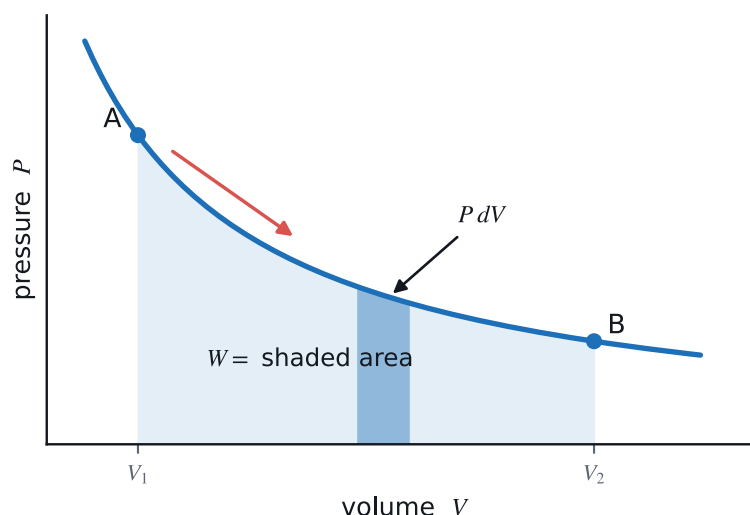


Figure 18. Work done by a gas is the area under the P - V curve. The narrow strip has area $P dV$.

So **the work done by a gas equals the area under its P - V curve**. Two consequences follow immediately:

- If the gas expands, $dV > 0$ and W is **positive** — the gas does work on the surroundings. If it is compressed, W is **negative**.
- Because different curves joining the same two points enclose different areas, the work depends on the path, not merely on the end states.

4.3 Heat and work; Internal energy and First law of thermodynamics

Heat Q is energy that crosses a boundary because of a temperature difference. **Work** W is energy that crosses because a force moves its point of application. Both are energy in transit; neither is stored in the gas.

What is stored is the **internal energy** U — the total kinetic energy of the random molecular motion plus the potential energy of intermolecular forces. For an ideal gas the intermolecular forces are zero, so

$$U = \frac{f}{2}nRT$$

where f is the number of degrees of freedom. The important consequence is that **for an ideal gas U depends only on temperature**: no temperature change means no change in internal energy, whatever happens to P and V .

Energy supplied as heat must either raise the internal energy or leave again as work. That is the first law.

FIRST LAW OF THERMODYNAMICS

The heat supplied to a system is equal to the increase in its internal energy plus the external work done by the system:

$$Q = \Delta U + W, \quad \text{or in differential form} \quad dQ = dU + dW$$

It is the law of conservation of energy applied to a thermodynamic system, and it asserts that internal energy is a state function.

Quantity	Positive when	Negative when
Q	heat is absorbed by the system	heat is rejected by the system
W	the gas expands (does work)	the gas is compressed
ΔU	temperature rises	temperature falls

SIGN CONVENTIONS DECIDE THE MARKS

NEB uses $Q = \Delta U + W$ with W as work done **by** the gas. Some books write $\Delta U = Q + W$ with W as work done **on** the gas. Both are correct physics, but mixing them changes the sign of your answer. Pick the NEB form, write it down first, and stick to it for the whole question.

WORKED EXAMPLE 4.1

Problem. A gas at a constant pressure of 2.0×10^5 Pa expands from $1.0 \times 10^{-3} \text{ m}^3$ to $3.0 \times 10^{-3} \text{ m}^3$ while absorbing 500 J of heat. Find the work done by the gas and the change in its internal energy.

Solution. At constant pressure the area under the P - V line is a rectangle:

$$W = P \Delta V = (2.0 \times 10^5)(3.0 \times 10^{-3} - 1.0 \times 10^{-3}) = 400 \text{ J}$$

From the first law, $\Delta U = Q - W = 500 - 400 = 100 \text{ J}$.

The internal energy rises by 100 J, so the gas also gets warmer.

4.4 Thermodynamic processes: Adiabatic, isochoric, isothermal and isobaric

Four special paths are worth memorising because in each one the first law collapses to a one-term equation.

Process	Held constant	Condition	First law becomes	P - V curve
Isochoric	volume	$dV = 0$	$Q = \Delta U$	vertical line
Isobaric	pressure	$dP = 0$	$Q = \Delta U + P\Delta V$	horizontal line
Isothermal	temperature	$\Delta U = 0$	$Q = W$	rectangular hyperbola
Adiabatic	no heat flow	$Q = 0$	$\Delta U = -W$	steeper hyperbola-like curve

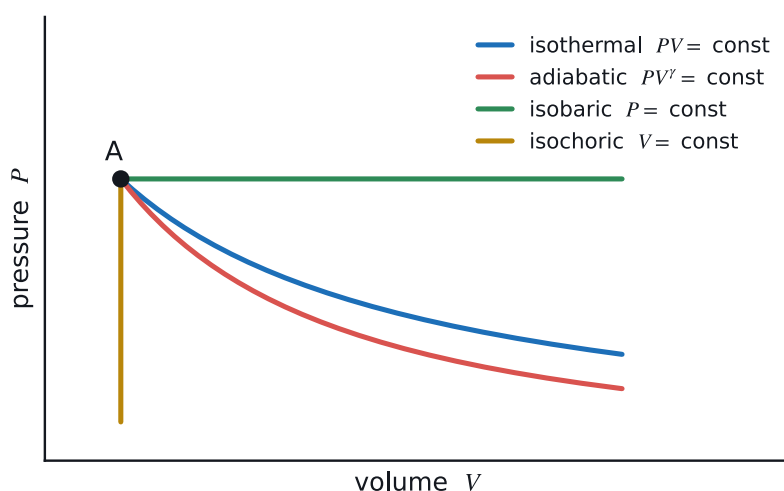


Figure 19. The four standard processes, all starting from the same state A. The adiabatic curve always falls more steeply than the isothermal.

A fifth case matters for engines: in a **cyclic process** the gas returns to its starting state, so $\Delta U = 0$ and $Q = W$. The net work equals the area enclosed by the loop — positive if the loop is traced clockwise, negative if anticlockwise.

Two more named changes appear in NEB questions. A **free expansion** is the expansion of a gas into a vacuum inside an insulated vessel: no heat enters ($Q = 0$) and there is nothing to push against ($W = 0$), so $\Delta U = 0$ and the temperature of an ideal gas is unchanged. A **throttling process** is a similar irreversible expansion through a porous plug.

4.5 Heat capacities of an ideal gas at constant pressure and volume and relation between them

The **molar heat capacity** of a gas is the heat needed to raise the temperature of one mole by one kelvin. A gas has two of them, because the answer depends on whether you let it expand.

piston clamped: V constant piston free: P constant

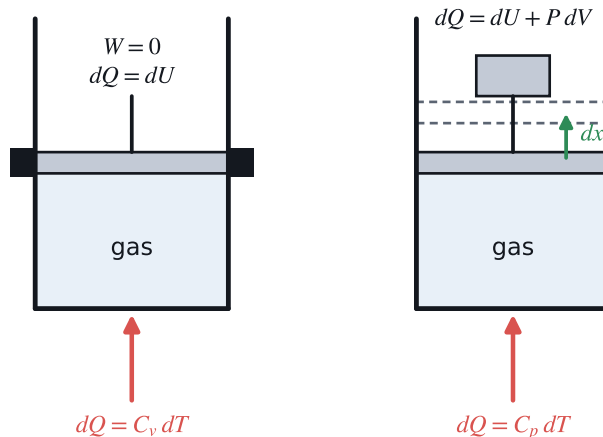


Figure 20. Heating one mole through dT at constant volume (left) and at constant pressure (right). The freely moving piston lets the gas do extra work $P dV$.

- C_v is measured with the piston clamped. No work is done, so all the heat raises the internal energy: $dQ = dU = C_v dT$ for one mole.
- C_p is measured with the piston free to move. The gas must also push the atmosphere back, so more heat is needed for the same rise in temperature.

Hence $C_p > C_v$ always. The exact difference is Mayer's relation.

MAYER'S RELATION, $C_p - C_v = R$

Take one mole of an ideal gas and heat it through dT .

At constant volume. $dV = 0$, so $dW = 0$ and the first law gives

$$dU = C_v dT$$

Because U depends only on T , this expression for dU is true for any process, not only the constant-volume one.

At constant pressure. Now $dQ = C_p dT$ and the gas does work $dW = P dV$, so the first law reads

$$C_p dT = C_v dT + P dV$$

Eliminate $P dV$. For one mole, $PV = RT$. Differentiating at constant P ,

$$P dV = R dT$$

Substituting,

$$C_p dT = C_v dT + R dT \implies C_p - C_v = R$$

In terms of specific heat capacities per kilogram, dividing by the molar mass M gives $c_p - c_v = R/M$.

The ratio $\gamma = C_p/C_v$ is fixed by the number of degrees of freedom f , since $C_v = \frac{f}{2}R$:

Gas	f	C_v (J mol ⁻¹ K ⁻¹)	C_p (J mol ⁻¹ K ⁻¹)	γ
Monatomic (He, Ar)	3	12.47	20.79	1.67
Diatomic (N ₂ , O ₂ , H ₂)	5	20.79	29.10	1.40
Polyatomic (CO ₂ , NH ₃)	6	24.94	33.26	1.33

WORKED EXAMPLE 4.2

Problem. For nitrogen gas $\gamma = 1.40$ and the molar mass is 28 g mol⁻¹. Calculate C_v , C_p and the two specific heat capacities per kilogram. Take $R = 8.314$ J mol⁻¹ K⁻¹.

Solution. From $C_p - C_v = R$ and $C_p = \gamma C_v$,

$$\gamma C_v - C_v = R \Rightarrow C_v = \frac{R}{\gamma - 1} = \frac{8.314}{0.40} = 20.79 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$C_p = \gamma C_v = 1.40 \times 20.79 = 29.10 \text{ J mol}^{-1} \text{ K}^{-1}$$

Dividing by $M = 0.028$ kg mol⁻¹:

$$c_v = \frac{20.79}{0.028} = 742 \text{ J kg}^{-1} \text{ K}^{-1}, \quad c_p = \frac{29.10}{0.028} = 1039 \text{ J kg}^{-1} \text{ K}^{-1}$$

Check: $c_p - c_v = 297 = R/M = 8.314/0.028$. Correct.

4.6 Isothermal and Adiabatic processes for an ideal gas

Isothermal process

The gas is kept at constant temperature, so $PV = \text{constant}$ (Boyle's law). In practice this needs thin, highly conducting walls and a very slow change, so that heat can flow in or out fast enough to hold T fixed. Since $\Delta U = 0$, all the heat absorbed reappears as work.

WORK DONE IN AN ISOTHERMAL CHANGE

For n moles, $P = nRT/V$ with T constant. Then

$$W = \int_{V_1}^{V_2} P \, dV = nRT \int_{V_1}^{V_2} \frac{dV}{V} = nRT [\ln V]_{V_1}^{V_2}$$

$$W = nRT \ln \left(\frac{V_2}{V_1} \right) = 2.303 \, nRT \log_{10} \left(\frac{V_2}{V_1} \right)$$

Since $P_1 V_1 = P_2 V_2$, this can also be written $W = nRT \ln(P_1/P_2)$. Here $Q = W$ and $\Delta U = 0$.

Adiabatic process

No heat enters or leaves: $Q = 0$. This needs thick insulating walls, or a change so fast that heat has no time to flow — the compression stroke of a diesel engine and the propagation of a sound wave are both

effectively adiabatic. The first law gives $\Delta U = -W$: an expanding gas does work at the cost of its own internal energy, and therefore **cools**.

THE ADIABATIC EQUATION $PV^\gamma = \text{CONSTANT}$

For one mole, $dU = C_v dT$ and $dW = P dV$. With $dQ = 0$ the first law gives

$$C_v dT + P dV = 0$$

From $PV = RT$, differentiating gives $P dV + V dP = R dT$, so $dT = (P dV + V dP)/R$. Substituting,

$$\frac{C_v}{R}(P dV + V dP) + P dV = 0$$

Multiply through by R and use $R = C_p - C_v$:

$$C_v V dP + (C_v + C_p - C_v)P dV = 0 \implies C_v V dP + C_p P dV = 0$$

Divide by $C_v PV$ and write $\gamma = C_p/C_v$:

$$\frac{dP}{P} + \gamma \frac{dV}{V} = 0$$

Integrating, $\ln P + \gamma \ln V = \text{constant}$, that is

$$PV^\gamma = \text{constant}$$

Using $PV = nRT$ to eliminate P or V gives the two companion forms

$$TV^{\gamma-1} = \text{constant}, \quad P^{1-\gamma} T^\gamma = \text{constant}$$

WORK DONE IN AN ADIABATIC CHANGE

Let $PV^\gamma = K$, so $P = KV^{-\gamma}$. Then

$$W = \int_{V_1}^{V_2} KV^{-\gamma} dV = K \left[\frac{V^{1-\gamma}}{1-\gamma} \right]_{V_1}^{V_2} = \frac{KV_2^{1-\gamma} - KV_1^{1-\gamma}}{1-\gamma}$$

But $K = P_1 V_1^\gamma = P_2 V_2^\gamma$, so $KV_2^{1-\gamma} = P_2 V_2$ and $KV_1^{1-\gamma} = P_1 V_1$. Hence

$$W = \frac{P_2 V_2 - P_1 V_1}{1-\gamma} = \frac{P_1 V_1 - P_2 V_2}{\gamma-1} = \frac{nR(T_1 - T_2)}{\gamma-1}$$

Differentiating the two curves shows why the adiabatic is steeper. For the isothermal, $P dV + V dP = 0$ gives slope $dP/dV = -P/V$; for the adiabatic, $dP/P = -\gamma dV/V$ gives slope $-\gamma P/V$. Since $\gamma > 1$, the adiabatic slope is γ times steeper at every point.

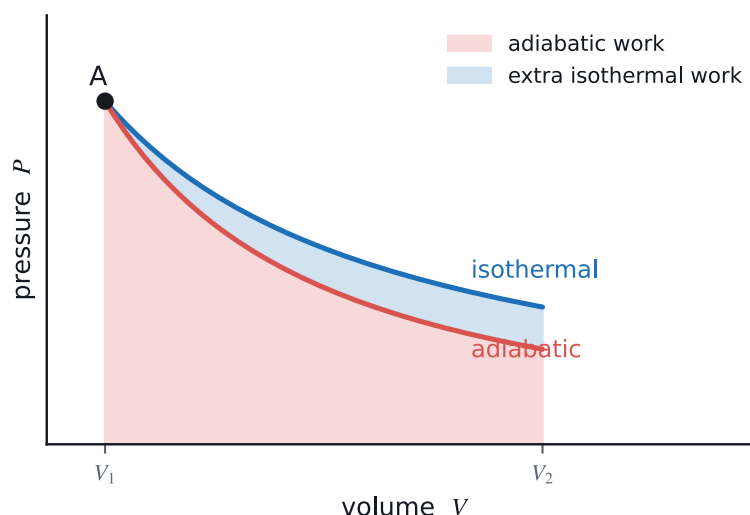


Figure 21. Expansion from A to the same final volume. The adiabatic path ends lower and encloses less area, so less work is done by the gas.

WORKED EXAMPLE 4.3

Problem. Two moles of an ideal gas at 300 K expand isothermally to three times the original volume. Find the work done by the gas and the heat absorbed. ($\ln 3 = 1.0986$.)

Solution.

$$W = nRT \ln \left(\frac{V_2}{V_1} \right) = 2 \times 8.314 \times 300 \times 1.0986$$

$$W = 4988.4 \times 1.0986 = 5.48 \times 10^3 \text{ J}$$

Because the temperature does not change, $\Delta U = 0$, so by the first law

$$Q = \Delta U + W = 0 + 5.48 \text{ kJ} = 5.48 \text{ kJ absorbed}$$

WORKED EXAMPLE 4.4

Problem. One mole of air ($\gamma = 1.40$) at 27 °C is compressed adiabatically to one-eighth of its volume. Find the final temperature and the work done on the gas. ($8^{0.4} = 2.297$.)

Solution. $T_1 = 300 \text{ K}$ and $V_1/V_2 = 8$. Using $TV^{\gamma-1} = \text{constant}$,

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 300 \times 8^{0.4} = 300 \times 2.297 = 689 \text{ K}$$

That is 416 °C — the reason diesel fuel ignites without a spark. The work done **by** the gas is

$$W = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{1 \times 8.314 \times (300 - 689)}{0.40} = -8.09 \times 10^3 \text{ J}$$

The negative sign means 8.09 kJ of work is done **on** the gas; since $Q = 0$, all of it becomes internal energy, $\Delta U = +8.09 \text{ kJ}$.

Chapter summary

- A system exchanges matter and energy with its surroundings across a boundary; open, closed and isolated systems differ in what may cross it. P , V , T and U are state functions; Q and W are path functions.
- Work done by a gas is $W = \int P \, dV$, the area under the P - V curve; positive for expansion, negative for compression.
- First law: $Q = \Delta U + W$, with Q positive for heat absorbed and W positive for expansion. For an ideal gas U depends only on T .
- Isochoric: $W = 0$, $Q = \Delta U$. Isobaric: $W = P\Delta V$. Isothermal: $\Delta U = 0$, $Q = W$. Adiabatic: $Q = 0$, $\Delta U = -W$. Cyclic: $\Delta U = 0$, $Q = W = \text{area of loop}$.
- Mayer's relation: $C_p - C_v = R$, so $C_v = R/(\gamma - 1)$ and $C_p = \gamma R/(\gamma - 1)$. Here $\gamma = 1.67$, 1.40 and 1.33 for mono-, di- and polyatomic gases.
- Isothermal: $PV = \text{constant}$ and $W = nRT \ln(V_2/V_1)$.
- Adiabatic: $PV^\gamma = \text{constant}$, $TV^{\gamma-1} = \text{constant}$, and $W = (P_1 V_1 - P_2 V_2)/(\gamma - 1) = nR(T_1 - T_2)/(\gamma - 1)$.
- At any common point the adiabatic curve is γ times steeper than the isothermal, so adiabatic expansion cools a gas.

Practice questions

Group A — Multiple choice (1 mark each)

1. In an isothermal expansion of an ideal gas, the heat absorbed is [1] (a) zero (b) equal to the work done by the gas (c) equal to the rise in internal energy (d) equal to $C_v \Delta T$
2. For one mole of an ideal gas, $C_p - C_v$ is equal to [1] (a) R (b) γR (c) R/γ (d) zero
3. At the same point on an indicator diagram, the slope of the adiabatic curve compared with that of the isothermal is [1] (a) equal (b) γ times greater in magnitude (c) $1/\gamma$ times (d) zero
4. In a complete cyclic process the change in internal energy of the gas is [1] (a) equal to Q (b) equal to W (c) zero (d) always negative
5. A gas expands freely into an evacuated insulated vessel. Its temperature [1] (a) rises (b) falls (c) stays the same (d) first rises, then falls

ANSWERS TO GROUP A

1. (b) — $\Delta U = 0$ at constant T , so the first law gives $Q = W$. **2.** (a) — Mayer's relation, $C_p - C_v = R$ for one mole. **3.** (b) — slopes are $-P/V$ and $-\gamma P/V$ respectively. **4.** (c) — U is a state function and the gas returns to its initial state. **5.** (c) — $Q = 0$ and $W = 0$, so $\Delta U = 0$; for an ideal gas that means $\Delta T = 0$.

Group B — Short answer (5 marks each)

1. Define internal energy. State the first law of thermodynamics, write it in differential form, and explain the sign convention used for Q and W . [5]
2. Derive the relation $C_p - C_v = R$ for one mole of an ideal gas, and explain physically why C_p must be greater than C_v . [5]

3. A gas expands at a constant pressure of 1.5×10^5 Pa from $2.0 \times 10^{-3} \text{ m}^3$ to $5.0 \times 10^{-3} \text{ m}^3$, absorbing 800 J of heat. Calculate the work done and the change in internal energy. [5]
4. One mole of oxygen at 27°C expands isothermally to four times its volume. Calculate the work done by the gas. ($\ln 4 = 1.386$) [5]

ANSWERS TO GROUP B

1. Internal energy is the total kinetic plus potential energy of the molecules of a system, measured in its own frame; it is a state function. First law: $Q = \Delta U + W$, i.e. $dQ = dU + dW$. Q is positive when heat is absorbed, W is positive when the gas expands, ΔU is positive when the temperature rises.

2. See the derivation in §4.5. Physically, at constant volume all the heat goes into internal energy, whereas at constant pressure the gas must additionally do work $P dV = R dT$ against the surroundings, so more heat is needed for the same dT .

3. $W = P\Delta V = 1.5 \times 10^5 \times (5.0 - 2.0) \times 10^{-3} = 450 \text{ J}$. Then $\Delta U = Q - W = 800 - 450 = 350 \text{ J}$.

4. $W = nRT \ln(V_2/V_1) = 1 \times 8.314 \times 300 \times 1.386 = 3.46 \times 10^3 \text{ J}$. Since $\Delta U = 0$, the gas also absorbs 3.46 kJ of heat.

Group C — Long answer (8 marks each)

1. (a) Show that for a quasi-static change the work done by a gas is $W = \int P dV$, and hence that it equals the area under the indicator diagram. [3] (b) Derive the adiabatic relation $PV^\gamma = \text{constant}$ for an ideal gas, and obtain an expression for the work done during an adiabatic expansion. [5]
2. A cylinder contains 0.5 mol of a diatomic ideal gas ($\gamma = 1.40$) at 300 K and 1.0×10^5 Pa. The gas is (i) heated at constant volume to 450 K, then (ii) expanded isothermally at 450 K until its pressure returns to 1.0×10^5 Pa. Calculate the heat supplied, the work done and the change in internal energy for each stage. ($\ln 1.5 = 0.4055$) [8]

ANSWER TO GROUP C QUESTION 2

Stage (i) — isochoric. $C_v = R/(\gamma - 1) = 8.314/0.40 = 20.79 \text{ J mol}^{-1} \text{K}^{-1}$.

$$W_1 = 0, \quad \Delta U_1 = nC_v \Delta T = 0.5 \times 20.79 \times 150 = 1559 \text{ J}$$

so $Q_1 = \Delta U_1 = 1.56 \times 10^3 \text{ J}$. The pressure rises to $P = 1.0 \times 10^5 \times (450/300) = 1.5 \times 10^5 \text{ Pa}$.

Stage (ii) — isothermal. The pressure falls from 1.5×10^5 to $1.0 \times 10^5 \text{ Pa}$, so $V_2/V_1 = P_1/P_2 = 1.5$.

$$W_2 = nRT \ln 1.5 = 0.5 \times 8.314 \times 450 \times 0.4055 = 758 \text{ J}$$

$\Delta U_2 = 0$ and $Q_2 = W_2 = 758 \text{ J}$.

Totals. $Q = 1559 + 758 = 2.32 \times 10^3 \text{ J}$, $W = 758 \text{ J}$, $\Delta U = 1559 \text{ J}$, which satisfies $Q = \Delta U + W$.

UNIT 5

Second Law of Thermodynamics

Heat and Thermodynamics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Thermodynamic systems and direction of thermodynamic processes
- Second law of thermodynamics
- Heat engines
- Internal combustion engines: Otto cycle, Diesel cycle, Carnot cycle
- Refrigerator
- Entropy and disorder (introduction only)

The first law says energy is conserved, but not which way things go. It would be perfectly happy if the water in a glass froze while the glass grew hot, or if a stone cooled slightly and leapt into the air. Nature refuses. The second law fixes the direction of every real process, and from it follow the limits on every engine and every refrigerator ever built.

WHAT THE EXAM WANTS FROM THIS UNIT

The two statements of the second law (Kelvin–Planck and Clausius), the heat engine formula $\eta = 1 - Q_2/Q_1$, the Carnot cycle with its derivation of $\eta = 1 - T_2/T_1$, and the coefficient of performance of a refrigerator. Numerical questions almost always reduce to those four results.

5.1 Thermodynamic systems and direction of thermodynamic processes

Every process you have watched runs one way only. Heat flows from hot tea to the cold cup, never back. A gas released from a cylinder fills the room and never gathers itself up again. A dropped ball bounces lower each time, but a warm floor never cools slightly and launches a ball upwards. Each reverse process would conserve energy perfectly, so the first law does not forbid it. Something else does.

REVERSIBLE AND IRREVERSIBLE PROCESSES

A **reversible process** is one that can be made to retrace its path exactly, so that both the system and its surroundings return to their initial states, leaving no trace anywhere. Any process that cannot is **irreversible**.

A process is reversible only if it is **quasi-static** (carried out infinitely slowly through a succession of equilibrium states) and free of all **dissipative effects** — friction, viscosity, electrical resistance, or heat flow across a finite temperature difference.

Reversible (idealised)	Irreversible (real)
infinitely slow, always in equilibrium	finite rate, passes through non-equilibrium states
no friction, viscosity or turbulence	dissipative forces always present
Carnot cycle	free expansion; burning fuel; rubbing hands

No real process is reversible. Reversibility is a limiting ideal, useful because it fixes the best performance any real machine could approach.

5.2 Second law of thermodynamics

There are two classical statements. They look quite different but each can be shown to imply the other.

KELVIN-PLANCK STATEMENT (ENGINE STATEMENT)

It is impossible to construct a device that, operating in a cycle, produces **no effect other than** the extraction of heat from a single reservoir and the performance of an equivalent amount of work.

CLAUSIUS STATEMENT (REFRIGERATOR STATEMENT)

It is impossible for any self-acting device, operating in a cycle, to transfer heat from a colder body to a hotter body **without any external work** being done on it.

In plain terms: Kelvin–Planck forbids a 100 % efficient engine, Clausius forbids a refrigerator that needs no power supply. The two are equivalent. Suppose a device violated Clausius and moved Q_2 from sink to source for free. Couple it to an ordinary engine that takes Q_1 from the source and rejects exactly Q_2 . The sink then gains and loses Q_2 each cycle, so it can be removed altogether, leaving an engine that draws $Q_1 - Q_2$ from one reservoir and converts it entirely into work — a violation of Kelvin–Planck.

"HEAT CANNOT BE FULLY CONVERTED INTO WORK" — READ THE SMALL PRINT

In an isothermal expansion of an ideal gas, $\Delta U = 0$, so $Q = W$: all the heat does become work. The second law is not violated because the gas has been left in an expanded state — the conversion was not the **sole** effect, and the process cannot be repeated in a cycle. The word "cycle" in both statements is doing real work; never drop it.

5.3 Heat engines

HEAT ENGINE

A heat engine is a device that converts heat into mechanical work continuously, by taking a working substance repeatedly around a cycle: it absorbs heat Q_1 from a **source** (hot reservoir) at T_1 , converts part of it into work W , and rejects the rest Q_2 to a **sink** (cold reservoir) at T_2 .

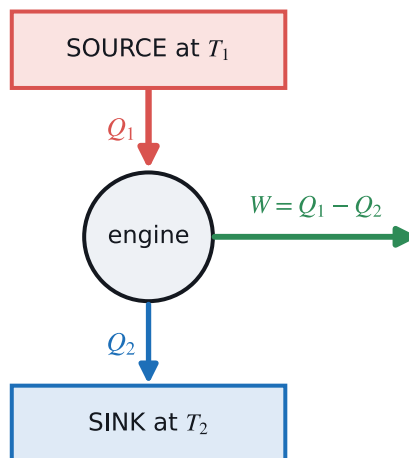


Figure 22. Energy flow in a heat engine. Of the heat Q_1 drawn from the source, only $W = Q_1 - Q_2$ leaves as work; the rest is dumped in the sink.

Because the working substance returns to its initial state every cycle, $\Delta U = 0$, and the first law gives

$$W = Q_1 - Q_2$$

The **thermal efficiency** is the fraction of the heat paid for that comes out as useful work:

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

Efficiency equals 1 only if $Q_2 = 0$, exactly what Kelvin–Planck forbids. **Every heat engine must have a sink**, so η is always below 100 %.

Engines are classified by where the fuel burns: outside the working cylinder in an **external combustion engine** (steam turbine), inside it in an **internal combustion engine** (petrol, diesel), where the burning mixture is itself the working substance.

WORKED EXAMPLE 5.1

Problem. A heat engine absorbs 2000 J of heat per cycle from a source at 500 K and rejects 1400 J to a sink at 300 K. Find the work done per cycle and the efficiency. Is such an engine possible?

Solution. Per cycle, $W = Q_1 - Q_2 = 2000 - 1400 = 600$ J, so

$$\eta = \frac{W}{Q_1} = \frac{600}{2000} = 0.30 = 30 \%$$

The best possible efficiency between these temperatures is that of a Carnot engine,

$$\eta_{\max} = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = 0.40 = 40 \%$$

Since $30 \% < 40 \%$, the engine does not violate the second law and is possible (though not reversible).

5.4 Internal combustion engines: Otto cycle, Diesel cycle, Carnot cycle

The Otto cycle — the ideal petrol engine

The four-stroke petrol engine is modelled by the **Otto cycle** — two adiabats and two isochorics.

Step	Process	What happens
1 → 2	adiabatic compression	piston compresses the fuel-air mixture; T rises
2 → 3	isochoric heat input Q_1	spark plug fires; pressure jumps at constant volume
3 → 4	adiabatic expansion	the power stroke; the gas drives the piston out
4 → 1	isochoric heat rejection Q_2	exhaust valve opens; pressure drops

With the **compression ratio** $r = V_1/V_2$, and $Q = nC_V\Delta T$ on both isochoric steps, the ideal efficiency is

$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}$$

It depends only on r and γ , not on how much fuel is burnt. Petrol engines are limited to $r \approx 8-10$; above that the mixture ignites on its own before the spark (knocking).

The Diesel cycle

A diesel engine compresses **air alone**, so there is no knocking limit and r can be 15–22. The compressed air becomes hot enough (Worked example 4.4) to ignite fuel sprayed in at the top of the stroke, so heat is added at roughly constant **pressure**:

$$\eta_{\text{Diesel}} = 1 - \frac{1}{\gamma} \frac{\rho^\gamma - 1}{\rho - 1}$$

where $\rho = V_3/V_2$ is the cut-off ratio. For the same r the diesel cycle is slightly less efficient than the Otto cycle, but because r can be made much larger the real diesel engine wins comfortably.

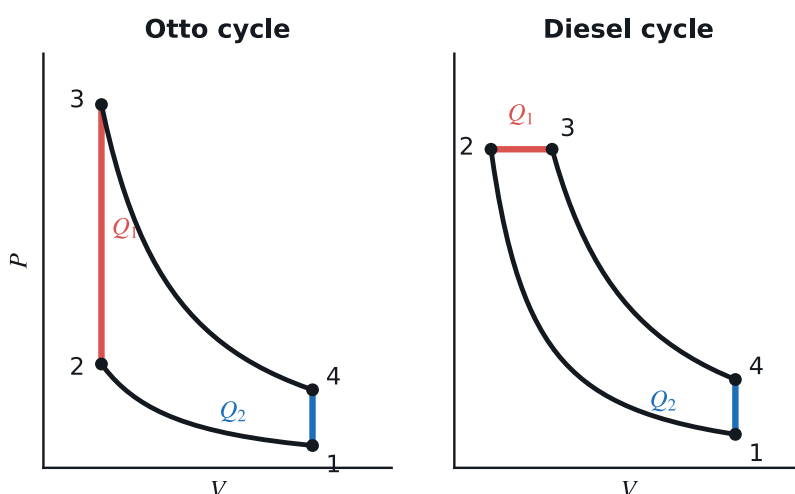


Figure 23. Indicator diagrams of the Otto cycle (heat added at constant volume) and the Diesel cycle (heat added at constant pressure). Both reject heat at constant volume. Drawn schematically: real compression ratios are larger.

The Carnot cycle

Sadi Carnot asked the deepest question: what is the greatest efficiency any engine working between two given temperatures can have? His answer is an imaginary, perfectly reversible engine running the cycle below.

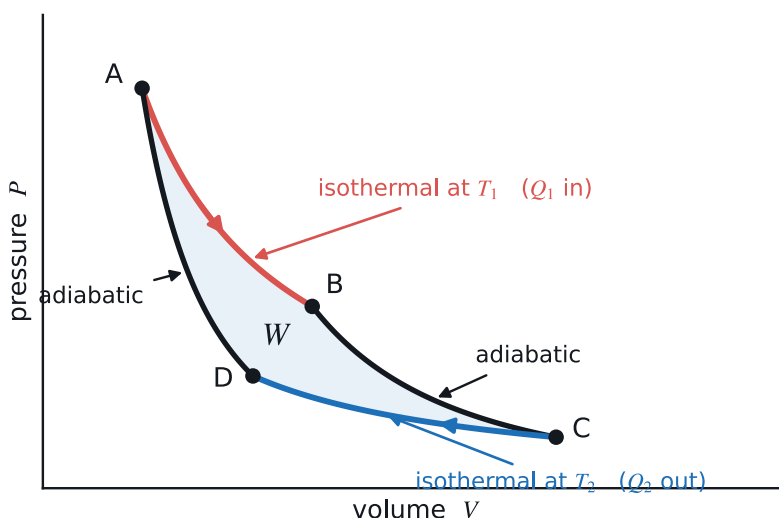


Figure 24. The Carnot cycle: isothermal expansion A to B at T_1 , adiabatic expansion B to C, isothermal compression C to D at T_2 , adiabatic compression D to A. The enclosed area is the net work W .

EFFICIENCY OF THE CARNOT ENGINE

Take n moles of an ideal gas round the cycle.

A → B, isothermal expansion at T_1 . $\Delta U = 0$, so the heat absorbed is

$$Q_1 = W_{AB} = nRT_1 \ln \left(\frac{V_2}{V_1} \right)$$

C → D, isothermal compression at T_2 . Similarly the heat rejected is

$$Q_2 = nRT_2 \ln \left(\frac{V_3}{V_4} \right)$$

The two adiabatics. Using $TV^{\gamma-1} = \text{constant}$ on B → C and on D → A,

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1}, \quad T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$$

Dividing one by the other,

$$\left(\frac{V_2}{V_1} \right)^{\gamma-1} = \left(\frac{V_3}{V_4} \right)^{\gamma-1} \implies \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Result. The two logarithms are therefore equal and cancel:

$$\frac{Q_2}{Q_1} = \frac{T_2}{T_1} \implies \eta = 1 - \frac{T_2}{T_1}$$

Notice that the result contains no property of the working substance — air, steam or helium give the same answer — and that it reaches 1 only if $T_2 = 0$ K, which is unattainable.

CARNOT'S THEOREM

(a) No engine working between two given temperatures can be more efficient than a reversible engine working between the same two temperatures. (b) All reversible engines working between the same two temperatures have the same efficiency, whatever the working substance.

WORKED EXAMPLE 5.2

Problem. A Carnot engine operates between a source at 227°C and a sink at 27°C , absorbing 6000 J per cycle. Find its efficiency, the work done and the heat rejected.

Solution. Convert to kelvin: $T_1 = 227 + 273 = 500$ K, $T_2 = 27 + 273 = 300$ K.

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = 0.40 = 40 \%$$

$$W = \eta Q_1 = 0.40 \times 6000 = 2400 \text{ J}$$

$$Q_2 = Q_1 - W = 6000 - 2400 = 3600 \text{ J}$$

Check: $Q_2/Q_1 = 3600/6000 = 0.60 = T_2/T_1$. Consistent.

WORKED EXAMPLE 5.3

Problem. A petrol engine working on the Otto cycle has a compression ratio of 8. Taking $\gamma = 1.40$, find its ideal efficiency. What would it become if the compression ratio were raised to 10? ($8^{0.4} = 2.297$, $10^{0.4} = 2.512$.)

Solution.

$$\eta = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{8^{0.4}} = 1 - \frac{1}{2.297} = 1 - 0.435 = 0.565 = 56.5 \%$$

For $r = 10$,

$$\eta = 1 - \frac{1}{2.512} = 1 - 0.398 = 0.602 = 60.2 \%$$

Real petrol engines reach only 25–30 %, because of friction, heat loss and incomplete combustion.

5.5 Refrigerator

A refrigerator is a heat engine run backwards. A compressor supplies work W ; the machine extracts heat Q_2 from the cold interior and dumps $Q_1 = Q_2 + W$ into the kitchen through the coils at the back. The Clausius statement is exactly the assertion that W cannot be zero.

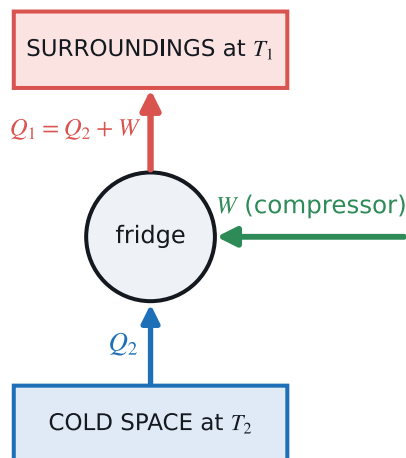


Figure 25. Energy flow in a refrigerator. Work W is supplied so that heat Q_2 can be pumped from the cold interior into the warmer surroundings.

The useful output is the heat removed from the cold space and the cost is the work supplied, so the performance figure is

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

called the **coefficient of performance** (COP). It is not an efficiency, because it is usually greater than 1: a good refrigerator moves four or five joules of heat per joule of electrical work. For a reversible (Carnot) refrigerator $Q_2/Q_1 = T_2/T_1$, so

$$\beta_{\max} = \frac{T_2}{T_1 - T_2}$$

Comparing with the engine formula gives $\beta = (1 - \eta)/\eta$. Note that β falls as the temperature difference grows: a deep freezer costs far more to run than a chiller.

In a domestic refrigerator the working substance is a volatile refrigerant. The **compressor** compresses its vapour, which condenses in the **condenser** coils at the back, giving out Q_1 to the kitchen; the liquid then passes through a **throttle valve**, where it expands and cools sharply, and evaporates in the **evaporator** coils inside the cabinet, absorbing Q_2 from the food. A **heat pump** is the same machine used for its hot end, with $\text{COP} = Q_1/W$.

WORKED EXAMPLE 5.4

Problem. A refrigerator maintains its freezer at -13°C in a kitchen at 27°C . Assuming ideal (reversible) operation, find its coefficient of performance, the work needed to remove 1300 J of heat, and the heat delivered to the kitchen.

Solution. $T_2 = -13 + 273 = 260\text{ K}$, $T_1 = 27 + 273 = 300\text{ K}$.

$$\beta = \frac{T_2}{T_1 - T_2} = \frac{260}{300 - 260} = \frac{260}{40} = 6.5$$

$$W = \frac{Q_2}{\beta} = \frac{1300}{6.5} = 200\text{ J}$$

$$Q_1 = Q_2 + W = 1300 + 200 = 1500 \text{ J}$$

So leaving the fridge door open warms the room by 200 J for every 1300 J moved, rather than cooling it.

5.6 Entropy and disorder (introduction only)

Clausius found a state function that turns the second law into an equation rather than a prohibition. For a reversible transfer of heat dQ at absolute temperature T , the change in **entropy** is

$$dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = \int \frac{dQ_{\text{rev}}}{T}$$

Its SI unit is J K^{-1} . Entropy is a **state function**: round any cycle it returns to its starting value. Check this on the Carnot cycle — the gas gains Q_1/T_1 and loses Q_2/T_2 , and these are equal since $Q_2/Q_1 = T_2/T_1$.

For a change at constant temperature (melting, boiling) the integral is trivial:

$$\Delta S = \frac{Q}{T} = \frac{mL}{T}$$

THE SECOND LAW AS AN INEQUALITY

In any process, the total entropy of the system together with its surroundings never decreases:

$$\Delta S_{\text{universe}} \geq 0$$

The equality holds only for a reversible process. Every real process increases the entropy of the universe.

Entropy measures **disorder** — more exactly, the number of microscopic arrangements of the molecules that look the same from outside. Ice has its molecules locked in a lattice (low entropy), water has them tumbling freely (higher), steam higher still. Heat flowing from hot to cold raises the total entropy, which is why it happens; the reverse would lower it, which is why it does not.

WORKED EXAMPLE 5.5

Problem. One kilogram of ice at 0°C melts completely into water at 0°C . Find the change in entropy of the ice. Take the latent heat of fusion as $3.34 \times 10^5 \text{ J kg}^{-1}$.

Solution. Melting happens at a constant temperature of $T = 0 + 273 = 273 \text{ K}$, and the heat absorbed is

$$Q = mL = 1 \times 3.34 \times 10^5 = 3.34 \times 10^5 \text{ J}$$

$$\Delta S = \frac{Q}{T} = \frac{3.34 \times 10^5}{273} = 1.22 \times 10^3 \text{ J K}^{-1}$$

The entropy increases, as it must: liquid water is more disordered than ice.

Chapter summary

- Real processes run one way only. A reversible process is quasi-static and free of all dissipative effects; no real process is reversible.
- Kelvin–Planck: no cyclic device can convert heat from a single reservoir entirely into work. Clausius: no self-acting cyclic device can move heat from a colder to a hotter body. The two statements are equivalent.
- Heat engine: $W = Q_1 - Q_2$ and $\eta = W/Q_1 = 1 - Q_2/Q_1$. A sink is compulsory, so $\eta < 1$ always.
- Otto cycle (petrol): two adiabatics and two isochorics, $\eta = 1 - r^{-(\gamma-1)}$. Diesel cycle: heat added isobarically, allowing a much larger r .
- Carnot cycle: two isothermals and two adiabatics, all reversible, with $\eta = 1 - T_2/T_1$ — the maximum possible for the given temperatures, and independent of the working substance (Carnot's theorem).
- Refrigerator: $Q_1 = Q_2 + W$, $\beta = Q_2/W = Q_2/(Q_1 - Q_2)$, and for the ideal case $\beta_{\max} = T_2/(T_1 - T_2) = (1 - \eta)/\eta$.
- Entropy $dS = dQ_{\text{rev}}/T$ is a state function measured in J K^{-1} , and for an isothermal change $\Delta S = Q/T$. The second law states $\Delta S_{\text{universe}} \geq 0$; entropy is a measure of disorder.

Practice questions

Group A — Multiple choice (1 mark each)

1. The efficiency of a Carnot engine working between 27°C and 327°C is [1] (a) 50 % (b) 25 % (c) 75 % (d) 92 %
2. The Clausius statement of the second law forbids [1] (a) an engine of 100 % efficiency (b) a self-acting device that moves heat from a cold body to a hot body (c) the conversion of work into heat (d) adiabatic expansion
3. In the ideal Otto cycle, heat is supplied at constant [1] (a) pressure (b) volume (c) temperature (d) entropy
4. The maximum coefficient of performance of a refrigerator working between 250 K and 300 K is [1] (a) 5 (b) 6 (c) 0.2 (d) 1.2
5. In every real (irreversible) process, the entropy of the universe [1] (a) decreases (b) remains constant (c) increases (d) becomes zero

ANSWERS TO GROUP A

- 1.** (a) — $T_1 = 600\text{ K}$, $T_2 = 300\text{ K}$, so $\eta = 1 - 300/600 = 0.5$. **2.** (b) — the Clausius prohibition; (a) is the Kelvin–Planck one. **3.** (b) — the spark fires while the piston is at the top, so the volume is fixed. **4.** (a) — $\beta_{\max} = T_2/(T_1 - T_2) = 250/50 = 5$. **5.** (c) — $\Delta S_{\text{universe}} > 0$ for irreversible changes.

Group B — Short answer (5 marks each)

1. State the Kelvin–Planck and Clausius statements of the second law, and show that a violation of one implies a violation of the other. [5]
2. What is a heat engine? Draw its energy-flow diagram, derive $\eta = 1 - Q_2/Q_1$, and explain why η can never be 100 %. [5]
3. A Carnot engine has an efficiency of 40 % when its sink is at 27°C . Find the temperature of the source. To what temperature must the source be raised to obtain an efficiency of 50 %, the sink being unchanged? [5]

4. Define the coefficient of performance of a refrigerator. A refrigerator with a COP of 5 extracts 600 J of heat per cycle from the cold chamber. Find the work done per cycle and the heat rejected to the surroundings. [5]

ANSWERS TO GROUP B

1. Statements and the equivalence argument as in §5.2.
2. $\Delta U = 0$ over a cycle gives $W = Q_1 - Q_2$, hence $\eta = W/Q_1 = 1 - Q_2/Q_1$. Here $\eta = 1$ needs $Q_2 = 0$, an engine with no sink, which Kelvin-Planck forbids.
3. $\eta = 1 - T_2/T_1$ gives $0.40 = 1 - 300/T_1$, so $T_1 = 300/0.60 = 500 \text{ K}$ (227°C). For $\eta = 0.50$, $T_1 = 300/0.50 = 600 \text{ K}$ (327°C) — a rise of 100 K.
4. COP is the ratio of the heat extracted from the cold body to the work supplied, $\beta = Q_2/W$. Here $W = Q_2/\beta = 600/5 = 120 \text{ J}$ and $Q_1 = Q_2 + W = 600 + 120 = 720 \text{ J}$.

Group C — Long answer (8 marks each)

1. Describe the Carnot cycle with the help of an indicator diagram, naming the four processes, and derive the expression $\eta = 1 - T_2/T_1$ for its efficiency. State Carnot's theorem. [8]
2. A petrol engine working on the ideal Otto cycle has a compression ratio of 9 and uses a gas with $\gamma = 1.40$. ($9^{0.4} = 2.408$) (a) Calculate its ideal efficiency. [3] (b) If it absorbs 2000 J of heat per cycle, find the work output and the heat rejected. [3] (c) A Carnot engine works between the same extreme temperatures, 1500 K and 300 K. Compare the two efficiencies and comment. [2]

ANSWER TO GROUP C QUESTION 2

- (a) $\eta = 1 - 1/r^{\gamma-1} = 1 - 1/9^{0.4} = 1 - 1/2.408 = 1 - 0.415 = 0.585$, that is 58.5 %.
- (b) $W = \eta Q_1 = 0.585 \times 2000 = 1.17 \times 10^3 \text{ J}$, and $Q_2 = Q_1 - W = 2000 - 1170 = 830 \text{ J}$.
- (c) $\eta_{\text{Carnot}} = 1 - 300/1500 = 0.80 = 80 \%$. The Otto cycle falls well short because it absorbs and rejects heat over a range of temperatures rather than at the two extremes.

UNIT 6

Wave motion

Waves and Optics • 2 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Progressive waves
- Mathematical description of a wave
- Stationary waves

Drop a stone in the Phewa lake and rings spread outwards, but a leaf floating on the surface only bobs up and down — it does not travel with the ring. That is the whole idea of a wave: the disturbance moves, carrying energy with it, while the matter stays put. This short unit sets up the language and the equation of a travelling wave, and then shows what happens when two such waves meet head-on. Everything in Units 7 to 9 is built on it.

WHAT THE EXAM WANTS FROM THIS UNIT

Be able to write the wave equation $y = a \sin(\omega t - kx)$ and read off a , f , λ and v from any given form of it; derive the stationary-wave equation $y = 2a \cos kx \sin \omega t$; and list the differences between progressive and stationary waves.

6.1 Progressive waves

PROGRESSIVE (TRAVELLING) WAVE

A progressive wave is a disturbance that travels continuously through a medium in a definite direction, transferring energy and momentum from one point to the next, **without any net transport of the medium itself**.

A mechanical wave needs a material medium with two properties: elasticity, to provide a restoring force, and inertia, to make the displaced particle overshoot. Light and radio waves are electromagnetic and need no medium; they are not treated in this unit.

Waves are classified by the direction in which the particles vibrate.

	Transverse wave	Longitudinal wave
Particle vibration	perpendicular to propagation	parallel to propagation
Form of the wave	crests and troughs	compressions and rarefactions
Travels through	solids and liquid surfaces	solids, liquids and gases
Can be polarised?	yes	no
Examples	wave on a string, ripple, light	sound in air, wave in a spring pushed lengthwise

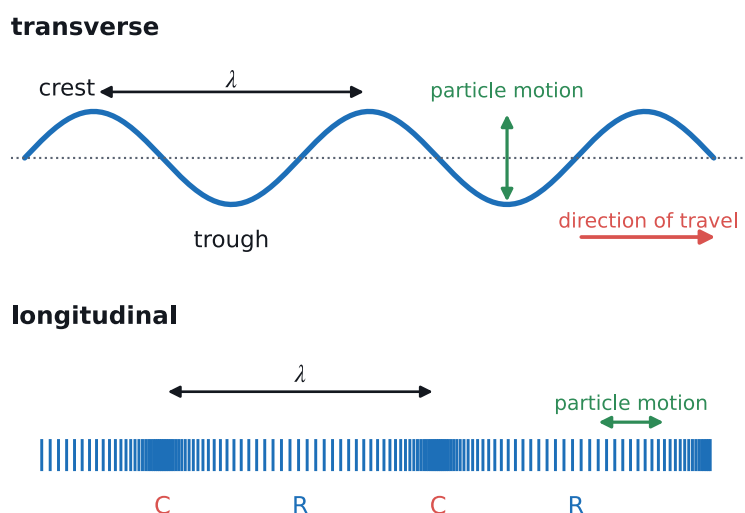


Figure 26. A transverse wave has crests and troughs; a longitudinal wave has compressions (C) and rarefactions (R). In both, one wavelength λ is the distance between successive identical points.

Five quantities describe any progressive wave:

- **Amplitude** a — the maximum displacement of a particle from its mean position.
- **Wavelength** λ — the distance between two successive particles in the same phase (crest to crest, or compression to compression).
- **Period** T — the time for one complete vibration of a particle.
- **Frequency** $f = 1/T$ — vibrations per second, in hertz.
- **Wave velocity** v — the speed at which the disturbance advances.

In one period the wave advances exactly one wavelength, so

$$v = \frac{\lambda}{T} = f\lambda$$

This **wave equation** holds for every wave, mechanical or electromagnetic. In a given medium v is fixed by the medium's elasticity and density, so changing the frequency of the source changes the wavelength, not the speed.

Two further facts are worth remembering: in a progressive wave **every** particle vibrates with the same amplitude and frequency, and each particle starts its vibration slightly later than the one before it — the phase lags steadily behind along the direction of travel.

6.2 Mathematical description of a wave

Let the particle at the origin perform SHM, $y = a \sin \omega t$. A wave moving in the $+x$ direction with speed v reaches a particle at distance x after a delay $t' = x/v$, so that particle simply repeats what the origin did x/v seconds earlier:

$$y = a \sin \omega \left(t - \frac{x}{v} \right)$$

THE STANDARD FORMS OF THE WAVE EQUATION

Putting $\omega = 2\pi/T$ and $v = \lambda/T$ into the expression above,

$$y = a \sin \frac{2\pi}{T} \left(t - \frac{x}{v} \right) = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

Defining the **angular frequency** $\omega = 2\pi/T$ and the **propagation constant** (or wave number) $k = 2\pi/\lambda$, this becomes

$$y = a \sin(\omega t - kx), \quad v = \frac{\omega}{k}$$

An equivalent form often quoted in NEB papers is

$$y = a \sin \frac{2\pi}{\lambda} (vt - x)$$

For a wave travelling in the $-x$ direction, replace x by $-x$: $y = a \sin(\omega t + kx)$.

The quantity $(\omega t - kx)$ is the **phase**. Two particles separated by a path difference Δx differ in phase by

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x$$

so a separation of one whole wavelength means a phase difference of 2π (the particles are in step), and a separation of $\lambda/2$ means π (exactly out of step).

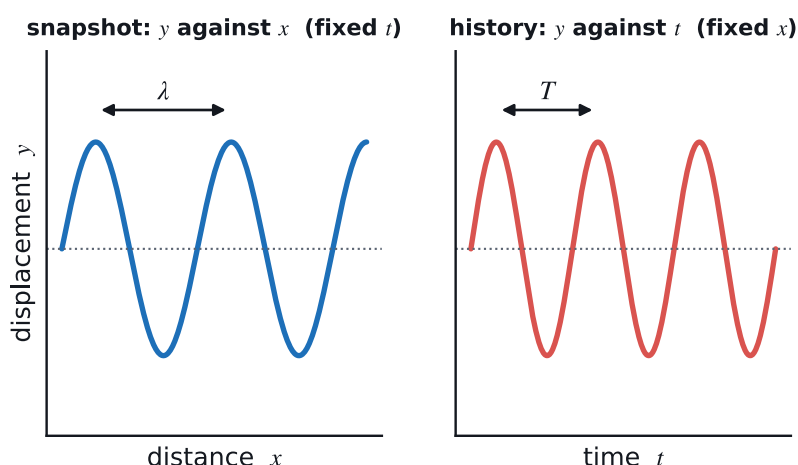


Figure 27. Left: a snapshot at one instant, where the repeat distance is the wavelength λ . Right: the motion of one particle in time, whose repeat interval is the period T .

Differentiating y with respect to time at a fixed x gives the **particle velocity**:

$$v_p = \frac{\partial y}{\partial t} = a\omega \cos(\omega t - kx), \quad (v_p)_{\max} = a\omega$$

PARTICLE VELOCITY IS NOT WAVE VELOCITY

The wave velocity $v = f\lambda$ is a constant property of the medium. The particle velocity $v_p = a\omega \cos(\omega t - kx)$ changes from instant to instant, is zero at the crests and troughs and greatest at the mean position. Differentiating y with respect to x instead gives the slope of the curve, and the two are linked by $v_p = -v (\partial y / \partial x)$.

WORKED EXAMPLE 6.1

Problem. A progressive wave is given by $y = 0.05 \sin(100\pi t - 0.5\pi x)$, all quantities in SI units. Find the amplitude, frequency, wavelength, wave speed and maximum particle velocity.

Solution. Comparing with $y = a \sin(\omega t - kx)$: $a = 0.05 \text{ m}$, $\omega = 100\pi \text{ rad s}^{-1}$, $k = 0.5\pi \text{ rad m}^{-1}$.

$$f = \frac{\omega}{2\pi} = \frac{100\pi}{2\pi} = 50 \text{ Hz}, \quad \lambda = \frac{2\pi}{k} = \frac{2\pi}{0.5\pi} = 4 \text{ m}$$

$$v = \frac{\omega}{k} = \frac{100\pi}{0.5\pi} = 200 \text{ m s}^{-1}$$

Check: $f\lambda = 50 \times 4 = 200 \text{ m s}^{-1}$. Finally

$$(v_p)_{\max} = a\omega = 0.05 \times 100\pi = 15.7 \text{ m s}^{-1}$$

WORKED EXAMPLE 6.2

Problem. A sound wave of frequency 500 Hz travels through air at 350 m s^{-1} . Find its wavelength and the phase difference between two points 0.175 m apart along the direction of travel.

Solution.

$$\lambda = \frac{v}{f} = \frac{350}{500} = 0.7 \text{ m}$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{0.7} \times 0.175 = 2\pi \times 0.25 = \frac{\pi}{2} \text{ rad}$$

That is 90° — the two points are a quarter of a wavelength apart, so one is at a crest when the other is at its mean position.

6.3 Stationary waves

When two progressive waves of the **same amplitude, frequency and speed** travel along the same line in **opposite directions**, they superpose to give a wave that does not travel at all. This is a **stationary** (or standing) wave. In practice the second wave is usually the reflection of the first from a fixed end — a plucked sarangi string, or the air column in a flute.

EQUATION OF A STATIONARY WAVE

Let the incident wave travelling in the $+x$ direction and the reflected wave travelling in the $-x$ direction be

$$y_1 = a \sin(\omega t - kx), \quad y_2 = a \sin(\omega t + kx)$$

By the principle of superposition the resultant displacement is $y = y_1 + y_2$. Using $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$ with $C = \omega t + kx$ and $D = \omega t - kx$,

$$y = 2a \cos kx \sin \omega t = \left[2a \cos \frac{2\pi x}{\lambda} \right] \sin \frac{2\pi t}{T}$$

This is **not** of the form $f(vt - x)$: the x and t dependences have separated, so nothing travels. Each particle performs SHM of period T , but with an amplitude

$$A = 2a \cos \frac{2\pi x}{\lambda}$$

that depends on where the particle is.

The amplitude A vanishes at some points and is maximum at others.

Nodes are points of permanently zero displacement, where $\cos(2\pi x/\lambda) = 0$:

$$\frac{2\pi x}{\lambda} = (2n + 1)\frac{\pi}{2} \implies x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$

Antinodes are points of maximum amplitude $2a$, where $\cos(2\pi x/\lambda) = \pm 1$:

$$x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots$$

So successive nodes are $\lambda/2$ apart, successive antinodes are $\lambda/2$ apart, and a node and the next antinode are $\lambda/4$ apart. Measuring the distance between two nodes is therefore the standard laboratory way of finding a wavelength.

If the origin is chosen at a node instead of an antinode, the same superposition gives $y = 2a \sin kx \cos \omega t$. Only the labelling has changed: the nodes are then at $x = 0, \lambda/2, \lambda, \dots$ and the antinodes midway between them. The spacings $\lambda/2$ and $\lambda/4$ are the same either way.

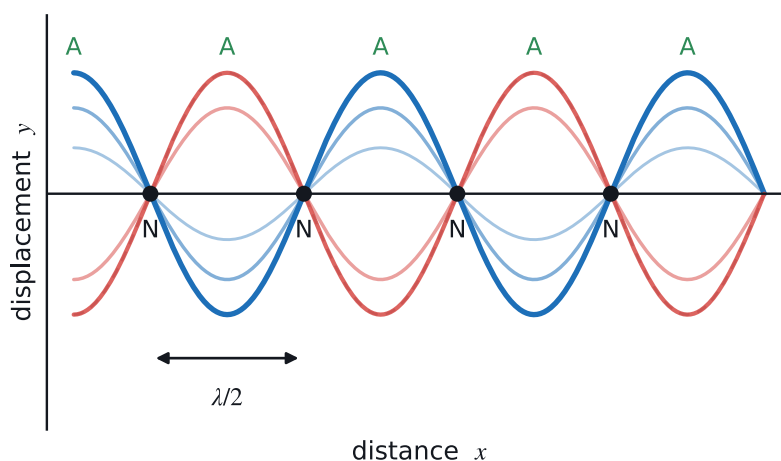


Figure 28. A stationary wave at several instants. The nodes N never move; the antinodes A swing between $+2a$ and $-2a$. Successive nodes are $\lambda/2$ apart.

	Progressive wave	Stationary wave
Disturbance	advances through the medium	confined between the boundaries
Energy	transported along the wave	not transported; only exchanged locally
Amplitude	same for every particle	varies from 0 at nodes to $2a$ at antinodes
Phase	changes continuously with x	same for all particles between two nodes; reverses across a node
Nodes and anti-nodes	none	present, spaced $\lambda/2$ apart
Particle at rest	no particle is permanently at rest	particles at the nodes never move

WORKED EXAMPLE 6.3

Problem. A stationary wave on a string is described by $y = 4\sin(0.5\pi x)\cos(200\pi t)$, where y and x are in centimetres and t in seconds. Find (a) the amplitude of the two component waves, (b) the wavelength, (c) the frequency, (d) the wave speed, and (e) the distance between successive nodes.

Solution. Compare with $y = 2a\sin kx\cos \omega t$.

(a) $2a = 4$ cm, so each component wave has amplitude $a = 2$ cm.

(b) $k = 0.5\pi$ cm $^{-1}$, so $\lambda = 2\pi/k = 2\pi/0.5\pi = 4$ cm.

(c) $\omega = 200\pi$ rad s $^{-1}$, so $f = \omega/2\pi = 100$ Hz.

(d) $v = f\lambda = 100 \times 0.04 = 4$ m s $^{-1}$.

(e) Successive nodes are $\lambda/2 = 2$ cm apart; here $\sin(0.5\pi x) = 0$ gives $x = 0, 2, 4, 6$ cm, confirming the spacing.

Chapter summary

- A progressive wave carries energy and momentum through a medium without carrying the medium. Transverse waves vibrate perpendicular to the direction of travel, longitudinal waves parallel to it.
- $v = f\lambda = \lambda/T$ for every wave; in a given medium v is fixed, so raising f shortens λ .
- Wave equation: $y = a\sin(\omega t - kx) = a\sin 2\pi(t/T - x/\lambda)$, with $\omega = 2\pi/T$, $k = 2\pi/\lambda$ and $v = \omega/k$. Use $+kx$ for a wave travelling in the $-x$ direction.
- Phase difference and path difference are linked by $\Delta\phi = (2\pi/\lambda)\Delta x$.
- Particle velocity $v_p = a\omega\cos(\omega t - kx)$, maximum $a\omega$; it is not the wave velocity.
- Two identical waves travelling in opposite directions superpose to give a stationary wave, $y = 2a\cos kx \sin \omega t$, whose amplitude $2a\cos(2\pi x/\lambda)$ depends on position.
- Nodes (zero amplitude) lie at $x = \lambda/4, 3\lambda/4, \dots$ and antinodes (amplitude $2a$) at $x = 0, \lambda/2, \lambda, \dots$. Successive nodes are $\lambda/2$ apart and a node is $\lambda/4$ from the nearest antinode.

Practice questions

Group A — Multiple choice (1 mark each)

1. The distance between two consecutive nodes of a stationary wave is [1] (a) λ (b) $\lambda/2$ (c) $\lambda/4$ (d) 2λ
2. For the wave $y = a\sin(\omega t - kx)$, the speed of the wave is [1] (a) ωk (b) ω/k (c) k/ω (d) $a\omega$
3. Two points in a progressive wave are $\lambda/4$ apart. Their phase difference is [1] (a) $\pi/4$ (b) $\pi/2$ (c) π (d) 2π
4. Sound waves travelling through air are [1] (a) transverse (b) longitudinal (c) electromagnetic (d) always stationary
5. In a stationary wave, the net energy transported across a node is [1] (a) maximum (b) zero (c) equal to $2a$ (d) proportional to f

ANSWERS TO GROUP A

- 1.** (b) — nodes occur at $x = \lambda/4, 3\lambda/4, \dots$, spaced $\lambda/2$ apart. **2.** (b) — $v = \omega/k = (2\pi/T)(\lambda/2\pi) = \lambda/T$. **3.** (b) — $\Delta\phi = (2\pi/\lambda)(\lambda/4) = \pi/2$. **4.** (b) — air has no shear elasticity, so it can carry only longitudinal waves. **5.** (b) — a node never moves, so no energy crosses it; energy is only exchanged between neighbouring nodes.

Group B — Short answer (5 marks each)

1. What is a progressive wave? Derive the expression $y = a\sin \frac{2\pi}{\lambda}(vt - x)$ for a wave travelling along the positive x direction. [5]
2. Distinguish between progressive and stationary waves, giving any five points of difference. [5]
3. A transverse wave is represented by $y = 0.03\sin(40\pi t - 0.2\pi x)$ in SI units. Find its amplitude, frequency, wavelength, speed and maximum particle velocity. [5]
4. A stationary wave is given by $y = 5\sin(0.4\pi x)\cos(100\pi t)$, with y and x in centimetres. Find the wavelength, frequency, wave speed and the distance between a node and the next antinode. [5]

ANSWERS TO GROUP B

1. See §6.2: the particle at x repeats the motion of the origin after a delay x/v , so $y = a \sin \omega(t - x/v)$; substituting $\omega = 2\pi/T$ and $v = \lambda/T$ gives the required form.
2. Any five rows of the comparison table in §6.3.
3. $a = 0.03 \text{ m}$; $\omega = 40\pi$, so $f = 20 \text{ Hz}$; $k = 0.2\pi$, so $\lambda = 2\pi/0.2\pi = 10 \text{ m}$; $v = \omega/k = 40\pi/0.2\pi = 200 \text{ m s}^{-1}$ (check: $f\lambda = 20 \times 10$); $(v_p)_{\max} = a\omega = 0.03 \times 40\pi = 3.77 \text{ m s}^{-1}$.
4. $k = 0.4\pi \text{ cm}^{-1}$, so $\lambda = 2\pi/0.4\pi = 5 \text{ cm}$; $\omega = 100\pi$, so $f = 50 \text{ Hz}$; $v = f\lambda = 50 \times 0.05 = 2.5 \text{ m s}^{-1}$; node to next antinode $= \lambda/4 = 1.25 \text{ cm}$.

Group C — Long answer (8 marks each)

1. Explain how a stationary wave is formed. Derive its equation by superposing two identical progressive waves travelling in opposite directions, and hence obtain the positions of the nodes and antinodes. Show that successive nodes are half a wavelength apart. [8]
2. A progressive wave of frequency 250 Hz and amplitude 2 cm travels along a stretched string at 400 m s^{-1} . (a) Find its wavelength and period. [2] (b) Find the phase difference and the time lag between two points 0.4 m apart. [3] (c) Write the equation of the wave, and find the maximum particle velocity. [3]

ANSWER TO GROUP C QUESTION 2

(a) $\lambda = v/f = 400/250 = 1.6 \text{ m}$ and $T = 1/f = 4 \times 10^{-3} \text{ s} = 4 \text{ ms}$.

(b) $\Delta\phi = (2\pi/\lambda)\Delta x = (2\pi/1.6)(0.4) = \pi/2 \text{ rad} = 90^\circ$. The time lag is $\Delta t = \Delta x/v = 0.4/400 = 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$, which is indeed $T/4$.

(c) $\omega = 2\pi f = 500\pi \text{ rad s}^{-1}$ and $k = 2\pi/\lambda = 2\pi/1.6 = 1.25\pi \text{ rad m}^{-1}$, so

$$y = 0.02 \sin(500\pi t - 1.25\pi x) \text{ m}$$

$(v_p)_{\max} = a\omega = 0.02 \times 500\pi = 10\pi = 31.4 \text{ m s}^{-1}$.

UNIT 7

Mechanical waves

Waves and Optics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Speed of wave motion; Velocity of sound in solid and liquid
- Velocity of sound in gas
- Laplace's correction
- Effect of temperature, pressure, humidity on velocity of sound

A mechanical wave carries energy through matter without carrying the matter along with it. Sound is the mechanical wave you meet every day, and this short unit answers one question about it: **what fixes its speed?** The answer turns out to depend on only two properties of the medium — how stiff it is and how heavy it is — plus, for a gas, one subtlety about heat that took physics a century to get right.

WHAT THE EXAMINER WANTS FROM THIS UNIT

Four things, over and over: the general formula $v = \sqrt{E/\rho}$ applied to a named medium; the derivation of **Newton's formula**; **Laplace's correction** and why it was needed; and a numerical using $v_t = v_0 \sqrt{T/273}$. Learn those four and the unit is finished.

7.1 Speed of wave motion; velocity of sound in solid and liquid

Every wave obeys the kinematic relation

$$v = f\lambda$$

but this only relates speed, frequency and wavelength — it does not say what the speed is. The speed is decided by the medium, not by the source. If you sound a horn louder or at a higher pitch, the sound still arrives at the same time.

A mechanical wave travels because a disturbed part of the medium pulls its neighbour back (elasticity) and because that neighbour resists being moved (inertia). Newton showed that these two combine as

$$v = \sqrt{\frac{E}{\rho}}$$

where ρ is the density of the medium and E is the **modulus of elasticity appropriate to the kind of deformation the wave produces**. Choosing the right modulus is the whole skill.

Medium and wave	Deformation	Modulus used	Speed
Long thin solid rod, longitudinal	stretch/squeeze along rod	Young's modulus Y	$v = \sqrt{Y/\rho}$
Liquid, longitudinal	change of volume	Bulk modulus B	$v = \sqrt{B/\rho}$
Gas, longitudinal	change of volume	Bulk modulus of gas	$v = \sqrt{B/\rho}$
Stretched string, transverse	change of shape	tension T , mass per length μ	$v = \sqrt{T/\mu}$

NEWTON'S GENERAL FORMULA FOR WAVE SPEED

The speed of a mechanical wave in a medium is the square root of the ratio of the relevant elastic modulus of the medium to its density: $v = \sqrt{E/\rho}$.

A liquid and a gas have no rigidity, so they cannot be sheared: **sound in a fluid is always longitudinal**. A solid resists both squeezing and shearing, so a solid carries both longitudinal and transverse waves — which is exactly why an earthquake such as Gorkha 2015 produces two body waves, the fast longitudinal P-wave and the slower transverse S-wave, and why seismologists at the National Seismological Centre can locate an epicentre from the gap between their arrivals.

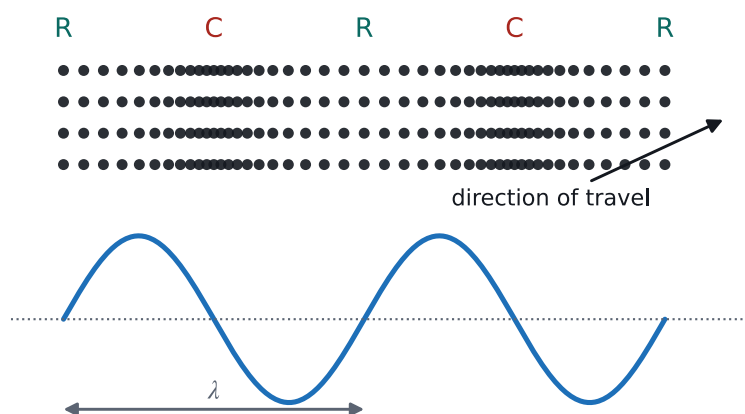


Figure 29. A longitudinal wave. The dotted rows show particles crowded at compressions (C) and thinned at rarefactions (R). The blue curve is the particle displacement y measured along the direction of travel.

Notice from the table that steel is about **eight times denser than water**, yet sound goes more than three times faster in steel. Density alone never decides the answer: Young's modulus of steel is roughly a hundred times the bulk modulus of water, and stiffness wins.

WORKED EXAMPLE 7.1

Problem. Find the speed of sound (a) in a steel rail, for which $Y = 2.0 \times 10^{11}$ Pa and $\rho = 7800$ kg m⁻³, and (b) in water, for which $B = 2.2 \times 10^9$ Pa and $\rho = 1000$ kg m⁻³.

Solution.

(a) A rail is a long thin solid, so use Young's modulus:

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2.0 \times 10^{11}}{7800}} = \sqrt{2.56 \times 10^7} = 5.06 \times 10^3 \text{ m s}^{-1}$$

(b) Water is a liquid, so use the bulk modulus:

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{2.2 \times 10^9}{1000}} = \sqrt{2.2 \times 10^6} = 1.48 \times 10^3 \text{ m s}^{-1}$$

So sound travels about **5060 m s⁻¹** in steel and **1480 m s⁻¹** in water, against roughly 340 m s⁻¹ in air. Put your ear to a rail and you hear the train long before the air carries the sound to you.

7.2 Velocity of sound in gas

For a gas the deformation is a change of volume, so the modulus needed is the bulk modulus

$$B = -\frac{\Delta P}{\Delta V/V}$$

The minus sign is there because an increase in pressure produces a decrease in volume, and it makes B positive.

Newton (1686) now had to decide **how** the gas is compressed. He argued that the compressions and rarefactions are slow enough, and the gas thin enough, for the heat generated in a compression to leak away at once, so the temperature of the gas stays constant. That is an **isothermal** change.

NEWTON'S FORMULA FOR THE VELOCITY OF SOUND IN A GAS

For an isothermal change of an ideal gas, Boyle's law gives

$$PV = \text{constant}$$

Differentiating the product,

$$P dV + V dP = 0 \implies -\frac{dP}{dV/V} = P$$

The left-hand side is precisely the bulk modulus, so the **isothermal bulk modulus is** $B_{iso} = P$. Substituting in $v = \sqrt{B/\rho}$:

$$v = \sqrt{\frac{P}{\rho}}$$

This is **Newton's formula**.

Test it at normal temperature and pressure, where $P = 1.013 \times 10^5 \text{ Pa}$ and $\rho = 1.293 \text{ kg m}^{-3}$ for air:

$$v = \sqrt{\frac{1.013 \times 10^5}{1.293}} = \sqrt{7.835 \times 10^4} = 280 \text{ m s}^{-1}$$

The measured value at 0 °C is about **332 m s⁻¹**. Newton's answer is too small by 52 m s⁻¹ — an error of about **16 %**. No experiment of that era was that bad, so the assumption, not the measurement, had to be wrong.

7.3 Laplace's correction

In 1816 Laplace located the faulty assumption. Sound of even the lowest audible frequency, 20 Hz, compresses a given layer of air 20 times a second; at 1000 Hz the compression lasts about half a millisecond. Air is also a very poor conductor of heat. In that time practically **no heat leaves a compression or enters a rarefaction**. The changes are therefore not isothermal but **adiabatic**.

LAPLACE'S CORRECTION

For an adiabatic change of an ideal gas,

$$PV^\gamma = \text{constant}, \quad \gamma = \frac{C_p}{C_v}$$

Differentiating the product,

$$\gamma PV^{\gamma-1} dV + V^\gamma dP = 0$$

Dividing throughout by $V^{\gamma-1}$ and rearranging,

$$-\frac{dP}{dV/V} = \gamma P \implies B_{adi} = \gamma P$$

The adiabatic bulk modulus is γ times the isothermal one. Hence the corrected speed is

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

For air $\gamma = 1.4$, so the corrected value at NTP is

$$v = \sqrt{1.4} \times 280 = 1.183 \times 280 = 331 \text{ m s}^{-1}$$

which agrees with experiment. The graph below shows why γ appears: an adiabatic curve through a given state is **steeper** than the isothermal curve through the same state, by exactly the factor γ , so the gas is harder to squeeze and the wave travels faster.

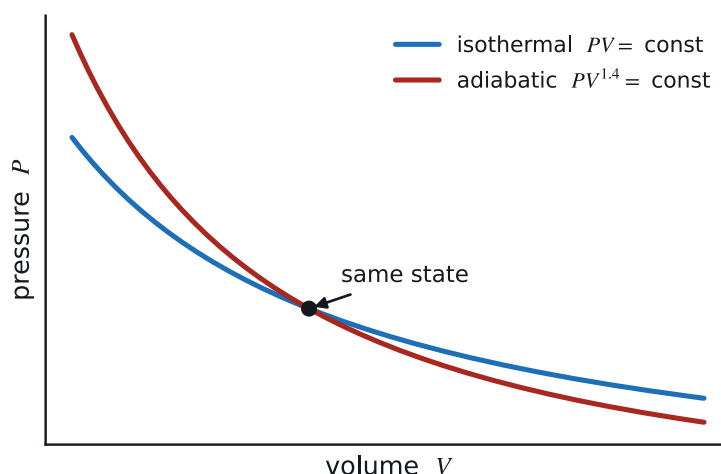


Figure 30. Isothermal ($PV = \text{const}$) and adiabatic ($PV^{1.4} = \text{const}$) curves through the same state. The adiabatic curve is steeper, so $B_{adi} = \gamma P > B_{iso} = P$.

	Newton	Laplace
Assumed process	isothermal	adiabatic
Reason given	heat escapes instantly	air is a bad conductor, changes are fast
Bulk modulus	$B = P$	$B = \gamma P$
Speed formula	$v = \sqrt{P/\rho}$	$v = \sqrt{\gamma P/\rho}$
Value for air at 0 °C	280 m s ⁻¹	331 m s ⁻¹
Agreement with experiment	16 % too low	agrees

LAPLACE DID NOT CHANGE THE FORMULA, ONLY THE MODULUS

A very common slip is to write "Laplace's correction is $v = \sqrt{\gamma P/\rho}$ because sound is a longitudinal wave". The longitudinal nature has nothing to do with it. The correction is only that the compressions are adiabatic, so the bulk modulus is γP and not P . Say that sentence in the exam and the mark is yours.

WORKED EXAMPLE 7.2

Problem. Calculate the velocity of sound in hydrogen at 0 °C, given $\gamma = 1.4$, $P = 1.013 \times 10^5$ Pa and density of hydrogen $= 0.09$ kg m⁻³. Compare it with the value in air.

Solution.

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{1.4 \times 1.013 \times 10^5}{0.09}} = \sqrt{1.576 \times 10^6} = 1.26 \times 10^3 \text{ m s}^{-1}$$

Since γ is the same for both diatomic gases, the ratio of speeds is

$$\frac{v_H}{v_{air}} = \sqrt{\frac{\rho_{air}}{\rho_H}} = \sqrt{\frac{1.293}{0.09}} = 3.79$$

Sound travels about **3.8 times faster in hydrogen** than in air, because hydrogen is far lighter while its stiffness γP is the same.

7.4 Effect of temperature, pressure, humidity on velocity of sound

To see which quantities really matter, eliminate ρ using the ideal gas equation. For n moles of gas of molar mass M in volume V , $\rho = nM/V$ and $PV = nRT$, so $P/\rho = RT/M$ and

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

This single result settles every case, because it contains **no pressure at all**.

Pressure — no effect

At a fixed temperature, doubling the pressure of a gas doubles its density too (Boyle's law), so the ratio P/ρ is unchanged and the speed is unchanged. This is why sound does not travel measurably faster at sea level in Biratnagar than high in Kathmandu valley for the same temperature — any difference you actually hear comes from the temperature difference, not the pressure.

Temperature — the one that matters

From $v = \sqrt{\gamma RT/M}$, at constant composition

$$v \propto \sqrt{T} \quad (T \text{ in kelvin})$$

So if v_0 is the speed at 0 °C (273 K) and v_t the speed at t °C,

$$\frac{v_t}{v_0} = \sqrt{\frac{273+t}{273}} \implies v_t = v_0 \sqrt{1 + \frac{t}{273}}$$

For small t , using the binomial approximation $(1+x)^{1/2} \approx 1 + x/2$,

$$v_t \approx v_0 \left(1 + \frac{t}{546}\right) = v_0 + \frac{v_0}{546}t$$

With $v_0 = 332 \text{ m s}^{-1}$ this gives $v_0/546 = 0.61$, so

$$v_t \approx 332 + 0.61 t \text{ m s}^{-1}$$

The velocity of sound in air increases by about 0.61 m s^{-1} for each 1°C rise in temperature. The graph shows how good the straight-line rule is over ordinary Nepali temperatures.

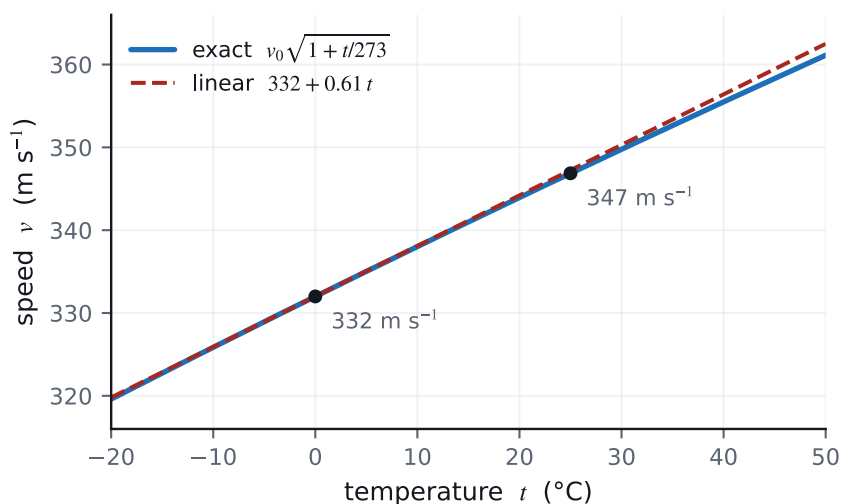


Figure 31. Speed of sound in air against temperature. The exact law $v = v_0 \sqrt{1 + t/273}$ and the rule of thumb $v \approx 332 + 0.61t$ differ by less than 1.5 m s^{-1} up to 50°C .

Humidity — a small increase

Water vapour has molar mass 18 g mol^{-1} while dry air averages about 28.8 g mol^{-1} . When water vapour is added to air it replaces some of the heavier nitrogen and oxygen molecules, so at the same pressure and temperature **moist air is less dense than dry air**. Since $v \propto 1/\sqrt{\rho}$, the speed rises. The effect is

small — a few metres per second between bone-dry and saturated air — but it is real, and it is one reason distant sounds seem to carry unusually well in the humid air just before a monsoon shower.

Factor changed	Effect on v	Reason
Pressure (at constant T)	none	P/ρ fixed by Boyle's law
Temperature	increases, $v \propto \sqrt{T}$	$v = \sqrt{\gamma RT/M}$
Humidity	increases slightly	moist air is less dense than dry air
Wind	adds vectorially	the medium itself is moving
Amplitude (loudness)	none	speed is a property of the medium
Frequency (pitch)	none	air is non-dispersive for sound

Wind does not change the speed relative to the air; it moves the air itself. If a wind of speed w blows at an angle θ to the direction in which the sound travels, the speed relative to the ground is

$$v_{\text{eff}} = v + w \cos \theta$$

Sound therefore carries further downwind than upwind.

WORKED EXAMPLE 7.3

Problem. The velocity of sound in air at 0 °C is 332 m s⁻¹. Find its value at 27 °C (a) exactly, and (b) using the linear rule.

Solution.

(a) $T = 273 + 27 = 300$ K, so

$$v_{27} = v_0 \sqrt{\frac{300}{273}} = 332 \times \sqrt{1.0989} = 332 \times 1.0483 = 348 \text{ m s}^{-1}$$

(b) $v_{27} \approx 332 + 0.61 \times 27 = 332 + 16.5 = 348.5 \text{ m s}^{-1}$.

The two agree to within 0.5 m s⁻¹, so the linear rule is perfectly safe for everyday temperatures.

WORKED EXAMPLE 7.4

Problem. A trekker claps her hands facing a cliff across a valley in Godavari and hears the echo 3.0 s later. The air temperature is 15 °C and the speed of sound at 0 °C is 332 m s⁻¹. How far away is the cliff?

Solution. First correct the speed for temperature:

$$v = 332 + 0.61 \times 15 = 332 + 9.2 = 341.2 \text{ m s}^{-1}$$

The sound travels to the cliff **and back**, so the path length is $2d$:

$$2d = vt = 341.2 \times 3.0 = 1023.6 \text{ m} \implies d = 512 \text{ m}$$

The cliff is about **512 m** away. Using the uncorrected 332 m s^{-1} would have given 498 m — an error of 14 m, which is why the temperature correction is worth doing.

Chapter summary

- Every wave obeys $v = f\lambda$, but the speed itself is set by the medium: $v = \sqrt{E/\rho}$, elasticity over inertia.
- Solid rod: $v = \sqrt{Y/\rho}$ ($\approx 5060 \text{ m s}^{-1}$ in steel). Liquid: $v = \sqrt{B/\rho}$ ($\approx 1480 \text{ m s}^{-1}$ in water). Fluids carry only longitudinal sound; solids carry longitudinal and transverse waves.
- **Newton's formula:** assuming isothermal compressions, $B_{iso} = P$ and $v = \sqrt{P/\rho}$, giving 280 m s^{-1} for air — 16 % too low.
- **Laplace's correction:** the compressions are adiabatic, so $B_{adi} = \gamma P$ and $v = \sqrt{\gamma P/\rho} = 331 \text{ m s}^{-1}$, which matches experiment.
- Eliminating density gives $v = \sqrt{\gamma RT/M}$: the speed depends on temperature and on the gas, never on the pressure.
- $v_t = v_0 \sqrt{1 + t/273} \approx v_0 + 0.61t$ for air: **+0.61 m s⁻¹ per °C**.
- Humidity raises v slightly (moist air is lighter); wind adds vectorially, $v_{eff} = v + w \cos \theta$; amplitude and frequency have no effect.

Practice questions

Group A — Multiple choice (1 mark each)

1. The velocity of sound in a gas is independent of [1] (a) temperature (b) pressure (c) density (d) nature of the gas
2. Laplace's correction to Newton's formula multiplies the velocity by [1] (a) γ (b) $\sqrt{\gamma}$ (c) $1/\gamma$ (d) γ^2
3. Sound travels fastest in [1] (a) vacuum (b) air (c) water (d) steel
4. If the absolute temperature of a gas is made four times, the velocity of sound in it becomes [1] (a) half (b) twice (c) four times (d) unchanged
5. The velocity of sound in moist air is [1] (a) greater than in dry air (b) less than in dry air (c) equal to that in dry air (d) zero

ANSWERS TO GROUP A

1. (b) — $v = \sqrt{\gamma RT/M}$ contains no P ; at constant T , P/ρ is fixed. **2.** (b) — the modulus changes from P to γP , so v changes by $\sqrt{\gamma}$. **3.** (d) — Young's modulus of steel is enormous compared with the moduli of fluids. **4.** (b) — $v \propto \sqrt{T}$, and $\sqrt{4} = 2$. **5.** (a) — water vapour ($M = 18$) is lighter than air ($M = 28.8$), so ρ falls and v rises.

Group B — Short answer (5 marks each)

1. Derive Newton's formula for the velocity of sound in a gas and state the assumption on which it rests. Why does it fail? [5]

- State and explain Laplace's correction. Show that it raises the calculated velocity in air at NTP from about 280 m s^{-1} to about 331 m s^{-1} . (Take $P = 1.013 \times 10^5 \text{ Pa}$, $\rho = 1.293 \text{ kg m}^{-3}$, $\gamma = 1.4$.) [5]
- Show that the velocity of sound in air increases by about 0.61 m s^{-1} for a rise of 1°C , given that it is 332 m s^{-1} at 0°C . [5]
- The velocity of sound in a certain liquid is 1500 m s^{-1} and its density is 1200 kg m^{-3} . Find the bulk modulus of the liquid. [5]
- Explain why sound travels faster in solids than in liquids, and faster in liquids than in gases, even though solids are the densest of the three. [5]

ANSWERS TO GROUP B

- Isothermal assumption, $PV = \text{const}$, $P dV + V dP = 0$, hence $B = P$ and $v = \sqrt{P/\rho}$. It gives 280 m s^{-1} against a measured 332 m s^{-1} — a 16 % error — because sound compressions are far too rapid, and air far too poor a conductor, for the heat to escape.
- Adiabatic: $PV^\gamma = \text{const}$ gives $B = \gamma P$, so $v = \sqrt{\gamma P/\rho}$. Numerically $\sqrt{1.013 \times 10^5 / 1.293} = 280 \text{ m s}^{-1}$, and $280 \times \sqrt{1.4} = 280 \times 1.183 = 331 \text{ m s}^{-1}$.
- $v_t = v_0(1 + t/273)^{1/2} \approx v_0(1 + t/546)$, so $v_t - v_0 = v_0 t/546 = (332/546)t = 0.61t$. For $t = 1^\circ \text{C}$ the increase is 0.61 m s^{-1} .
- $v = \sqrt{B/\rho} \Rightarrow B = \rho v^2 = 1200 \times (1500)^2 = 1200 \times 2.25 \times 10^6 = 2.7 \times 10^9 \text{ Pa}$.
- Speed is $\sqrt{E/\rho}$. Going gas $\rightarrow$ liquid $\rightarrow$ solid the density rises by a factor of roughly 10^3 and then 8, but the elastic modulus rises by a far larger factor (about 10^4 from air to water and about 10^2 again from water to steel). Stiffness therefore wins over inertia.

Group C — Long answer (8 marks each)

- (a) Starting from $v = \sqrt{E/\rho}$, derive Newton's formula for the speed of sound in a gas and explain why it disagrees with experiment. [4] (b) State Laplace's correction, derive the corrected formula, and hence show that $v = \sqrt{\gamma RT/M}$. Use this to explain the effect of pressure, temperature and humidity on the speed of sound. [4]
- (a) A gun is fired in a valley at Pokhara and the echo from a cliff is heard after 2.5 s on a day when the temperature is 20°C . Taking the speed of sound at 0°C as 332 m s^{-1} , find the distance of the cliff. [4] (b) The same gun is fired when the temperature has fallen to 5°C . By how much does the echo time change? [4]

ANSWER TO GROUP C QUESTION 2

(a) $v_{20} = 332 + 0.61 \times 20 = 332 + 12.2 = 344.2 \text{ m s}^{-1}$. Path length $= 2d = 344.2 \times 2.5 = 860.5 \text{ m}$, so $d = 430 \text{ m}$.

(b) $v_5 = 332 + 0.61 \times 5 = 335.05 \text{ m s}^{-1}$. The cliff has not moved, so $2d = 860.5 \text{ m}$ still and

$$t = \frac{860.5}{335.05} = 2.57 \text{ s}$$

The echo now takes about 0.07 s longer — a small but measurable delay, and a reminder that any echo or resonance-tube measurement of v must record the air temperature.

UNIT 8

Wave in pipes and strings

Waves and Optics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Stationary waves in closed and open pipes
- Harmonics and overtones in closed and open organ pipes
- End correction in pipes
- Velocity of transverse waves along a stretched string
- Vibration of string and overtones
- Laws of vibration of fixed string

A flute made from Kathmandu bamboo, a sarangi string, a bottle you blow across — all of them make a musical note the same way. A wave is trapped between two boundaries, reflects back and forth, and only those wavelengths that fit the boundaries survive. This unit works out exactly which wavelengths fit, for air columns and for stretched strings, and turns each answer into a frequency you can calculate.

THE ONE QUESTION EVERY PROBLEM ASKS

Which whole number of quarter- or half-wavelengths fits between the two ends? Fix the node/antinode at each end from the physics, count the loops, read off λ , and use $f = v/\lambda$. Everything in this unit is that one step.

8.1 Stationary waves in closed and open pipes

When a progressive wave reaches the end of a pipe it is reflected. The reflected wave and the incident wave, of equal frequency and amplitude but opposite direction, superpose to give a **stationary (standing) wave** with fixed **nodes** (zero displacement) and **antinodes** (maximum displacement). Adjacent nodes are $\lambda/2$ apart; a node and the next antinode are $\lambda/4$ apart.

What decides where the nodes are is the **boundary condition** at each end:

End of pipe	Can the air move?	Can the pressure change?	Displacement	Pressure
Closed (rigid)	no	yes	node	antinode
Open (to atmosphere)	yes	no	antinode	node

At a closed end the rigid surface stops the air moving, so the displacement must be zero — a displacement node. At an open end the air is in contact with the atmosphere, whose pressure it cannot change, so the pressure variation is zero — a pressure node, which is the same place as a displacement antinode.

DISPLACEMENT NODE = PRESSURE ANTINODE

The two descriptions are always a quarter wavelength out of step. A closed end is a displacement node but a pressure antinode. If a question says "pressure antinode at the closed end" it is saying exactly the same thing as "displacement node". Decide which quantity you are drawing before you draw.

A **closed pipe** (closed organ pipe) is closed at one end and open at the other. An **open pipe** (open organ pipe) is open at both ends. A flute or bansuri behaves as an open pipe; a stopped organ pipe and a test-tube behave as closed pipes.

8.2 Harmonics and overtones in closed and open organ pipes

Closed organ pipe

The closed end must be a node and the open end an antinode, so the shortest column that fits is a quarter of a wavelength.

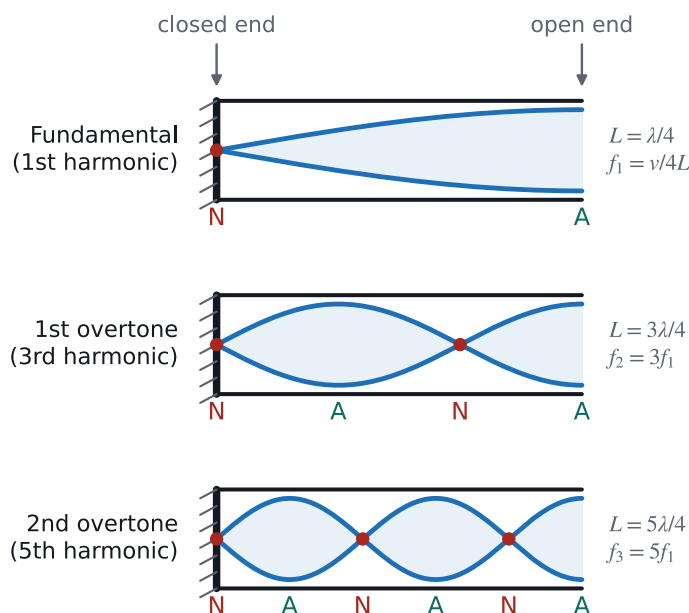


Figure 32. The first three modes of a closed organ pipe of length L . The closed end is always a displacement node (N), the open end an antinode (A). Only odd harmonics are possible.

Reading the modes off the diagram, the possible lengths are $L = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$, i.e.

$$L = (2n - 1)\frac{\lambda_n}{4} \implies \lambda_n = \frac{4L}{2n - 1}$$

and since $f = v/\lambda$,

$$f_n = (2n - 1)\frac{v}{4L} = (2n - 1)f_1, \quad f_1 = \frac{v}{4L}$$

So a closed pipe produces **only the odd harmonics** $f_1 : 3f_1 : 5f_1 : \dots$. The even harmonics are missing.

Open organ pipe

Both ends are antinodes, so the shortest column is half a wavelength.

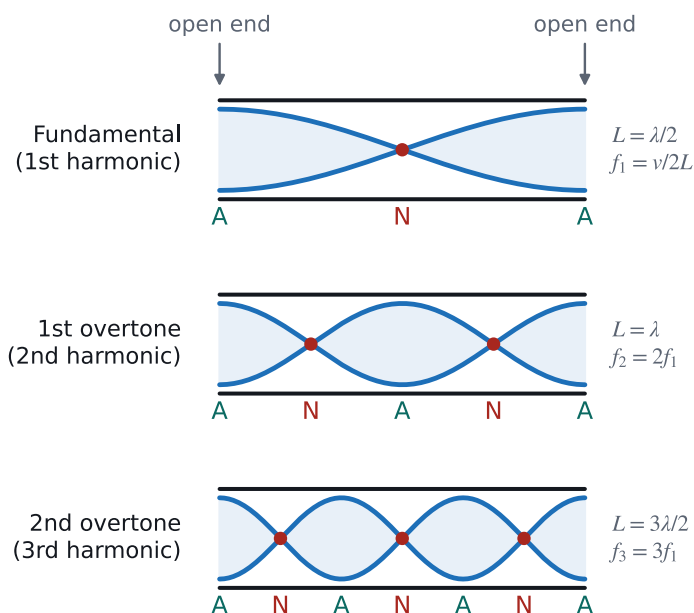


Figure 33. The first three modes of an open organ pipe. Both ends are antinodes (A), so all harmonics — odd and even — are present.

Here $L = n\lambda_n/2$, so $\lambda_n = 2L/n$ and

$$f_n = n\frac{v}{2L} = nf_1, \quad f_1 = \frac{v}{2L}$$

	Closed pipe	Open pipe
Fundamental	$f_1 = v/4L$	$f_1 = v/2L$
Harmonics present	odd only: 1, 3, 5, ...	all: 1, 2, 3, ...
1st overtone	3rd harmonic, $3f_1$	2nd harmonic, $2f_1$
2nd overtone	5th harmonic, $5f_1$	3rd harmonic, $3f_1$
Quality of note	poorer (few harmonics)	richer (many harmonics)
Same length L	note one octave lower	note one octave higher

HARMONIC OR OVERTONE?

A **harmonic** is any whole-number multiple of the fundamental. An **overtone** is any tone actually produced above the fundamental, counted in order. So for a closed pipe the 1st overtone is the 3rd harmonic — the 2nd harmonic does not exist. Confusing the two words loses marks in almost every paper.

WORKED EXAMPLE 8.1

Problem. A closed organ pipe is 25 cm long. Taking the speed of sound as 340 m s^{-1} , find its fundamental frequency and the frequencies of its first two overtones. What would the fundamental be if the same pipe were open at both ends?

Solution. Closed pipe, $L = 0.25 \text{ m}$:

$$f_1 = \frac{v}{4L} = \frac{340}{4 \times 0.25} = \frac{340}{1.00} = 340 \text{ Hz}$$

Only odd harmonics occur, so the first overtone is the 3rd harmonic and the second overtone is the 5th harmonic:

$$f_2 = 3f_1 = 1020 \text{ Hz}, \quad f_3 = 5f_1 = 1700 \text{ Hz}$$

If the same pipe were open at both ends,

$$f'_1 = \frac{v}{2L} = \frac{340}{0.50} = 680 \text{ Hz}$$

exactly twice as high — one octave above.

8.3 End correction in pipes

The air just outside an open end still takes part in the vibration, so the antinode does not sit exactly at the rim: it lies a small distance **outside**. That distance is the **end correction** e . Rayleigh showed that for a pipe of internal radius r ,

$$e = 0.6 r$$

The pipe therefore behaves as though it were longer than it measures:

$$\text{closed pipe: } f_1 = \frac{v}{4(L+e)}, \quad \text{open pipe: } f_1 = \frac{v}{2(L+2e)}$$

(an open pipe has two open ends, so it gets two corrections). End correction matters most for short, wide pipes.

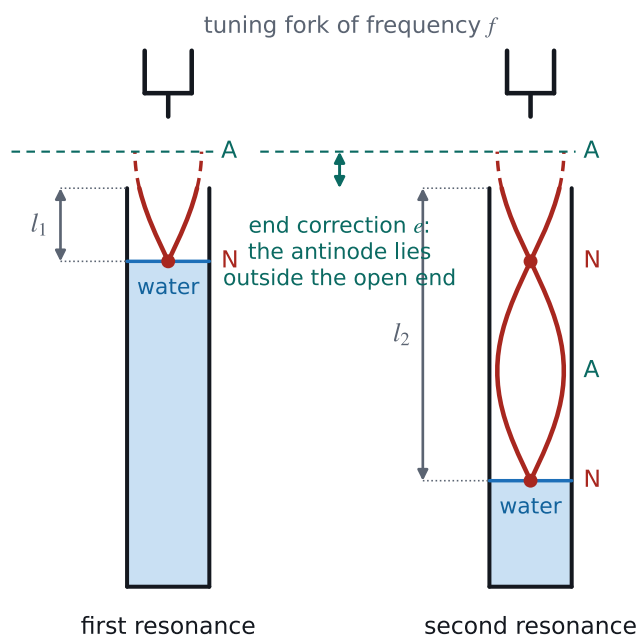


Figure 34. Resonance-tube experiment. The air column resonates at l_1 and again at l_2 . The antinode lies a distance e above the rim (e is exaggerated here).

ELIMINATING THE END CORRECTION IN A RESONANCE TUBE

Lower the water level below a sounding tuning fork of frequency f and the air column resonates first at length l_1 and next at l_2 . Including the correction,

$$l_1 + e = \frac{\lambda}{4}, \quad l_2 + e = \frac{3\lambda}{4}$$

Subtracting the first from the second removes e altogether:

$$l_2 - l_1 = \frac{\lambda}{2} \implies \lambda = 2(l_2 - l_1)$$

$$v = f\lambda = 2f(l_2 - l_1)$$

Multiplying the first by 3 and subtracting the second instead gives e :

$$3l_1 + 3e = l_2 + e \implies e = \frac{l_2 - 3l_1}{2}$$

This is why the resonance tube is the standard school experiment for measuring the velocity of sound: the awkward unknown e cancels.

WORKED EXAMPLE 8.2

Problem. In a resonance-tube experiment a tuning fork of frequency 480 Hz gives resonance when the air column is 17.0 cm long and again when it is 53.0 cm long. Find the velocity of sound in air and the end correction.

Solution. $l_1 = 0.170$ m, $l_2 = 0.530$ m.

$$\lambda = 2(l_2 - l_1) = 2(0.530 - 0.170) = 0.720 \text{ m}$$

$$v = f\lambda = 480 \times 0.720 = 345.6 \text{ m s}^{-1}$$

For the end correction,

$$e = \frac{l_2 - 3l_1}{2} = \frac{0.530 - 0.510}{2} = 0.010 \text{ m} = 1.0 \text{ cm}$$

Since $e = 0.6r$, the internal radius of the tube is $r = 1.0/0.6 = 1.7 \text{ cm}$.

8.4 Velocity of transverse waves along a stretched string

A stretched string carries **transverse** waves. Two quantities control their speed: the tension T that pulls a displaced element back, and the mass per unit length (linear density) $\mu = m/L$ that resists the pull.

SPEED OF A TRANSVERSE WAVE ON A STRING

Let a pulse travel with speed v along a string of tension T and linear density μ . View it from a frame moving with the pulse: the pulse stands still and the string streams backwards through it with speed v .

Take a short element at the crest of the pulse. It is very nearly an arc of a circle of radius R subtending a small angle $\Delta\theta$ at the centre, so its length is $R\Delta\theta$ and its mass is

$$\Delta m = \mu R \Delta\theta$$

The tension T pulls tangentially at each end of the arc. The tangential directions make angles $\Delta\theta/2$ with the horizontal, so the horizontal components cancel and the vertical components add, giving a resultant towards the centre of curvature:

$$F = 2T \sin \frac{\Delta\theta}{2} \approx 2T \cdot \frac{\Delta\theta}{2} = T \Delta\theta$$

In this frame the element moves along the arc with speed v , so F is the centripetal force:

$$T \Delta\theta = \Delta m \frac{v^2}{R} = \mu R \Delta\theta \cdot \frac{v^2}{R} = \mu v^2 \Delta\theta$$

Cancelling $\Delta\theta$,

$$v^2 = \frac{T}{\mu} \implies v = \sqrt{\frac{T}{\mu}}$$

Check the dimensions: $[T]/[\mu] = (\text{M L T}^{-2})/(\text{M L}^{-1}) = \text{L}^2 \text{T}^{-2}$, whose square root is a velocity. If the wire has radius r and material density ρ then $\mu = \pi r^2 \rho$, so a thicker or denser wire carries slower waves.

8.5 Vibration of string and overtones

A sonometer wire is clamped at both ends, so **both ends must be nodes**. The string is therefore a whole number of half-wavelength loops long:

$$L = n \frac{\lambda_n}{2} \implies \lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3, \dots$$

Combining with $f = v/\lambda$ and $v = \sqrt{T/\mu}$,

$$f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Unlike a closed pipe, a string gives **every** harmonic. That is why a plucked string sounds far richer than a stopped pipe of the same pitch.

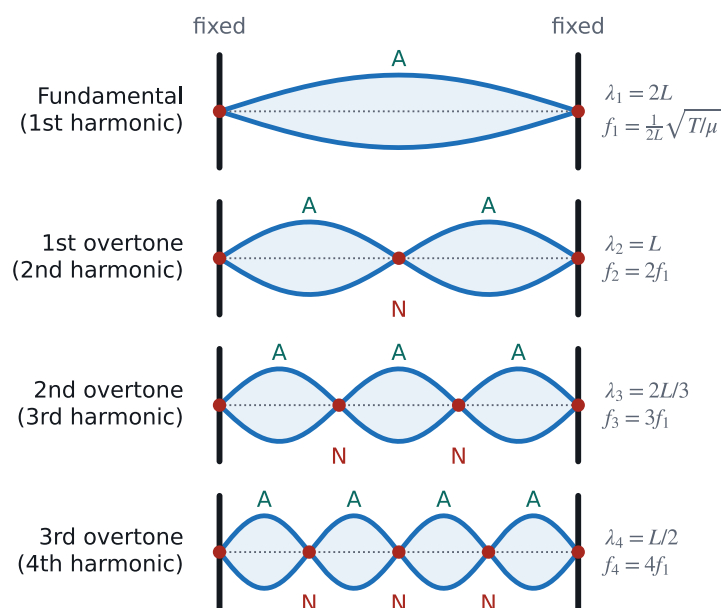


Figure 35. The first four modes of a string fixed at both ends. Every harmonic is possible, and the n th mode has n loops, $n + 1$ nodes and n antinodes.

WORKED EXAMPLE 8.3

Problem. A sonometer wire 50 cm long has a mass per unit length of 1.0 g m^{-1} and is stretched by a tension of 160 N. Find the speed of transverse waves on it, its fundamental frequency, and the frequency of its second overtone.

Solution. $\mu = 1.0 \text{ g m}^{-1} = 1.0 \times 10^{-3} \text{ kg m}^{-1}$, $L = 0.50 \text{ m}$, $T = 160 \text{ N}$.

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{160}{1.0 \times 10^{-3}}} = \sqrt{1.6 \times 10^5} = 400 \text{ m s}^{-1}$$

$$f_1 = \frac{v}{2L} = \frac{400}{2 \times 0.50} = 400 \text{ Hz}$$

A string gives all harmonics, so the second overtone is the 3rd harmonic:

$$f_3 = 3f_1 = 1200 \text{ Hz}$$

8.6 Laws of vibration of fixed string

Writing $f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$ and holding two factors fixed at a time gives the three laws, all verifiable on a sonometer.

Law	Statement	Form
Law of length	$f \propto 1/L$ when T and μ are constant	$fL = \text{constant}$
Law of tension	$f \propto \sqrt{T}$ when L and μ are constant	$f/\sqrt{T} = \text{constant}$
Law of mass	$f \propto 1/\sqrt{\mu}$ when L and T are constant	$f\sqrt{\mu} = \text{constant}$

Because $\mu = \pi r^2 \rho$ for a wire of radius r and density ρ , the third law is often split into a **law of diameter** ($f \propto 1/D$) and a **law of density** ($f \propto 1/\sqrt{\rho}$). Putting $\mu = \pi D^2 \rho/4$ in the fundamental gives the compact working form

$$f_1 = \frac{1}{LD} \sqrt{\frac{T}{\pi \rho}}$$

RATIO METHOD BEATS SUBSTITUTION

Almost every exam numerical on these laws is a ratio question. Write $\frac{f_2}{f_1} = \frac{L_1}{L_2} \sqrt{\frac{T_2}{T_1}} \sqrt{\frac{\mu_1}{\mu_2}}$ and substitute only the quantities that changed. You never need v or μ in absolute terms.

WORKED EXAMPLE 8.4

Problem. A sonometer wire sounds its fundamental at 256 Hz under a tension of 100 N. (a) To what value must the tension be raised to make it sound 320 Hz at the same length? (b) Instead, keeping the tension at 100 N, where should a bridge be placed on the 60 cm wire to obtain 320 Hz?

Solution.

(a) By the law of tension, $f \propto \sqrt{T}$ at constant L and μ :

$$\frac{T_2}{T_1} = \left(\frac{f_2}{f_1}\right)^2 = \left(\frac{320}{256}\right)^2 = (1.25)^2 = 1.5625$$

$$T_2 = 1.5625 \times 100 = 156.25 \text{ N}$$

(b) By the law of length, $f \propto 1/L$ at constant T and μ :

$$L_2 = L_1 \frac{f_1}{f_2} = 60 \times \frac{256}{320} = 60 \times 0.8 = 48 \text{ cm}$$

The bridge must be set 48 cm from the fixed end — that is, the vibrating length must be shortened by 12 cm.

Chapter summary

- A stationary wave forms when incident and reflected waves superpose. Closed end → displacement **node**; open end → displacement **antinode**. Adjacent nodes are $\lambda/2$ apart.
- **Closed pipe:** $L = (2n - 1)\lambda/4$, so $f_n = (2n - 1)v/4L$ — odd harmonics only. The 1st overtone is the 3rd harmonic.
- **Open pipe:** $L = n\lambda/2$, so $f_n = nv/2L$ — all harmonics. An open pipe sounds one octave above a closed pipe of the same length and gives a richer note.
- **End correction** $e = 0.6r$: closed pipe $f_1 = v/4(L + e)$, open pipe $f_1 = v/2(L + 2e)$.
- Resonance tube: $v = 2f(l_2 - l_1)$ and $e = (l_2 - 3l_1)/2$.
- Transverse waves on a string: $v = \sqrt{T/\mu}$, derived from the centripetal force on a curved element.
- String fixed at both ends: $f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$, all harmonics present.
- Laws of a vibrating string: $f \propto 1/L$, $f \propto \sqrt{T}$, $f \propto 1/\sqrt{\mu}$.

Practice questions

Group A — Multiple choice (1 mark each)

1. A closed organ pipe of length L produces harmonics of frequency [1] (a) $nv/2L$ (b) $(2n - 1)v/4L$ (c) $nv/4L$ (d) $(2n - 1)v/2L$
2. The first overtone of a closed pipe is its [1] (a) 2nd harmonic (b) 3rd harmonic (c) 4th harmonic (d) 5th harmonic
3. If the tension in a stretched string is made four times, its fundamental frequency becomes [1] (a) half (b) twice (c) four times (d) unchanged
4. The end correction of a cylindrical pipe of internal radius r is about [1] (a) $0.3r$ (b) $0.6r$ (c) r (d) $1.2r$
5. At the open end of an organ pipe there is always a [1] (a) displacement node (b) pressure antinode (c) displacement antinode (d) point of zero velocity

ANSWERS TO GROUP A

1. (b) — only odd multiples of $v/4L$ fit a node-to-antinode column. **2.** (b) — the 2nd harmonic is forbidden, so the first tone above the fundamental is $3f_1$. **3.** (b) — $f \propto \sqrt{T}$ and $\sqrt{4} = 2$. **4.** (b) — Rayleigh's result, $e = 0.6r$. **5.** (c) — the air is free to move there but cannot change the atmospheric pressure.

Group B — Short answer (5 marks each)

1. Show that a closed organ pipe produces only odd harmonics while an open pipe of the same length produces all harmonics, and hence compare their fundamental frequencies. [5]
2. What is meant by end correction? In a resonance-tube experiment, show that $v = 2f(l_2 - l_1)$ and $e = (l_2 - 3l_1)/2$. [5]
3. State the three laws of vibration of a stretched string and write the single formula from which all three follow. [5]
4. The third harmonic of a closed pipe has the same frequency as the first overtone of an open pipe 50 cm long. Find the length of the closed pipe. [5]

5. A closed pipe 20 cm long has an internal radius of 2.0 cm. Taking $v = 340 \text{ m s}^{-1}$, find its fundamental frequency with and without the end correction. [5]

ANSWERS TO GROUP B

1. Closed: node at one end, antinode at the other, so $L = (2n - 1)\lambda/4$ and $f_n = (2n - 1)v/4L$ — odd harmonics. Open: antinodes at both ends, so $L = n\lambda/2$ and $f_n = nv/2L$ — all harmonics. For the same L , $f_1(\text{open}) = 2f_1(\text{closed})$.

4. The first overtone of an open pipe is its 2nd harmonic, $f = 2 \times \frac{v}{2L_o} = \frac{v}{L_o}$. The third harmonic of a closed pipe is $f = \frac{3v}{4L_c}$. Equating the two,

$$\frac{3v}{4L_c} = \frac{v}{L_o} \implies L_c = \frac{3L_o}{4} = \frac{3 \times 0.50}{4} = 0.375 \text{ m} = 37.5 \text{ cm}$$

5. Without correction, $f_1 = v/4L = 340/(4 \times 0.20) = 425 \text{ Hz}$. With correction, $e = 0.6 \times 0.020 = 0.012 \text{ m}$, so

$$f_1 = \frac{340}{4(0.200 + 0.012)} = \frac{340}{0.848} = 401 \text{ Hz}$$

The correction lowers the predicted note by 24 Hz — about a semitone, which is easily heard.

Group C — Long answer (8 marks each)

- (a) Derive the expression $v = \sqrt{T/\mu}$ for the speed of a transverse wave on a stretched string. [4] (b) Hence obtain the frequencies of the harmonics of a string fixed at both ends and state the laws of vibration of a fixed string. [4]
- A wire 1.0 m long has a mass per unit length of $5.0 \times 10^{-3} \text{ kg m}^{-1}$ and is stretched by a tension of 200 N. (a) Find the speed of transverse waves and the fundamental frequency. [3] (b) What tension would raise the fundamental to 150 Hz? [3] (c) With the original tension restored, where must a bridge be placed to give a fundamental of 250 Hz? [2]

ANSWER TO GROUP C QUESTION 2

(a) $v = \sqrt{T/\mu} = \sqrt{200/(5.0 \times 10^{-3})} = \sqrt{4.0 \times 10^4} = 200 \text{ m s}^{-1}$, and

$$f_1 = \frac{v}{2L} = \frac{200}{2 \times 1.0} = 100 \text{ Hz}$$

(b) $f \propto \sqrt{T}$, so $T' = T(f'/f_1)^2 = 200 \times (150/100)^2 = 200 \times 2.25 = 450 \text{ N}$.

(c) $f \propto 1/L$, so $L' = Lf_1/f' = 1.0 \times (100/250) = 0.40 \text{ m}$. The bridge must be placed 40 cm from one end.

UNIT 9

Acoustic phenomena

Waves and Optics • 5 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Sound waves: Pressure amplitude
- Characteristics of sound: Intensity, loudness, quality and pitch
- Doppler's effect

Everything you hear — the pitch of a temple bell, the roar of a bus on the Ring Road, the rising whine of an ambulance as it passes — is a pressure wave arriving at your eardrum. This unit turns the everyday words loud, shrill and sweet into measurable physical quantities, and then works out what happens to a note when the source or the listener is moving.

THREE SEPARATE IDEAS, THREE SEPARATE SYMBOLS

Do not let **intensity** (I , in W m^{-2}), **intensity level** (β , in dB) and **loudness** (L , a sensation) blur together. Intensity is energy per second per square metre; intensity level is a logarithmic comparison with a standard; loudness is what your ear reports. The exam tests the difference.

9.1 Sound waves: pressure amplitude

A sound wave can be described in two equivalent ways: by the **displacement** of the air particles, or by the **excess pressure** — the small amount by which the pressure rises above or falls below the steady atmospheric value.

Let a plane progressive wave travel along x with displacement

$$y = a \sin(\omega t - kx)$$

Consider a thin layer of air of original thickness Δx . When the wave passes, its two faces move by different amounts, so its new thickness is $\Delta x + (\partial y / \partial x) \Delta x$. The volume strain is therefore $\partial y / \partial x$, and by the definition of the bulk modulus B the excess pressure is

$$\Delta P = -B \frac{\partial y}{\partial x}$$

PRESSURE AMPLITUDE OF A SOUND WAVE

Differentiating $y = a \sin(\omega t - kx)$ with respect to x ,

$$\frac{\partial y}{\partial x} = -k a \cos(\omega t - kx)$$

so

$$\Delta P = B a \cos(\omega t - kx)$$

The **pressure amplitude** is the largest value of ΔP :

$$P_0 = B a$$

For a sound wave $v = \sqrt{B/\rho}$, so $B = \rho v^2$, and $k = \omega/v$. Substituting,

$$P_0 = \rho v^2 \cdot a \cdot \frac{\omega}{v} = \rho v \omega a = 2\pi \rho v f a$$

Two consequences matter. First, ΔP contains $\cos$ where y contains $\sin$: **the pressure wave is a quarter of a wavelength out of step with the displacement wave.** Where the displacement is zero the pressure change is greatest, and vice versa. Second, for a given displacement amplitude the pressure amplitude grows in proportion to the frequency.

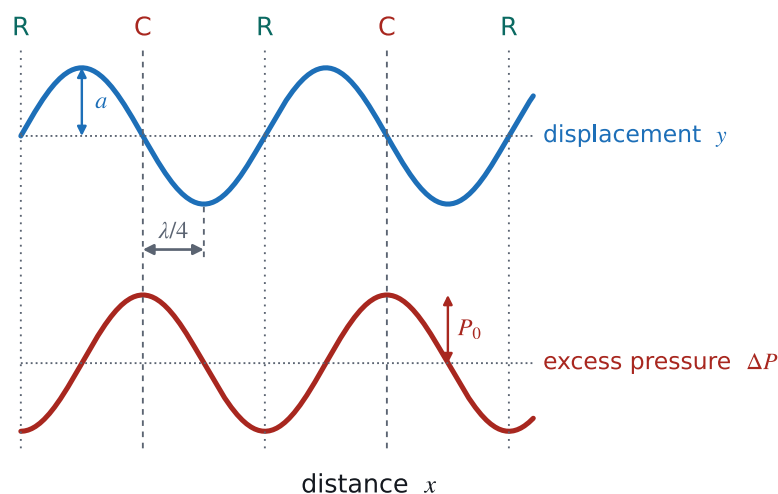


Figure 36. Displacement y and excess pressure ΔP for the same sound wave at one instant. They are $\lambda/4$ out of step: at a compression (C) the displacement is zero but ΔP is maximum.

The ear is astonishingly sensitive. At 1 kHz the faintest audible sound has a root-mean-square pressure of only about 2×10^{-5} Pa — one five-thousand-millionth of atmospheric pressure — which corresponds to a displacement amplitude of roughly 10^{-11} m, smaller than the diameter of an atom.

9.2 Characteristics of sound: intensity, loudness, quality and pitch

A musical note has three characteristics you can hear — loudness, pitch and quality — and each is governed by a measurable physical quantity.

Intensity

INTENSITY OF SOUND

The intensity of sound at a point is the energy carried by the wave per second through unit area held perpendicular to the direction of propagation. Its SI unit is the watt per square metre (W m^{-2}).

For a plane progressive wave the intensity works out to

$$I = \frac{1}{2} \rho v \omega^2 a^2 = 2\pi^2 \rho v f^2 a^2 = \frac{P_0^2}{2\rho v}$$

So intensity depends on the **square** of the amplitude and the **square** of the frequency, and on the medium through ρv . For a point source radiating power P uniformly in all directions, the energy at distance r is spread over a sphere of area $4\pi r^2$:

$$I = \frac{P}{4\pi r^2} \quad \text{so} \quad I \propto \frac{1}{r^2}$$

This **inverse-square law** is why you must stand close to hear a soft speaker.

Intensity level and the decibel

The ear can cope with intensities spanning a factor of 10^{12} , so a logarithmic scale is used. Taking the standard reference $I_0 = 10^{-12} \text{ W m}^{-2}$ (roughly the threshold of hearing at 1 kHz), the **intensity level** is

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right) \text{ decibel (dB)}$$

Sound	$I \text{ (W m}^{-2}\text{)}$	$\beta \text{ (dB)}$
Threshold of hearing	10^{-12}	0
Rustling leaves, soft whisper	10^{-10}	20
Quiet room at night	10^{-8}	40
Normal conversation	10^{-6}	60
Busy Kathmandu street	10^{-4}	80
Motorcycle horn at 1 m	10^{-1}	110
Threshold of pain	1	120

THE 3 DB AND 10 DB SHORTCUTS

Doubling the intensity adds $10 \log_{10} 2 = 3.0 \text{ dB}$; multiplying it by ten adds exactly 10 dB. Doubling the distance from a point source quarters the intensity, so the level falls by $10 \log_{10} 4 = 6.0 \text{ dB}$. These three numbers answer most decibel questions without a calculator.

Loudness

Loudness is the sensation produced in the ear. It grows with intensity, but not in proportion to it: by the **Weber-Fechner law** the sensation is proportional to the logarithm of the stimulus,

$$L = k \log_{10} I$$

Loudness also depends on frequency — the human ear is most sensitive between about 1 kHz and 5 kHz, so a 100 Hz note and a 3 kHz note of the same intensity do not sound equally loud. Loudness is measured in phon or sone, intensity level in decibels; they are not the same thing.

INTENSITY IS PHYSICAL, LOUDNESS IS PHYSIOLOGICAL

Two sounds of equal intensity can have very different loudness if their frequencies differ, and a sound can be perfectly measurable yet completely inaudible (an ultrasonic whistle). Never write "loudness = $10 \log(I/I_0)$ " — that is the intensity level.

Pitch

Pitch is the sensation of how high or low a note is, and it is decided by **frequency**: the higher the frequency, the higher the pitch. A normal human ear responds from about 20 Hz to 20 000 Hz. Below 20 Hz is **infrasonic** (earthquake waves, elephant calls); above 20 kHz is **ultrasonic** (bat and dolphin echolocation, medical scanning).

Quality (timbre)

Two instruments playing the same note at the same loudness still sound different. A real instrument produces the fundamental plus a set of overtones, and the **number of overtones and their relative intensities** decide the shape of the resulting waveform. That shape is the **quality** or **timbre**.

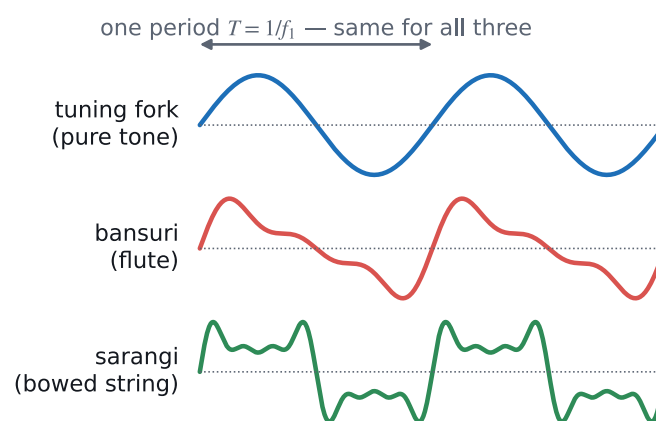


Figure 37. Three sounds of the same pitch and the same loudness but different quality. The waveform repeats at the same period; only the harmonic mixture differs.

Characteristic	Physical quantity	Depends on
Loudness	intensity	square of amplitude, $1/r^2$, sensitivity of ear
Pitch	frequency	frequency of the fundamental
Quality	waveform	number and relative intensity of overtones

WORKED EXAMPLE 9.1

Problem. A sound wave of frequency 1000 Hz in air has a displacement amplitude of 1.0×10^{-8} m. Taking $\rho = 1.29 \text{ kg m}^{-3}$ and $v = 340 \text{ m s}^{-1}$, find (a) the pressure amplitude, (b) the intensity, and (c) the intensity level in decibels.

Solution.

(a) $\omega = 2\pi f = 2\pi \times 1000 = 6.283 \times 10^3 \text{ rad s}^{-1}$ and $\rho v = 1.29 \times 340 = 438.6 \text{ kg m}^{-2} \text{ s}^{-1}$.

$$P_0 = \rho v \omega a = 438.6 \times 6.283 \times 10^3 \times 1.0 \times 10^{-8} = 2.76 \times 10^{-2} \text{ Pa}$$

(b)

$$I = \frac{P_0^2}{2\rho v} = \frac{(2.76 \times 10^{-2})^2}{2 \times 438.6} = \frac{7.60 \times 10^{-4}}{877.2} = 8.66 \times 10^{-7} \text{ W m}^{-2}$$

(c)

$$\beta = 10 \log_{10} \frac{8.66 \times 10^{-7}}{10^{-12}} = 10 \log_{10} (8.66 \times 10^5) = 10 \times 5.94 = 59.4 \text{ dB}$$

That is a little under 60 dB — the level of ordinary conversation — produced by air particles moving only a hundredth of a micrometre.

WORKED EXAMPLE 9.2

Problem. A small loudspeaker radiates 2.0 W of sound power uniformly in all directions. Find the intensity and the intensity level at 10 m, and the intensity level at 20 m.

Solution. At $r = 10 \text{ m}$,

$$I = \frac{P}{4\pi r^2} = \frac{2.0}{4\pi(10)^2} = \frac{2.0}{1256.6} = 1.59 \times 10^{-3} \text{ W m}^{-2}$$

$$\beta = 10 \log_{10} \frac{1.59 \times 10^{-3}}{10^{-12}} = 10 \log_{10} (1.59 \times 10^9) = 92.0 \text{ dB}$$

Doubling the distance to 20 m quarters the intensity, so the level falls by $10 \log_{10} 4 = 6.0 \text{ dB}$:

$$\beta' = 92.0 - 6.0 = 86.0 \text{ dB}$$

9.3 Doppler's effect

DOPPLER EFFECT

The apparent change in the frequency of a wave observed when the source, the observer, or both are in motion relative to the medium is called the Doppler effect. The frequency emitted by the source never changes; only the frequency received changes.

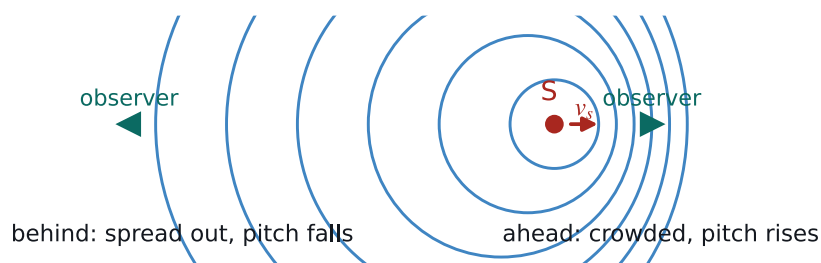


Figure 38. Wavefronts from a source S moving to the right. Ahead of the source the wavelength is squeezed to $(v - v_s)/f$ and the pitch rises; behind it the wavelength is stretched to $(v + v_s)/f$ and the pitch falls.

THE DOPPLER FORMULA

Let the source emit sound of frequency f , let v be the speed of sound in still air, v_s the speed of the source and v_o the speed of the observer.

Case 1: source moving towards a stationary observer. In one period $T = 1/f$ the wave front advances vT but the source itself advances $v_s T$. The next wave front is therefore emitted from a point closer to the observer, and the waves are packed into a shorter wavelength:

$$\lambda' = vT - v_s T = \frac{v - v_s}{f}$$

The observer is at rest, so the waves reach her at speed v and

$$f' = \frac{v}{\lambda'} = \frac{v}{v - v_s} f$$

Case 2: observer moving towards a stationary source. Now the wavelength is unchanged, $\lambda = v/f$, but the observer runs into the waves, so their speed relative to her is $v + v_o$:

$$f' = \frac{v + v_o}{\lambda} = \frac{v + v_o}{v} f$$

General case. Combining the two,

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f$$

GETTING THE SIGNS RIGHT

Use $f' = \frac{v+v_o}{v-v_s}f$ always, and fix the signs by asking only one question: **does this motion bring them closer?**

- Observer moving **towards** the source: v_o is **positive**; moving away, negative.
- Source moving **towards** the observer: v_s is **positive**; moving away, negative.

Any approach makes the numerator bigger or the denominator smaller, so $f' > f$. If your answer disagrees with that check, you have a sign wrong.

Situation	Apparent frequency
Source approaches, observer at rest	$f' = \frac{v}{v-v_s}f$
Source recedes, observer at rest	$f' = \frac{v}{v+v_s}f$
Observer approaches, source at rest	$f' = \frac{v+v_o}{v}f$
Observer recedes, source at rest	$f' = \frac{v-v_o}{v}f$
Both approach each other	$f' = \frac{v+v_o}{v-v_s}f$
Both recede from each other	$f' = \frac{v-v_o}{v+v_s}f$

When there is no Doppler effect. If the source and observer move with the same velocity in the same direction, their separation is constant and $f' = f$. If the motion is perpendicular to the line joining them — the instant a train passes you broadside — there is no shift either. And a steady **wind** produces no change for a stationary source and observer, because it increases the speed of sound and the wavelength in exactly the same proportion.

Applications. Traffic police speed guns and weather radar; SONAR for measuring the speed of a submarine or shoal of fish; Doppler ultrasound (echocardiography) to measure the speed of blood in an artery; and, in astronomy, the **red shift** of light from distant galaxies, which is the evidence that the universe is expanding.

WORKED EXAMPLE 9.3

Problem. A train approaching a station sounds a whistle of frequency 640 Hz while moving at 20 m s^{-1} . Taking the speed of sound as 340 m s^{-1} , find the frequency heard by a stationary passenger on the platform (a) as the train approaches and (b) after it has passed. What is the change in frequency as the train goes by?

Solution.

(a) Source approaching, observer at rest, so $v_s = +20 \text{ m s}^{-1}$:

$$f' = \frac{v}{v-v_s}f = \frac{340}{340-20} \times 640 = \frac{340}{320} \times 640 = 680 \text{ Hz}$$

(b) Source receding, so $v_s = -20 \text{ m s}^{-1}$:

$$f'' = \frac{v}{v + v_s} f = \frac{340}{360} \times 640 = 604.4 \text{ Hz}$$

The pitch drops by $680 - 604.4 = 75.6 \text{ Hz}$ as the train passes — a fall of more than a musical tone, which is exactly what you hear.

WORKED EXAMPLE 9.4

Problem. Two vehicles approach each other on the Prithvi Highway. A jeep sounding a horn of frequency 400 Hz travels at 20 m s^{-1} ; a motorcycle comes towards it at 15 m s^{-1} . Taking $v = 340 \text{ m s}^{-1}$, find the frequency heard by the motorcyclist.

Solution. Both are approaching, so both signs are positive: $v_o = +15 \text{ m s}^{-1}$ and $v_s = +20 \text{ m s}^{-1}$.

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f = \left(\frac{340 + 15}{340 - 20} \right) \times 400 = \frac{355}{320} \times 400$$

$$f' = 1.1094 \times 400 = 443.8 \text{ Hz}$$

The horn sounds about 44 Hz sharper than it really is.

Chapter summary

- Excess pressure $\Delta P = -B \partial y / \partial x$; the pressure amplitude is $P_0 = Bak = \rho v \omega a = 2\pi \rho v f a$.
- The pressure wave is $\lambda/4$ out of phase with the displacement wave: a displacement node is a pressure antinode.
- Intensity $I = \frac{1}{2} \rho v \omega^2 a^2 = P_0^2 / 2\rho v$, in W m^{-2} ; for a point source $I = P / 4\pi r^2$, so $I \propto 1/r^2$.
- Intensity level $\beta = 10 \log_{10} (I/I_0)$ dB with $I_0 = 10^{-12} \text{ W m}^{-2}$. Doubling I adds 3 dB; doubling the distance subtracts 6 dB.
- Loudness (Weber–Fechner, $L = k \log I$) is the sensation; pitch is set by frequency; quality is set by the overtones present, i.e. by the waveform.
- Doppler effect: $f' = \frac{v + v_o}{v - v_s} f$, with v_o and v_s counted positive when the motion brings source and observer closer.
- There is no Doppler shift for motion perpendicular to the line of sight, for equal velocities in the same direction, or for a steady wind.

Practice questions

Group A — Multiple choice (1 mark each)

1. The pressure amplitude of a sound wave of displacement amplitude a and angular frequency ω is [1] (a) $\rho v a$ (b) $\rho v \omega a$ (c) $\rho v \omega^2 a$ (d) $\rho \omega a$
2. The displacement wave and the pressure wave in air differ in phase by [1] (a) 0 (b) $\pi/4$ (c) $\pi/2$ (d) π
3. If the intensity of a sound is doubled, its intensity level increases by about [1] (a) 2 dB (b) 3 dB (c) 6 dB (d) 10 dB
4. The quality of a musical note depends on [1] (a) frequency (b) amplitude (c) number and intensity of overtones (d) speed of sound

5. A source and an observer move with the same velocity in the same direction. The apparent frequency is [1] (a) greater than f (b) less than f (c) equal to f (d) zero

ANSWERS TO GROUP A

1. (b) — $P_0 = B a k$ and $B = \rho v^2$, $k = \omega/v$, giving $\rho v \omega a$. 2. (c) — $\Delta P \propto \cos$ where $y \propto \sin$, a quarter cycle apart. 3. (b) — $10 \log_{10} 2 = 3.0$ dB. 4. (c) — the harmonic mixture fixes the waveform, and the waveform is the quality. 5. (c) — the separation does not change, so no wavefronts are gained or lost.

Group B — Short answer (5 marks each)

1. Show that the pressure amplitude of a sound wave is $P_0 = \rho v \omega a$ and explain why the pressure wave leads the displacement wave by a quarter of a wavelength. [5]
2. Distinguish between intensity, intensity level and loudness of sound, giving the unit of each. [5]
3. The intensity level of a sound is 70 dB. Find its intensity, and the new intensity level if the intensity is increased 100 times. [5]
4. Explain how the pitch and the quality of a note are related to the physical properties of the sound wave. Why do a bansuri and a sarangi sounding the same note sound different? [5]
5. A whistle of frequency 500 Hz is sounded on a car moving away from a stationary listener at 25 m s^{-1} . Taking $v = 340 \text{ m s}^{-1}$, find the frequency heard. [5]

ANSWERS TO GROUP B

1. With $y = a \sin(\omega t - kx)$, $\Delta P = -B \frac{\partial y}{\partial x} = B a k \cos(\omega t - kx)$, so $P_0 = B a k$. Using $B = \rho v^2$ and $k = \omega/v$ gives $P_0 = \rho v \omega a$. The cosine leads the sine by $\pi/2$, which in space is a quarter wavelength.
3. $\beta = 10 \log_{10} (I/I_0) = 70$ gives $I/I_0 = 10^7$, so $I = 10^7 \times 10^{-12} = 1.0 \times 10^{-5} \text{ W m}^{-2}$. Multiplying the intensity by 100 = 10^2 adds $10 \log_{10} 10^2 = 20$ dB, giving $\beta' = 90$ dB.
5. The source recedes, so $v_s = -25 \text{ m s}^{-1}$:

$$f' = \frac{v}{v + 25} f = \frac{340}{365} \times 500 = 465.8 \text{ Hz}$$

The whistle sounds about 34 Hz flatter than it really is.

Group C — Long answer (8 marks each)

1. (a) State the Doppler effect and derive an expression for the apparent frequency when both the source and the observer are in motion along the line joining them. [5] (b) Give two situations in which there is relative motion but no Doppler shift, and state two practical applications of the effect. [3]
2. A point source emits sound uniformly with a power of 0.50 W. (a) Find the intensity and the intensity level at a distance of 5.0 m. [4] (b) At what distance does the intensity level fall to 60 dB? [4] (Take $I_0 = 10^{-12} \text{ W m}^{-2}$.)

ANSWER TO GROUP C QUESTION 2

(a)

$$I = \frac{P}{4\pi r^2} = \frac{0.50}{4\pi(5.0)^2} = \frac{0.50}{314.16} = 1.59 \times 10^{-3} \text{ W m}^{-2}$$

$$\beta = 10 \log_{10} \frac{1.59 \times 10^{-3}}{10^{-12}} = 10 \log_{10} (1.59 \times 10^9) = 92.0 \text{ dB}$$

(b) A level of 60 dB means $I' = 10^6 I_0 = 1.0 \times 10^{-6} \text{ W m}^{-2}$. From $I' = P/4\pi r'^2$,

$$r'^2 = \frac{P}{4\pi I'} = \frac{0.50}{4\pi \times 1.0 \times 10^{-6}} = 3.979 \times 10^4 \text{ m}^2$$

$$r' = 199 \text{ m}$$

So the level drops from 92 dB to 60 dB — a factor of about 1600 in intensity — between 5 m and about 200 m, exactly as the inverse-square law requires.

UNIT 10

Nature and propagation of light

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Huygen's principle
- Reflection and Refraction according to wave theory

Ray optics treats light as a straight line and gets mirrors and lenses right, but it cannot say why light bends, or what happens when it meets an obstacle only a few thousandths of a millimetre across. Wave optics can. This unit replaces the ray by the **wavefront**, states Huygens' principle, and uses it to rebuild the two laws you already know — reflection and refraction — from scratch. Everything in the next three units (interference, diffraction, polarization) stands on the geometry you learn here.

WHAT THE EXAMINER ASKS FROM THIS UNIT

Two things, almost every year: state Huygens' principle and explain the construction, and prove the laws of reflection or Snell's law using Huygens' construction. Both are pure marks if you can draw the diagram correctly, so practise the diagram until you can produce it from memory in ninety seconds.

10.1 Huygen's principle

A source of light sends energy out in all directions. At any instant, the points that the disturbance has just reached all started vibrating at the same moment, so they are all vibrating **in step**.

WAVEFRONT

A **wavefront** is the locus of all points of a medium that are vibrating in the same phase. The perpendicular drawn to a wavefront, in the direction in which the wave advances, is a **ray**. The speed with which a wavefront advances is the speed of the wave in that medium.

Close to a point source the wavefronts are spheres; close to a long thin slit they are cylinders. Very far from any source a small patch of the sphere is indistinguishable from a flat plane, so sunlight reaching the Earth arrives as **plane wavefronts**. A converging lens turns an incoming plane wavefront into a spherical wavefront that shrinks onto the focus.

Type of wavefront	Shape	Source	Amplitude falls as
Spherical	sphere	point source	$1/r$
Cylindrical	cylinder	linear source (slit)	$1/\sqrt{r}$
Plane	plane	very distant source	constant

Statement of the principle

Christiaan Huygens (1678) gave a rule for advancing a wavefront from one instant to the next. It has two parts:

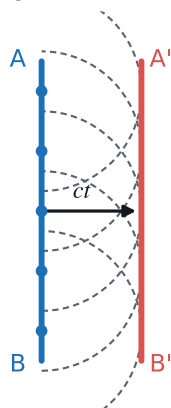
1. **Every point on a wavefront acts as a fresh source of secondary wavelets**, which spread out in all forward directions with the speed of the wave in that medium.
2. **The new wavefront**, after a time t , is the surface that touches all these secondary wavelets — their forward **envelope** or common tangent surface.

If the wave speed in the medium is v , every secondary wavelet drawn from the old wavefront is a sphere of radius

$$r = vt$$

So to advance a wavefront you take a compass, set it to vt , strike arcs from a row of points on the old wavefront, and draw the tangent.

(a) plane wavefront



(b) spherical wavefront

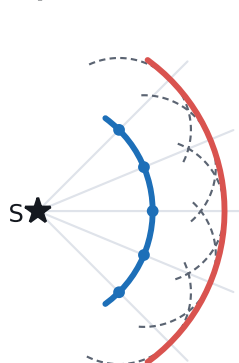


Figure 39. Huygens' construction. (a) A plane wavefront AB advances to the plane A'B'. (b) A spherical wavefront from S advances to a larger sphere. In each case the new wavefront is the forward envelope of secondary wavelets of radius ct .

Why there is no backward wave

Drawn honestly, the secondary wavelets are complete spheres, so they also have a backward envelope — a wavefront travelling back towards the source. We never observe one. Huygens simply asserted that the wavelets have zero intensity in the backward direction. Fresnel and later Kirchhoff showed the amplitude of a secondary wavelet actually varies with the angle θ measured from the forward direction as the **obliquity factor**

$$K = \frac{1}{2}(1 + \cos \theta)$$

which equals 1 straight ahead and 0 straight backwards. That is why only the forward envelope survives. For NEB purposes it is enough to say: secondary wavelets are effective only in the forward direction.

WAVELETS ARE NOT RAYS

A secondary wavelet is a tiny spherical wave, not a ray. Drawing straight arrows from points on the wavefront instead of arcs earns no marks — the whole proof depends on the arcs and on their common tangent.

10.2 Reflection and Refraction according to wave theory

Reflection at a plane surface

Let a plane wavefront AB travel through a medium in which light has speed c and strike a plane mirror. Take the moment when the end A has just touched the mirror at A while the other end is still at B. Let the wavefront meet the mirror again at C after a further time t .

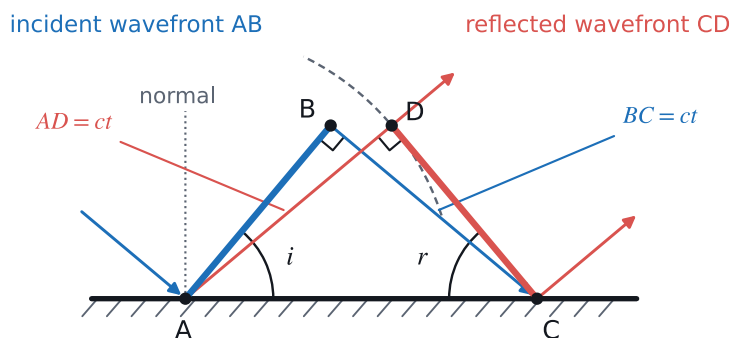


Figure 40. Reflection of a plane wavefront. While B travels to C, the secondary wavelet from A grows to radius ct , and CD is the reflected wavefront. Since $BC = AD$, the angle of incidence equals the angle of reflection.

While B travels the distance BC , every point of the mirror between A and C has in turn started a secondary wavelet. The wavelet from A, which started earliest, has had the full time t to grow, so it is a hemisphere of radius $AD = ct$. The wavelet from C has just started and has zero radius. The tangent drawn from C to the wavelet centred on A is the **reflected wavefront** CD.

THE LAWS OF REFLECTION

In the right-angled triangles ABC and ADC :

- $BC = ct$ is the distance travelled by B in the medium, and $AD = ct$ is the radius of the wavelet from A. Hence $BC = AD$.
- AC is common, and angle $ABC = \text{angle } ADC = 90^\circ$ (a wavefront is always perpendicular to its ray).

From triangle ABC : $\sin i = \frac{BC}{AC}$, and from triangle ADC : $\sin r = \frac{AD}{AC}$.

$$\frac{\sin i}{\sin r} = \frac{BC}{AD} = \frac{ct}{ct} = 1 \implies i = r$$

That is the **second law of reflection**. Also, the incident wavefront, the mirror and the reflected wavefront are all built on the same plane of the diagram, so the incident ray, the reflected ray and the normal are coplanar — the **first law**.

Refraction at a plane surface

Now let the same plane wavefront AB fall on the flat surface separating medium 1 (speed v_1) from a denser medium 2 (speed $v_2 < v_1$). Again A touches the surface first, and B reaches C after time t , so $BC = v_1 t$. In that same time the wavelet that started at A has spread **into medium 2** and has grown only to radius $AD = v_2 t$, which is smaller. The tangent CD from C to this wavelet is the refracted wavefront, and it is tilted towards the normal.

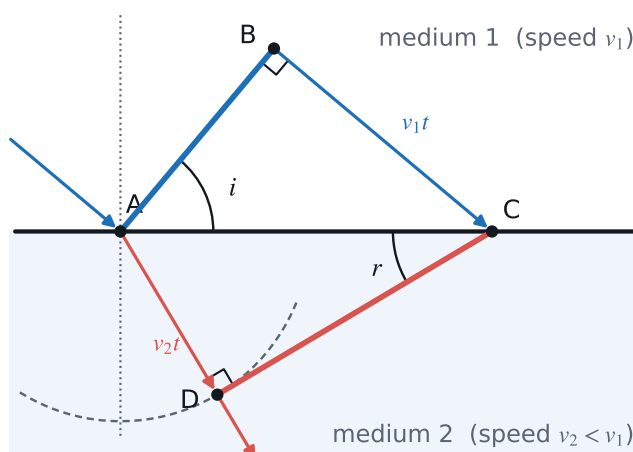


Figure 41. Refraction of a plane wavefront. In time t the wavefront covers $BC = v_1 t$ in the rarer medium while the wavelet from A grows to only $AD = v_2 t$ in the denser medium, so the refracted wavefront CD bends towards the normal.

SNELL'S LAW FROM HUYGENS' CONSTRUCTION

From the right-angled triangle ABC , with the angle at A equal to i :

$$\sin i = \frac{BC}{AC} = \frac{v_1 t}{AC}$$

From the right-angled triangle ADC , with the angle at C equal to r :

$$\sin r = \frac{AD}{AC} = \frac{v_2 t}{AC}$$

Dividing, the common factors AC and t cancel:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = {}_1n_2$$

The right-hand side depends only on the two media, not on i . So the ratio $\sin i / \sin r$ is a constant — **Snell's law**.

Writing the absolute refractive index of a medium as $n = c/v$, we get $v_1 = c/n_1$ and $v_2 = c/n_2$, so

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \implies n_1 \sin i = n_2 \sin r$$

Frequency, wavelength and optical path

The secondary wavelets in medium 2 are started by the vibrations arriving from medium 1, so they must vibrate at the same rate. **Frequency does not change on refraction** — it is fixed by the source. Since $v = f\lambda$ and v falls,

$$\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1}, \quad \lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Light slows down and its wavelength shrinks, but its colour (frequency) is unchanged. A path of geometrical length x inside a medium of index n holds the same number of waves as a length nx in vacuum, so we define the **optical path**

$$\Delta_{\text{opt}} = n x$$

This single idea is used constantly in the next two units: whenever light passes through glass or water, replace the real distance by the optical path.

Prediction	Newton's corpuscular theory	Huygens' wave theory
Nature of light	stream of tiny particles	disturbance spreading as wavefronts
Speed in water compared with air	greater	smaller
Reflection	explained	explained
Refraction	explained, wrong speed	explained, correct speed
Interference, diffraction, polarization	not explained	explained
Experimental verdict	disproved by Foucault (1850)	confirmed

Foucault's rotating-mirror measurement in 1850 showed that light travels more slowly in water than in air. That single result killed the corpuscular theory and established the wave picture used in Units 11–13.

WORKED EXAMPLE 10.1

Problem. Light strikes a glass block of refractive index 1.5 at an angle of incidence of 60° . Take $c = 3 \times 10^8 \text{ m s}^{-1}$ and a vacuum wavelength of 600 nm. Find (a) the angle of refraction, (b) the speed of light in the glass, (c) the wavelength in the glass.

Solution.

(a) $\sin r = \frac{\sin 60^\circ}{1.5} = \frac{0.8660}{1.5} = 0.5774$, so $r = 35.3^\circ$.

(b) $v = \frac{c}{n} = \frac{3 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$.

(c) $\lambda_g = \frac{\lambda_0}{n} = \frac{600}{1.5} = 400 \text{ nm}$.

The frequency stays at $f = c/\lambda_0 = 5.0 \times 10^{14} \text{ Hz}$ in both media.

WORKED EXAMPLE 10.2

Problem. A beam of frequency 5.0×10^{14} Hz passes from air into a liquid in which its wavelength is 375 nm. Find the refractive index of the liquid and the speed of light in it.

Solution. Frequency is unchanged, so in the liquid

$$v = f\lambda_m = (5.0 \times 10^{14})(375 \times 10^{-9}) = 1.875 \times 10^8 \text{ m s}^{-1}$$

$$n = \frac{c}{v} = \frac{3 \times 10^8}{1.875 \times 10^8} = 1.6$$

Check with wavelengths: $\lambda_0 = c/f = 600 \text{ nm}$, and $\lambda_0/\lambda_m = 600/375 = 1.6$. The two routes agree.

WORKED EXAMPLE 10.3

Problem. A glass plate of thickness 2.0 mm and refractive index 1.5 is placed in a beam of light of vacuum wavelength 500 nm. Find (a) the optical path through the plate, (b) the number of waves contained in the plate, (c) the extra time the light takes compared with travelling the same 2.0 mm in vacuum.

Solution.

(a) $\Delta_{\text{opt}} = nx = 1.5 \times 2.0 \times 10^{-3} = 3.0 \times 10^{-3} \text{ m}$.

(b) The number of waves is the optical path divided by the **vacuum** wavelength:

$$N = \frac{nx}{\lambda_0} = \frac{3.0 \times 10^{-3}}{500 \times 10^{-9}} = 6000$$

(c) Time in glass = nx/c , time in vacuum = x/c , so the delay is

$$\Delta t = \frac{(n-1)x}{c} = \frac{0.5 \times 2.0 \times 10^{-3}}{3 \times 10^8} = 3.3 \times 10^{-12} \text{ s}$$

Chapter summary

- A **wavefront** is the locus of points vibrating in the same phase; a **ray** is the normal to the wavefront. Wavefronts may be spherical, cylindrical or plane.
- **Huygens' principle:** every point of a wavefront is a source of secondary wavelets of radius vt ; the new wavefront is their forward envelope. Backward wavelets are suppressed by the obliquity factor $K = \frac{1}{2}(1 + \cos \theta)$.
- **Reflection:** $BC = AD = ct$ gives $\sin i / \sin r = 1$, so $i = r$, and the incident ray, reflected ray and normal are coplanar.
- **Refraction:** $BC = v_1 t$, $AD = v_2 t$ give $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$, i.e. $n_1 \sin i = n_2 \sin r$.
- On refraction the **frequency is unchanged**; the speed and wavelength both fall by the factor n : $v = c/n$ and $\lambda_m = \lambda_0/n$.
- **Optical path** = $n \times$ geometrical path; it is the vacuum distance holding the same number of waves.

- The wave theory predicts $v_{\text{water}} < v_{\text{air}}$, confirmed by Foucault in 1850; the corpuscular theory predicted the opposite and failed.

Practice questions

Group A — Multiple choice (1 mark each)

1. A wavefront is a surface on which all points have the same [1] (a) amplitude (b) phase (c) wavelength (d) intensity
2. When light travels from air into glass, the quantity that does **not** change is [1] (a) speed (b) wavelength (c) frequency (d) direction
3. Secondary wavelets in Huygens' construction travel with [1] (a) the speed of the source (b) the speed of light in that medium (c) infinite speed (d) half the wave speed
4. According to the wave theory of light, the speed of light in water is [1] (a) greater than in air (b) less than in air (c) equal to that in air (d) zero
5. The optical path of a ray covering 4 cm in a medium of index 1.5 is [1] (a) 2.67 cm (b) 4 cm (c) 6 cm (d) 9 cm

ANSWERS TO GROUP A

1. (b) — a wavefront is by definition a surface of constant phase. **2.** (c) — frequency is set by the source and is not altered by the medium. **3.** (b) — each wavelet expands at the wave speed $v = c/n$ of that medium. **4.** (b) — the wave theory needs $v_2 < v_1$ to bend light towards the normal; Foucault confirmed it. **5.** (c) — optical path $= nx = 1.5 \times 4 = 6$ cm.

Group B — Short answer (5 marks each)

1. Define a wavefront and state Huygens' principle. Using it, show how a plane wavefront and a spherical wavefront advance. [5]
2. Using Huygens' construction, prove the laws of reflection of light at a plane surface. [5]
3. Light of wavelength 600 nm in vacuum enters a medium of refractive index 1.25. Find its speed, frequency and wavelength in the medium. [5]
4. Explain why the frequency of light does not change on refraction while its wavelength does. Hence define optical path. [5]

ANSWERS TO GROUP B

1. Outline: wavefront = locus of points in the same phase; each point emits secondary wavelets of radius vt ; new wavefront = forward envelope. A plane wavefront gives a plane envelope, a spherical one a larger sphere (Figure 1).

3. $v = c/n = 3 \times 10^8 / 1.25 = 2.4 \times 10^8 \text{ m s}^{-1}$. Frequency is unchanged: $f = c/\lambda_0 = 3 \times 10^8 / 600 \times 10^{-9} = 5.0 \times 10^{14} \text{ Hz}$. Wavelength: $\lambda_m = v/f = 2.4 \times 10^8 / 5.0 \times 10^{14} = 4.8 \times 10^{-7} \text{ m} = 480 \text{ nm}$ (equivalently $600/1.25$).

4. Outline: the vibrations in medium 2 are driven by those arriving from medium 1, so they repeat at the same rate — f is fixed. Since $v = f\lambda$ and v falls to c/n , the wavelength falls to λ_0/n . Optical

path $= n \times$ geometrical path is the equivalent vacuum distance containing the same number of waves.

Group C — Long answer (8 marks each)

- (a) State Huygens' principle and explain the meaning of a secondary wavelet. [3] (b) Using Huygens' construction, derive Snell's law of refraction and hence show that the refractive index of a medium equals c/v . [5]
- A ray of light is incident at 45° on a parallel-sided glass slab of refractive index 1.5 and thickness 6.0 cm. Find (a) the angle of refraction, (b) the speed of light inside the slab, (c) the time taken to cross the slab, and (d) the lateral displacement of the emergent ray. [8]

ANSWER TO GROUP C QUESTION 2

(a) $\sin r = \sin 45^\circ / 1.5 = 0.7071 / 1.5 = 0.4714$, so $r = 28.1^\circ$.

(b) $v = c/n = 3 \times 10^8 / 1.5 = 2.0 \times 10^8 \text{ m s}^{-1}$.

(c) Path length inside the slab $= t / \cos r = 6.0 / 0.8819 = 6.80 \text{ cm}$, so

$$\text{time} = \frac{6.80 \times 10^{-2}}{2.0 \times 10^8} = 3.4 \times 10^{-10} \text{ s}$$

(d) Lateral displacement

$$d = \frac{t \sin(i - r)}{\cos r} = \frac{6.0 \times \sin 16.9^\circ}{0.8819} = \frac{6.0 \times 0.2902}{0.8819} = 1.97 \text{ cm}$$

UNIT 11

Interference

Waves and Optics • 2 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Phenomenon of Interference: Coherent sources
- Young's double slit experiment

Two torches pointed at the same wall never produce dark bands. Yet light from a single lamp, split into two and recombined, produces a ladder of bright and dark stripes. That is **interference**, and it was Thomas Young's demonstration of it in 1801 that finally proved light is a wave. This short unit sets out the conditions under which interference can be seen, and works through the one experiment the NEB asks about every year — Young's double slit.

THE TWO THINGS THIS UNIT IS WORTH

A derivation of the fringe width $\beta = \lambda D/d$ from the path difference, and numericals built on it. Get the path difference $\Delta = yd/D$ right and every question in the unit follows from it.

11.1 Phenomenon of Interference: Coherent sources

When two waves arrive at the same point, the **principle of superposition** says the resultant displacement is the vector sum of the individual displacements. For two waves of the same frequency and amplitudes a_1, a_2 arriving with a phase difference ϕ , the resultant amplitude is

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi}$$

Since intensity is proportional to the square of the amplitude, $I \propto A^2$,

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos\phi$$

The last term is the **interference term**. It is positive where the waves arrive in step and negative where they arrive out of step.

INTERFERENCE OF LIGHT

Interference is the modification of intensity produced by the superposition of two or more light waves. Where the resultant intensity is a maximum the interference is **constructive**; where it is a minimum it is **destructive**.

A path difference Δ corresponds to a phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta$$

so the conditions can be written either way:

	Path difference Δ	Phase difference ϕ	Resultant
Constructive (bright)	$n\lambda$	$2n\pi$	$A = a_1 + a_2, I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$
Destructive (dark)	$(2n - 1)\lambda/2$	$(2n - 1)\pi$	$A = a_1 - a_2 , I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$

with $n = 0, 1, 2, \dots$ For two equal sources of intensity I_0 each, $I_{\max} = 4I_0$ and $I_{\min} = 0$, and the general result is

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

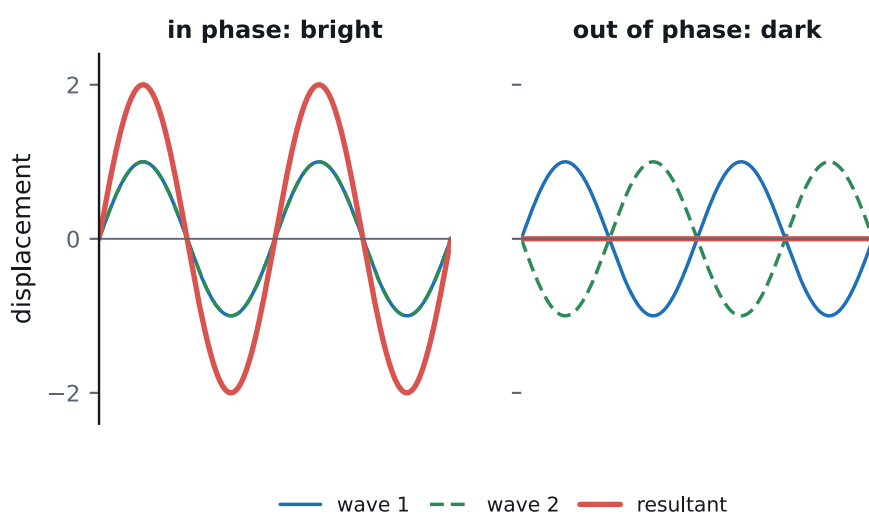


Figure 42. Superposition of two waves of equal amplitude. Left: in phase, the crests add and the amplitude doubles. Right: a phase difference of π (path difference $\lambda/2$) cancels them completely.

Coherent sources

The stripes only stay put if ϕ at each point stays the same from one instant to the next.

COHERENT SOURCES

Two sources are **coherent** if they emit light of the same frequency (and ideally the same amplitude) with a **constant phase difference** that does not change with time.

An ordinary lamp emits light in bursts lasting about 10^{-8} s, each burst starting with a random phase. Two separate lamps therefore have a phase difference that reshuffles about 10^8 times a second — far faster than the eye or any detector can follow — so $\cos \phi$ averages to zero and

$$I = I_1 + I_2$$

The intensities simply add and no fringes are seen. The way round this is to **derive both beams from a single source**, so that whatever phase jump occurs happens in both beams together and cancels out of ϕ . Two methods are used:

- **Division of wavefront** — two parts of the same wavefront are used: Young's double slit, Fresnel's biprism, Lloyd's mirror.
- **Division of amplitude** — one beam is partly reflected and partly transmitted: thin films, soap bubbles, the colours on a layer of oil on a Kathmandu street after rain, Newton's rings.

Lasers are naturally coherent over long distances, which is why a laser pointer makes the double-slit experiment easy to show in a school laboratory.

INTERFERENCE DOES NOT DESTROY ENERGY

At a dark fringe the intensity is zero, but energy is not lost — it has been **redistributed** to the bright fringes, where the intensity is $4I_0$ instead of the $2I_0$ you would get without interference. Averaged over one fringe spacing, the mean of $4I_0 \cos^2(\phi/2)$ is exactly $2I_0$, so energy is conserved.

11.2 Young's double slit experiment

Light from a monochromatic source falls on a narrow slit S . The cylindrical wavefront spreading from S reaches two more narrow slits S_1 and S_2 , placed symmetrically a small distance d apart. Because S_1 and S_2 are carved out of the **same** wavefront, they are coherent. They act as two line sources, and the overlapping light falls on a screen a distance D away.

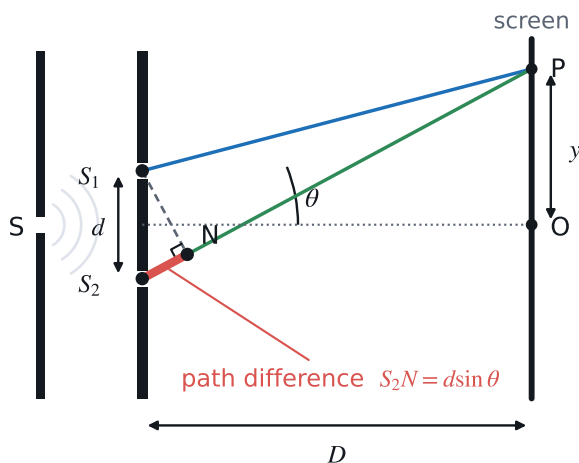


Figure 43. Young's double slit. S_1 and S_2 are coherent sources a distance d apart; the screen is at a distance D . For a point P at height y the wave from S_2 travels the extra distance $S_2N = d \sin \theta \approx yd/D$. The slit separation is greatly exaggerated.

Path difference

Take O as the foot of the perpendicular from the midpoint of S_1S_2 to the screen, and let P be a point at height y above O . From the right-angled triangles,

$$(S_2P)^2 = D^2 + \left(y + \frac{d}{2}\right)^2, \quad (S_1P)^2 = D^2 + \left(y - \frac{d}{2}\right)^2$$

Subtracting,

$$(S_2P)^2 - (S_1P)^2 = 2yd \implies (S_2P - S_1P)(S_2P + S_1P) = 2yd$$

In practice D is about a metre while y and d are under a millimetre, so both distances are very nearly D and $S_2P + S_1P \approx 2D$. Hence the path difference is

$$\Delta = S_2P - S_1P = \frac{yd}{D} \quad (= d \sin \theta \approx d\theta)$$

Positions of the fringes

Bright fringes need $\Delta = n\lambda$:

$$\frac{y_n d}{D} = n\lambda \implies y_n = \frac{n\lambda D}{d}, \quad n = 0, 1, 2, \dots$$

Dark fringes need $\Delta = (2n - 1)\lambda/2$:

$$y_n = \frac{(2n - 1)\lambda D}{2d}, \quad n = 1, 2, 3, \dots$$

FRINGE WIDTH

The **fringe width** β is the distance between two consecutive bright (or two consecutive dark) fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d}$$

The same subtraction on the dark-fringe formula gives the same answer, so bright and dark fringes are **equally spaced** and the pattern is a set of uniformly spaced, equally bright straight lines parallel to the slits. Rearranged, $\lambda = \beta d/D$ — this is how Young measured the wavelength of light.

The intensity across the screen follows from $\phi = 2\pi\Delta/\lambda$:

$$I = 4I_0 \cos^2 \left(\frac{\pi y d}{\lambda D} \right)$$

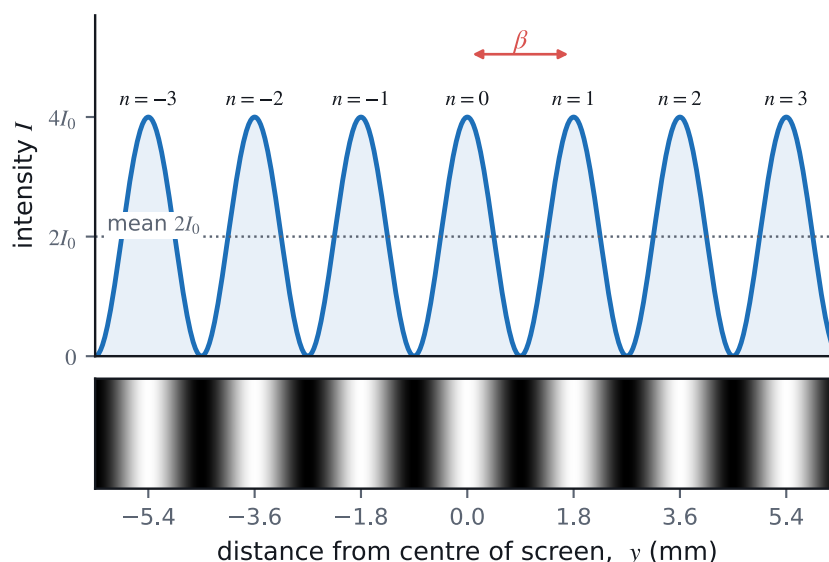


Figure 44. Intensity across the screen in Young's experiment, for $\lambda = 600 \text{ nm}$, $d = 0.5 \text{ mm}$, $D = 1.5 \text{ m}$. The fringes are evenly spaced with $\beta = \lambda D/d = 1.8 \text{ mm}$ and every maximum has the same height $4I_0$.

What changes the fringes

Change	Effect on $\beta = \lambda D/d$
Longer wavelength (red instead of blue)	β increases
Slits moved further apart (d up)	β decreases
Screen moved back (D up)	β increases
Whole apparatus in water, index n	$\lambda \rightarrow \lambda/n$, so $\beta \rightarrow \beta/n$
One slit covered	fringes vanish, uniform illumination
White light used	central fringe white, coloured fringes on either side, violet nearest the centre

WHY THE CENTRAL FRINGE IS WHITE

At O the path difference is zero **for every wavelength**, so all colours are bright there and the central fringe is white. Away from O, $\beta \propto \lambda$, so violet fringes crowd nearer the centre and red spread out further, giving a few coloured fringes before they overlap into white.

WORKED EXAMPLE 11.1

Problem. In a Young's double slit experiment the slits are 0.50 mm apart and the screen is 1.5 m away. Light of wavelength 600 nm is used. Find (a) the fringe width, (b) the distance of the 3rd bright fringe from the centre, (c) the distance of the 3rd dark fringe from the centre.

Solution.

(a) $\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1.5}{0.50 \times 10^{-3}} = 1.8 \times 10^{-3} \text{ m} = 1.8 \text{ mm}$.

(b) $y_3 = 3\beta = 3 \times 1.8 = 5.4 \text{ mm}$.

(c) The 3rd dark fringe has $\Delta = (2 \times 3 - 1)\lambda/2 = 2.5\lambda$, so $y = 2.5\beta = 4.5 \text{ mm}$.

WORKED EXAMPLE 11.2

Problem. In an interference experiment the distance between the 2nd and the 7th bright fringes on the same side of the centre is 9.0 mm. The slit separation is 0.50 mm and the screen is 1.5 m away. Find the wavelength of the light used.

Solution. Between the 2nd and 7th bright fringes there are $7 - 2 = 5$ fringe widths, so

$$5\beta = 9.0 \text{ mm} \implies \beta = 1.8 \text{ mm}$$

$$\lambda = \frac{\beta d}{D} = \frac{1.8 \times 10^{-3} \times 0.50 \times 10^{-3}}{1.5} = 6.0 \times 10^{-7} \text{ m} = 600 \text{ nm}$$

This is orange-red light.

WORKED EXAMPLE 11.3

Problem. The apparatus of Example 11.1 is now completely immersed in water of refractive index $4/3$. Find the new fringe width. What would the fringe width become if, in air, D were doubled and d halved?

Solution. Inside water the wavelength becomes $\lambda_w = \lambda/n$, so

$$\beta_w = \frac{\lambda_w D}{d} = \frac{\beta}{n} = \frac{1.8}{4/3} = 1.8 \times \frac{3}{4} = 1.35 \text{ mm}$$

With $D \rightarrow 2D$ and $d \rightarrow d/2$ in air, $\beta \rightarrow 4\beta = 7.2 \text{ mm}$.

WORKED EXAMPLE 11.4

Problem. Two coherent sources have intensities in the ratio 4 : 1. Find (a) the ratio of their amplitudes, (b) the ratio $I_{\max} : I_{\min}$. (c) For two **equal** sources, what fraction of $I_{\max}$ is the intensity at a point where the path difference is $\lambda/3$?

Solution.

(a) $I \propto a^2$, so $a_1/a_2 = \sqrt{4/1} = 2$.

(b) $\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(2+1)^2}{(2-1)^2} = \frac{9}{1}$.

(c) $\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$, so

$$I = 4I_0 \cos^2\left(\frac{\pi}{3}\right) = 4I_0(0.5)^2 = I_0 = \frac{I_{\max}}{4}$$

Chapter summary

- Superposition gives $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$, with $\phi = 2\pi\Delta/\lambda$.
- Bright when $\Delta = n\lambda$; dark when $\Delta = (2n - 1)\lambda/2$. For equal sources $I = 4I_0 \cos^2(\phi/2)$, with $I_{\max} = 4I_0$, $I_{\min} = 0$.
- $I_{\max} : I_{\min} = (\sqrt{I_1} + \sqrt{I_2})^2 : (\sqrt{I_1} - \sqrt{I_2})^2$.
- **Coherent sources** have the same frequency and a constant phase difference; they must be obtained from one source, by division of wavefront or of amplitude.
- In Young's experiment $\Delta = yd/D$, bright fringes at $y_n = n\lambda D/d$, dark at $y_n = (2n - 1)\lambda D/2d$.
- Fringe width $\beta = \lambda D/d$; all fringes are equally spaced and equally bright. In a medium of index n , β shrinks to β/n .
- Energy is redistributed, not destroyed: the average intensity across the pattern is $2I_0$, exactly what the two beams carry.

Practice questions

Group A — Multiple choice (1 mark each)

- Two sources are coherent if they have the same frequency and [1] (a) a constant phase difference (b) zero phase difference (c) the same amplitude only (d) the same intensity only
- In Young's double slit experiment the fringe width is [1] (a) $\lambda d/D$ (b) $\lambda D/d$ (c) Dd/λ (d) λ/Dd
- At a point where the path difference is $\lambda/2$, the intensity due to two equal coherent sources of intensity I_0 each is [1] (a) $4I_0$ (b) $2I_0$ (c) I_0 (d) zero
- If the whole double-slit apparatus is immersed in water, the fringe width [1] (a) increases (b) decreases (c) is unchanged (d) becomes zero
- In white-light interference, the central fringe is [1] (a) violet (b) red (c) white (d) dark

ANSWERS TO GROUP A

- 1.** (a) — the phase difference must be constant in time; it need not be zero. **2.** (b) — $\beta = \lambda D/d$. **3.** (d) — $\Delta = \lambda/2$ gives $\phi = \pi$ and $I = 4I_0 \cos^2(\pi/2) = 0$. **4.** (b) — the wavelength shrinks to λ/n , so β shrinks to β/n . **5.** (c) — zero path difference at the centre for every wavelength, so all colours are bright.

Group B — Short answer (5 marks each)

- What are coherent sources? Explain why two independent bulbs of the same colour cannot produce a steady interference pattern. [5]
- Derive an expression for the fringe width in Young's double slit experiment. [5]
- In a double slit experiment $d = 0.28$ mm, $D = 1.4$ m and $\lambda = 600$ nm. Find the fringe width and the distance of the 4th dark fringe from the central bright fringe. [5]
- Show that interference of light does not violate the principle of conservation of energy. [5]

ANSWERS TO GROUP B

- 1.** Outline: coherent = same frequency, constant phase difference. An ordinary source emits wave trains of about 10^{-8} s with random phase, so the phase difference between two independent lamps changes about 10^8 times per second; $\cos \phi$ averages to zero, giving $I = I_1 + I_2$ and no fringes.

3. $\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1.4}{0.28 \times 10^{-3}} = 3.0 \times 10^{-3} \text{ m} = 3.0 \text{ mm}$. The 4th dark fringe is at $y = \frac{(2 \times 4 - 1)\lambda D}{2d} = 3.5\beta = 10.5 \text{ mm}$.

4. Outline: $I = 4I_0 \cos^2(\phi/2)$; the average of $\cos^2$ over a fringe is $\frac{1}{2}$, so the mean intensity is $2I_0$ — exactly the sum of the two beams. The energy missing from the dark fringes reappears in the bright ones.

Group C — Long answer (8 marks each)

- (a) Describe Young's double slit experiment with a labelled diagram. [3] (b) Derive expressions for the positions of the bright and dark fringes and hence for the fringe width. State two ways in which the fringe width can be increased. [5]
- In a Young's double slit experiment, light of wavelength 589 nm falls on two slits 1.0 mm apart and the fringes are observed on a screen 1.0 m away. Find (a) the fringe width, (b) the distance between the 3rd bright fringe on one side and the 3rd dark fringe on the other side of the centre, and (c) the new fringe width if the source is changed to one of wavelength 442 nm. [8]

ANSWER TO GROUP C QUESTION 2

(a) $\beta = \frac{\lambda D}{d} = \frac{589 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}} = 5.89 \times 10^{-4} \text{ m} = 0.589 \text{ mm}$.

(b) 3rd bright fringe: $y_1 = 3\beta = 1.767 \text{ mm}$ on one side. 3rd dark fringe: $y_2 = 2.5\beta = 1.4725 \text{ mm}$ on the other side. Since they are on opposite sides, the separation is $1.767 + 1.4725 = 3.24 \text{ mm}$.

(c) $\beta' = \frac{442 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}} = 0.442 \text{ mm}$ — blue light gives narrower fringes.

UNIT 12

Diffraction

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Diffraction from a single slit
- Diffraction pattern of image; Diffraction grating
- Resolving power of optical instruments

Stand inside a room in Bhaktapur with the shutters almost closed and the sunlight squeezing through does not stop at a sharp edge — it fans out. Light **bends round obstacles and spreads after passing through narrow openings**. This is **diffraction**, and it is the direct consequence of Huygens' secondary wavelets interfering with one another. It sets a hard limit on what any telescope or microscope can see, and it is the working principle of the diffraction grating, the most accurate instrument we have for measuring wavelength.

THE ONE INEQUALITY BEHIND THE WHOLE UNIT

Diffraction is only noticeable when the width of the obstacle or aperture is **comparable with the wavelength**. Sound ($\lambda \approx 1$ m) bends round a doorway easily; light ($\lambda \approx 5 \times 10^{-7}$ m) needs a slit a fraction of a millimetre wide before the spreading can be seen.

12.1 Diffraction from a single slit

DIFFRACTION

Diffraction is the bending of waves round the edges of an obstacle or aperture, and the resulting redistribution of intensity, when the size of the obstacle or aperture is comparable with the wavelength of the wave.

Two classes are distinguished:

	Fresnel diffraction	Fraunhofer diffraction
Source and screen	at finite distance from the slit	effectively at infinity
Incident wavefront	spherical or cylindrical	plane
Lenses needed	none	collimating lens and focusing lens
Mathematics	difficult	simple

Everything in this unit is **Fraunhofer** diffraction: parallel light in, parallel light out, brought to a focus by a converging lens.

Condition for the minima

Let a plane wavefront of wavelength λ fall normally on a slit AB of width a . By Huygens' principle every point of the slit sends out secondary wavelets. Consider those that leave at an angle θ to the original direction; the lens brings them to a single point P on the screen. The wavelet from the bottom edge B travels farther than the one from the top edge A by

$$\Delta = a \sin \theta$$

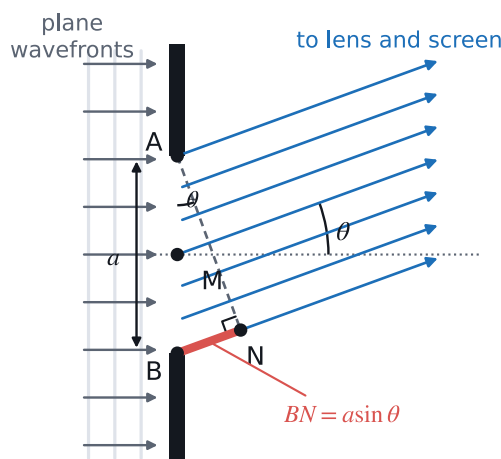


Figure 45. Fraunhofer diffraction at a single slit of width a . All wavelets leaving at angle θ meet at one point of the screen. The path difference between the edges is $BN = a \sin \theta$; when this equals λ the slit splits into two halves that cancel in pairs.

POSITIONS OF THE MINIMA

Imagine the slit divided into **two halves**, AM and MB, each of width $a/2$. Pair off a point in the top half with the point exactly $a/2$ below it in the bottom half. The path difference for such a pair is $\frac{a}{2} \sin \theta$.

Every pair cancels if that equals half a wavelength:

$$\frac{a}{2} \sin \theta = \frac{\lambda}{2} \implies a \sin \theta = \lambda$$

Every wavelet from the top half is then cancelled by its partner from the bottom half, and P is **dark**. Dividing the slit into four equal strips instead gives $a \sin \theta = 2\lambda$, into six strips $a \sin \theta = 3\lambda$, and so on. The general condition for a **minimum** is

$$a \sin \theta = n\lambda, \quad n = \pm 1, \pm 2, \pm 3, \dots$$

Note that $n = 0$ is excluded: at $\theta = 0$ every wavelet arrives in phase and the centre is the brightest point of the pattern.

The **secondary maxima** lie roughly halfway between the minima, at

$$a \sin \theta \approx \left(n + \frac{1}{2}\right)\lambda$$

and they are very weak — the first is only about 4.7% of the central intensity, the second about 1.7%. The full intensity distribution is

$$I = I_0 \left(\frac{\sin \beta}{\beta} \right)^2, \quad \beta = \frac{\pi a \sin \theta}{\lambda}$$

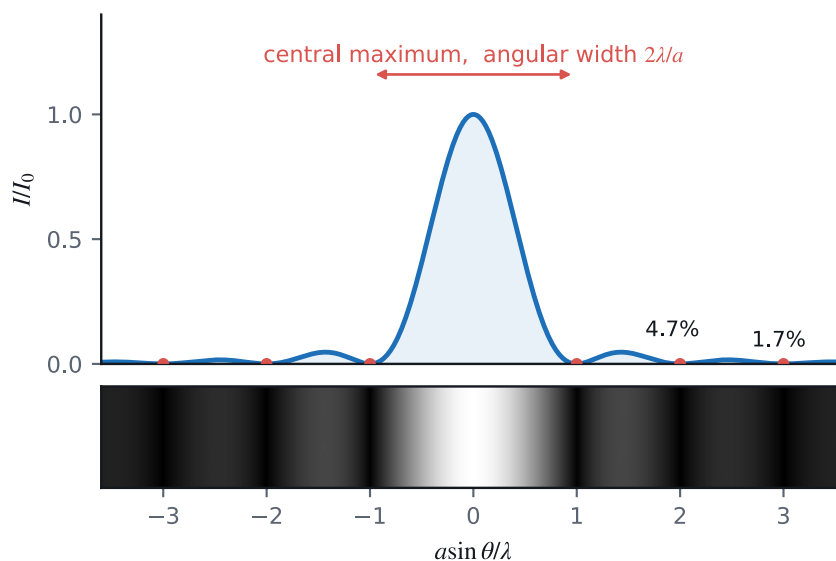


Figure 46. Intensity in single-slit diffraction against $a \sin \theta / \lambda$. Minima fall at $\pm 1, \pm 2, \pm 3 \dots$; the central maximum is twice as wide as the others and carries most of the light.

Width of the central maximum

The first minima lie at $\sin \theta = \pm \lambda / a$. For small angles $\sin \theta \approx \theta$, so the **angular half-width** of the central maximum is λ / a and its **angular width** is

$$2\theta_1 = \frac{2\lambda}{a}$$

On a screen a distance D from the slit (or in the focal plane of a lens of focal length f), the linear width of the central bright band is

$$w = \frac{2\lambda D}{a} \quad \text{or} \quad w = \frac{2\lambda f}{a}$$

NARROWER SLIT, WIDER PATCH

Students expect a narrower slit to give a narrower patch of light. The opposite is true: $w \propto 1/a$, so **halving the slit width doubles the spread**. Only when $a \gg \lambda$ does the patch shrink to the geometrical shadow of the slit, which is why ray optics works for everyday apertures.

WORKED EXAMPLE 12.1

Problem. A slit 0.20 mm wide is illuminated normally by light of wavelength 600 nm. The diffraction pattern is observed on a screen 1.5 m away. Find (a) the angular position of the first minimum, (b) the linear width of the central maximum, (c) the distance of the second minimum from the centre.

Solution.

$$(a) \sin \theta_1 = \frac{\lambda}{a} = \frac{600 \times 10^{-9}}{0.20 \times 10^{-3}} = 3.0 \times 10^{-3}, \text{ so } \theta_1 = 3.0 \times 10^{-3} \text{ rad} = 0.17^\circ.$$

$$(b) w = \frac{2\lambda D}{a} = 2 \times 3.0 \times 10^{-3} \times 1.5 = 9.0 \times 10^{-3} \text{ m} = 9.0 \text{ mm}.$$

$$(c) \sin \theta_2 = 2\lambda/a = 6.0 \times 10^{-3}, \text{ so } y_2 = D \tan \theta_2 \approx 1.5 \times 6.0 \times 10^{-3} = 9.0 \text{ mm}.$$

12.2 Diffraction pattern of image; Diffraction grating

The image of a slit

Geometrical optics says the image of an illuminated slit is a sharp bright rectangle. What is actually seen in the focal plane of the lens is a **broad central bright band** flanked by progressively fainter bands separated by dark lines — the pattern of the previous figure. Every optical image is really a diffraction pattern; a "point" image is never a point.

Interference and diffraction are both superposition effects, but the NEB expects you to be able to separate them:

Interference (Young)	Diffraction (single slit)
Superposition of waves from two separate coherent sources	Superposition of secondary wavelets from different parts of the same wavefront
Fringes of equal width	Central band twice as wide as the rest
All bright fringes of equal intensity	Intensity falls off rapidly away from the centre
Dark fringes are perfectly dark	Minima are dark but the pattern never fully repeats
Bright: $\Delta = n\lambda$	Dark: $a \sin \theta = n\lambda$

Diffraction grating

A **diffraction grating** is a glass or plastic plate ruled with a very large number of equidistant parallel lines; the ruled lines are opaque and the gaps between them act as slits. If there are N lines per metre, the distance from the centre of one slit to the centre of the next is the **grating element**

$$d = \frac{1}{N} = e + b$$

where e is the width of a transparent slit and b that of an opaque ruling. A good grating has 5000–15000 lines per centimetre, so d is a couple of micrometres.

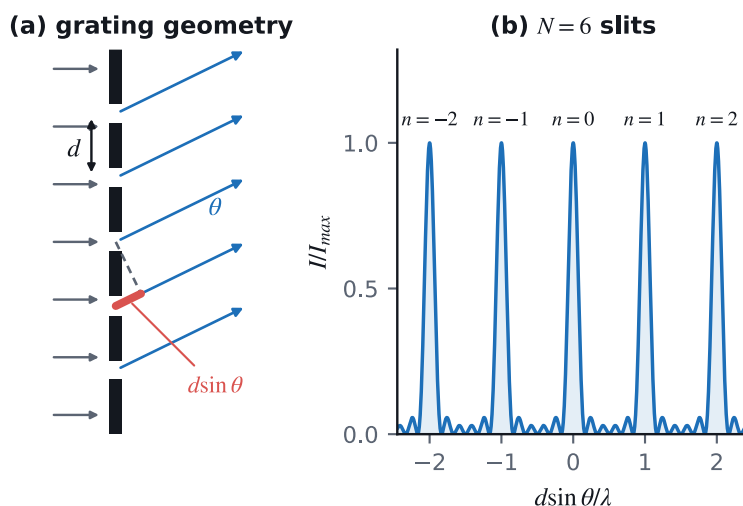


Figure 47. (a) Plane waves on a grating of element d ; adjacent slits contribute a path difference $d \sin \theta$. (b) Intensity from $N = 6$ slits: the principal maxima at $d \sin \theta = n\lambda$ are far sharper than two-slit fringes.

THE GRATING EQUATION

Plane monochromatic light falls normally on the grating. Take the parallel rays that leave every slit at an angle θ ; a lens brings them to one point. The path difference between the wavelets from any two **adjacent** slits is the same, namely $d \sin \theta$.

All N slits then reinforce one another only when this common path difference is a whole number of wavelengths:

$$d \sin \theta = n\lambda, \quad n = 0, 1, 2, 3, \dots$$

This is the **grating equation**, and n is the order of the spectrum. Because N is huge (tens of thousands), the slightest departure from the condition throws the contributions of distant slits out of step, so the maxima are extremely sharp — which is what makes a grating a precision wavelength-measuring instrument.

Two consequences follow:

- Since $\sin \theta \leq 1$, the **highest order** obtainable is $n_{\max} = d/\lambda$ (rounded down to a whole number).
- For a given order, $\sin \theta \propto \lambda$, so white light is spread into a spectrum with **violet closest to the centre and red furthest out** — the opposite way round from a prism, where red is deviated least.

WORKED EXAMPLE 12.2

Problem. A plane diffraction grating has 5000 lines per centimetre. Light of wavelength 500 nm falls normally on it. Find the angles of the first, second and third order maxima, and the highest order that can be observed.

Solution. The grating element is

$$d = \frac{1 \text{ cm}}{5000} = \frac{10^{-2}}{5000} = 2.0 \times 10^{-6} \text{ m}$$

From $d \sin \theta = n\lambda$, $\sin \theta = n\lambda/d = n(0.25)$.

- $n = 1$: $\sin \theta = 0.25$, $\theta_1 = 14.5^\circ$
- $n = 2$: $\sin \theta = 0.50$, $\theta_2 = 30.0^\circ$
- $n = 3$: $\sin \theta = 0.75$, $\theta_3 = 48.6^\circ$

The highest order needs $\sin \theta \leq 1$, i.e. $n \leq d/\lambda = 4$. But $n = 4$ gives $\sin \theta = 1$, i.e. $\theta = 90^\circ$ — light grazing along the grating, which cannot be collected. So only **three** orders are actually seen on each side of the centre.

12.3 Resolving power of optical instruments

Because every image is a diffraction patch, two nearby stars produce two overlapping patches. If the patches overlap too much they merge into one and the stars cannot be told apart. Lord Rayleigh gave the practical dividing line.

RAYLEIGH CRITERION

Two point objects are said to be **just resolved** when the central maximum of the diffraction pattern of one falls exactly on the first minimum of the pattern of the other.

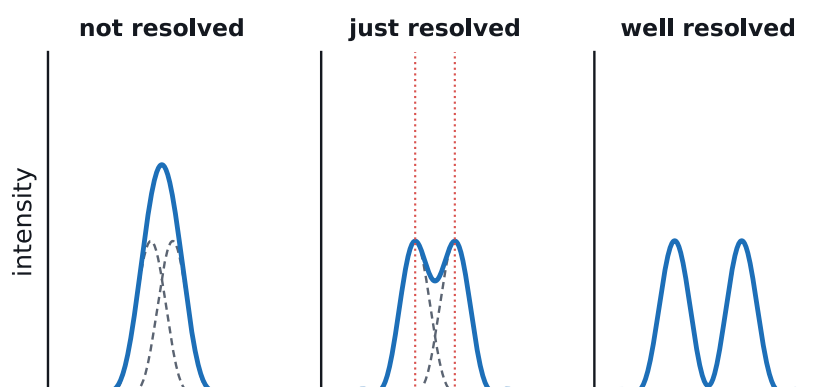


Figure 48. Rayleigh's criterion for a circular aperture. The dashed curves are the two individual diffraction patterns and the solid curve is what the eye actually receives. At the Rayleigh separation the combined curve has a dip of about 26% and the pair is just resolved.

Telescope

For a circular aperture of diameter a the first dark ring of the diffraction pattern lies at an angle $1.22\lambda/a$ (the factor 1.22 comes from the circular shape; for a slit it would be 1). So two stars are just resolved when their angular separation is

$$\theta_{\min} = \frac{1.22\lambda}{a}$$

and the **resolving power** of the telescope, defined as the reciprocal of the limit of resolution, is

$$\text{R.P.} = \frac{1}{\theta_{\min}} = \frac{a}{1.22\lambda}$$

A bigger objective therefore resolves better and collects more light — which is why research telescopes are built with mirrors metres across.

Microscope

For a microscope the useful quantity is the smallest separation x between two points in the object that can still be distinguished:

$$x_{\min} = \frac{1.22\lambda}{2\mu\sin\theta}, \quad \text{R.P.} = \frac{1}{x_{\min}} = \frac{2\mu\sin\theta}{1.22\lambda}$$

Here θ is the half-angle of the cone of light entering the objective and μ is the refractive index of the medium between object and objective. The product $\mu\sin\theta$ is the **numerical aperture**. Two ways of improving a microscope follow at once: use shorter wavelengths (blue light, or electrons in an electron microscope), and raise μ by putting a drop of cedar oil ($\mu \approx 1.5$) between the slide and the objective — oil immersion.

For a **grating** the corresponding quantity is the ability to separate two close wavelengths, and it works out to $\lambda/d\lambda = nN$, where N is the total number of lines actually illuminated and n is the order.

WORKED EXAMPLE 12.3

Problem. A telescope objective has a diameter of 100 mm. Taking the mean wavelength of light as 550 nm, find (a) its limit of resolution and (b) the smallest distance between two objects on the Moon, 3.84×10^8 m away, that this telescope could just resolve.

Solution.

(a) $\theta_{\min} = \frac{1.22\lambda}{a} = \frac{1.22 \times 550 \times 10^{-9}}{0.100} = 6.71 \times 10^{-6} \text{ rad}$

(b) For small angles the linear separation is $x = r\theta_{\min}$:

$$x = 3.84 \times 10^8 \times 6.71 \times 10^{-6} = 2.58 \times 10^3 \text{ m}$$

so about 2.6 km — a small telescope cannot show anything on the Moon smaller than a large town.

WORKED EXAMPLE 12.4

Problem. An oil-immersion microscope objective works with light of wavelength 500 nm, a half-angle of 60° and oil of refractive index 1.5. Find the smallest separation it can resolve.

Solution. The numerical aperture is $\mu\sin\theta = 1.5 \times \sin 60^\circ = 1.5 \times 0.866 = 1.299$.

$$x_{\min} = \frac{1.22\lambda}{2\mu\sin\theta} = \frac{1.22 \times 500 \times 10^{-9}}{2 \times 1.299} = \frac{6.10 \times 10^{-7}}{2.598} = 2.35 \times 10^{-7} \text{ m}$$

That is 235 nm — smaller than a bacterium, but far larger than a virus, which is why viruses need an electron microscope.

Chapter summary

- **Diffraction** is the bending of light round edges, noticeable only when the aperture is comparable with λ . Fraunhofer diffraction uses plane waves.

- Single slit **minima**: $a \sin \theta = n\lambda$ for $n = \pm 1, \pm 2, \dots$; secondary maxima near $a \sin \theta = (n + \frac{1}{2})\lambda$, of relative intensity 4.7%, 1.7%, ...
- Intensity: $I = I_0(\sin \beta / \beta)^2$ with $\beta = \pi a \sin \theta / \lambda$.
- Central maximum: angular width $2\lambda/a$, linear width $2\lambda D/a$ — it gets **wider** as the slit gets narrower.
- Grating element $d = 1/N$; **grating equation** $d \sin \theta = n\lambda$; highest order $n_{\max} = d/\lambda$; in each order violet is deviated least.
- **Rayleigh criterion**: just resolved when one central maximum sits on the other's first minimum.
- Telescope: $\theta_{\min} = 1.22\lambda/a$, R.P. = $a/1.22\lambda$. Microscope: $x_{\min} = 1.22\lambda/2\mu \sin \theta$. Grating: $\lambda/d\lambda = nN$.

Practice questions

Group A — Multiple choice (1 mark each)

1. In Fraunhofer diffraction at a single slit, the condition for the n th minimum is [1] (a) $a \sin \theta = n\lambda$ (b) $a \sin \theta = (2n + 1)\lambda/2$ (c) $d \sin \theta = n\lambda$ (d) $a \cos \theta = n\lambda$
2. The width of the central maximum in single-slit diffraction [1] (a) increases with slit width (b) decreases with slit width (c) does not depend on the slit width (d) is always zero
3. A grating has 2000 lines per centimetre. Its grating element is [1] (a) $2 \mu\text{m}$ (b) $5 \mu\text{m}$ (c) $0.5 \mu\text{m}$ (d) $20 \mu\text{m}$
4. The resolving power of a telescope can be increased by [1] (a) increasing the wavelength (b) increasing the aperture (c) decreasing the aperture (d) increasing the eyepiece magnification
5. Diffraction is most easily observed when the size of the aperture is [1] (a) much larger than λ (b) comparable with λ (c) exactly zero (d) infinite

ANSWERS TO GROUP A

1. (a) — pairing the two halves of the slit gives $a \sin \theta = n\lambda$ for **darkness**. **2.** (b) — $w = 2\lambda D/a$, so w and a are inversely related. **3.** (b) — $d = 1 \text{ cm}/2000 = 5 \times 10^{-4} \text{ cm} = 5 \mu\text{m}$. **4.** (b) — R.P. = $a/1.22\lambda$, so a larger objective resolves better. **5.** (b) — bending is appreciable only when the aperture is of the order of the wavelength.

Group B — Short answer (5 marks each)

1. What is diffraction of light? Distinguish between interference and diffraction. [5]
2. Derive the condition for minima in Fraunhofer diffraction at a single slit and hence obtain the width of the central maximum. [5]
3. Light of wavelength 500 nm falls normally on a slit of width 0.10 mm . The pattern is seen on a screen 1.0 m away. Find the distance between the first minima on the two sides of the centre. [5]
4. What is a diffraction grating? Obtain the grating equation and explain why the maxima are sharp. [5]

ANSWERS TO GROUP B

1. Outline: diffraction = bending of light at an aperture comparable with λ , caused by superposition of secondary wavelets from one wavefront; interference involves two separate coherent

sources. Compare fringe width, intensity of successive maxima and the darkness of the minima (see the table in §12.2).

3. $\sin \theta_1 = \frac{\lambda}{a} = \frac{500 \times 10^{-9}}{1.0 \times 10^{-4}} = 5.0 \times 10^{-3}$. Distance of one first minimum from the centre is $y_1 = D\theta_1 = 1.0 \times 5.0 \times 10^{-3} = 5.0 \text{ mm}$, so the separation of the two first minima (the width of the central maximum) is $2y_1 = 10 \text{ mm} = 1.0 \text{ cm}$.

Group C — Long answer (8 marks each)

- (a) Explain Fraunhofer diffraction at a single slit and derive the positions of the minima. [5] (b) Sketch the intensity distribution and state the Rayleigh criterion for resolution. [3]
- A diffraction grating has 6000 lines per centimetre and is illuminated normally by light of wavelength 600 nm. Find (a) the grating element, (b) the angles of the first and second order maxima, (c) the highest order that can be seen, and (d) the smallest wavelength difference the grating can resolve in the second order if 8000 lines are illuminated. [8]

ANSWER TO GROUP C QUESTION 2

(a) $d = \frac{10^{-2}}{6000} = 1.667 \times 10^{-6} \text{ m} = 1.667 \text{ } \mu\text{m}$.

(b) $\sin \theta = n\lambda/d$ with $\lambda/d = 600 \times 10^{-9}/1.667 \times 10^{-6} = 0.36$. For $n = 1$: $\sin \theta_1 = 0.36$, $\theta_1 = 21.1^\circ$. For $n = 2$: $\sin \theta_2 = 0.72$, $\theta_2 = 46.1^\circ$.

(c) $n_{\max} = d/\lambda = 1/0.36 = 2.78$, so the highest observable order is $n = 2$.

(d) $\frac{d\lambda}{d\lambda} = nN = 2 \times 8000 = 16000$, so

$$d\lambda = \frac{600}{16000} = 0.0375 \text{ nm}$$

UNIT 13

Polarization

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Phenomenon of polarization
- Brewster's law; transverse nature of light
- Polaroid

Interference and diffraction prove that light is a **wave**, but they cannot say whether it is a longitudinal wave like sound or a transverse one. Polarization settles that question. Only a transverse wave can be filtered so that its vibrations survive in one direction and not another, and light can. This unit explains what polarized light is, how it is produced, and the two formulae the NEB asks for: Brewster's law and Malus's law.

WHAT POLARIZATION TELLS US

Sound cannot be polarized; light can. That single experimental fact proves that light vibrations are **perpendicular to the direction of travel** — light is a transverse wave. In an electromagnetic wave it is the electric field vector that we mean by "the vibration".

13.1 Phenomenon of polarization

In a light wave from an ordinary source — a lamp, the Sun, a burning stick — the electric field vibrates in a direction at right angles to the direction of travel, but that direction changes randomly millions of times a second as one atom after another emits its short wave train. Such light is **unpolarized**.

POLARIZATION OF LIGHT

Polarization is the process of confining the vibrations of the electric vector of a light wave to a single plane. Light in which the vibrations are confined to one plane is called **plane polarized** (or linearly polarized) light.

Two planes are named, and students regularly confuse them:

Plane	Definition
Plane of vibration	the plane containing the direction of vibration and the direction of propagation
Plane of polarization	the plane through the direction of propagation and perpendicular to the plane of vibration

The two planes are always at 90° to each other. No vibration takes place in the plane of polarization.

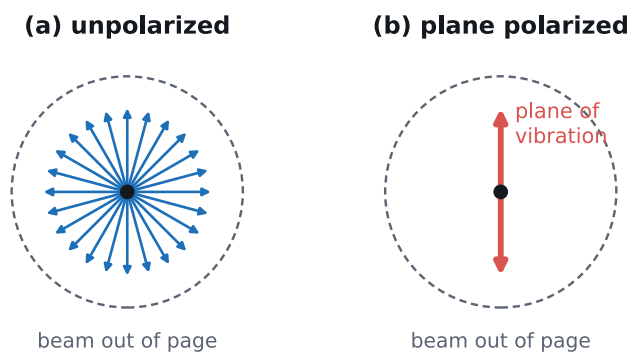


Figure 49. End-on view, looking straight along the beam. (a) Unpolarized light: the electric vector takes every direction in the plane of the page, at random. (b) Plane polarized light: the vibration is confined to one direction.

In diagrams drawn side-on, vibrations **in** the plane of the page are shown as short double-headed arrows ($\updownarrow$) and vibrations **perpendicular** to the page as dots ($\bullet$). Unpolarized light is drawn with both, plane polarized light with only one of the two.

Ways of producing polarized light

Method	How it works	Example
Reflection	at the polarizing angle the reflected light is fully plane polarized	glare from a wet road or from Phewa Lake
Refraction (pile of plates)	the transmitted beam becomes progressively polarized after many glass plates	pile-of-plates polarizer
Selective absorption (dichroism)	one component is absorbed, the other transmitted	Polaroid sheet
Double refraction	a calcite crystal splits the beam into two oppositely polarized rays	Nicol prism
Scattering	light scattered at 90° from air molecules is plane polarized	the blue sky at right angles to the Sun

Double refraction. When unpolarized light enters a calcite (Iceland spar) or quartz crystal it splits into two refracted rays: the **ordinary ray**, which obeys Snell's law, and the **extraordinary ray**, which does not. The two rays are plane polarized in mutually perpendicular planes. A Nicol prism is a calcite crystal cut and cemented with Canada balsam so that the ordinary ray is totally internally reflected away and only the plane polarized extraordinary ray emerges.

Partial polarization. In most of these processes the emerging light is a mixture. The **degree of polarization** is

$$P = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

where $I_{\max}$ and $I_{\min}$ are the largest and smallest intensities observed as an analyser is rotated through a full turn. $P = 1$ for fully plane polarized light and $P = 0$ for unpolarized light.

13.2 Brewster's law; transverse nature of light

When unpolarized light strikes a transparent surface, both the reflected and the refracted beams are **partially** polarized. In 1811 David Brewster found that at one particular angle of incidence the reflected beam is **completely** plane polarized, with its vibrations perpendicular to the plane of incidence. That angle is the **polarizing angle** or **Brewster angle**, θ_p .

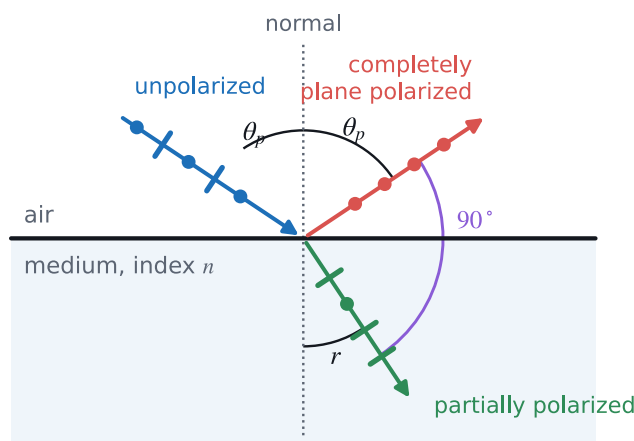


Figure 50. Polarization by reflection. At the polarizing angle θ_p the reflected beam is completely plane polarized (dots = vibrations perpendicular to the page) and the reflected and refracted beams are exactly 90° apart, so $\theta_p + r = 90^\circ$ and $n = \tan \theta_p$.

BREWSTER'S LAW

At the polarizing angle the reflected and refracted rays are found experimentally to be at right angles to each other. From the figure the three angles at O must add to a straight angle:

$$\theta_p + 90^\circ + r = 180^\circ \implies r = 90^\circ - \theta_p$$

Apply Snell's law for light going from air into the medium:

$$n = \frac{\sin \theta_p}{\sin r} = \frac{\sin \theta_p}{\sin(90^\circ - \theta_p)} = \frac{\sin \theta_p}{\cos \theta_p}$$

$$n = \tan \theta_p$$

This is **Brewster's law**: the tangent of the polarizing angle of a medium is equal to its refractive index.

The physical reason is neat. The refracted wave sets the electrons of the medium vibrating along its own electric vector, at right angles to the refracted ray. A vibrating charge radiates nothing **along** its own line of vibration. At the Brewster angle the reflected direction is exactly along that line for the component in the plane of incidence, so that component simply cannot be reflected. Only the component perpendicular to the plane of incidence survives — and the reflected light is fully plane polarized.

Because n varies slightly with wavelength (dispersion), θ_p is not quite the same for every colour: for crown glass it is about 56.3° for yellow light and a fraction of a degree larger for violet.

BREWSTER'S ANGLE IS NOT THE CRITICAL ANGLE

$\theta_p = \tan^{-1} n$ applies when light goes from a **rarer** medium into a denser one, and the reflected beam is the polarized one. The critical angle $C = \sin^{-1} (1/n)$ applies going the other way, from denser to rarer. For $n = 1.5$, $\theta_p = 56.3^\circ$ but $C = 41.8^\circ$ — different angles, different phenomena.

Transverse nature of light

Take two Polaroid sheets, P and A. Send light through P; the beam that emerges looks slightly dimmer but otherwise normal. Now rotate A about the beam as axis. The transmitted intensity rises and falls, and twice in every complete turn it drops to **zero**.

If light were a longitudinal wave, its vibrations would be along the direction of travel, and a rotation of A about that same axis could make no difference whatsoever — the wave has no "sideways" direction to be picky about. The fact that rotating A extinguishes the beam shows the vibrations must lie **across** the beam. Polarization is therefore the only optical phenomenon that establishes the **transverse** nature of light waves.

WORKED EXAMPLE 13.1

Problem. Find the polarizing angle for a glass of refractive index 1.5, and the corresponding angle of refraction. Verify that the reflected and refracted rays are perpendicular.

Solution. By Brewster's law,

$$\tan \theta_p = n = 1.5 \implies \theta_p = \tan^{-1} (1.5) = 56.3^\circ$$

$$r = 90^\circ - \theta_p = 90^\circ - 56.3^\circ = 33.7^\circ$$

Check with Snell's law: $\frac{\sin 56.3^\circ}{\sin 33.7^\circ} = \frac{0.832}{0.555} = 1.50$ — correct. The angle between the reflected ray and the refracted ray is $180^\circ - 56.3^\circ - 33.7^\circ = 90^\circ$.

13.3 Polaroid

A **Polaroid** is a thin commercial sheet that transmits light vibrating in one direction only. Edwin Land made the first one in 1928 by embedding tiny aligned needle-shaped crystals of **herapathite** (quinine iodosulphate) in nitrocellulose — the J-type sheet. Modern **H-Polaroid** is made from a sheet of polyvinyl alcohol that is stretched so that its long molecules all lie parallel, then soaked in iodine. The iodine-doped chains conduct along their length, so they absorb the component of the electric field parallel to the chains and transmit the perpendicular component. The direction that is passed is called the **transmission axis** or pass axis.

When two Polaroids are used, the first is the **polarizer** and the second the **analyser**.

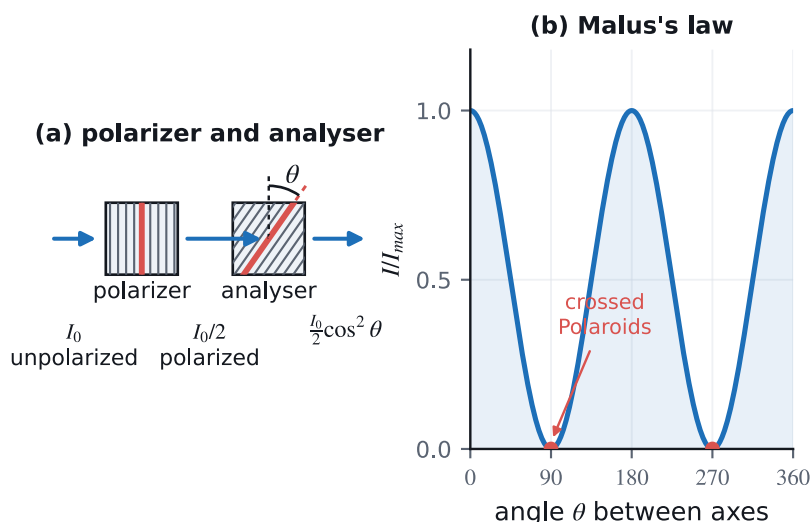


Figure 51. (a) Unpolarized light of intensity I_0 passes a polarizer (axis vertical) and emerges plane polarized with intensity $I_0/2$; the analyser, whose axis makes an angle θ with it, transmits $(I_0/2)\cos^2 \theta$. (b) Malus's law: transmitted intensity against the angle between the two axes.

MALUS'S LAW

Let plane polarized light of amplitude A fall on an analyser whose transmission axis makes an angle θ with the plane of vibration of the incoming light. Resolve A into two components:

- $A\cos\theta$ **along** the transmission axis — this is transmitted;
- $A\sin\theta$ **perpendicular** to it — this is absorbed.

Intensity is proportional to the square of the amplitude, so if I_0 is the intensity of the light reaching the analyser,

$$I = k(A\cos\theta)^2 = kA^2\cos^2\theta$$

$$I = I_0\cos^2\theta$$

This is **Malus's law**. It gives $I = I_0$ for parallel axes ($\theta = 0^\circ$) and $I = 0$ for **crossed** axes ($\theta = 90^\circ$).

THE FIRST POLAROID ALWAYS HALVES THE INTENSITY

Malus's law applies to light that is **already** plane polarized. When unpolarized light of intensity I_0 falls on the first Polaroid, θ takes every value at random and the average of $\cos^2\theta$ is $\frac{1}{2}$, so the emergent intensity is $I_0/2$ **whatever the orientation of the sheet**. Only then does $\cos^2\theta$ apply, at the second sheet.

Uses of Polaroids

- **Sunglasses.** Glare reflected from water or a wet road is largely horizontally polarized, so the lenses are cut with a vertical transmission axis and the glare is removed.
- **Photography.** A rotatable polarizing filter darkens a blue sky and kills reflections from glass and water.

- **LCD screens** in calculators, watches, laptops and mobile phones work by rotating polarized light between two crossed Polaroids.
- **3-D cinema**, where the two eyes receive images polarized at right angles.
- **Photoelastic stress analysis**: a plastic model of a bridge or a machine part placed between crossed Polaroids shows coloured fringes wherever it is strained.
- **Controlling light intensity** smoothly in optical instruments and in laboratory work, using two Polaroids in series.

WORKED EXAMPLE 13.2

Problem. Unpolarized light of intensity I_0 falls on a polarizer. An analyser is placed behind it with its axis at 60° to that of the polarizer. Find the intensity of the emergent light.

Solution. After the polarizer the light is plane polarized with intensity

$$I_1 = \frac{I_0}{2}$$

By Malus's law, the analyser transmits

$$I_2 = I_1 \cos^2 60^\circ = \frac{I_0}{2} \left(\frac{1}{2}\right)^2 = \frac{I_0}{8} = 0.125 I_0$$

WORKED EXAMPLE 13.3

Problem. Three Polaroids are placed one behind the other. The first and the third are crossed. The middle one makes an angle of 30° with the first. Unpolarized light of intensity I_0 enters. Find the intensity emerging from the third sheet.

Solution. After sheet 1: $I_1 = I_0/2$.

Sheet 2 is at 30° to sheet 1:

$$I_2 = \frac{I_0}{2} \cos^2 30^\circ = \frac{I_0}{2} (0.866)^2 = 0.375 I_0$$

Sheet 3 is at 90° to sheet 1, hence at $90^\circ - 30^\circ = 60^\circ$ to sheet 2:

$$I_3 = 0.375 I_0 \times \cos^2 60^\circ = 0.375 I_0 \times 0.25 = 0.09375 I_0 = \frac{3I_0}{32}$$

Note that with the middle sheet removed nothing at all would get through — adding a third Polaroid increases the transmitted light.

WORKED EXAMPLE 13.4

Problem. The polarizing angle of a certain transparent medium is 60° . Find (a) its refractive index, (b) the speed of light in it, and (c) its critical angle. Take $c = 3 \times 10^8 \text{ m s}^{-1}$.

Solution.

(a) $n = \tan 60^\circ = \sqrt{3} = 1.732.$

$$(b) \nu = \frac{c}{n} = \frac{3 \times 10^8}{1.732} = 1.73 \times 10^8 \text{ m s}^{-1}.$$

$$(c) \sin C = \frac{1}{n} = \frac{1}{1.732} = 0.5774, \text{ so } C = 35.3^\circ.$$

Chapter summary

- **Unpolarized** light vibrates in all directions perpendicular to the ray; **plane polarized** light vibrates in one plane only. The plane of polarization is perpendicular to the plane of vibration.
- Polarized light is produced by reflection, refraction through a pile of plates, selective absorption (Polaroid), double refraction (Nicol prism) and scattering.
- Degree of polarization $P = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$.
- **Brewster's law:** $n = \tan \theta_p$, with $\theta_p + r = 90^\circ$; the reflected and refracted rays are then perpendicular and the reflected beam is fully plane polarized.
- Brewster's angle ($\tan^{-1} n$, rarer to denser) is **not** the critical angle ($\sin^{-1} (1/n)$, denser to rarer).
- **Malus's law:** $I = I_0 \cos^2 \theta$. Unpolarized light through the first Polaroid always gives $I_0/2$; crossed Polaroids give zero.
- Polarization occurs for transverse waves only, so it proves that light is a transverse wave. Sound, being longitudinal, cannot be polarized.

Practice questions

Group A — Multiple choice (1 mark each)

1. Polarization of light establishes that light waves are [1] (a) longitudinal (b) transverse (c) stationary (d) mechanical
2. Brewster's law is stated as [1] (a) $n = \sin \theta_p$ (b) $n = \tan \theta_p$ (c) $n = \cot \theta_p$ (d) $n = \cos \theta_p$
3. At the polarizing angle, the angle between the reflected and the refracted ray is [1] (a) 0° (b) 45° (c) 90° (d) 180°
4. Unpolarized light of intensity I_0 passes through a single Polaroid. The emergent intensity is [1] (a) I_0 (b) $I_0/2$ (c) $I_0/4$ (d) zero
5. Two Polaroids are crossed. The intensity of the light coming out is [1] (a) I_0 (b) $I_0/2$ (c) $I_0/4$ (d) zero

ANSWERS TO GROUP A

1. (b) — only a transverse wave has a vibration direction across the ray that can be selected. 2. (b) — $n = \tan \theta_p$. 3. (c) — that perpendicularity is what makes $\theta_p + r = 90^\circ$. 4. (b) — averaging $\cos^2 \theta$ over all directions gives $\frac{1}{2}$. 5. (d) — $\theta = 90^\circ$ gives $I = I_0 \cos^2 90^\circ = 0$.

Group B — Short answer (5 marks each)

1. What is plane polarized light? How does it differ from unpolarized light? Explain how plane polarized light can be produced by reflection. [5]
2. State and prove Brewster's law, and show that at the polarizing angle the reflected and refracted rays are perpendicular to each other. [5]
3. State Malus's law and derive it. Unpolarized light of intensity I_0 passes through two Polaroids whose axes are at 30° . Find the emergent intensity. [5]

4. What is a Polaroid? Describe how it is made and state four of its uses. [5]

ANSWERS TO GROUP B

1. Outline: plane polarized = vibrations in one plane only; unpolarized = vibrations in all directions perpendicular to the ray. At the polarizing angle $\theta_p = \tan^{-1} n$ the reflected beam contains only the component perpendicular to the plane of incidence, so it is fully plane polarized.

3. Derivation as in §13.3. Numerically, after the first Polaroid $I_1 = I_0/2$; after the second,

$$I_2 = \frac{I_0}{2} \cos^2 30^\circ = \frac{I_0}{2} (0.75) = 0.375 I_0 = \frac{3I_0}{8}$$

4. Outline: a sheet of stretched iodine-treated polyvinyl alcohol whose long aligned molecules absorb the electric vector parallel to them and transmit the perpendicular component. Uses: sunglasses, camera filters, LCD displays, 3-D glasses, photoelastic stress analysis, intensity control.

Group C — Long answer (8 marks each)

- (a) Explain the phenomenon of polarization by reflection and derive Brewster's law. [4] (b) Explain how polarization proves the transverse nature of light. [2] (c) The polarizing angle of a medium is 58° . Calculate its refractive index and the speed of light in it. [2]
- (a) State and derive Malus's law. [4] (b) Unpolarized light of intensity I_0 is passed through three Polaroids. The first and the third are crossed and the second makes an angle θ with the first. Show that the emergent intensity is $\frac{I_0}{8} \sin^2 2\theta$, and find the value of θ for which it is maximum. [4]

ANSWERS TO GROUP C

1. (c) $n = \tan 58^\circ = 1.60$, and $v = c/n = 3 \times 10^8 / 1.60 = 1.87 \times 10^8 \text{ m s}^{-1}$.

2. (b) After the first Polaroid, $I_1 = I_0/2$. After the second, $I_2 = \frac{I_0}{2} \cos^2 \theta$. The third is at $(90^\circ - \theta)$ to the second, so

$$I_3 = \frac{I_0}{2} \cos^2 \theta \cos^2 (90^\circ - \theta) = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

Using $2 \sin \theta \cos \theta = \sin 2\theta$,

$$I_3 = \frac{I_0}{2} \cdot \frac{\sin^2 2\theta}{4} = \frac{I_0}{8} \sin^2 2\theta$$

This is maximum when $\sin 2\theta = 1$, i.e. $\theta = 45^\circ$, giving $I_3 = I_0/8$.

UNIT 14

Electrical circuits

Electricity and Magnetism • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Kirchhoff's law
- Wheatstone bridge circuit; Meter bridge
- Potentiometer: Comparison of e.m.f.; measurement of internal resistance of a cell
- Super conductors; Perfect conductors
- Conversion of galvanometer into voltmeter and ammeter; Ohmmeter
- Joule's law

Ohm's law alone cannot solve a circuit with two cells in different branches, or a bridge, or a network that is neither series nor parallel. This unit gives you the two rules that solve any network — Kirchhoff's laws — and then applies them to the three instruments the NEB paper asks about every year: the Wheatstone bridge, the metre bridge and the potentiometer. It ends with what current actually does in a resistor: it heats it, according to Joule's law.

WHAT THE EXAM WANTS FROM THIS UNIT

Group B almost always carries one numerical from the metre bridge or the potentiometer, and one galvanometer-conversion sum. Group C likes "derive the balance condition of a Wheatstone bridge" and "describe how a potentiometer measures the internal resistance of a cell". Learn the four boxed formulas and you can walk through most of them.

14.1 Kirchhoff's law

A **junction** (or node) is a point where three or more conductors meet. A **loop** is any closed conducting path. Kirchhoff gave one rule for each.

KIRCHHOFF'S TWO LAWS

First law (junction law). The algebraic sum of the currents meeting at a junction is zero:

$$\sum I = 0$$

Currents flowing into the junction are taken positive, those flowing out negative. This is a statement of **conservation of charge** — charge cannot pile up at a point.

Second law (loop law). Around any closed loop, the algebraic sum of the e.m.f.s equals the algebraic sum of the potential drops IR :

$$\sum E = \sum IR$$

This is a statement of **conservation of energy** — a unit charge taken once round a loop returns to the same potential.

Sign rules that never fail. Choose a direction to walk round the loop, then:

Element met while walking round	Term to write
Resistor traversed along the assumed current	$-IR$
Resistor traversed against the assumed current	$+IR$
Cell entered at $-$ and left at $+$	$+E$
Cell entered at $+$ and left at $-$	$-E$

Set the algebraic sum to zero. If a current comes out negative, its true direction is opposite to the one you assumed — the magnitude is still right, so **never go back and redraw**.

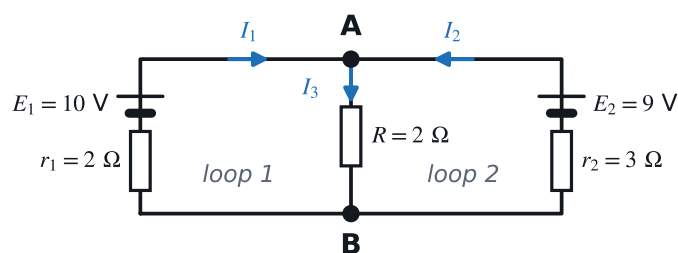


Figure 52. Two-cell network. Junction A gives $I_3 = I_1 + I_2$; the two loops give two more equations.

WORKED EXAMPLE 14.1

Problem. In the network above, $E_1 = 10 \text{ V}$ with $r_1 = 2 \Omega$, $E_2 = 9 \text{ V}$ with $r_2 = 3 \Omega$, and $R = 2 \Omega$. Find the current in each branch.

Solution. Let I_1 and I_2 flow towards junction A through the two cells, and I_3 flow from A to B through R .

Junction law at A: $I_3 = I_1 + I_2$.

Loop 1 (left cell and R), walking with the currents:

$$E_1 = I_1 r_1 + I_3 R \Rightarrow 10 = 2I_1 + 2(I_1 + I_2) = 4I_1 + 2I_2$$

Loop 2 (right cell and R):

$$E_2 = I_2 r_2 + I_3 R \Rightarrow 9 = 3I_2 + 2(I_1 + I_2) = 2I_1 + 5I_2$$

From the first, $I_2 = 5 - 2I_1$. Substituting in the second, $2I_1 + 5(5 - 2I_1) = 9$, so $-8I_1 = -16$ and

$$I_1 = 2 \text{ A}, \quad I_2 = 5 - 4 = 1 \text{ A}, \quad I_3 = 3 \text{ A}$$

Check: the potential at A is $I_3 R = 6 \text{ V}$; the left cell drops $10 - 6 = 4 \text{ V}$ across 2Ω (giving 2 A) and the right cell drops $9 - 6 = 3 \text{ V}$ across 3Ω (giving 1 A). Consistent.

DO NOT APPLY THE LOOP LAW TO AN OPEN BRANCH

Every loop equation must be written round a **complete conducting path**. A branch containing an ideal voltmeter or a galvanometer with no current carries $I = 0$, so its IR term vanishes — but the branch's e.m.f. and the potential difference across it do **not** vanish. This is exactly how a balanced bridge works.

14.2 Wheatstone bridge circuit; Meter bridge

Four resistors P, Q, R, S form a quadrilateral $ABCD$. A cell is across one diagonal AC and a galvanometer across the other, BD . The bridge is **balanced** when the galvanometer shows no deflection.

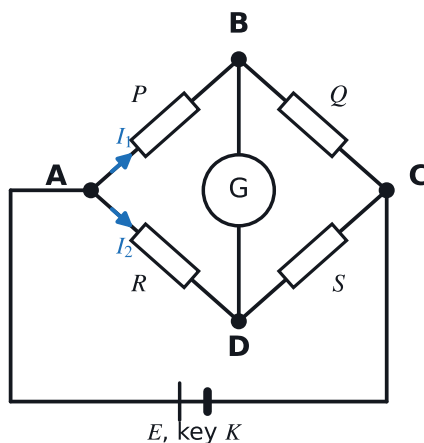


Figure 53. Wheatstone bridge. At balance $V_B = V_D$, the galvanometer reads zero and $P/Q = R/S$.

BALANCE CONDITION OF THE WHEATSTONE BRIDGE

At balance no current flows through the galvanometer, so by the junction law the current I_1 in P also flows through Q , and the current I_2 in R also flows through S .

Since $I_g = 0$, points B and D are at the **same potential**. Hence

- along ABD : $V_A - V_B = V_A - V_D$, so $I_1 P = I_2 R$ (1)

- along BCD : $V_B - V_C = V_D - V_C$, so $I_1 Q = I_2 S$ (2)

Dividing (1) by (2):

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S} \implies \frac{P}{Q} = \frac{R}{S}$$

If three resistances are known the fourth follows. Note that the condition does **not** involve E , r or the galvanometer resistance — this is why the bridge is a null method and is so accurate.

The metre bridge

The metre bridge is the Wheatstone bridge built from a single uniform wire. A manganin or constantan wire exactly 100 cm long is stretched along a metre scale between copper strips. The unknown resistance X goes in the left gap and a resistance box R in the right gap. A jockey slides along the wire until the galvanometer reads zero at a balance length l from the left end.

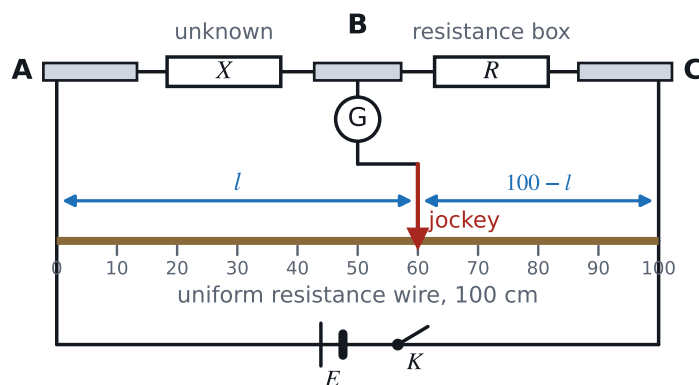


Figure 54. Metre bridge. The jockey is moved until G reads zero at length l ; then $X = R l / (100 - l)$.

If the wire has resistance σ per centimetre, the two parts of the wire have resistances σl and $\sigma(100 - l)$. Applying the balance condition,

$$\frac{X}{R} = \frac{\sigma l}{\sigma(100 - l)} \implies X = R \frac{l}{100 - l}$$

KEEP THE BALANCE POINT NEAR THE MIDDLE

The fractional error in X is smallest when $l \approx 50$ cm, so choose R in the box to bring the null point between about 40 cm and 60 cm. Repeating the reading with X and R **interchanged** and averaging cancels the error caused by the end resistances of the copper strips.

WORKED EXAMPLE 14.2

Problem. In a metre bridge the unknown resistance X is in the left gap and $R = 4 \Omega$ in the right gap. The balance point is 60 cm from the left end. Find X , and the new balance point when X and R are interchanged.

Solution.

$$X = R \frac{l}{100-l} = 4 \times \frac{60}{40} = 6 \, \Omega$$

On interchanging, $R = 4 \, \Omega$ is now on the left, so

$$\frac{4}{6} = \frac{l'}{100-l'} \Rightarrow 400 - 4l' = 6l' \Rightarrow l' = 40 \, \text{cm}$$

The balance point simply moves to $100 - 60 = 40 \, \text{cm}$, as symmetry demands.

14.3 Potentiometer: Comparison of e.m.f.; measurement of internal resistance of a cell

A potentiometer is a long uniform wire (usually 4 m or 10 m, wound in 1 m strips) carrying a steady current from a **driver cell** of e.m.f. E_0 . The potential falls uniformly along it, so the potential difference across a length l is $V = kl$, where the **potential gradient**

$$k = \frac{V_{AB}}{L} \quad (\text{volt per metre})$$

is constant as long as the driver current is steady.

WHY A POTENTIOMETER BEATS A VOLTMETER

At the balance point **no current is drawn from the cell being tested**. The cell therefore has no internal drop Ir , and the potentiometer measures the true **e.m.f.**, not the terminal p.d. A voltmeter always draws some current, so it always reads slightly low. A potentiometer behaves like a voltmeter of infinite resistance.

Comparison of e.m.f. of two cells

Connect the two cells E_1 and E_2 through a two-way key so that either can be put in the galvanometer branch, with their positive terminals joined to the same end A as the driver cell. If they balance at lengths l_1 and l_2 ,

$$E_1 = kl_1, \quad E_2 = kl_2 \Rightarrow \frac{E_1}{E_2} = \frac{l_1}{l_2}$$

Internal resistance of a cell

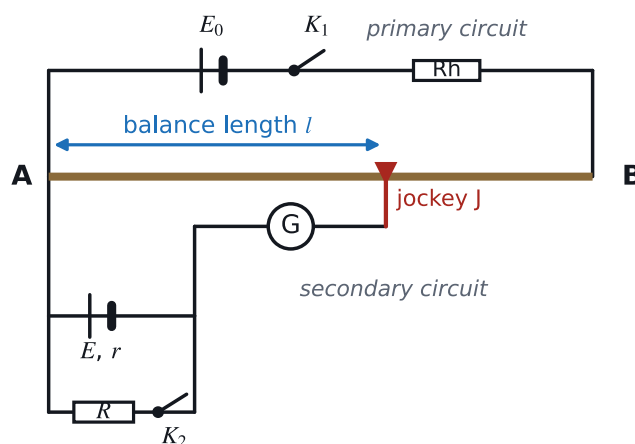


Figure 55. Potentiometer set up to find the internal resistance of the cell E . With K_2 open the balance length is l_1 ; with K_2 closed it falls to l_2 .

INTERNAL RESISTANCE FROM TWO BALANCE LENGTHS

Step 1 — key K_2 open. No current is drawn from the test cell, so the potentiometer balances against its full e.m.f.:

$$E = kl_1$$

Step 2 — key K_2 closed. The cell now drives a current through the known resistance R . At the new balance point the potentiometer matches the terminal p.d.:

$$V = kl_2, \quad V = \frac{ER}{R+r}$$

Step 3 — divide.

$$\frac{E}{V} = \frac{l_1}{l_2} = \frac{R+r}{R} \implies \frac{l_1}{l_2} - 1 = \frac{r}{R}$$

$$r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

THE BALANCE POINT MUST EXIST

If the driver cell's e.m.f. E_0 is less than the e.m.f. being measured, the potential drop over the whole wire is too small and the galvanometer deflects the same way everywhere — there is no null point. Always take $E_0 > E$, and check that the positive terminals of both cells are connected to the **same** end of the wire. One-sided deflection at both ends is the classic symptom of a reversed cell.

WORKED EXAMPLE 14.3

Problem. A cell balances at $l_1 = 240$ cm on a potentiometer. When a resistance of $5\ \Omega$ is connected across its terminals the balance length falls to $l_2 = 200$ cm. Find the internal resistance of the cell. If the potential gradient is $2.0\ \text{mV cm}^{-1}$, find the e.m.f. as well.

Solution.

$$r = R\left(\frac{l_1 - l_2}{l_2}\right) = 5\left(\frac{240 - 200}{200}\right) = 5 \times 0.2 = 1 \, \Omega$$

$$E = kl_1 = (2.0 \times 10^{-3} \, \text{V cm}^{-1})(240 \, \text{cm}) = 0.48 \, \text{V}$$

14.4 Super conductors; Perfect conductors

The resistivity of a normal metal falls as it is cooled, because the lattice ions vibrate less and scatter electrons less. But it never reaches zero: impurities and defects leave a **residual resistivity** even at 0 K.

Certain materials behave completely differently. Below a sharp **critical temperature** T_c their resistivity drops abruptly to zero — not "very small", but zero, as far as any measurement can tell. Kamerlingh Onnes discovered this in mercury in 1911 at $T_c = 4.2 \, \text{K}$.

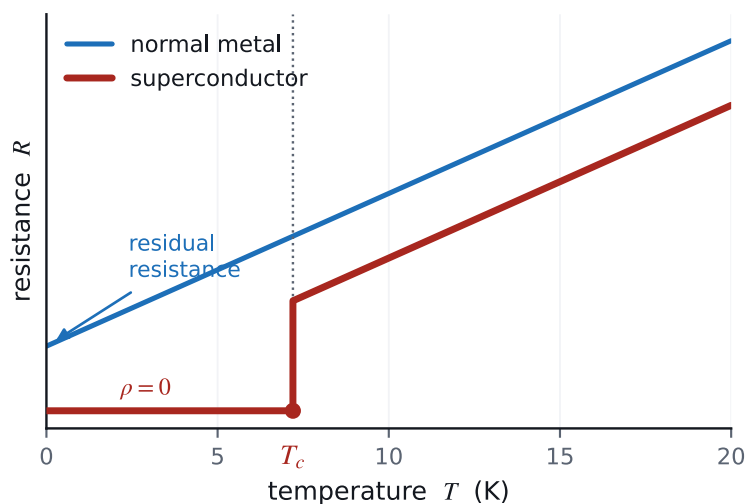


Figure 56. Resistance against temperature. A normal metal (blue) keeps a residual resistance; a superconductor (red) drops to exactly zero at T_c .

SUPERCONDUCTOR

A superconductor is a material whose electrical resistivity becomes exactly zero below a critical temperature T_c , and which at the same time expels magnetic flux from its interior, so that $B = 0$ inside it (the **Meissner effect**).

Perfect conductor versus superconductor. A perfect conductor is an idealisation: a material with $\rho = 0$ but with no other special property. The distinction is not academic, and examiners like it:

	Perfect conductor (hypothetical)	Superconductor (real)
Resistivity	zero	zero
Defined by	$\rho = 0$ only	$\rho = 0$ and $B = 0$ inside
Cooled in a magnetic field	traps the flux already inside it, because $d\vec{B}/dt = 0$	expels the flux completely (Meissner effect)
Final state	depends on the history of cooling and magnetising	same state whatever the order — a true thermodynamic phase
Exists?	no	yes, below T_c

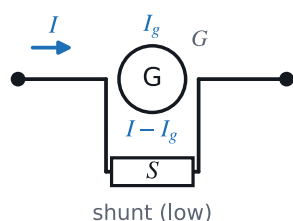
A superconductor is destroyed by too large a magnetic field (the **critical field** B_c , which falls to zero at T_c) or by too large a current (the **critical current**). Both limits matter in real magnets.

High-temperature superconductors. In 1986–87 ceramic oxides such as $\text{YBa}_2\text{Cu}_3\text{O}_7$ were found to superconduct up to about 92 K — above the boiling point of liquid nitrogen (77 K), which is cheap. Uses: MRI and NMR magnets, particle-accelerator magnets, maglev trains, SQUID magnetometers, and loss-free power cables.

14.5 Conversion of galvanometer into voltmeter and ammeter; Ohmmeter

A moving-coil galvanometer has a resistance G and gives full-scale deflection for a small current I_g (typically a few milliamperes). It is the movement inside every analogue meter; only the resistor bolted to it changes.

ammeter: shunt in parallel



voltmeter: high R in series

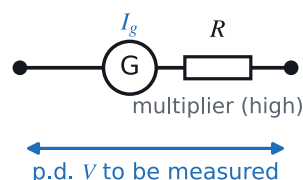


Figure 57. Left: a low shunt S in parallel makes an ammeter. Right: a high resistance R in series makes a voltmeter.

SHUNT FOR AN AMMETER, MULTIPLIER FOR A VOLTMETER

Ammeter. To read up to a current I , a low resistance S (the **shunt**) is joined in parallel with the galvanometer so that only I_g passes through the coil and $(I - I_g)$ bypasses it. The p.d. across the parallel pair is the same:

$$I_g G = (I - I_g) S \implies S = \frac{I_g G}{I - I_g}$$

The resistance of the finished ammeter is $R_A = \frac{GS}{G+S}$, which is very small — an ammeter must be connected **in series** and must not disturb the current. An ideal ammeter has zero resistance.

Voltmeter. To read up to a p.d. V , a high resistance R (the **multiplier**) is joined in series, chosen so that the current is exactly I_g when the p.d. is V :

$$V = I_g(G + R) \implies R = \frac{V}{I_g} - G$$

The resistance of the finished voltmeter is $G + R$, which is very large — a voltmeter is connected **in parallel** and must draw as little current as possible. An ideal voltmeter has infinite resistance.

The ohmmeter

An ohmmeter is a galvanometer, a cell of e.m.f. E and an adjustable resistor R in series, with two test leads. First the leads are touched together (zero external resistance) and R is adjusted so that the deflection is exactly full scale:

$$I_g = \frac{E}{G + R}$$

With an unknown resistance X between the leads the current becomes

$$I = \frac{E}{G + R + X}$$

so each deflection corresponds to one value of X . Two features follow at once and are often asked:

- the scale is **non-uniform** (cramped at large X), because I is not proportional to X ;
- the scale is **reversed**: zero ohms is at the full-scale end (right) and infinite ohms at the zero-current end (left).

WORKED EXAMPLE 14.4

Problem. A galvanometer of resistance $G = 50 \, \Omega$ gives full-scale deflection for $I_g = 2 \, \text{mA}$. How would you convert it into (a) an ammeter reading 0–5 A, and (b) a voltmeter reading 0–10 V? Find the resistance of each finished instrument.

Solution.

(a) $I = 5 \, \text{A}$, $I_g = 2 \times 10^{-3} \, \text{A}$:

$$S = \frac{I_g G}{I - I_g} = \frac{(2 \times 10^{-3})(50)}{5 - 0.002} = \frac{0.100}{4.998} = 0.0200 \, \Omega$$

A shunt of about $0.02 \, \Omega$ is connected **in parallel**. Meter resistance

$$R_A = \frac{GS}{G + S} = \frac{50 \times 0.0200}{50.02} = 0.0200 \, \Omega$$

(b) $V = 10 \, \text{V}$:

$$R = \frac{V}{I_g} - G = \frac{10}{2 \times 10^{-3}} - 50 = 5000 - 50 = 4950 \, \Omega$$

This is connected **in series**; the voltmeter's resistance is $G + R = 5000 \, \Omega$, i.e. $500 \, \Omega$ per volt.

14.6 Joule's law

When a current passes through a conductor, the ordered drift energy the electrons gain from the field is handed to the lattice in collisions, and the conductor warms up.

JOULE'S LAW OF HEATING

In time t a charge $q = It$ moves through a potential difference V . The work done by the source on this charge is

$$W = qV = VIt$$

None of it appears as kinetic energy, because the drift velocity stays constant; all of it appears as heat. Using $V = IR$,

$$H = I^2 R t = VIt = \frac{V^2}{R} t$$

and the rate of heating — the power dissipated — is $P = H/t = I^2 R = V^2/R$.

The three statements usually quoted as **Joule's laws** are just the three proportionalities hidden in $H = I^2 R t$:

1. $H \propto I^2$ when R and t are constant;
2. $H \propto R$ when I and t are constant;
3. $H \propto t$ when I and R are constant.

If the heat is wanted in calories, divide by the mechanical equivalent of heat $J = 4.2 \text{ J cal}^{-1}$: $H = I^2 R t / J$ calories.

WHICH FORMULA FOR RESISTORS IN SERIES AND IN PARALLEL?

For resistors **in series** the current I is common, so use $H = I^2 R t$ and the **largest** resistance gets hottest. For resistors **in parallel** the voltage V is common, so use $H = V^2 t / R$ and the **smallest** resistance gets hottest. Picking the wrong form reverses the answer — this is a favourite Group A trap.

Where it is used. Heaters, immersion rods, electric irons and incandescent lamps use a high-resistivity alloy (nichrome, $\rho \approx 1.1 \times 10^{-6} \text{ } \Omega \cdot \text{m}$) with a high melting point. A **fuse** does the opposite job: a short piece of low-melting tin-lead alloy that melts and breaks the circuit when $I^2 R$ heating becomes dangerous. Electrical energy is sold in **kilowatt-hours**: $1 \text{ kWh} = 1000 \times 3600 = 3.6 \times 10^6 \text{ J}$.

WORKED EXAMPLE 14.5

Problem. An electric kettle is marked "1500 W, 230 V". It is used to heat 1.0 kg of water from 20 °C to 100 °C. If 80 % of the electrical energy reaches the water, how long does it take? Take the specific heat capacity of water as $4200 \text{ J kg}^{-1} \text{ K}^{-1}$. Also find the resistance of the element.

Solution. Heat required by the water:

$$Q = mc \Delta\theta = (1.0)(4200)(100 - 20) = 3.36 \times 10^5 \text{ J}$$

Useful power delivered $= 0.80 \times 1500 = 1200 \text{ W}$, so

$$t = \frac{Q}{P_{\text{useful}}} = \frac{3.36 \times 10^5}{1200} = 280 \text{ s} = 4 \text{ min } 40 \text{ s}$$

Resistance of the element at its working temperature:

$$R = \frac{V^2}{P} = \frac{230^2}{1500} = \frac{52900}{1500} = 35.3 \text{ } \Omega$$

Chapter summary

- Kirchhoff's junction law $\sum I = 0$ expresses conservation of charge; the loop law $\sum E = \sum IR$ expresses conservation of energy.
- A Wheatstone bridge is balanced when $I_g = 0$, and then $P/Q = R/S$. The condition is independent of the cell and of the galvanometer.
- Metre bridge: $X = R l / (100 - l)$, with l measured from the end next to X . Keep l near 50 cm and interchange X and R to cancel end errors.
- Potentiometer: $V = kl$ with $k = V_{AB}/L$. Comparison of e.m.f. $E_1/E_2 = l_1/l_2$; internal resistance $r = R(l_1 - l_2)/l_2$. At balance the cell supplies no current, so the true e.m.f. is measured.
- A superconductor has $\rho = 0$ below T_c **and** expels magnetic flux (Meissner effect); a merely perfect conductor would trap the flux instead.
- Galvanometer conversions: ammeter $S = I_g G / (I - I_g)$ in parallel; voltmeter $R = V/I_g - G$ in series. An ohmmeter has a reversed, non-uniform scale.
- Joule's law: $H = I^2 R t = V I t = V^2 t / R$; power $P = I^2 R = V^2 / R$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Practice questions

Group A — Multiple choice (1 mark each)

1. Kirchhoff's junction law is a consequence of the conservation of [1] (a) energy (b) charge (c) momentum (d) magnetic flux
2. In a balanced Wheatstone bridge, the current through the galvanometer is [1] (a) maximum (b) zero (c) half the main current (d) equal to the main current
3. A potentiometer measures the e.m.f. of a cell more accurately than a voltmeter because at balance it [1] (a) has a very low resistance (b) draws no current from the cell (c) uses a longer wire (d) is calibrated in volts
4. To convert a galvanometer into an ammeter, we connect a [1] (a) high resistance in series (b) high resistance in parallel (c) low resistance in series (d) low resistance in parallel
5. Two resistors of $2 \text{ } \Omega$ and $4 \text{ } \Omega$ are joined in parallel across a battery. The ratio of the heat produced in them in a given time is [1] (a) $1:2$ (b) $2:1$ (c) $1:4$ (d) $4:1$
6. Below its critical temperature a superconductor [1] (a) has infinite resistance (b) becomes an insulator (c) expels magnetic flux from its interior (d) traps all the flux inside it

ANSWERS TO GROUP A

1. (b) — charge cannot accumulate at a junction, so what flows in must flow out. **2.** (b) — balance is defined by the null (zero) galvanometer deflection. **3.** (b) — with no current drawn there is no internal drop I_r , so the reading is the true e.m.f. **4.** (d) — the shunt $S = I_g G / (I - I_g)$ is small and goes in parallel. **5.** (b) — in parallel V is common, so $H = V^2 t / R \propto 1/R$, giving $H_2 : H_4 = \frac{1}{4} : \frac{1}{2}$ inverted, i.e. 2 : 1. **6.** (c) — the Meissner effect; that, plus $\rho = 0$, defines superconductivity.

Group B — Short answer (5 marks each)

1. State Kirchhoff's laws and name the conservation principle behind each. Using them, obtain the balance condition of a Wheatstone bridge. [5]
2. In a metre bridge the unknown resistance X is in the left gap and a $6\ \Omega$ resistance in the right gap. The balance point is 40 cm from the left end. Find X and the new balance length if the two are interchanged. [5]
3. A cell balances against 150 cm of a potentiometer wire. When a $4\ \Omega$ resistor is connected across the cell, the balance length becomes 120 cm. Calculate the internal resistance of the cell. [5]
4. A galvanometer of resistance $100\ \Omega$ gives full-scale deflection for $1\ \text{mA}$. Find the resistance needed to convert it into (a) an ammeter reading up to $1\ \text{A}$ and (b) a voltmeter reading up to $5\ \text{V}$. [5]
5. Distinguish between a perfect conductor and a superconductor. Why is a superconducting magnet cooled below T_c before the field is switched on? [5]

ANSWERS TO GROUP B

- 1.** Junction law $\sum I = 0$ (conservation of charge); loop law $\sum E = \sum IR$ (conservation of energy). At balance $I_g = 0$, so I_1 flows through P and Q and I_2 through R and S , and $V_B = V_D$. Then $I_1 P = I_2 R$ and $I_1 Q = I_2 S$; dividing gives $P/Q = R/S$.
- 2.** $X = R l / (100 - l) = 6 \times 40 / 60 = 4\ \Omega$. Interchanged, the left gap holds $6\ \Omega$, so $6/4 = l' / (100 - l')$, giving $600 - 6l' = 4l'$ and $l' = 60\ \text{cm}$.
- 3.** $r = R \left(\frac{l_1 - l_2}{l_2} \right) = 4 \left(\frac{150 - 120}{120} \right) = 4 \times 0.25 = 1\ \Omega$.
- 4.** (a) $S = \frac{I_g G}{I - I_g} = \frac{(10^{-3})(100)}{1 - 0.001} = \frac{0.1}{0.999} = 0.100\ \Omega$ in parallel. (b) $R = \frac{V}{I_g} - G = \frac{5}{10^{-3}} - 100 = 5000 - 100 = 4900\ \Omega$ in series.
- 5.** A perfect conductor only has $\rho = 0$; a superconductor also has $B = 0$ inside it. If a perfect conductor were cooled in a field it would trap that flux, because $d\vec{B}/dt = 0$ inside a resistanceless material; a superconductor expels the flux (Meissner effect) whatever the order of cooling and magnetising. In practice the magnet is cooled first so that the whole winding is superconducting before any current — and hence any $I^2 R$ heating at a normal spot — can appear; a normal region would heat, spread, and "quench" the magnet.

Group C — Long answer (8 marks each)

- 1.** (a) Draw a labelled diagram of a potentiometer arranged to compare the e.m.f.s of two cells, and derive $E_1/E_2 = l_1/l_2$. [4] (b) A Daniell cell of e.m.f. $1.08\ \text{V}$ balances at $216\ \text{cm}$. A second cell balances at $300\ \text{cm}$ on the same wire. Find the e.m.f. of the second cell and the potential gradient of the wire. [4]

2. (a) State and derive Joule's law of heating, and explain why the element of a heater glows red while its copper connecting leads stay cool. [4] (b) An electric heater marked "1000 W, 220 V" is used for 5 hours a day for 30 days. Find its resistance, the current it draws, and the monthly cost of running it at NPR 10 per unit. [4]

ANSWERS TO GROUP C

1. (b) Since $E \propto l$ for the same wire and driver current,

$$E_2 = E_1 \frac{l_2}{l_1} = 1.08 \times \frac{300}{216} = 1.50 \text{ V}$$

Potential gradient $k = E_1/l_1 = 1.08/216 = 5.0 \times 10^{-3} \text{ V cm}^{-1} = 0.50 \text{ V m}^{-1}$.

2. (a) $W = qV = VIt = I^2Rt$, all of which appears as heat since the drift velocity is unchanged. In a series path the current is the same everywhere, so $H \propto R$: the nichrome element has a resistance of tens of ohms and a high melting point, while the thick copper leads have a resistance of a fraction of an ohm, so almost all the heat is produced in the element.

(b) $R = V^2/P = 220^2/1000 = 48.4 \text{ } \Omega$; $I = P/V = 1000/220 = 4.55 \text{ A}$. Energy = $1.0 \text{ kW} \times 5 \text{ h} \times 30 = 150 \text{ kWh}$, so the cost is $150 \times 10 = \text{NPR } 1500$ per month.

UNIT 15

Thermoelectric effects

Electricity and Magnetism • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Seebeck effect; Thermocouples
- Peltier effect: Variation of thermoelectric e.m.f. with temperature; Thermopile

Heat and electricity are usually separate chapters, but at the junction of two different metals they are directly linked: a temperature difference produces an e.m.f., and a current produces a temperature difference. These are the Seebeck and Peltier effects. They are the basis of the thermocouple thermometer used in every industrial furnace, of the thermopile that detects heat radiation, and of the solid-state coolers in portable fridges.

WHAT THE EXAM WANTS FROM THIS UNIT

Three things, every year: the direction of the thermoelectric current (Seebeck series), the parabola $E = \alpha\theta + \frac{1}{2}\beta\theta^2$ with its neutral and inversion temperatures, and the table of differences between Peltier and Joule heating. The numericals are almost always about θ_n and θ_i .

15.1 Seebeck effect; Thermocouples

SEEBECK EFFECT

When two wires of **different** metals are joined to form a closed circuit and the two junctions are kept at **different temperatures**, an e.m.f. is set up in the circuit and a current flows round it. The effect is called the Seebeck or thermoelectric effect, the e.m.f. is the **thermo-e.m.f.**, and the pair of metals with its two junctions is a **thermocouple**.

Thomas Seebeck discovered it in 1821. The thermo-e.m.f. is small — of the order of tens of microvolts per degree — and it depends on only two things:

1. the **nature of the two metals**, and
2. the **temperature difference** between the two junctions.

It does not depend on the thickness of the wires, the length of the wires, or the shape of the junctions.

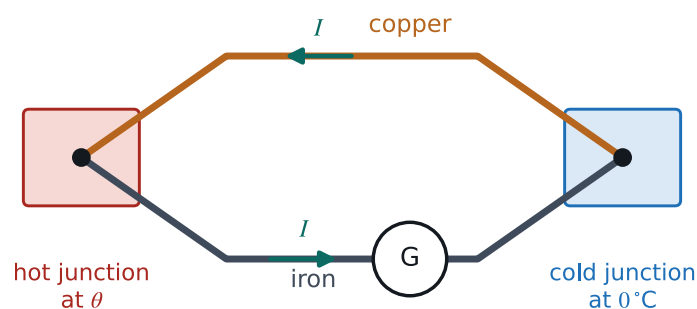


Figure 58. A copper-iron thermocouple. The current flows from iron to copper through the cold junction, and from copper to iron through the hot junction.

Direction of the thermoelectric current: the thermoelectric series

Arrange metals in the **Seebeck (thermoelectric) series**:

Sb, Fe, Cd, Zn, Ag, Au, Cr, Sn, Pb, Hg, Mn, Cu, Pt, Co, Ni, Bi

Two rules follow:

- **Direction.** In the thermocouple the current flows through the **cold junction** from the metal that comes earlier in the series to the metal that comes later. (Through the hot junction it therefore flows the other way.)
- **Magnitude.** The further apart the two metals are in the series, the larger the thermo-e.m.f. Antimony and bismuth stand at the two ends, so an Sb–Bi couple gives the biggest e.m.f. of all — which is why thermopiles are built from it.

TWO MNEMONICS WORTH MEMORISING

"A-B-C" — Antimony to Bismuth through the **Cold** junction. **"hot coffee"** — Copper to **Fe** (iron) through the **hot** junction, and therefore iron to copper through the cold junction.

Why it happens. The free-electron concentration is different in the two metals. Where they touch, electrons diffuse from the metal of higher concentration to the metal of lower concentration until a **contact potential difference** stops the flow. This contact p.d. depends on temperature. If both junctions are at the same temperature the two contact p.d.s are equal and opposite and cancel round the loop; if the junctions are at different temperatures they no longer cancel, and the difference is the thermo-e.m.f.

Thermocouple thermometers

To use a thermocouple as a thermometer, keep the cold junction in melting ice (a fixed 0°C reference) and put the hot junction where the temperature is wanted. The galvanometer or potentiometer reading is converted to temperature from a calibration curve.

Thermocouple	Useful range
Copper-constantan	$-200\text{ }^{\circ}\text{C}$ to $400\text{ }^{\circ}\text{C}$
Iron-constantan	$-200\text{ }^{\circ}\text{C}$ to $750\text{ }^{\circ}\text{C}$
Chromel-alumel	$-200\text{ }^{\circ}\text{C}$ to $1100\text{ }^{\circ}\text{C}$
Platinum-platinum/rhodium	$0\text{ }^{\circ}\text{C}$ to $1600\text{ }^{\circ}\text{C}$

Advantages: very wide range; the junction is tiny, so its heat capacity is small and it responds in a fraction of a second; it measures the temperature at a point; the reading can be taken hundreds of metres away, which is why the kilns of a brick factory or a cement plant in Hetauda are monitored this way.

Disadvantages: the e.m.f. is small, so a sensitive instrument is needed; the calibration is **not linear**; a reference junction must be maintained.

A THERMOCOUPLE WITH BOTH JUNCTIONS IN THE SAME BATH READS NOTHING

The Seebeck e.m.f. depends on the difference of junction temperatures. If the "cold" junction is allowed to drift to the temperature of the hot one, the e.m.f. falls to zero even though the furnace is white hot. Always check the reference junction first.

15.2 Peltier effect: Variation of thermoelectric e.m.f. with temperature; Thermopile

The Peltier effect

PELTIER EFFECT

When an electric current is passed through the junction of two dissimilar metals, heat is **evolved** at one junction and **absorbed** at the other. If the direction of the current is reversed, the two junctions exchange roles. This is the converse of the Seebeck effect.

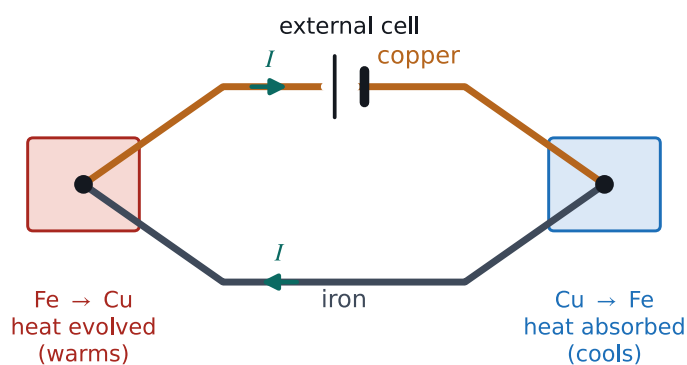


Figure 59. Peltier effect. An external cell drives the current; where it passes from copper to iron the junction cools, where it passes from iron to copper the junction warms.

The **Peltier coefficient** π of a junction is the heat absorbed or evolved when unit charge crosses it. Since a current I carries a charge $q = It$ in time t ,

$$H = \pi q = \pi It$$

so π is measured in joule per coulomb, i.e. **volts**; for metal junctions it is of the order of millivolts.

PELTIER HEATING IS NOT JOULE HEATING

Peltier heat	Joule heat
Only at a junction of two different metals	In any conductor
$H = \pi It$ — proportional to I	$H = I^2 Rt$ — proportional to I^2
Reversible: reversing I turns evolution into absorption	Irreversible: heat is evolved whichever way I flows
Can cool a junction below room temperature	Always warms the conductor

A junction that absorbs Peltier heat is one at which the current flows in the direction it would take through the **hot** junction of a Seebeck circuit of the same pair — for copper and iron, from copper to iron. The reason is simply that a working thermocouple is a heat engine: it takes in heat at the hot junction and rejects it at the cold one.

Thomson effect (for completeness). Heat is also absorbed or evolved when a current flows along a single conductor whose two ends are at different temperatures. It is positive for copper, silver and zinc, negative for iron, cobalt and nickel, and zero for lead — which is why lead is taken as the reference metal.

Variation of thermo-e.m.f. with temperature

Keep the cold junction at 0°C and raise the hot junction to θ . Experiment shows that the thermo-e.m.f. is not proportional to θ ; it follows a parabola:

$$E = \alpha\theta + \frac{1}{2}\beta\theta^2$$

where α and β are constants of the pair of metals. For most couples α is positive and β is small and **negative**, so the curve rises, turns over and comes back down.

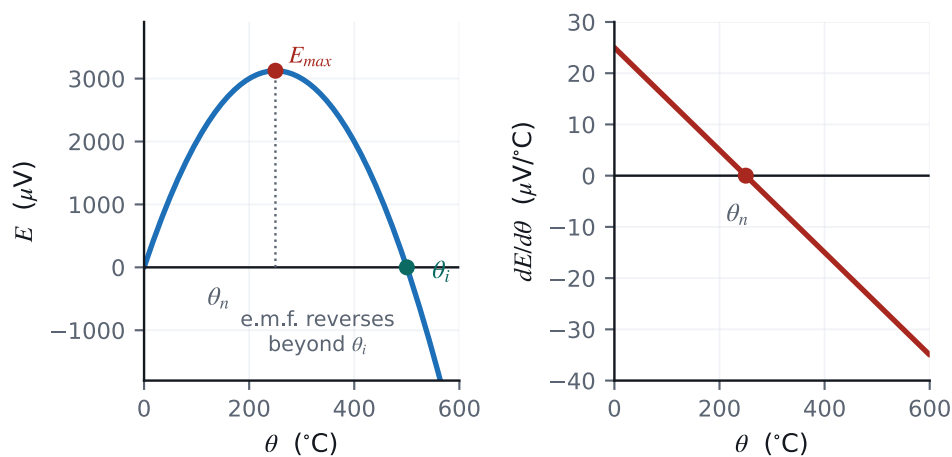


Figure 60. Left: thermo-e.m.f. against hot-junction temperature for $\alpha = 25 \text{ } \mu\text{V/}^{\circ}\text{C}$, $\beta = -0.1 \text{ } \mu\text{V/}^{\circ}\text{C}^2$. Right: the thermoelectric power $dE/d\theta$ falls linearly and changes sign at θ_n .

NEUTRAL TEMPERATURE AND TEMPERATURE OF INVERSION

Thermoelectric power. Differentiating $E = \alpha\theta + \frac{1}{2}\beta\theta^2$,

$$S = \frac{dE}{d\theta} = \alpha + \beta\theta$$

This is the **thermoelectric power** (or Seebeck coefficient) — the e.m.f. per degree. It falls linearly as θ rises.

Neutral temperature θ_n . The e.m.f. is a maximum where $dE/d\theta = 0$:

$$\alpha + \beta\theta_n = 0 \implies \theta_n = -\frac{\alpha}{\beta}$$

θ_n contains only α and β , so it depends **only on the pair of metals** — not on the temperature of the cold junction. The maximum e.m.f. is $E_{\max} = -\alpha^2/2\beta$.

Temperature of inversion θ_i . Beyond θ_n the e.m.f. falls, and at some higher temperature it becomes zero and then **reverses** direction. Putting $E = 0$ with the cold junction at 0°C :

$$\theta(\alpha + \frac{1}{2}\beta\theta) = 0 \implies \theta_i = -\frac{2\alpha}{\beta} = 2\theta_n$$

If the cold junction is at θ_c instead of 0°C , the neutral temperature stays put but the inversion temperature shifts, because θ_n always lies **midway** between θ_c and θ_i :

$$\theta_i = 2\theta_n - \theta_c$$

θ_n IS FIXED, θ_i IS NOT

A very common error is to say that both temperatures depend on the cold junction. Only θ_i does. Raise the cold junction by 10°C and θ_i falls by 10°C , while θ_n does not move at all.

WORKED EXAMPLE 15.1

Problem. For a thermocouple with its cold junction at 0°C , the thermo-e.m.f. in microvolts is $E = 25\theta - 0.05\theta^2$. Find (a) α and β , (b) the neutral temperature, (c) the temperature of inversion, (d) the maximum e.m.f., (e) the thermoelectric power at 100°C , and (f) the two hot-junction temperatures at which $E = 3000\ \mu\text{V}$.

Solution.

(a) Comparing with $E = \alpha\theta + \frac{1}{2}\beta\theta^2$: $\alpha = 25\ \mu\text{V } ^\circ\text{C}^{-1}$ and $\frac{1}{2}\beta = -0.05$, so $\beta = -0.1\ \mu\text{V } ^\circ\text{C}^{-2}$.

(b) $\theta_n = -\alpha/\beta = -25/(-0.1) = 250^\circ\text{C}$.

(c) $\theta_i = 2\theta_n - \theta_c = 2(250) - 0 = 500^\circ\text{C}$.

(d) $E_{\text{max}} = 25(250) - 0.05(250)^2 = 6250 - 3125 = 3125\ \mu\text{V} = 3.125\ \text{mV}$.

(e) $S = \alpha + \beta\theta = 25 - 0.1(100) = 15\ \mu\text{V } ^\circ\text{C}^{-1}$.

(f) $25\theta - 0.05\theta^2 = 3000 \Rightarrow \theta^2 - 500\theta + 60000 = 0$,

$$\theta = \frac{500 \pm \sqrt{250000 - 240000}}{2} = \frac{500 \pm 100}{2}$$

so $\theta = 200^\circ\text{C}$ or 300°C — one on each side of θ_n , as the parabola requires.

WORKED EXAMPLE 15.2

Problem. The neutral temperature of a copper-iron thermocouple is 270°C . Find its temperature of inversion when the cold junction is kept at (a) 0°C and (b) 20°C . If $\alpha = 13.5\ \mu\text{V } ^\circ\text{C}^{-1}$, find β and the e.m.f. at $\theta = 100^\circ\text{C}$ with the cold junction at 0°C .

Solution.

(a) $\theta_i = 2\theta_n - \theta_c = 540 - 0 = 540^\circ\text{C}$.

(b) $\theta_i = 540 - 20 = 520^\circ\text{C}$.

From $\theta_n = -\alpha/\beta$:

$$\beta = -\frac{\alpha}{\theta_n} = -\frac{13.5}{270} = -0.05\ \mu\text{V } ^\circ\text{C}^{-2}$$

$$E = 13.5(100) + \frac{1}{2}(-0.05)(100)^2 = 1350 - 250 = 1100\ \mu\text{V} = 1.10\ \text{mV}$$

WORKED EXAMPLE 15.3

Problem. The Peltier coefficient of a junction is $12\ \text{mV}$ and its resistance is $0.50\ \Omega$. A current of $2.0\ \text{A}$ is passed for 5.0 minutes. Find (a) the Peltier heat and (b) the Joule heat, and comment.

Solution. $t = 5.0 \times 60 = 300\ \text{s}$.

(a) $H_P = \pi It = (12 \times 10^{-3})(2.0)(300) = 7.2\ \text{J}$

(b) $H_J = I^2 Rt = (2.0)^2(0.50)(300) = 600\ \text{J}$

The Joule heat is about 83 times larger, and it is always evolved. Reversing the current leaves the 600 J of Joule heating unchanged but turns the 7.2 J from evolved to absorbed. This is why practical Peltier coolers use semiconductor pellets, whose Peltier coefficient is hundreds of times larger than that of a metal junction while their resistance stays small.

The thermopile

A single thermocouple gives only microvolts. A **thermopile** is a large number of thermocouples — usually antimony-bismuth, because that pair gives the largest e.m.f. — joined **in series**, so that their e.m.f.s add.

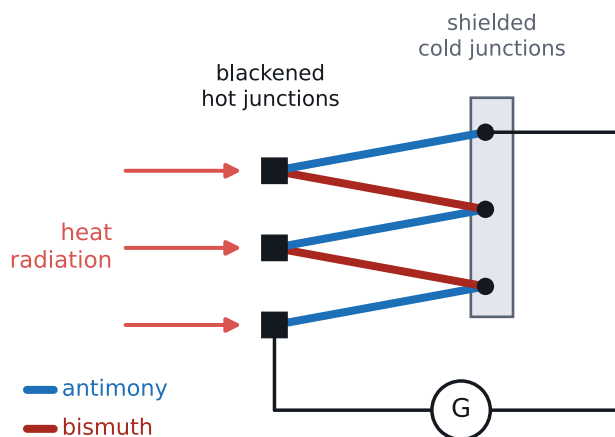


Figure 61. A thermopile: antimony and bismuth bars in series. The front junctions are blackened and face the radiation; the back junctions are shielded and stay cool.

Alternate junctions are **blackened** and exposed at the front of a polished metal cone; the other set is shielded inside the case and stays at room temperature. Radiation falling on the cone is absorbed by the black faces, raises their temperature, and the series e.m.f. drives a current through a sensitive galvanometer.

A thermopile is astonishingly sensitive: it will register the heat of a person standing a few metres away, or of the Moon. Its uses include measuring radiant heat and comparing the heating effects of different sources, radiation pyrometers for furnaces, infrared detectors and burglar alarms, and — running the Peltier effect backwards — portable thermoelectric refrigerators and the radioisotope thermoelectric generators that power deep-space probes.

WORKED EXAMPLE 15.4

Problem. A copper-constantan thermocouple has a thermoelectric power of $42 \mu\text{V } ^\circ\text{C}^{-1}$ near room temperature. The galvanometer used with it can just detect $2.1 \mu\text{V}$. What is the smallest temperature difference it can measure? What e.m.f. would a thermopile of 40 such couples in series give for a temperature difference of $0.50 ^\circ\text{C}$?

Solution. Smallest detectable difference:

$$\Delta\theta = \frac{\Delta E}{S} = \frac{2.1 \mu\text{V}}{42 \mu\text{V } ^\circ\text{C}^{-1}} = 0.050 ^\circ\text{C}$$

For 40 couples in series the e.m.f.s add:

$$E = NS \Delta\theta = 40 \times 42 \times 0.50 = 840 \mu\text{V} = 0.84 \text{ mV}$$

— comfortably measurable, which is exactly the point of a thermopile.

Chapter summary

- **Seebeck effect:** two dissimilar metals, two junctions at different temperatures, an e.m.f. appears. It depends only on the metals and on the temperature difference.
- Seebeck series Sb, Fe, Cd, Zn, Ag, Au, Cr, Sn, Pb, Hg, Mn, Cu, Pt, Co, Ni, Bi: the current flows from the earlier metal to the later one through the **cold** junction, and the further apart the metals, the greater the e.m.f.
- **Peltier effect:** a current through a junction evolves heat at one junction and absorbs it at the other; $H = \pi It$, proportional to I and **reversible**, unlike Joule heat $I^2 R t$.
- $E = \alpha\theta + \frac{1}{2}\beta\theta^2$ with the cold junction at 0°C ; thermoelectric power $S = dE/d\theta = \alpha + \beta\theta$.
- Neutral temperature $\theta_n = -\alpha/\beta$ (where E is maximum, fixed by the metals alone); temperature of inversion $\theta_i = 2\theta_n - \theta_c$ (where E returns to zero and reverses).
- $E_{\max} = -\alpha^2/2\beta$, reached at θ_n .
- A **thermopile** is many Sb–Bi couples in series, alternate junctions blackened, used to detect and measure heat radiation.

Practice questions

Group A — Multiple choice (1 mark each)

1. The Seebeck effect converts [1] (a) electrical energy into heat (b) heat into electrical energy (c) chemical energy into heat (d) magnetic energy into heat
2. The neutral temperature of a thermocouple depends on [1] (a) the temperature of the cold junction (b) the nature of the two metals only (c) the current in the circuit (d) the length of the wires
3. The Peltier heat developed at a junction is proportional to [1] (a) I^2 (b) I (c) $\sqrt{I}$ (d) $1/I$
4. The neutral temperature of a thermocouple is 270°C . With the cold junction at 10°C the temperature of inversion is [1] (a) 260°C (b) 280°C (c) 530°C (d) 550°C
5. In a thermopile the thermocouples are joined [1] (a) in parallel (b) in series (c) in a bridge (d) back to back
6. Antimony and bismuth are chosen for a thermopile because [1] (a) they are cheap (b) they melt at a high temperature (c) they lie at opposite ends of the thermoelectric series (d) they are black

ANSWERS TO GROUP A

1. (b) — a temperature difference produces an e.m.f. **2.** (b) — $\theta_n = -\alpha/\beta$ contains only constants of the metal pair. **3.** (b) — $H = \pi It$; the I^2 law belongs to Joule heating. **4.** (c) — $\theta_i = 2\theta_n - \theta_c = 540 - 10 = 530^\circ\text{C}$. **5.** (b) — in series so that the small e.m.f.s add. **6.** (c) — the widest separation in the series gives the largest thermo-e.m.f.

Group B — Short answer (5 marks each)

1. State the Seebeck effect. What is the thermoelectric series, and how is it used to find the direction of the thermoelectric current? State the direction in a copper-iron thermocouple. [5]
2. Starting from $E = \alpha\theta + \frac{1}{2}\beta\theta^2$, define the thermoelectric power, and obtain expressions for the neutral temperature and the temperature of inversion. [5]
3. For a thermocouple with its cold junction at 0°C the thermo-e.m.f. in microvolts is $E = 16\theta - 0.02\theta^2$. Find α , β , the neutral temperature, the temperature of inversion, the maximum e.m.f. and the thermoelectric power at 100°C . [5]
4. Distinguish between Peltier heating and Joule heating on four counts. A junction of Peltier coefficient 10 mV carries 3.0 A for 200 s ; find the Peltier heat. [5]
5. Give four advantages and two disadvantages of a thermocouple thermometer over a mercury-in-glass thermometer. [5]

ANSWERS TO GROUP B

1. See §15.1. Series: Sb, Fe, Cd, Zn, Ag, Au, Cr, Sn, Pb, Hg, Mn, Cu, Pt, Co, Ni, Bi. Current flows from the earlier metal to the later one through the cold junction; since Fe comes before Cu, the current flows **from iron to copper through the cold junction** and from copper to iron through the hot junction.
3. $\alpha = 16\text{ }\mu\text{V }^\circ\text{C}^{-1}$; $\frac{1}{2}\beta = -0.02$ so $\beta = -0.04\text{ }\mu\text{V }^\circ\text{C}^{-2}$. $\theta_n = -\alpha/\beta = 16/0.04 = 400^\circ\text{C}$; $\theta_i = 2\theta_n = 800^\circ\text{C}$; $E_{\max} = 16(400) - 0.02(400)^2 = 6400 - 3200 = 3200\text{ }\mu\text{V} = 3.2\text{ mV}$; $S = 16 - 0.04(100) = 12\text{ }\mu\text{V }^\circ\text{C}^{-1}$.
4. Table in §15.2. $H = \pi It = (10 \times 10^{-3})(3.0)(200) = 6.0\text{ J}$.
5. Advantages: very wide range; small heat capacity so fast response; measures temperature at a point; reading can be taken remotely (and recorded electrically). Disadvantages: the e.m.f. is very small and needs a sensitive instrument; the calibration is non-linear and a reference junction must be maintained.

Group C — Long answer (8 marks each)

1. (a) Describe the Seebeck effect and sketch the variation of thermo-e.m.f. with the temperature of the hot junction. Define and locate the neutral temperature and the temperature of inversion on your graph. [4] (b) A copper-iron thermocouple has $\alpha = 13.5\text{ }\mu\text{V }^\circ\text{C}^{-1}$ and $\beta = -0.05\text{ }\mu\text{V }^\circ\text{C}^{-2}$ with the cold junction at 0°C . Find the neutral temperature, the temperature of inversion, the maximum e.m.f., and the e.m.f. when the hot junction is at 200°C . [4]
2. (a) State the Peltier effect and explain, with a diagram, how it differs from Joule heating. [4] (b) Describe the construction and working of a thermopile and give two of its uses. A thermopile of 50 antimony-bismuth couples, each of thermoelectric power $100\text{ }\mu\text{V }^\circ\text{C}^{-1}$, is connected to a galvanometer of total circuit resistance $25\text{ }\Omega$. Find the current when the junctions differ by 0.40°C . [4]

ANSWERS TO GROUP C

1. (b) $\theta_n = -\alpha/\beta = 13.5/0.05 = 270^\circ\text{C}$; $\theta_i = 2(270) = 540^\circ\text{C}$;

$$E_{\max} = 13.5(270) - 0.025(270)^2 = 3645 - 1822.5 = 1822.5\text{ }\mu\text{V} \approx 1.82\text{ mV}$$

$$E(200) = 13.5(200) - 0.025(200)^2 = 2700 - 1000 = 1700 \mu\text{V} = 1.70 \text{ mV}$$

2. (b) Total e.m.f. $E = NS\Delta\theta = 50 \times 100 \times 0.40 = 2000 \mu\text{V} = 2.0 \times 10^{-3} \text{ V}$. Hence

$$I = \frac{E}{R} = \frac{2.0 \times 10^{-3}}{25} = 8.0 \times 10^{-5} \text{ A} = 80 \mu\text{A}$$

which a sensitive galvanometer reads easily.

UNIT 16

Magnetic field

Electricity and Magnetism • 9 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Magnetic field lines and magnetic flux; Oersted's experiment
- Force on moving charge; Force on a conductor
- Force and Torque on rectangular coil; Moving coil galvanometer
- Hall effect
- Magnetic field of a moving charge
- Biot and Savart law and application to circular coil, long straight conductor, long solenoid
- Ampere's law and applications to long straight conductor, straight solenoid, toroidal solenoid
- Force between two parallel conductors carrying current; definition of ampere

Before 1820 electricity and magnetism were two separate sciences. Then Oersted noticed a compass needle twitch beside a current-carrying wire, and within ten years Biot, Savart and Ampère had turned that twitch into a complete theory: a moving charge makes a magnetic field, and a magnetic field pushes a moving charge. This unit is the longest in the electricity section because every motor, loud-speaker, galvanometer and Hall sensor you will ever meet is an application of those two sentences.

HOW THIS UNIT IS EXAMINED

Two derivations appear again and again in Group C: the field of a long straight conductor (Biot-Savart **or** Ampère) and the torque on a rectangular coil leading to the moving-coil galvanometer. Group B favours $r = mv/qB$, the solenoid, and the force between parallel wires with the definition of the ampere. Learn the direction rules until they are automatic — half the marks lost in this unit are lost on a wrong direction, not a wrong number.

16.1 Magnetic field lines and magnetic flux; Oersted's experiment

A **magnetic field** is the region around a magnet or a current in which a magnetic force can be detected. It is a vector field $\vec{B}$, called the **magnetic flux density**, and its SI unit is the **tesla** (T), where $1 \text{ T} = 1 \text{ N A}^{-1} \text{ m}^{-1} = 1 \text{ Wb m}^{-2}$.

A **magnetic field line** is a line drawn so that its tangent at every point gives the direction of $\vec{B}$ there. Their properties:

- They run from N to S **outside** a magnet and from S to N **inside** it, so every line is a **closed loop** — unlike electric field lines, which begin and end on charges.
- The number of lines per unit area crossing a surface at right angles measures the strength of $\vec{B}$: crowded lines mean a strong field.
- **Two field lines never intersect.** If they did, the field would have two directions at one point, which is impossible.

- Lines are in tension along their length and repel sideways — which is why unlike poles attract and like poles repel.

MAGNETIC FLUX

The magnetic flux through a surface of area A placed in a uniform field $\vec{B}$ is the product of the area and the component of $\vec{B}$ normal to it:

$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

where θ is the angle between $\vec{B}$ and the normal to the surface. The SI unit is the **weber** (Wb); $1 \text{ Wb} = 1 \text{ T m}^2$. Flux is a **scalar**.

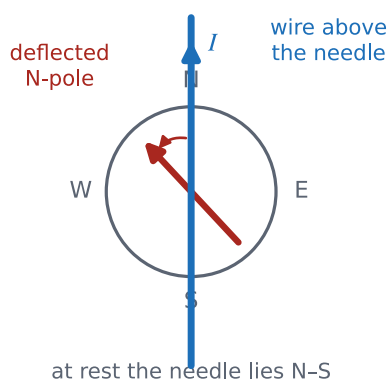
Because every field line is a closed loop, every line that enters a closed surface must leave it again. Hence **Gauss's law for magnetism**:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

The net magnetic flux through any closed surface is zero: isolated magnetic poles (monopoles) do not exist.

Oersted's experiment

(a) Oersted 1820 — plan view



(b) field of a long straight wire

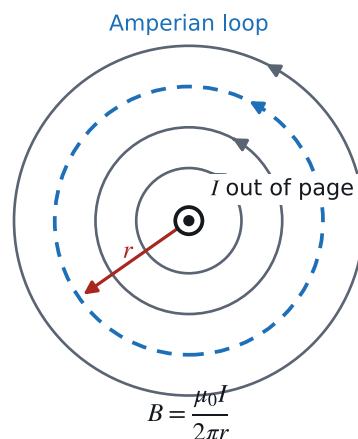


Figure 62. Left: Oersted's experiment — with the current flowing south to north in a wire above the needle, the N-pole deflects west. Right: the field of a long straight wire is a set of concentric circles; the dashed circle of radius r is the Amperian loop of §16.7.

Hans Christian Oersted placed a straight wire **above** a pivoted compass needle and parallel to it, i.e. along the magnetic north–south line. He found:

1. with no current, the needle rested along N–S;
2. with a current, the needle turned through a large angle and settled almost perpendicular to the wire;
3. reversing the current reversed the deflection;
4. the deflection increased with the current and decreased as the wire was moved further away.

Conclusion: an electric current produces a magnetic field around it.

SNOW AND THE RIGHT-HAND GRIP RULE

SNOW — current **S**outh to **N**orth in a wire **O**ver the needle sends the N-pole **W**est.

Right-hand grip rule — grip the wire with the right hand, thumb along the conventional current; the curled fingers give the direction of $\vec{B}$. This is the rule to use for every straight-wire problem.

16.2 Force on moving charge; Force on a conductor

A charge q moving with velocity $\vec{v}$ in a field $\vec{B}$ experiences

$$\vec{F} = q \vec{v} \times \vec{B}, \quad F = qvB \sin \theta$$

where θ is the angle between $\vec{v}$ and $\vec{B}$.

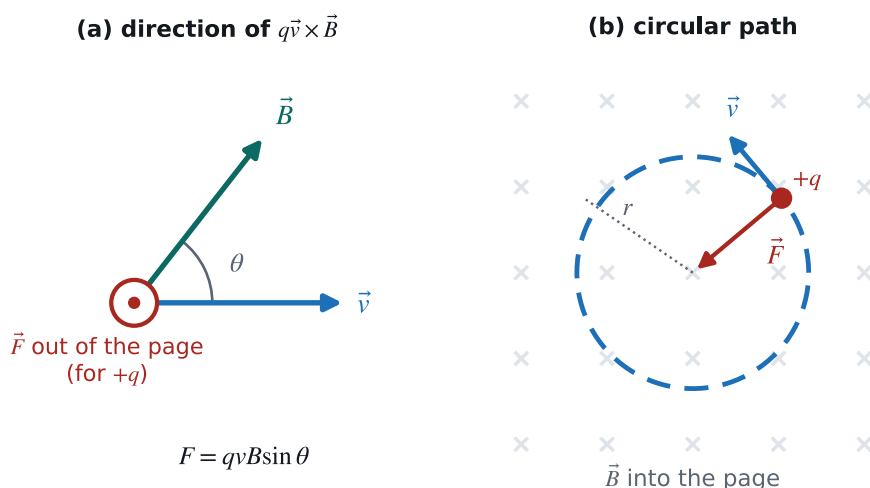


Figure 63. Left: $\vec{F} = q\vec{v} \times \vec{B}$ is perpendicular to the plane of $\vec{v}$ and $\vec{B}$. Right: a positive charge projected at right angles to a uniform field moves in a circle of radius $r = mv/qB$.

THE TESLA

The magnetic flux density is 1 tesla if a charge of 1 coulomb moving at 1 m s^{-1} **at right angles** to the field experiences a force of 1 newton.

Three consequences follow straight from $F = qvB \sin \theta$:

- If $\vec{v}$ is **parallel** or antiparallel to $\vec{B}$ ($\theta = 0$ or 180°) the force is **zero** — the particle goes straight on.
- The force is greatest when $\vec{v} \perp \vec{B}$, and then $F = qvB$.
- $\vec{F}$ is always perpendicular to $\vec{v}$, so the magnetic force **does no work**. It can change the direction of motion but never the speed or the kinetic energy.

Direction. Use **Fleming's left-hand rule** for the force on a positive charge (or a current): stretch the first three fingers of the left hand mutually perpendicular — **Fore**finger = **F**ield, **Cent**re finger = **C**urrent, **Th**umb = **T**hrust. For a negative charge, reverse the answer.

CIRCULAR MOTION OF A CHARGE IN A UNIFORM FIELD

If a charge enters at right angles to a uniform field, the force qvB stays perpendicular to $\vec{v}$ and constant in magnitude, so it acts as a centripetal force and the path is a circle:

$$qvB = \frac{mv^2}{r} \implies r = \frac{mv}{qB} = \frac{p}{qB}$$

The period is the circumference divided by the speed:

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}, \quad f = \frac{qB}{2\pi m}$$

Notice that T and f do **not** contain v : fast particles simply travel round bigger circles in the same time. This is the principle of the cyclotron. If the velocity has a component along $\vec{B}$ as well, that component is unaffected and the path becomes a **helix** of pitch $p = v \cos \theta \cdot T$.

WORKED EXAMPLE 16.1

Problem. A proton ($m = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C) enters a uniform magnetic field of 0.50 T at right angles with a speed of 2.0×10^6 m s⁻¹. Find the radius of its path, the period and the frequency of revolution.

Solution.

$$r = \frac{mv}{qB} = \frac{(1.67 \times 10^{-27})(2.0 \times 10^6)}{(1.6 \times 10^{-19})(0.50)} = \frac{3.34 \times 10^{-21}}{8.0 \times 10^{-20}} = 4.18 \times 10^{-2} \text{ m} = 4.18 \text{ cm}$$

$$T = \frac{2\pi m}{qB} = \frac{2\pi(1.67 \times 10^{-27})}{8.0 \times 10^{-20}} = \frac{1.049 \times 10^{-26}}{8.0 \times 10^{-20}} = 1.31 \times 10^{-7} \text{ s}$$

$$f = \frac{1}{T} = 7.62 \times 10^6 \text{ Hz} = 7.62 \text{ MHz}$$

Force on a current-carrying conductor

A current is a stream of moving charges, so a wire in a magnetic field feels a force.

$$F = BIL \sin \theta \quad \text{FROM} \quad F = qvB \sin \theta$$

Take a straight conductor of length L and cross-sectional area A containing n free electrons per unit volume, each of charge q , drifting with speed v_d at an angle θ to $\vec{B}$.

Force on **one** carrier: $f = qv_d B \sin \theta$.

Number of carriers in the length L : $N = nAL$.

Total force:

$$F = (nAL)(qv_d B \sin \theta) = (nAqv_d) LB \sin \theta$$

But the current is $I = nAqv_d$, so

$$F = BIL\sin\theta, \quad \vec{F} = I \vec{L} \times \vec{B}$$

The force is maximum (BIL) when the conductor is perpendicular to the field and zero when it lies along the field.

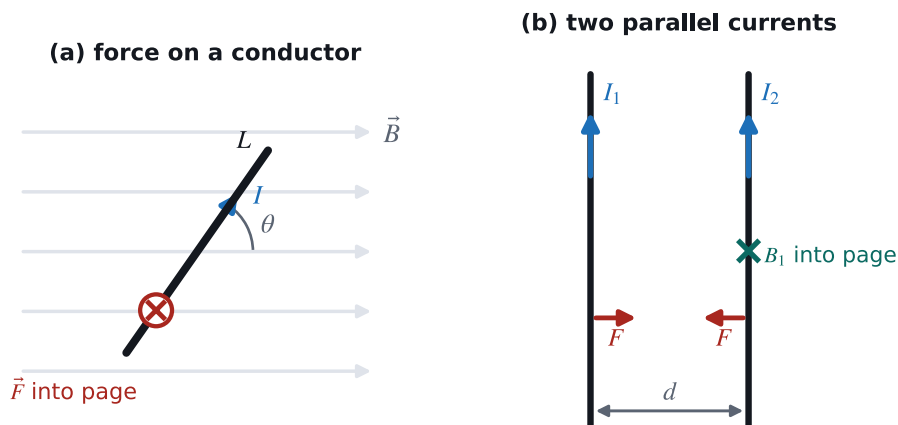


Figure 64. Left: a conductor at angle θ to the field feels $F = BIL\sin\theta$, here directed into the page. Right: parallel currents in the same direction attract, with $F/L = \mu_0 I_1 I_2 / 2\pi d$.

WORKED EXAMPLE 16.2

Problem. A straight wire of length 20 cm carries a current of 5.0 A and is placed in a uniform field of 0.40 T so that it makes 30° with the field. Find the force on it, and the force if the wire is turned perpendicular to the field.

Solution.

$$F = BIL\sin\theta = (0.40)(5.0)(0.20)\sin 30^\circ = (0.40)(5.0)(0.20)(0.50) = 0.20 \text{ N}$$

Turned perpendicular, $\sin\theta = 1$:

$$F_{\max} = BIL = (0.40)(5.0)(0.20) = 0.40 \text{ N}$$

16.3 Force and Torque on rectangular coil; Moving coil galvanometer

Consider a rectangular coil of N turns, length l and breadth b , carrying a current I in a uniform field $\vec{B}$ parallel to the plane of the coil.

TORQUE ON A RECTANGULAR COIL

The two sides of length b lie **along** the field, so the forces on them are zero or are equal, opposite and along the same line — they produce no turning effect.

Each side of length l is **perpendicular** to $\vec{B}$, so it carries a force

$$F = NBIL$$

The two forces are equal and opposite but act along **different** lines: they form a **couple**. If α is the angle between the plane of the coil and $\vec{B}$, the perpendicular distance between the two lines of action is $b \cos \alpha$, so

$$\tau = F \times b \cos \alpha = NBil \ b \cos \alpha = NBI A \cos \alpha$$

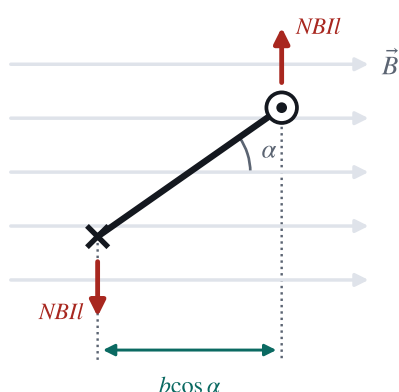
with $A = lb$ the area of the coil. Writing $\theta = 90^\circ - \alpha$ for the angle between the **normal** to the coil and $\vec{B}$,

$$\tau = NBI A \sin \theta$$

Defining the **magnetic dipole moment** of the coil as $m = NIA$ (unit $A \ m^2$), this is simply $\tau = mB \sin \theta$.

- τ is **maximum** $= NBI A$ when the plane of the coil contains $\vec{B}$ ($\alpha = 0$, $\theta = 90^\circ$).
- τ is **zero** when the plane is perpendicular to $\vec{B}$ ($\theta = 0$) — the coil is then in stable equilibrium.
- The result is independent of the shape of the coil: any plane coil of area A gives $\tau = NBI A \sin \theta$.

(a) couple on a coil (plan view)



(b) moving-coil galvanometer

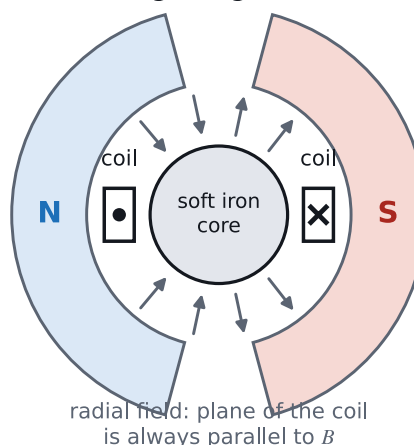


Figure 65. Left: plan view of a coil whose plane makes angle α with $\vec{B}$; the two forces $NBIl$ form a couple of arm $b \cos \alpha$. Right: cross-section of a moving-coil galvanometer — the concave poles and the soft-iron core make the field radial.

Moving coil galvanometer

Construction. A rectangular coil of many turns of fine insulated copper wire is wound on a light non-magnetic aluminium frame. It hangs by a thin **phosphor-bronze suspension strip** between the **concave** pole pieces of a strong permanent horseshoe magnet, with a cylindrical **soft-iron core** fixed inside it (not touching). The suspension strip carries the current in and provides the restoring couple; a fine spring below takes the current out. A small mirror on the strip and a lamp-and-scale arrangement read the deflection.

Why the field must be radial. The concave pole faces plus the soft-iron core make the field in the gap point everywhere along a radius of the cylinder. So however far the coil turns, its plane always stays **parallel** to $\vec{B}$, i.e. $\theta = 90^\circ$ and $\sin \theta = 1$ always.

WHY THE GALVANOMETER SCALE IS LINEAR

With a radial field the deflecting couple is $NBIA$ whatever the deflection. At equilibrium it is balanced by the restoring couple $c\varphi$ of the suspension, where c is the torsional constant (couple per unit twist):

$$NBIA = c\varphi \implies \varphi = \left(\frac{NBA}{c}\right)I$$

Since N , B , A and c are all constants, $\varphi \propto I$: the scale is **uniform**. Without a radial field we would get $\varphi \propto I \sin \theta$ and a badly cramped scale.

Quantity	Definition	Expression
Current sensitivity	deflection per unit current	$\varphi/I = NBA/c$
Voltage sensitivity	deflection per unit p.d.	$\varphi/V = NBA/cG$

To make a galvanometer more sensitive: increase N , A and B , and decrease c (hence phosphor bronze, which has a very small torsional constant). The soft-iron core does double duty — it makes the field radial and concentrates the flux, increasing B .

HIGH CURRENT SENSITIVITY DOES NOT MEAN HIGH VOLTAGE SENSITIVITY

Doubling the number of turns N doubles NBA/c , but it also roughly doubles the coil resistance G , so the voltage sensitivity NBA/cG hardly changes. An instrument can be very sensitive to current and poor at detecting small p.d.s.

WORKED EXAMPLE 16.3

Problem. A rectangular coil of 100 turns and dimensions $5.0 \text{ cm} \times 4.0 \text{ cm}$ carries a current of 2.0 A in a uniform field of 0.20 T . Find the torque when (a) the plane of the coil is parallel to the field and (b) the plane makes 60° with the field.

Solution. Area $A = 0.050 \times 0.040 = 2.0 \times 10^{-3} \text{ m}^2$.

(a) Plane parallel to $\vec{B}$ means $\alpha = 0$, so $\cos \alpha = 1$:

$$\tau = NBIA = (100)(0.20)(2.0)(2.0 \times 10^{-3}) = 0.080 \text{ N m}$$

(b) $\alpha = 60^\circ$, so $\cos 60^\circ = 0.50$:

$$\tau = NBIA \cos \alpha = 0.080 \times 0.50 = 0.040 \text{ N m}$$

16.4 Hall effect

HALL EFFECT

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, a potential difference is set up across the conductor in the direction perpendicular to **both** the current and the field. This transverse p.d. is the **Hall voltage** V_H .

THE HALL VOLTAGE

Let a slab of thickness t (measured along $\vec{B}$) and width w carry a current I of carriers of charge q , number density n and drift speed v_d , with $\vec{B}$ perpendicular to I .

Step 1. Each carrier feels a magnetic force qv_dB pushing it sideways, so charge piles up on one face and the opposite charge is left on the other.

Step 2. The separated charge sets up a transverse **Hall field** E_H . Equilibrium is reached when the electric force balances the magnetic force:

$$qE_H = qv_dB \implies E_H = v_dB$$

Step 3. The Hall voltage across the width w is

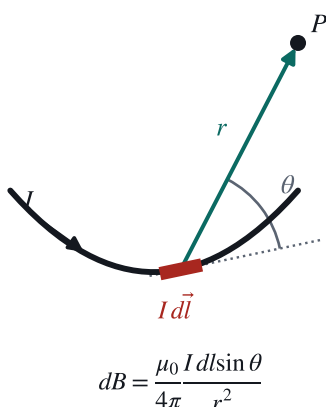
$$V_H = E_H w = v_d B w$$

Step 4. From $I = nqv_dA = nqv_d(wt)$ we get $v_d = I/(nqwt)$, so

$$V_H = \frac{IB}{nqt}$$

The **Hall coefficient** is defined as $R_H = 1/nq$, giving $V_H = R_H IB/t$.

(a) Biot-Savart law



(b) Hall effect

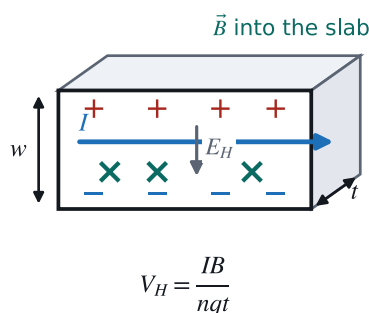


Figure 66. Left: the Biot-Savart geometry — the element $I d\vec{l}$, the distance r to P and the angle θ between them. Right: Hall effect — positive carriers are pushed to the upper face, making it positive; t is measured along $\vec{B}$.

Uses of the Hall effect.

- The **sign** of V_H gives the **sign of the charge carriers** — this is how we know that conduction in an n-type semiconductor is by electrons and in a p-type by holes.
- From $R_H = 1/nq$ we get the **carrier concentration** n , and hence the drift velocity.
- A calibrated slab is a **Hall probe**, the standard instrument for measuring magnetic flux density.
- Hall sensors count the revolutions of motors and wheels and detect whether a phone's flip cover is closed.

WORKED EXAMPLE 16.4

Problem. A copper strip of thickness 0.50 mm carries a current of 10 A in a magnetic field of 1.2 T applied perpendicular to its face. If copper has $n = 8.5 \times 10^{28} \text{ m}^{-3}$ free electrons, find the Hall voltage and the Hall coefficient.

Solution.

$$nq = (8.5 \times 10^{28})(1.6 \times 10^{-19}) = 1.36 \times 10^{10} \text{ C m}^{-3}$$

$$V_H = \frac{IB}{nqt} = \frac{(10)(1.2)}{(1.36 \times 10^{10})(0.50 \times 10^{-3})} = \frac{12}{6.8 \times 10^6} = 1.76 \times 10^{-6} \text{ V} = 1.76 \mu\text{V}$$

$$R_H = \frac{1}{nq} = \frac{1}{1.36 \times 10^{10}} = 7.35 \times 10^{-11} \text{ m}^3\text{C}^{-1}$$

The Hall voltage in a metal is tiny because n is enormous. In a semiconductor n is about 10^{16} times smaller, so V_H is millions of times larger — which is why Hall probes are made of semiconductors.

16.5 Magnetic field of a moving charge

A single charge q moving with velocity $\vec{v}$ sets up a magnetic field at a point P whose position vector from the charge is $\vec{r}$:

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r^2}, \quad \vec{B} = \frac{\mu_0}{4\pi} \frac{q \vec{v} \times \hat{r}}{r^2}$$

where θ is the angle between $\vec{v}$ and $\vec{r}$, and

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}, \quad \frac{\mu_0}{4\pi} = 10^{-7} \text{ T m A}^{-1}$$

is the **permeability of free space**. Points to note:

- $B = 0$ along the line of motion ($\theta = 0$ or 180°) and is greatest in the plane through the charge perpendicular to $\vec{v}$.
- $\vec{B}$ is perpendicular to both $\vec{v}$ and $\vec{r}$; the lines are circles centred on the line of motion — the right-hand grip rule again.
- A stationary charge ($v = 0$) produces **no** magnetic field, only an electric one.

For example, an electron moving at $2.0 \times 10^7 \text{ m s}^{-1}$ produces, at a point 1.0 nm away at right angles to its motion,

$$B = 10^{-7} \times \frac{(1.6 \times 10^{-19})(2.0 \times 10^7)}{(1.0 \times 10^{-9})^2} = \frac{3.2 \times 10^{-19}}{1.0 \times 10^{-18}} = 0.32 \text{ T}$$

— an enormous field, which is why atomic-scale magnetism is so strong.

Adding up the fields of all the carriers in a short length dl of wire, with $I = nAqv_d$, converts this formula into the Biot–Savart law of the next section: $q\vec{v} \rightarrow I \vec{dl}$.

16.6 Biot and Savart law and application to circular coil, long straight conductor, long solenoid

BIOT-SAVART LAW

The magnetic flux density dB at a point P due to a small element of length dl of a conductor carrying current I is

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

where r is the distance of P from the element and θ is the angle between the element and the line joining it to P . The direction of $d\vec{B}$ is that of $\vec{dl} \times \hat{r}$, i.e. perpendicular to the plane containing the element and P .

So $dB \propto I$, $dB \propto dl$, $dB \propto \sin \theta$ and $dB \propto 1/r^2$. The total field is found by integrating over the whole conductor.

(a) Long straight conductor

Let P be at perpendicular distance a from a straight wire. Taking an element at distance l from the foot of the perpendicular and measuring the angle ϕ from that perpendicular, integration gives the standard result

$$B = \frac{\mu_0 I}{4\pi a} (\sin \phi_1 + \sin \phi_2)$$

where ϕ_1 and ϕ_2 are the angles subtended at P by the two ends. For an **infinitely long** wire $\phi_1 = \phi_2 = 90^\circ$, so

$$B = \frac{\mu_0 I}{2\pi a}$$

and for a semi-infinite wire ending at the foot of the perpendicular, $B = \mu_0 I / 4\pi a$.

(b) Circular coil — field at the centre

FIELD AT THE CENTRE OF A CIRCULAR COIL

Take a circular coil of radius a carrying current I . Every element dl of the coil is perpendicular to the radius drawn to the centre, so $\theta = 90^\circ$ and $\sin \theta = 1$, and every element is the same distance a

from the centre. Also, every element contributes $d\vec{B}$ in the **same** direction — along the axis — so the magnitudes simply add:

$$B = \oint dB = \frac{\mu_0}{4\pi} \frac{I}{a^2} \oint dl = \frac{\mu_0}{4\pi} \frac{I}{a^2} (2\pi a)$$

$$B = \frac{\mu_0 I}{2a} \quad \text{and for } N \text{ turns} \quad B = \frac{\mu_0 NI}{2a}$$

At a point on the **axis** at distance x from the centre the same integration gives

$$B = \frac{\mu_0 NI a^2}{2(a^2 + x^2)^{3/2}}$$

which reduces to $\mu_0 NI/2a$ at the centre ($x=0$) and falls off as $1/x^3$ far away.

(c) Long solenoid

For a solenoid with n turns per unit length, integrating the on-axis coil formula over all the turns gives, at an axial point,

$$B = \frac{1}{2} \mu_0 n I (\sin \alpha_1 + \sin \alpha_2)$$

where α_1 and α_2 are the semi-angles subtended by the two ends. Two limits matter:

Position	Angles	Field
Middle of a very long solenoid	$\alpha_1 = \alpha_2 = 90^\circ$	$B = \mu_0 n I$
At one end of a long solenoid	$\alpha_1 = 0, \alpha_2 = 90^\circ$	$B = \frac{1}{2} \mu_0 n I$

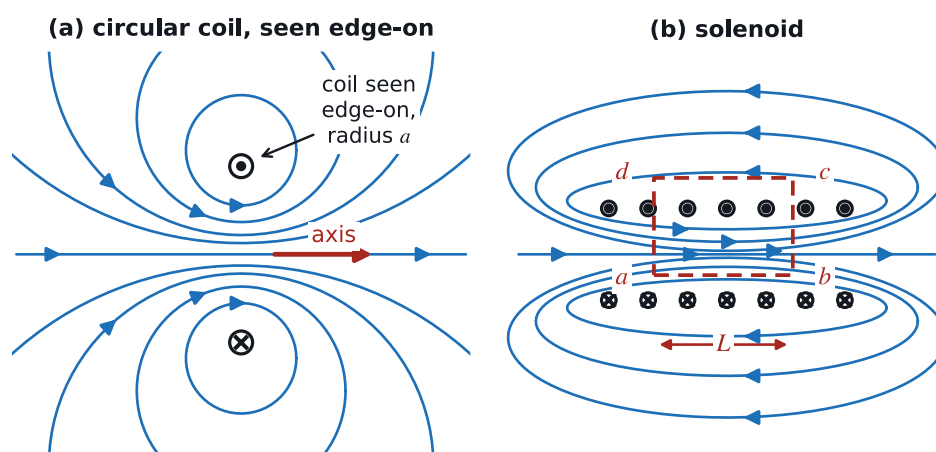


Figure 67. Field lines computed from the currents themselves. Left: a circular coil seen edge-on, the field through the centre is $\mu_0 I/2a$. Right: a solenoid — the field inside is uniform and the rectangle $abcd$ is the Amperian loop of §16.7.

WORKED EXAMPLE 16.5

Problem. (a) A circular coil of 50 turns and radius 10 cm carries a current of 2.0 A. Find the flux density at its centre. (b) A long straight wire carries 10 A. Find the flux density at a point 5.0 cm from it.

Solution.

(a)

$$B = \frac{\mu_0 NI}{2a} = \frac{(4\pi \times 10^{-7})(50)(2.0)}{2(0.10)} = \frac{1.2566 \times 10^{-4}}{0.20} = 6.28 \times 10^{-4} \text{ T}$$

(b)

$$B = \frac{\mu_0 I}{2\pi a} = \frac{(2 \times 10^{-7})(10)}{0.050} = \frac{2.0 \times 10^{-6}}{0.050} = 4.0 \times 10^{-5} \text{ T}$$

(Using $\mu_0/2\pi = 2 \times 10^{-7}$ saves a line of arithmetic every time.)

$2\pi a$ OR $2a$?

$B = \mu_0 I / 2\pi a$ is the field of a **straight** wire at distance a ; $B = \mu_0 I / 2a$ is the field at the **centre of a circular coil** of radius a . They differ by a factor of π . Check which geometry the question describes before you write the formula.

16.7 Ampere's law and applications to long straight conductor, straight solenoid, toroidal solenoid

AMPÈRE'S CIRCUITAL LAW

The line integral of $\vec{B}$ around any **closed** path (an Amperian loop) equals μ_0 times the **net current** threading that path:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

Currents are counted positive if they are related to the sense of travel round the loop by the right-hand rule.

Ampère's law is to magnetism what Gauss's law is to electrostatics: always true, but only useful where the symmetry lets you take B outside the integral.

(a) Long straight conductor

Choose a circle of radius r centred on the wire, in the plane perpendicular to it. By symmetry B has the same magnitude everywhere on it and is everywhere tangential, so $\vec{B} \cdot d\vec{l} = B dl$:

$$\oint \vec{B} \cdot d\vec{l} = B \oint dl = B(2\pi r) = \mu_0 I \implies B = \frac{\mu_0 I}{2\pi r}$$

in two lines, where Biot–Savart needed an integration. **Inside** a solid cylindrical conductor of radius R carrying a uniform current, the loop of radius $r < R$ encloses only the fraction Ir^2/R^2 , giving

$$B = \frac{\mu_0 I r}{2\pi R^2} \quad (r < R)$$

so B rises linearly from zero on the axis to $\mu_0 I / 2\pi R$ at the surface, then falls as $1/r$ outside.

(b) Straight solenoid

$B = \mu_0 n I$ **INSIDE A LONG SOLENOID**

Take the rectangular Amperian loop $abcd$ of figure 5(b), with side $ab = L$ lying along the axis inside the solenoid and side cd well outside it. Split the integral into four parts:

- along ab (inside, $\vec{B}$ parallel to $d\vec{l}$): contributes BL ;
- along bc and da : $\vec{B}$ is perpendicular to $d\vec{l}$ where the path is inside, and $B \approx 0$ outside, so each contributes **zero**;
- along cd (outside a long solenoid, where $B \approx 0$): contributes **zero**.

So $\oint \vec{B} \cdot d\vec{l} = BL$. The number of turns threading the loop is nL , each carrying I , so $I_{enc} = nLI$ and

$$BL = \mu_0 n L I \implies B = \mu_0 n I$$

The result contains neither L nor the position of the loop inside the solenoid: **the field inside a long solenoid is uniform** and depends only on the turns per unit length and the current.

(c) Toroidal solenoid

A toroid is a solenoid bent into a ring of N turns. Take a circular Amperian loop of radius r running along the middle of the core. By symmetry B is constant and tangential along it, and the loop is threaded by all N turns:

$$B(2\pi r) = \mu_0 N I \implies B = \frac{\mu_0 N I}{2\pi r}$$

Writing $n = N/2\pi r$ for the turns per unit length along the core recovers $B = \mu_0 n I$. For a loop drawn **inside the hole** or **outside the toroid**, the enclosed current is zero, so $B = 0$ there: a toroid confines its field completely inside the windings — which is why transformer and inductor cores are toroidal.

WORKED EXAMPLE 16.6

Problem. A solenoid 50 cm long has 500 turns and carries 3.0 A. Find the flux density (a) at its centre and (b) at one end. (c) The same wire is wound into a toroid of mean radius 15 cm with 500 turns and the same current; find the flux density along the mean circle.

Solution. Turns per unit length $n = 500/0.50 = 1000 \text{ m}^{-1}$.

(a)

$$B = \mu_0 n I = (4\pi \times 10^{-7})(1000)(3.0) = 3.77 \times 10^{-3} \text{ T}$$

(b) At the end the field is half that value: $B = 1.88 \times 10^{-3} \text{ T}$.

(c)

$$B = \frac{\mu_0 NI}{2\pi r} = \frac{(2 \times 10^{-7})(500)(3.0)}{0.15} = \frac{3.0 \times 10^{-4}}{0.15} = 2.0 \times 10^{-3} \text{ T}$$

16.8 Force between two parallel conductors carrying current; definition of ampere

Two long parallel wires a distance d apart carry currents I_1 and I_2 .

FORCE PER UNIT LENGTH BETWEEN PARALLEL CURRENTS

Wire 1 produces at the position of wire 2 a field

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

which is perpendicular to wire 2. A length L of wire 2 therefore feels

$$F = B_1 I_2 L = \frac{\mu_0 I_1 I_2 L}{2\pi d} \implies \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

By Newton's third law wire 1 feels an equal and opposite force. Applying Fleming's left-hand rule shows that

- **currents in the same direction attract**, and
- **currents in opposite directions repel**

LIKE CURRENTS ATTRACT — THE OPPOSITE OF CHARGES

Two like charges repel, but two like currents attract. Students mix these up constantly. Remember it as: parallel wires pull together, anti-parallel wires push apart.

THE AMPERE

One **ampere** is that steady current which, when maintained in each of two infinitely long straight parallel conductors of negligible circular cross-section placed 1 metre apart in vacuum, produces a force of 2×10^{-7} newton per metre of length between them.

Check it: with $I_1 = I_2 = 1 \text{ A}$ and $d = 1 \text{ m}$,

$$\frac{F}{L} = \frac{(4\pi \times 10^{-7})(1)(1)}{2\pi(1)} = 2 \times 10^{-7} \text{ N m}^{-1}$$

A modern footnote. Since 20 May 2019 the ampere has been defined instead by fixing the elementary charge at $e = 1.602176634 \times 10^{-19} \text{ C}$, so that 1 A is the current corresponding to $1/e$ elementary charges per second. As a result μ_0 is now a measured quantity, though it

still equals $4\pi \times 10^{-7} \text{ T m A}^{-1}$ to within a few parts in 10^{10} . NEB papers still ask for the force-based definition above.

WORKED EXAMPLE 16.7

Problem. Two long parallel wires 10 cm apart carry currents of 5.0 A and 10 A in the same direction. Find the force per metre between them, and the total force on a 2.0 m length. State whether it is attractive or repulsive. Where on the line joining them is the resultant field zero?

Solution.

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(2 \times 10^{-7})(5.0)(10)}{0.10} = \frac{1.0 \times 10^{-5}}{0.10} = 1.0 \times 10^{-4} \text{ N m}^{-1}$$

On 2.0 m; $F = 2.0 \times 10^{-4} \text{ N}$, **attractive**, because the currents are in the same direction.

Between the wires the two fields oppose. At a distance x from the 5 A wire,

$$\frac{\mu_0(5.0)}{2\pi x} = \frac{\mu_0(10)}{2\pi(0.10 - x)} \Rightarrow 5(0.10 - x) = 10x \Rightarrow 0.50 = 15x$$

so $x = 0.0333 \text{ m} = 3.33 \text{ cm}$ from the 5 A wire.

Chapter summary

- Magnetic flux $\Phi = BA \cos \theta$ (weber); field lines are closed loops, so $\oint \vec{B} \cdot d\vec{A} = 0$ — there are no magnetic monopoles.
- Oersted: a current produces a magnetic field. Direction by the right-hand grip rule (or SNOW).
- Force on a moving charge $\vec{F} = q\vec{v} \times \vec{B}$, $F = qvB \sin \theta$; it does no work. Perpendicular entry gives a circle with $r = mv/qB$, $T = 2\pi m/qB$, independent of speed.
- Force on a conductor $F = BIL \sin \theta$; direction by Fleming's left-hand rule.
- Torque on a coil $\tau = NBI A \sin \theta = NBI A \cos \alpha$, with dipole moment $m = NIA$. A radial field makes $\sin \theta = 1$, so a moving-coil galvanometer has $\phi = (NBA/c)I$ and a uniform scale.
- Hall effect: $V_H = IB/nqt$ with $R_H = 1/nq$; it gives the sign and the concentration of the carriers.
- Biot–Savart: $dB = (\mu_0/4\pi)(I dl \sin \theta/r^2)$. Results: straight wire $\mu_0 I/2\pi a$; centre of a coil $\mu_0 NI/2a$; on the axis $\mu_0 NIa^2/2(a^2 + x^2)^{3/2}$; long solenoid $\mu_0 nI$ (half that at an end).
- Ampère: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$, which gives the straight wire, the solenoid ($\mu_0 nI$) and the toroid ($\mu_0 NI/2\pi r$, zero outside) in a few lines each.
- Parallel wires: $F/L = \mu_0 I_1 I_2/2\pi d$; like currents attract. This force defines the ampere as $2 \times 10^{-7} \text{ N m}^{-1}$ for 1 A at 1 m.

Practice questions

Group A — Multiple choice (1 mark each)

1. The SI unit of magnetic flux is [1] (a) tesla (b) weber (c) henry (d) ampere-metre
2. A charged particle moves parallel to a uniform magnetic field. The magnetic force on it is [1] (a) qvB (b) zero (c) $qvB \sin 45^\circ$ (d) maximum

- A charged particle enters a uniform magnetic field at right angles to it. Its path is [1] (a) a straight line (b) a parabola (c) a circle (d) a helix
- The magnetic flux density at the centre of a circular coil of N turns and radius a carrying a current I is [1] (a) $\mu_0 NI/2\pi a$ (b) $\mu_0 NI/2a$ (c) $\mu_0 NI/4\pi a$ (d) $\mu_0 NIa$
- Two long parallel wires carry currents in the same direction. They [1] (a) attract each other (b) repel each other (c) exert no force (d) rotate about each other
- The magnetic flux density inside a very long solenoid of n turns per metre carrying current I is [1] (a) $\mu_0 nI/2$ (b) $\mu_0 nI$ (c) $\mu_0 nI/2\pi$ (d) zero
- The field of a moving-coil galvanometer is made radial so that [1] (a) the coil does not rotate (b) the deflection is proportional to the current (c) the coil resistance is reduced (d) the torque is zero
- The Hall voltage of a slab of thickness t is proportional to [1] (a) t (b) $1/t$ (c) t^2 (d) $\sqrt{t}$

ANSWERS TO GROUP A

1. (b) — $\Phi = BA \cos \theta$, measured in weber = T m². 2. (b) — $\sin 0^\circ = 0$, so $F = qvB \sin \theta = 0$. 3. (c) — the force stays perpendicular to v and constant, so it is centripetal. 4. (b) — $B = \mu_0 NI/2a$; the 2π form belongs to a straight wire. 5. (a) — like currents attract. 6. (b) — Ampère's law on a rectangular loop gives $B = \mu_0 nI$. 7. (b) — with $\sin \theta = 1$ always, $\phi = (NBA/c)I$, a uniform scale. 8. (b) — $V_H = IB/nqt$.

Group B — Short answer (5 marks each)

- Define magnetic flux and state its SI unit. Describe Oersted's experiment and state the conclusion drawn from it. [5]
- Starting from the force on a single moving charge, derive the expression $F = BIL \sin \theta$ for the force on a current-carrying conductor in a magnetic field. [5]
- An electron ($m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C) moves at 3.0×10^7 m s⁻¹ at right angles to a magnetic field of 0.20 T. Find the radius of its path and its period of revolution. [5]
- State the Biot-Savart law and use it to find the magnetic flux density at the centre of a circular coil of N turns and radius a carrying current I . [5]
- Explain the Hall effect and show that $V_H = IB/nqt$. A metal strip of thickness 1.0 mm with $n = 5.0 \times 10^{28}$ m⁻³ carries 20 A in a field of 1.5 T; find the Hall voltage. [5]
- Two long parallel wires 20 cm apart carry currents of 10 A and 15 A in **opposite** directions. Find the force per metre between them and state its nature. Hence define the ampere. [5]

ANSWERS TO GROUP B

3.

$$r = \frac{mv}{eB} = \frac{(9.1 \times 10^{-31})(3.0 \times 10^7)}{(1.6 \times 10^{-19})(0.20)} = \frac{2.73 \times 10^{-23}}{3.2 \times 10^{-20}} = 8.53 \times 10^{-4} \text{ m} = 0.853 \text{ mm}$$

$$T = \frac{2\pi m}{eB} = \frac{2\pi(9.1 \times 10^{-31})}{3.2 \times 10^{-20}} = 1.79 \times 10^{-10} \text{ s}$$

5. Derivation as in §16.4. Numerically, $nq = (5.0 \times 10^{28})(1.6 \times 10^{-19}) = 8.0 \times 10^9$, so

$$V_H = \frac{IB}{nqt} = \frac{(20)(1.5)}{(8.0 \times 10^9)(1.0 \times 10^{-3})} = \frac{30}{8.0 \times 10^6} = 3.75 \times 10^{-6} \text{ V} = 3.75 \mu\text{V}$$

6.

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(2 \times 10^{-7})(10)(15)}{0.20} = \frac{3.0 \times 10^{-5}}{0.20} = 1.5 \times 10^{-4} \text{ N m}^{-1}$$

The currents are opposite, so the force is **repulsive**. Definition of the ampere: see §16.8.

Group C — Long answer (8 marks each)

- (a) State the Biot–Savart law and use it to derive the magnetic flux density at a perpendicular distance a from a long straight conductor carrying a current I . [5] (b) A long straight wire carries 20 A. Find the flux density 10 cm from it. A circular coil of 100 turns and radius 10 cm carries 2.0 A; find the flux density at its centre. [3]
- (a) Derive an expression for the torque on a rectangular coil carrying a current in a uniform magnetic field. Explain, with a diagram, why the field of a moving-coil galvanometer is made radial and show that its scale is uniform. [5] (b) A galvanometer coil of 200 turns and area $2.0 \times 10^{-4} \text{ m}^2$ hangs in a radial field of 0.20 T. The torsional constant of the suspension is $1.0 \times 10^{-6} \text{ N m}$ per degree. Find the deflection produced by 5.0 mA and the current sensitivity. [3]
- (a) State Ampère's circuital law and apply it to find the magnetic flux density inside a long straight solenoid and inside a toroidal solenoid. [5] (b) A toroid of mean radius 15 cm has 500 turns and carries 3.0 A. Find the flux density along its mean circle and the flux density outside the toroid. [3]

ANSWERS TO GROUP C

1. (b) Straight wire:

$$B = \frac{\mu_0 I}{2\pi a} = \frac{(2 \times 10^{-7})(20)}{0.10} = 4.0 \times 10^{-5} \text{ T}$$

Circular coil:

$$B = \frac{\mu_0 NI}{2a} = \frac{(4\pi \times 10^{-7})(100)(2.0)}{2(0.10)} = \frac{2.513 \times 10^{-4}}{0.20} = 1.26 \times 10^{-3} \text{ T}$$

2. (b)

$$\tau = NBIA = (200)(0.20)(5.0 \times 10^{-3})(2.0 \times 10^{-4}) = 4.0 \times 10^{-5} \text{ N m}$$

$$\varphi = \frac{\tau}{c} = \frac{4.0 \times 10^{-5}}{1.0 \times 10^{-6}} = 40 \text{ degrees}$$

Current sensitivity $= \frac{NBA}{c} = \frac{(200)(0.20)(2.0 \times 10^{-4})}{1.0 \times 10^{-6}} = 8.0 \times 10^3 \text{ degrees per ampere, i. e. } 8 \text{ degrees per milliampere.}$

3. (b)

$$B = \frac{\mu_0 NI}{2\pi r} = \frac{(2 \times 10^{-7})(500)(3.0)}{0.15} = \frac{3.0 \times 10^{-4}}{0.15} = 2.0 \times 10^{-3} \text{ T}$$

Outside the toroid (and in the central hole) the Amperian loop encloses no net current, so $B = 0$.

UNIT 17

Magnetic properties of materials

Electricity and Magnetism • 5 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Magnetic field lines and magnetic flux
- Flux density in magnetic material; Relative permeability; Susceptibility
- Hysteresis
- Dia-, para- and ferro-magnetic materials

Unit 16 treated magnetic fields in empty space. Real coils, however, are wound on iron, and the iron changes everything: the same current produces a field a thousand times stronger, and the material remembers what you did to it. This unit introduces the two field vectors $\vec{B}$ and $\vec{H}$, the numbers that link them (permeability and susceptibility), the loop that describes magnetic memory (hysteresis), and the three-way classification of all matter as diamagnetic, paramagnetic or ferromagnetic.

WHAT THE EXAMINER WANTS HERE

Three items dominate: the derivation $\mu_r = 1 + \chi_m$, a labelled hysteresis loop with retentivity and coercivity defined, and the soft-iron-versus-steel comparison table. Numericals are almost always "find H , then B , then M " for a solenoid or ring with a core. Keep B (tesla) and H (ampere per metre) strictly apart — mixing their units is the single commonest error in this unit.

17.1 Magnetic field lines and magnetic flux

A **magnetic field line** is a curve whose tangent at every point gives the direction of the magnetic flux density $\vec{B}$ at that point. Because there are no magnetic monopoles, every field line closes on itself: outside a magnet it runs N to S, inside the magnet it continues S to N.

MAGNETIC FLUX

The magnetic flux through a surface of area A in a uniform field is

$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

where θ is the angle between $\vec{B}$ and the normal to the surface. Flux is a **scalar**; its SI unit is the **weber** (Wb), and $1 \text{ Wb} = 1 \text{ T m}^2$.

Rearranged, $B = \Phi/A$ for a surface held perpendicular to the field, which is why B is called the **magnetic flux density** — flux per unit area. Since every line entering a closed surface must leave it,

$$\oint \vec{B} \cdot d\vec{A} = 0$$

This is **Gauss's law for magnetism**, and it holds inside matter as well as in vacuum. Its practical consequence is that flux is continuous: if a magnetic circuit narrows, the flux stays the same and B rises.

Magnetic materials are studied with a **toroidal ring** (a Rowland ring) because in a toroid the flux is trapped entirely inside the core, with no leakage and no end effects. The whole of the winding's magnetising action is then felt by the specimen.

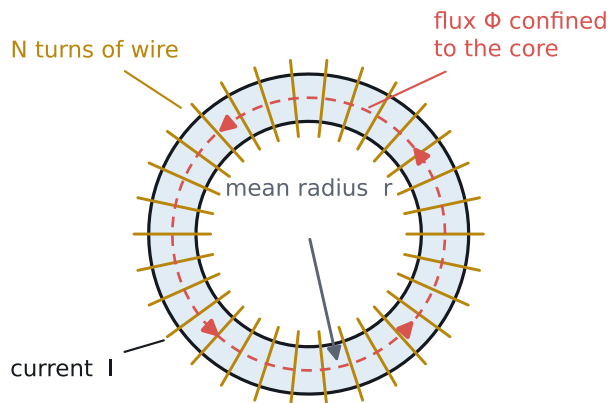


Figure 68. Rowland ring: N turns wound on a toroidal specimen of mean radius r . The flux Φ is confined to the core, and the magnetising field is $H = NI/2\pi r$.

WORKED EXAMPLE 17.1

Problem. A closed iron ring of cross-sectional area 4 cm^2 and mean circumference 0.50 m carries a flux of $6 \times 10^{-4} \text{ Wb}$. The relative permeability of the iron is 1200 . Find (a) the flux density in the core and (b) the ampere-turns needed on the winding.

Solution.

(a) $A = 4 \times 10^{-4} \text{ m}^2$, so

$$B = \frac{\Phi}{A} = \frac{6 \times 10^{-4}}{4 \times 10^{-4}} = 1.5 \text{ T}$$

(b) Inside the core $B = \mu_0 \mu_r H$, hence

$$H = \frac{B}{\mu_0 \mu_r} = \frac{1.5}{(4\pi \times 10^{-7})(1200)} = \frac{1.5}{1.508 \times 10^{-3}} = 995 \text{ A m}^{-1}$$

For a ring, $H = NI/l$ with $l = 0.50 \text{ m}$, so

$$NI = Hl = 995 \times 0.50 \approx 5.0 \times 10^2 \text{ ampere-turns}$$

17.2 Flux density in magnetic material; Relative permeability; Susceptibility

When a specimen is placed inside a current-carrying solenoid, two separate causes produce the field inside it:

1. the **free current** in the winding, described by the **magnetising field** $\vec{H}$, and

2. the **atomic currents** inside the specimen itself, described by the **intensity of magnetisation** $\vec{M}$.

MAGNETISING FIELD AND MAGNETISATION

The **magnetising field** H of a long solenoid or ring is the ampere-turns per metre of its winding, $H = nI = NI/l$. The **intensity of magnetisation** M is the net magnetic dipole moment per unit volume of the specimen, $M = m/V$. Both are measured in **ampere per metre (A m⁻¹)**, and both are independent of the material's response.

The resultant flux density inside the specimen is the sum of the two contributions:

$$B = \mu_0(H + M)$$

THE RELATION $\mu_r = 1 + \chi_m$

For a material that responds linearly, the magnetisation produced is proportional to the magnetising field. The constant of proportionality is the **magnetic susceptibility**:

$$\chi_m = \frac{M}{H}$$

Since M and H have the same unit, χ_m is a **pure number**. Substituting $M = \chi_m H$,

$$B = \mu_0(H + \chi_m H) = \mu_0(1 + \chi_m)H$$

The **absolute permeability** of the medium is defined by $\mu = B/H$, and the **relative permeability** by $\mu_r = \mu/\mu_0$. Comparing,

$$\mu = \mu_0(1 + \chi_m) \implies \mu_r = 1 + \chi_m$$

Quantity	Symbol	Defining relation	SI unit
Magnetic flux	Φ	$BA \cos \theta$	weber (Wb)
Flux density	B	Φ/A	tesla (T)
Magnetising field	H	NI/l	A m ⁻¹
Magnetisation	M	m/V	A m ⁻¹
Susceptibility	χ_m	M/H	none
Permeability	μ	B/H	T m A ⁻¹ (= H m ⁻¹)
Relative permeability	μ_r	μ/μ_0	none

Here $\mu_0 = 4\pi \times 10^{-7}$ T m A⁻¹ is the permeability of free space; for vacuum $\chi_m = 0$ and $\mu_r = 1$ exactly.

***B* AND *H* ARE NOT THE SAME THING**

H is fixed by the current you pass through the winding; *B* is what the material gives you back. Changing the core changes *B* but not *H*. Writing "*H* = 1.5 T" or "*B* = 2000 A m⁻¹" costs marks even if the arithmetic is right.

WORKED EXAMPLE 17.2

Problem. A solenoid 50 cm long has 500 turns and carries 2.0 A. An iron rod fills the solenoid, and the flux density measured inside is 1.6 T. Calculate *H*, μ , μ_r , *M* and χ_m .

Solution. Turns per metre: $n = 500/0.50 = 1000 \text{ m}^{-1}$.

$$H = nI = 1000 \times 2.0 = 2.0 \times 10^3 \text{ A m}^{-1}$$

$$\mu = \frac{B}{H} = \frac{1.6}{2.0 \times 10^3} = 8.0 \times 10^{-4} \text{ T m A}^{-1}$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{8.0 \times 10^{-4}}{4\pi \times 10^{-7}} = 637$$

From $B = \mu_0(H + M)$,

$$M = \frac{B}{\mu_0} - H = \frac{1.6}{4\pi \times 10^{-7}} - 2000 = 1.273 \times 10^6 - 2 \times 10^3 = 1.27 \times 10^6 \text{ A m}^{-1}$$

$$\chi_m = \frac{M}{H} = \frac{1.27 \times 10^6}{2.0 \times 10^3} = 636$$

As a check, $\mu_r = 1 + \chi_m = 637$, which agrees.

17.3 Hysteresis

Take an unmagnetised ferromagnetic ring and raise *H* from zero. *B* climbs along the **initial magnetisation curve** *Oa* and then flattens: all the domains are aligned and the specimen has reached **saturation**. Now reduce *H* back to zero. *B* does not retrace the curve — it falls along *ab* and stops at a non-zero value *B_r*. The specimen has become a magnet on its own.

HYSTERESIS, RETENTIVITY AND COERCIVITY

Hysteresis is the lagging of the flux density *B* behind the magnetising field *H* when a ferromagnetic specimen is taken through a cycle of magnetisation.

Retentivity (remanence) *B_r* is the flux density that remains in the specimen when the magnetising field is reduced to zero.

Coercivity *H_c* is the reverse magnetising field needed to reduce the retained flux density to zero.

Continuing to reverse *H* drives the specimen to saturation the other way, and bringing *H* forward again closes the curve. The closed figure *abcdefa* is the **hysteresis loop**.

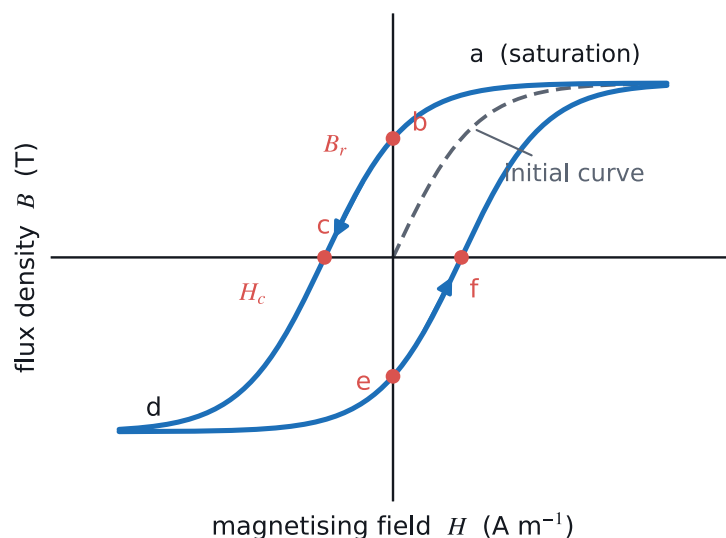


Figure 69. A complete hysteresis loop. Oa is the initial magnetisation curve, $Ob = B_r$ is the retentivity and $Oc = H_c$ the coercivity.

Energy lost per cycle

THE LOOP AREA IS THE ENERGY DISSIPATED PER UNIT VOLUME PER CYCLE

Take a ring of mean length l , cross-section A and N turns. When the flux changes, a back e.m.f. $e = N \frac{d\Phi}{dt}$ opposes the source, so in time dt the source does work

$$dW = eI dt = NI d\Phi = NI A dB$$

But $H = NI/l$, so $NI = Hl$ and

$$dW = H l A dB = (Al) H dB$$

Since Al is the volume V of the specimen, the work done per unit volume in one complete cycle is

$$\frac{W}{V} = \oint H dB = \text{area enclosed by the } B-H \text{ loop}$$

The unit of this area is $(A m^{-1})(T) = J m^{-3}$. Because the loop is traversed once per cycle and the work is not recovered, it appears as heat. For a core of volume V driven at frequency f ,

$$P_{\text{hyst}} = (\text{loop area}) \times V \times f$$

Soft and hard magnetic materials

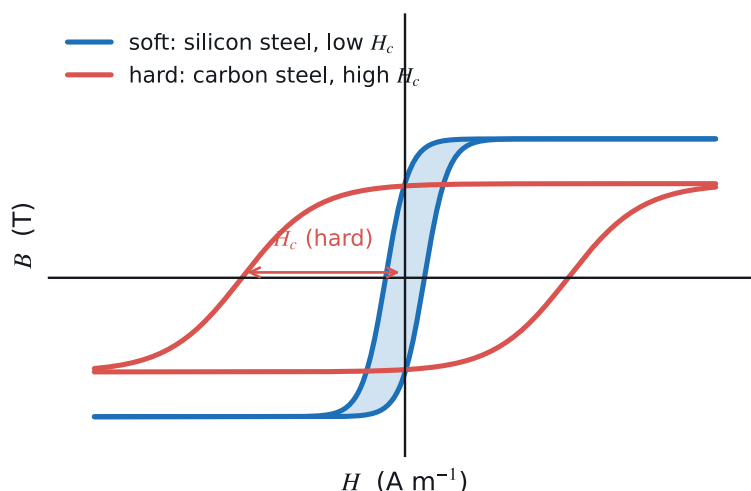


Figure 70. Loops for a soft magnetic material (narrow, small area) and a hard one (broad, large B_r and H_c). Loop area is the energy wasted as heat per cubic metre per cycle.

Property	Soft magnetic material	Hard magnetic material
Loop shape	narrow, tall, small area	broad, large area
Retentivity B_r	high	high
Coercivity H_c	very low	very high
Susceptibility, μ_r	very large	moderate
Energy loss per cycle	small	large
Magnetisation	easily magnetised and demagnetised	hard to demagnetise
Examples	soft iron, silicon steel, mu-metal, ferrites	carbon steel, cobalt steel, alnico, ferrite magnets
Used for	transformer and motor cores, electromagnets, relays	permanent magnets, loudspeakers, compass needles, magnetic latches

CHOOSING A CORE IN ONE LINE

Anything that is magnetised and demagnetised many times a second wants a **small loop area**, so use a soft material. Anything that must stay magnetised wants a **large coercivity**, so use a hard material.

WORKED EXAMPLE 17.3

Problem. The hysteresis loop of a transformer core material encloses an area equivalent to 250 J m^{-3} per cycle. The core has volume 1200 cm^3 and the transformer runs at 50 Hz . Find the energy lost per cycle and the hysteresis power loss.

Solution. $V = 1200 \text{ cm}^3 = 1200 \times 10^{-6} = 1.2 \times 10^{-3} \text{ m}^3$.

Energy per cycle:

$$W = 250 \times 1.2 \times 10^{-3} = 0.30 \text{ J}$$

Power lost (50 cycles each second):

$$P = Wf = 0.30 \times 50 = 15 \text{ W}$$

Low-loss silicon steel is used precisely to keep this figure small.

17.4 Dia-, para- and ferro-magnetic materials

Every electron in an atom is both orbiting and spinning, and each motion is a tiny current loop with a magnetic moment. How those moments add up decides the class of the material.

Diamagnetic materials. The atomic moments cancel exactly, so the atom has **no permanent** magnetic moment. An applied field changes the electron orbits slightly (Lenz's law at atomic scale) and induces a moment opposing the field. Hence M is antiparallel to H , χ_m is small and **negative** (about -10^{-5}), and μ_r is slightly less than 1. Field lines are pushed out of the specimen, and a rod suspended in a non-uniform field moves from strong to weak field and sets **perpendicular** to it. Examples: bismuth, copper, water, gold, mercury, nitrogen, sodium chloride. Superconductors are perfect diamagnets with $\chi_m = -1$.

Paramagnetic materials. The atoms do have permanent moments, but thermal agitation keeps them randomly oriented. An applied field aligns them slightly, so M is parallel to H with χ_m small and **positive** (about 10^{-5} to 10^{-3}) and μ_r slightly greater than 1. A rod moves from weak to strong field and sets **parallel** to it. Alignment is destroyed by heat, so susceptibility obeys **Curie's law**:

$$\chi_m = \frac{C}{T}$$

where C is the Curie constant. Examples: aluminium, platinum, manganese, chromium, oxygen, copper sulphate solution.

Ferromagnetic materials. Neighbouring atomic moments lock parallel to each other over regions called **domains**, each containing about 10^{17} – 10^{21} atoms. In an unmagnetised sample the domains point in random directions and cancel. A field makes favourably oriented domains **grow** at the expense of the others and finally rotates them all into line — which is why M saturates. χ_m is large and positive (10^2 to 10^5), μ_r runs into thousands, field lines crowd strongly into the specimen, and the material shows hysteresis. Examples: iron, cobalt, nickel, gadolinium and alloys such as alnico.

Above a critical temperature, the **Curie temperature** T_C , thermal energy destroys the domain alignment and a ferromagnet becomes an ordinary paramagnet obeying the **Curie-Weiss law**:

$$\chi_m = \frac{C}{T - T_C} \quad (T > T_C)$$

For iron $T_C = 1043 \text{ K}$ (770°C), for cobalt 1394 K and for nickel 631 K .

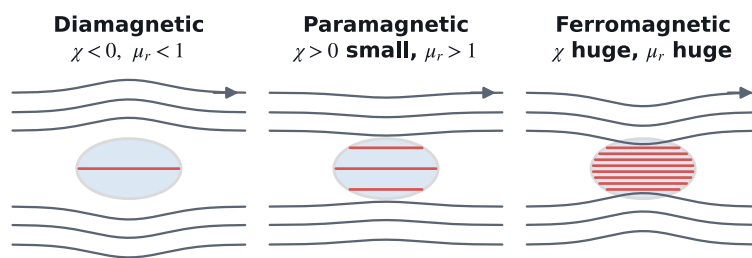


Figure 71. Field lines near a specimen. They are expelled by a diamagnetic, drawn in slightly by a paramagnetic and concentrated strongly by a ferromagnetic material.

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Atomic moment	zero	permanent, random	permanent, aligned in domains
χ_m	small, negative ($\approx -10^{-5}$)	small, positive ($10^{-5} - 10^{-3}$)	large, positive ($10^2 - 10^5$)
μ_r	slightly < 1	slightly > 1	$\gg 1$ ($10^2 - 10^5$)
M vs H	linear, negative slope	linear, small positive slope	non-linear, saturates
Field lines	expelled	slightly concentrated	strongly concentrated
In non-uniform field	moves to weaker field	moves to stronger field	moves strongly to stronger field
Effect of temperature	almost none	$\chi_m = C/T$	ferromagnetic below T_C , paramagnetic above
Hysteresis	no	no	yes
Examples	Bi, Cu, Au, H ₂ O, N ₂	Al, Pt, Mn, O ₂ , CuSO ₄	Fe, Co, Ni, Gd, alnico

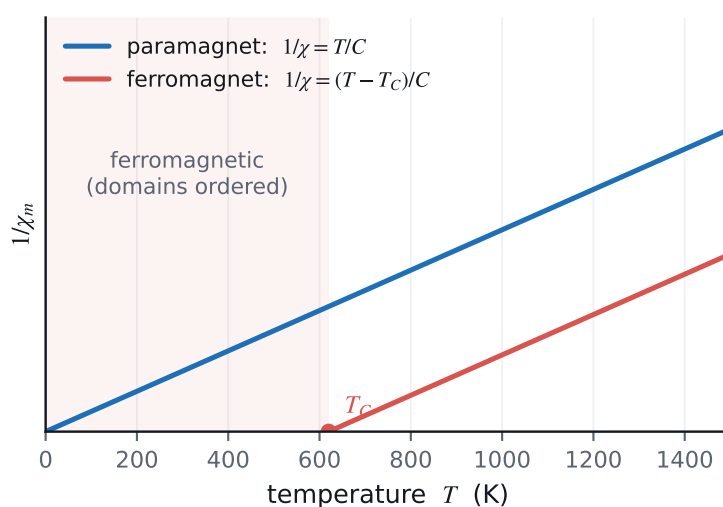


Figure 72. Plotting $1/\chi_m$ against temperature straightens both laws: a paramagnet gives a line through the origin, a ferromagnet above T_C a line cutting the axis at T_C .

WORKED EXAMPLE 17.4

Problem. A paramagnetic salt has susceptibility 2.4×10^{-4} at 300 K. (a) Find its susceptibility at 200 K. (b) At 200 K the salt is placed in a magnetising field of $1.6 \times 10^4 \text{ A m}^{-1}$. Find its magnetisation and the flux density inside it.

Solution.

(a) Curie's law gives $\chi_m T = C$, a constant, so $\chi_1 T_1 = \chi_2 T_2$:

$$\chi_2 = \chi_1 \frac{T_1}{T_2} = 2.4 \times 10^{-4} \times \frac{300}{200} = 3.6 \times 10^{-4}$$

(b) $M = \chi_m H = 3.6 \times 10^{-4} \times 1.6 \times 10^4 = 5.76 \text{ A m}^{-1}$.

$$B = \mu_0(H + M) = 4\pi \times 10^{-7} (1.6 \times 10^4 + 5.76) = 1.257 \times 10^{-6} \times 16006$$

$$B = 2.01 \times 10^{-2} \text{ T}$$

The magnetisation contributes only 0.04% of the total — a paramagnet barely changes the field, unlike the iron of Worked example 17.2.

"HIGHER PERMEABILITY MEANS STRONGER MAGNET"

No. Permeability measures how easily a material carries flux when a field is applied. Soft iron has an enormous μ_r but makes a hopeless permanent magnet, because its coercivity is tiny and it loses its magnetism as soon as the current is switched off.

Chapter summary

- Magnetic flux $\Phi = BA \cos \theta$ (weber); $B = \Phi/A$ is the flux density (tesla); $\oint \vec{B} \cdot d\vec{A} = 0$ since field lines are closed loops.
- The magnetising field is $H = NI/l$ (A m^{-1}); magnetisation is $M = m/V$ (A m^{-1}); inside matter $B = \mu_0(H + M)$.
- $\chi_m = M/H$, $\mu = B/H$, $\mu_r = \mu/\mu_0$ and $\mu_r = 1 + \chi_m$. Both χ_m and μ_r are dimensionless.
- Hysteresis is the lag of B behind H . Retentivity B_r is the flux density left when $H = 0$; coercivity H_c is the reverse field that makes $B = 0$.
- Energy dissipated per unit volume per cycle equals the loop area, $\oint H dB$ (J m^{-3}); the hysteresis power loss is (area) $\times V \times f$.
- Soft materials (small H_c , narrow loop) are for transformer cores and electromagnets; hard materials (large H_c , broad loop) are for permanent magnets.
- Diamagnetics: χ_m small negative, $\mu_r < 1$, repelled. Paramagnetics: χ_m small positive, $\mu_r > 1$, $\chi_m = C/T$. Ferromagnetics: domains, $\chi_m \gg 1$, saturation and hysteresis, and $\chi_m = C/(T - T_C)$ above the Curie temperature.

Practice questions

Group A — Multiple choice (1 mark each)

1. The SI unit of magnetic susceptibility is [1] (a) tesla (b) A m^{-1} (c) weber (d) it has no unit
2. For a paramagnetic substance [1] (a) $\chi_m < 0$, $\mu_r < 1$ (b) $\chi_m > 0$, $\mu_r > 1$ (c) $\chi_m \gg 1$ (d) $\mu_r = 1$
3. The area of a B-H hysteresis loop represents [1] (a) retentivity (b) coercivity (c) energy lost per unit volume per cycle (d) permeability
4. Soft iron rather than steel is used for a transformer core because it has [1] (a) high coercivity (b) low coercivity and small loop area (c) low permeability (d) high density
5. Above its Curie temperature, iron becomes [1] (a) diamagnetic (b) paramagnetic (c) non-magnetic (d) more strongly ferromagnetic

ANSWERS TO GROUP A

1. (d) — $\chi_m = M/H$, and M and H share the unit A m^{-1} , so the ratio is a pure number. **2.** (b) — the moments align weakly with the field, so M is positive and $\mu_r = 1 + \chi_m > 1$. **3.** (c) — the loop area is $\oint H dB$, whose unit is J m^{-3} . **4.** (b) — the core is remagnetised 100 times a second, so the loop area (energy wasted) must be small. **5.** (b) — thermal agitation destroys domain alignment, leaving randomly oriented permanent moments.

Group B — Short answer (5 marks each)

1. Define magnetic flux and magnetic flux density, and state their SI units. Distinguish clearly between $\vec{B}$ and $\vec{H}$. [5]
2. Define magnetic susceptibility and relative permeability, and derive the relation $\mu_r = 1 + \chi_m$. [5]
3. A solenoid with 400 turns per metre carries a current of 3.0 A and is filled with an iron core of relative permeability 800. Calculate H , B , M and χ_m . [5]
4. Explain, with reference to the hysteresis loop, why soft iron is used in electromagnets while steel is used for permanent magnets. [5]
5. State Curie's law. A paramagnetic sample has $\chi_m = 3.6 \times 10^{-4}$ at 200 K. Find its susceptibility at 300 K. [5]

ANSWERS TO GROUP B

3. $H = nI = 400 \times 3.0 = 1.2 \times 10^3 \text{ A m}^{-1}$. $B = \mu_0 \mu_r H = (4\pi \times 10^{-7})(800)(1200) = 1.21 \text{ T}$. $\chi_m = \mu_r - 1 = 799$. $M = \chi_m H = 799 \times 1200 = 9.59 \times 10^5 \text{ A m}^{-1}$.

4. Outline: an electromagnet must switch on and off, so it needs a **small coercivity** (demagnetises the instant the current stops) and a **small loop area** (little heat wasted per cycle) — soft iron. A permanent magnet must keep its magnetism against stray fields and knocks, so it needs a **large coercivity** and large retentivity — steel. Sketch the narrow and broad loops side by side.

5. Curie's law: for a paramagnetic material $\chi_m = C/T$, i.e. the susceptibility is inversely proportional to the absolute temperature. $\chi_2 = \chi_1 T_1/T_2 = 3.6 \times 10^{-4} \times 200/300 = 2.4 \times 10^{-4}$.

Group C — Long answer (8 marks each)

1. (a) What is hysteresis? Draw a complete hysteresis loop for a ferromagnetic material and define retentivity and coercivity on it. [4] (b) Show that the energy dissipated per unit volume per cycle equals the area of the loop. A core of volume 0.010 m^3 whose loop area is 300 J m^{-3} is used at 50 Hz ; find the power lost as heat. [4]
2. Compare diamagnetic, paramagnetic and ferromagnetic materials with respect to susceptibility, relative permeability, behaviour of field lines, behaviour in a non-uniform field and effect of temperature, giving two examples of each. Hence explain what happens to a ferromagnetic material as it is heated through its Curie temperature. [8]

ANSWERS TO GROUP C

1. (a) Hysteresis is the lagging of B behind H over a magnetising cycle; B_r (retentivity) is the intercept on the B -axis, H_c (coercivity) the intercept on the H -axis. See the labelled loop in §17.3.

(b) For a ring of volume $V = Al$ and N turns, the source does work $dW = eI dt = NI d\Phi = NIA dB = (Al)H dB$, so $W/V = \oint H dB = \text{loop area}$. Numerically, energy per cycle $= 300 \times 0.010 = 3.0 \text{ J}$, and

$$P = Wf = 3.0 \times 50 = 150 \text{ W}$$

2. Outline: reproduce the comparison table of §17.4 with two examples each (Bi and Cu; Al and Pt; Fe and Ni). For the Curie point: below T_C the exchange interaction holds neighbouring atomic moments parallel inside domains, giving huge χ_m , saturation and hysteresis. As T rises, thermal energy disrupts this order; at $T = T_C$ the domain structure collapses completely. Above T_C the atoms keep their individual moments but point randomly, so the material is paramagnetic with $\chi_m = C/(T - T_C)$, a value smaller by several orders of magnitude. For iron $T_C = 1043 \text{ K}$; a red-hot iron nail is not attracted by a magnet.

UNIT 18

Electromagnetic Induction

Electricity and Magnetism • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Faraday's laws; Induced electric fields
- Lenz's law; Motional electromotive force
- A.C. generators; Eddy currents
- Self-inductance and mutual inductance
- Energy stored in an inductor
- Transformer

In 1831 Michael Faraday showed that a changing magnetic field drives a current. That discovery is why power generated at Kulekhani can be stepped up to 220 kV, carried across the country and stepped back down for your house. Two laws run the whole unit: Faraday's, which gives how big the induced e.m.f. is, and Lenz's, which gives which way it drives current.

HOW THIS UNIT IS EXAMINED

Group C asks for one of three derivations: motional e.m.f. $e = Blv$, the generator equation $e = NBA\omega \sin \omega t$, or the transformer turns-ratio with its losses. Group B likes solenoid self-inductance, $U = \frac{1}{2}LI^2$ and numericals on $e = -N \frac{d\Phi}{dt}$. Never drop the minus sign — it is Lenz's law.

18.1 Faraday's laws; Induced electric fields

Move a magnet towards a coil joined to a galvanometer and the needle deflects. Stop the magnet and the needle returns to zero even though the magnet is still there. Move it away and the needle deflects the other way. What matters is not the flux but the **rate of change** of flux.

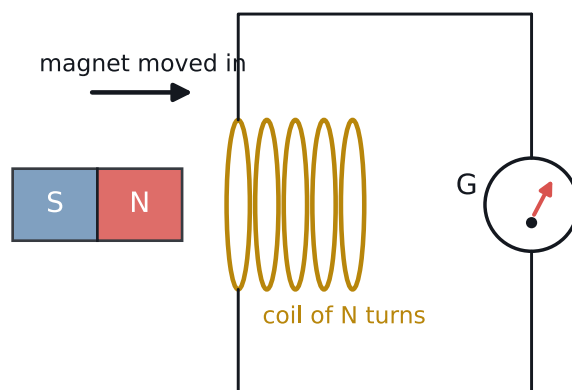


Figure 73. Faraday's experiment. Only while the flux linked with the coil is changing does the galvanometer deflect; a faster movement gives a bigger deflection.

FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

First law. Whenever the magnetic flux linked with a circuit changes, an e.m.f. is induced in it; the e.m.f. lasts only as long as the change lasts.

Second law. The magnitude of the induced e.m.f. is directly proportional to the rate of change of flux linkage:

$$e = -N \frac{d\Phi}{dt} = -\frac{d(N\Phi)}{dt}$$

The product $N\Phi$ is called the **flux linkage**, measured in weber-turns.

Since $\Phi = BA \cos \theta$, the linkage changes if B changes, if A changes, or if the coil turns. All three are used in real machines.

If the circuit has total resistance R , the induced current is $I = e/R$ and the **charge** that flows in a change is independent of how fast the change happens:

$$q = \int I dt = \int \frac{N}{R} \frac{d\Phi}{dt} dt = \frac{N \Delta\Phi}{R}$$

Induced electric fields. A stationary coil in a changing field has charges that are not moving, so no magnetic force acts on them. Something else must push them — a changing magnetic field creates an **electric field** in the space around it, even where there is no wire:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

This induced field is **non-conservative**: its line integral round a closed loop is not zero, so no potential can be defined for it.

WORKED EXAMPLE 18.1

Problem. A coil of 200 turns and area 300 cm^2 is placed with its plane perpendicular to a magnetic field. The field increases uniformly from 0.20 T to 0.60 T in 0.40 s . The coil has a resistance of 12Ω . Find (a) the induced e.m.f., (b) the induced current and (c) the charge that circulates.

Solution. $A = 300 \text{ cm}^2 = 3.0 \times 10^{-2} \text{ m}^2$ and $\theta = 0$, so $\Phi = BA$.

(a)

$$|e| = NA \frac{dB}{dt} = 200 \times 3.0 \times 10^{-2} \times \frac{0.60 - 0.20}{0.40} = 200 \times 0.03 \times 1.0 = 6.0 \text{ V}$$

(b)

$$I = \frac{|e|}{R} = \frac{6.0}{12} = 0.50 \text{ A}$$

(c)

$$q = \frac{N \Delta\Phi}{R} = \frac{200 \times (3.0 \times 10^{-2} \times 0.40)}{12} = \frac{2.4}{12} = 0.20 \text{ C}$$

18.2 Lenz's law; Motional electromotive force

LENZ'S LAW

The induced current always flows in such a direction that it **opposes the change in flux that produced it**.

In the figure below, the N-pole of a magnet approaches a coil. The flux through the coil towards the right is increasing, so the induced current flows so as to make the near face of the coil a **north pole**, which repels the approaching magnet. Pull the magnet away and the current reverses, the near face becomes a **south pole**, and the coil tries to hold the magnet back.

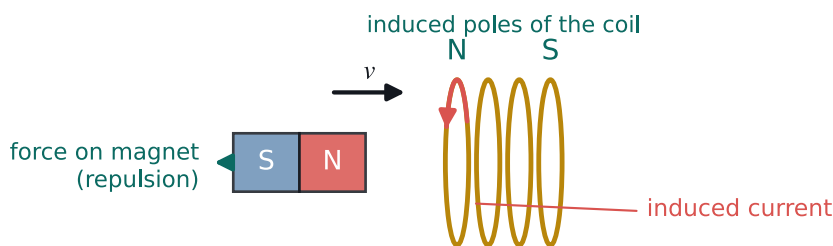


Figure 74. Lenz's law. An approaching N-pole induces an anticlockwise current (seen from the magnet) that makes the near face a north pole, and the coil repels the magnet.

Lenz's law is **conservation of energy in disguise**. If the induced current helped the magnet, the magnet would accelerate by itself while the coil also got hot — energy from nothing. Because the coil pushes back, the work you do moving the magnet reappears as electrical energy. That is what the minus sign in $e = -N d\Phi/dt$ says.

Motional e.m.f.

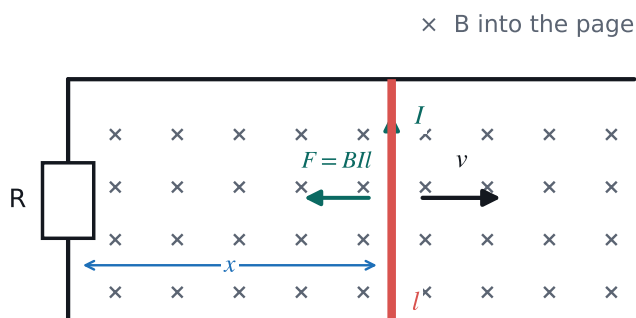


Figure 75. A rod of length l sliding at speed v on rails in a field B into the page. The flux enclosed grows, so an e.m.f. Blv drives a current, and the field then opposes the motion with a force BIl .

MOTIONAL E.M.F. $e = Blv$

A conducting rod of length l slides with speed v on frictionless rails in a uniform field B perpendicular to the plane of the rails. After time t the enclosed area is $A = lx$, so the flux is $\Phi = Blx$. Then

$$e = -\frac{d\Phi}{dt} = -Bl\frac{dx}{dt} = -Blv$$

so the magnitude of the e.m.f. is $e = Blv$.

By the force on the charges. A free charge in the rod moves at speed v , so it feels a force qvB along the rod. Charge separates until the electric field balances it, $qE = qvB$, giving $E = vB$ and $e = El = Blv$ — the same answer.

Energy check. The induced current is $I = Blv/R$, and the field exerts a retarding force $F = BI l = B^2 l^2 v/R$ on the rod. The mechanical power needed to keep it moving is

$$P = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R$$

which is exactly the electrical power dissipated. No energy is created.

If the rod moves at an angle θ to the field, only the perpendicular component counts and $e = Blv \sin \theta$.

WORKED EXAMPLE 18.2

Problem. A rod of length 0.50 m slides on rails at 4.0 m s^{-1} in a uniform field of 0.80 T perpendicular to the plane of the rails. The circuit resistance is 2.0Ω . Find the e.m.f., the current, the force needed to keep the rod moving uniformly, and the power supplied. Verify that the power supplied equals the heat produced.

Solution.

$$e = Blv = 0.80 \times 0.50 \times 4.0 = 1.6 \text{ V}$$

$$I = \frac{e}{R} = \frac{1.6}{2.0} = 0.80 \text{ A}$$

$$F = BI l = 0.80 \times 0.80 \times 0.50 = 0.32 \text{ N}$$

$$P = Fv = 0.32 \times 4.0 = 1.28 \text{ W}$$

Heat produced: $I^2 R = (0.80)^2 \times 2.0 = 1.28 \text{ W}$. The two agree, as Lenz's law requires.

THE MINUS SIGN IS NOT DECORATION

$e = -N d\Phi/dt$. The sign carries the whole physical content of Lenz's law, so state the signed form first and take magnitudes only afterwards.

18.3 A.C. generators; Eddy currents

An **a.c. generator (dynamo)** converts mechanical energy into electrical energy by rotating a coil in a magnetic field. Its parts are an **armature** (a coil of N turns wound on a soft-iron core), a **field magnet**, two **slip rings** and two carbon **brushes**.

THE GENERATOR EQUATION

Let the coil of N turns and area A rotate with constant angular velocity ω in a uniform field B . If the normal to the coil makes angle $\theta = \omega t$ with $\vec{B}$, the flux linked is

$$\Phi = BA \cos \omega t$$

By Faraday's law,

$$e = -N \frac{d\Phi}{dt} = -NBA \frac{d}{dt}(\cos \omega t) = NBA\omega \sin \omega t$$

So $e = e_0 \sin \omega t$ with **peak e.m.f.** $e_0 = NBA\omega = 2\pi f NBA$.

The e.m.f. is maximum when the coil's plane is **parallel** to the field (lines are cut fastest) and zero when it is **perpendicular**, even though the flux is then greatest. A split-ring commutator in place of the slip rings gives a d.c. generator.

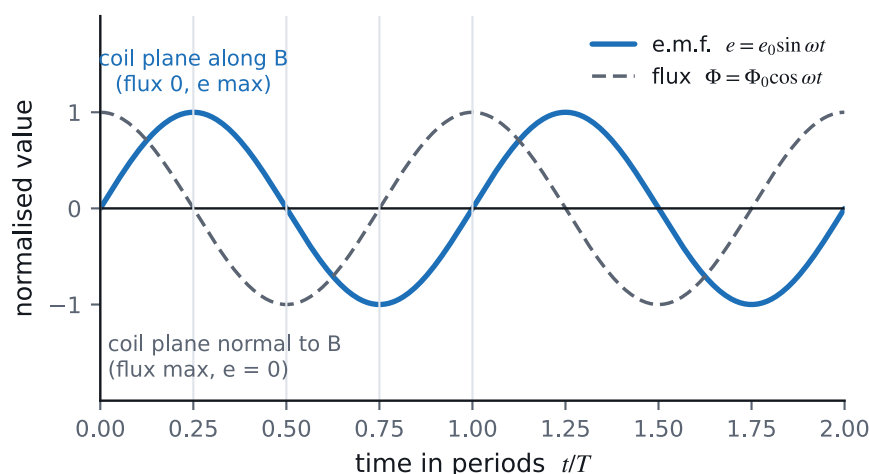


Figure 76. Output of an a.c. generator, $e = NBA\omega \sin \omega t$. The e.m.f. is zero when the flux is maximum and maximum when the flux is zero.

Eddy currents

When a **solid** block of metal sits in a changing magnetic field, the induced e.m.f. drives swirling closed loops of current through the metal. These are **eddy currents**. The resistance of a solid block is tiny, so the currents are large, the I^2R heating is wasteful, and by Lenz's law they oppose the motion that produced them.

Eddy currents as a nuisance	Eddy currents put to work
Heat and energy loss in transformer and motor cores	Induction furnace for melting metals
Damping and heating in dynamo armatures	Induction cooker / induction heating
Reduced efficiency of chokes	Electromagnetic braking in trains and treadmills
—	Dead-beat galvanometers and speedometers
—	Induction-type energy meters

They are reduced by **laminating** the core: varnished silicon-steel sheets stacked with their planes parallel to the flux. This breaks the large eddy loops into many small high-resistance ones, so the loss falls sharply.

18.4 Self-inductance and mutual inductance

Self-inductance. A changing current in a coil changes its own flux, which induces an e.m.f. in the same coil opposing the change — a **back e.m.f.** For a coil with no ferromagnetic core, the flux linkage is proportional to the current:

$$N\Phi = LI \quad \text{and} \quad e = -N\frac{d\Phi}{dt} = -L\frac{dI}{dt}$$

SELF-INDUCTANCE AND THE HENRY

The **self-inductance** L of a coil is the flux linkage per unit current, $L = N\Phi/I$; equivalently it is the back e.m.f. per unit rate of change of current. Its SI unit is the **henry (H)**: a coil has a self-inductance of one henry if a current changing at 1 A s^{-1} induces a back e.m.f. of 1 V . Thus $1 \text{ H} = 1 \text{ Wb A}^{-1} = 1 \text{ V s A}^{-1}$.

SELF-INDUCTANCE OF A LONG SOLENOID

For a solenoid of length l , cross-sectional area A and N turns carrying a current I , the interior field is $B = \mu_0 nI$ with $n = N/l$. The flux through one turn is $\Phi = BA = \mu_0 nIA$, so the flux linkage is

$$N\Phi = \mu_0 n^2 l A I \quad \Rightarrow \quad L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}$$

With a core of relative permeability μ_r this becomes $L = \mu_0 \mu_r N^2 A / l$.

Notice L depends only on the **geometry** of the coil and the core material — never on the current.

Mutual inductance. If a changing current I_1 in coil 1 links flux through a nearby coil 2, an e.m.f. appears in coil 2:

$$N_2\Phi_2 = MI_1, \quad e_2 = -M\frac{dI_1}{dt}$$

M is the **mutual inductance**, also measured in henry. For two coaxial solenoids of the same length l and area A ,

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

It is reciprocal ($M_{12} = M_{21}$) and related to the self-inductances by

$$M = k\sqrt{L_1 L_2}$$

where the **coefficient of coupling** k lies between 0 and 1; $k \rightarrow 1$ for coils wound on a common closed iron core, as in a transformer.

18.5 Energy stored in an inductor

$$U = \frac{1}{2}LI^2$$

While the current in an inductor is growing, the source must work against the back e.m.f. $e = -L \frac{dI}{dt}$. In time dt the work done is

$$dW = -eI dt = LI \frac{dI}{dt} dt = LI dI$$

Integrating from zero current to the final current I ,

$$W = \int_0^I LI dI = \frac{1}{2}LI^2$$

This work is not dissipated — it is stored in the magnetic field, and it comes back when the current is switched off:

$$U = \frac{1}{2}LI^2$$

Energy density. For a solenoid, $L = \mu_0 n^2 Al$ and $B = \mu_0 nI$, so $I = B/\mu_0 n$ and

$$U = \frac{1}{2} \mu_0 n^2 Al \cdot \frac{B^2}{\mu_0^2 n^2} = \frac{B^2}{2\mu_0} (Al)$$

Since Al is the volume, the energy stored per unit volume of the field is

$$u = \frac{B^2}{2\mu_0}$$

WORKED EXAMPLE 18.3

Problem. An air-cored solenoid is 50 cm long, has a cross-sectional area of 4.0 cm^2 and 1000 turns. Find (a) its self-inductance, (b) the energy stored when it carries 2.0 A, and (c) the average induced e.m.f. if this current is switched off in 10 ms.

Solution. $l = 0.50 \text{ m}$, $A = 4.0 \times 10^{-4} \text{ m}^2$, $N = 1000$.

(a)

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7})(1000)^2(4.0 \times 10^{-4})}{0.50} = 1.01 \times 10^{-3} \text{ H}$$

so $L \approx 1.0 \text{ mH}$.

(b)

$$U = \frac{1}{2}LI^2 = \frac{1}{2}(1.01 \times 10^{-3})(2.0)^2 = 2.0 \times 10^{-3} \text{ J} = 2.0 \text{ mJ}$$

(c)

$$|e| = L \frac{\Delta I}{\Delta t} = (1.01 \times 10^{-3}) \times \frac{2.0}{10 \times 10^{-3}} = 0.20 \text{ V}$$

18.6 Transformer

A **transformer** changes an alternating voltage to another value at the same frequency, by **mutual induction**. A primary of N_p turns and a secondary of N_s turns share a laminated soft-iron core, so nearly all the primary's flux also threads the secondary.

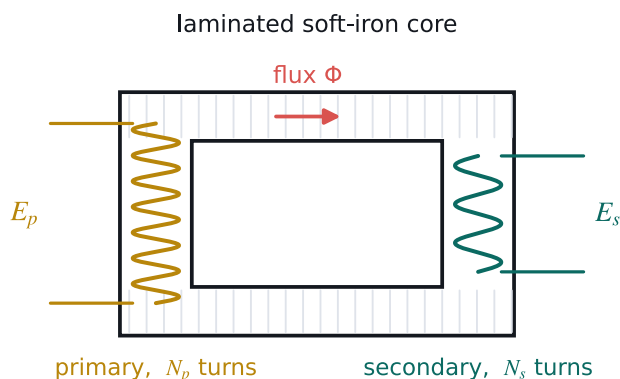


Figure 77. A step-down transformer. Both coils share the same flux Φ in the laminated soft-iron core, so $E_s/E_p = N_s/N_p$.

THE TURNS-RATIO RELATION

Let Φ be the flux through each turn of the core at time t . The same $d\Phi/dt$ acts on both coils, so

$$E_p = -N_p \frac{d\Phi}{dt}, \quad E_s = -N_s \frac{d\Phi}{dt}$$

Dividing,

$$\frac{E_s}{E_p} = \frac{N_s}{N_p} = k$$

where k is the **turns ratio**. If $k > 1$ the transformer steps **up**; if $k < 1$ it steps **down**.

For an ideal (100% efficient) transformer, input power equals output power:

$$E_p I_p = E_s I_s \quad \Rightarrow \quad \frac{I_s}{I_p} = \frac{N_p}{N_s} = \frac{1}{k}$$

So voltage is stepped up only at the cost of current, and vice versa.

Energy loss	Cause	Minimised by
Copper loss	I^2R heating in the windings	thick, low-resistance copper wire
Eddy-current loss	induced currents in the core	laminated, varnished core sheets
Hysteresis loss	repeated remagnetisation of the core	soft core with a narrow loop (silicon steel)
Flux leakage	flux that misses the secondary	interleaved windings on a closed core
Humming (magnetostriction)	core vibrating at $2f$	tight clamping of laminations

Efficiency is $\eta = P_{\text{out}}/P_{\text{in}}$, typically 96–99%. High-voltage transmission pays because the same power at ten times the voltage needs one-tenth the current and so wastes one-hundredth the I^2R heat in the line.

WORKED EXAMPLE 18.4

Problem. A step-down transformer at a substation in Hetauda converts 11 kV to 220 V. The primary has 5000 turns, and the secondary supplies 20 A. Find (a) the number of secondary turns, (b) the output power, and (c) the primary current if the efficiency is 80%.

Solution.

(a)

$$N_s = N_p \frac{E_s}{E_p} = 5000 \times \frac{220}{11000} = 100 \text{ turns}$$

(b)

$$P_{\text{out}} = E_s I_s = 220 \times 20 = 4400 \text{ W} = 4.4 \text{ kW}$$

(c)

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{4400}{0.80} = 5500 \text{ W}$$

$$I_p = \frac{P_{\text{in}}}{E_p} = \frac{5500}{11000} = 0.50 \text{ A}$$

The 1100 W difference is lost as heat in the copper and the core.

Chapter summary

- Faraday: an e.m.f. appears whenever flux linkage changes, $e = -N d\Phi/dt$; the charge circulated is $q = N\Delta\Phi/R$, independent of the time taken.
- A changing $\vec{B}$ produces a non-conservative induced electric field, $\oint \vec{E} \cdot d\vec{l} = -d\Phi/dt$.
- Lenz: the induced current opposes the change producing it — a statement of energy conservation, carried by the minus sign.
- Motional e.m.f. $e = Blv$ (or $Blv \sin \theta$); the retarding force is $B^2 l^2 v/R$ and $Fv = I^2 R$.

- A.C. generator: $\Phi = BA \cos \omega t$ gives $e = NBA \omega \sin \omega t$ with peak value $e_0 = NBA \omega$.
- Eddy currents are reduced by laminating cores; they are used in induction furnaces, induction cookers and electromagnetic brakes.
- $L = N\Phi/I$, $e = -L \, dI/dt$, solenoid $L = \mu_0 \mu_r N^2 A/l$; $M = k \sqrt{L_1 L_2}$ and $e_2 = -M \, dI_1/dt$. Unit: henry.
- Energy stored $U = \frac{1}{2} LI^2$; energy density $u = B^2/2\mu_0$.
- Transformer: $E_s/E_p = N_s/N_p$ and, if ideal, $I_s/I_p = N_p/N_s$. Losses are copper, eddy-current, hysteresis, leakage and humming.

Practice questions

Group A — Multiple choice (1 mark each)

1. Lenz's law is a direct consequence of the conservation of [1] (a) charge (b) momentum (c) energy (d) magnetic flux
2. The SI unit of self-inductance, the henry, is equivalent to [1] (a) Wb A^{-1} (b) Wb A (c) A Wb^{-1} (d) V A^{-1}
3. Eddy currents in a transformer core are reduced by [1] (a) using thicker wire (b) laminating the core (c) using a steel core (d) increasing the frequency
4. In a step-up transformer, compared with the primary the secondary has [1] (a) more turns and larger current (b) more turns and smaller current (c) fewer turns and larger current (d) fewer turns and smaller current
5. A coil of self-inductance L carries a current I . The energy stored is [1] (a) LI^2 (b) $\frac{1}{2}LI^2$ (c) $\frac{1}{2}L^2I$ (d) LI

ANSWERS TO GROUP A

1. (c) — if the induced current aided the change, energy would be created from nothing. **2.** (a) — from $L = N\Phi/I$, henry = weber per ampere. **3.** (b) — laminations break the large eddy loops into small high-resistance ones. **4.** (b) — $E_s/E_p = N_s/N_p$ and $E_p I_p = E_s I_s$, so a higher voltage means a lower current. **5.** (b) — from $W = \int LI \, dI$.

Group B — Short answer (5 marks each)

1. State Faraday's laws of electromagnetic induction and define magnetic flux linkage. Show that the charge induced in a circuit is independent of the time in which the flux change occurs. [5]
2. Derive an expression for the motional e.m.f. induced in a rod of length l moving with velocity v perpendicular to a uniform field B , and show that the mechanical power supplied equals the heat produced. [5]
3. A coil of 100 turns and area $0.050 \, \text{m}^2$ lies with its plane perpendicular to a uniform field of $0.20 \, \text{T}$. It is turned through 180° in $0.10 \, \text{s}$. Calculate the average induced e.m.f. [5]
4. Define self-inductance and derive an expression for the self-inductance of a long air-cored solenoid. [5]
5. What are eddy currents? Give two useful applications and two ways in which they are a nuisance, and state how they are minimised. [5]

ANSWERS TO GROUP B

1. Outline: state both laws, define flux linkage $N\Phi$ (weber-turns). For the charge, $I = e/R = (N/R)(d\Phi/dt)$, so $q = \int I dt = N\Delta\Phi/R$ — it contains no t , so a slow change gives a small current for a long time and a fast change a large current for a short time, with the same total charge.

3. Initially $\Phi_1 = BA\cos 0^\circ = 0.20 \times 0.050 = 0.010$ Wb; after turning, $\Phi_2 = BA\cos 180^\circ = -0.010$ Wb. So $\Delta\Phi = -0.020$ Wb and

$$|e| = N \frac{|\Delta\Phi|}{\Delta t} = 100 \times \frac{0.020}{0.10} = 20 \text{ V}$$

5. Outline: circulating induced currents set up in the body of a conductor in a changing field. Useful: induction furnace, induction cooker, electromagnetic braking, dead-beat galvanometer. Nuisance: heating and loss in transformer cores and motor armatures. Minimised by laminating the core with varnished silicon-steel sheets parallel to the flux.

Group C — Long answer (8 marks each)

- (a) State Lenz's law and describe an experiment to verify it. Show that it follows from the principle of conservation of energy. [4] (b) Describe the construction of an a.c. generator and derive an expression for the instantaneous e.m.f. it produces. [4]
- (a) Explain the principle and working of a transformer and derive the relation between the voltages and the numbers of turns. List its energy losses and how each is reduced. [4] (b) A transformer steps 240 V down to 12 V. The primary has 1200 turns, and the secondary delivers 5.0 A to a lamp. Assuming the transformer is 90% efficient, find the number of secondary turns, the output power and the primary current. [4]

ANSWERS TO GROUP C

1. Outline. (a) State the law; use the magnet-and-coil experiment of §18.2 — pushing the N-pole in makes the near face a north pole (repulsion), withdrawing it makes that face a south pole (attraction). Energy argument: an aiding current would accelerate the magnet and heat the coil, creating energy from nothing. (b) Armature, field magnet, slip rings, brushes; $\Phi = BA\cos \omega t$ gives $e = NBA\omega \sin \omega t$, peak $e_0 = NBA\omega$.

2. (b) $N_s = N_p(E_s/E_p) = 1200 \times (12/240) = 60$ turns.

$$P_{\text{out}} = 12 \times 5.0 = 60 \text{ W}$$

$$P_{\text{in}} = \frac{60}{0.90} = 66.7 \text{ W}$$

$$I_p = \frac{66.7}{240} = 0.28 \text{ A}$$

UNIT 19

Alternating Currents

Electricity and Magnetism • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Peak and rms value of AC current and voltage
- AC through a resistor, a capacitor and an inductor
- Phasor diagram
- Series circuits containing combination of resistance, capacitance and inductance
- Series resonance; quality factor
- Power in AC circuits: power factor

The supply from the Nepal Electricity Authority is a sine wave of 230 V, 50 Hz — its direction reverses one hundred times every second. With a steady current, a circuit is described by one number, the resistance. With an alternating current, a capacitor and an inductor also obstruct the flow, and they do it out of step with the voltage. This unit gives you the tools for that: r.m.s. values, reactance, phasors, impedance, resonance and the power factor.

WHAT THE EXAMINER WANTS HERE

Almost every paper contains a series LCR numerical — find Z , then I , then the phase angle and the power. Group C asks you to derive the impedance from a phasor diagram and then discuss resonance and the quality factor. Learn the two mnemonics for phase (**CIVIL**: in a **C**apacitor **I** leads **V**, **V** leads **I** in an **L**) and never add reactances to resistances arithmetically.

19.1 Peak and rms value of AC current and voltage

An alternating current reverses direction periodically. The commonest form is sinusoidal:

$$i = I_0 \sin \omega t, \quad v = V_0 \sin \omega t$$

where I_0 and V_0 are the **peak** (maximum or amplitude) values, $\omega = 2\pi f$ is the angular frequency and $f = 1/T$.

The **average value over a full cycle is zero**, because the current spends half the cycle positive and half negative. Averaging over half a cycle gives a useful number:

$$I_{av} = \frac{2I_0}{\pi} = 0.637 I_0$$

But the quantity that a meter reads, and the one that matters for heating, is the **root-mean-square** value.

ROOT-MEAN-SQUARE VALUE

The r.m.s. value of an alternating current is that steady direct current which would produce the same heating effect in a given resistor in the same time.

$$I_{rms} = I_0 / \sqrt{2}$$

The instantaneous power dissipated in a resistance R is $p = i^2 R$. Over one period the mean square current is

$$\overline{i^2} = \frac{1}{T} \int_0^T I_0^2 \sin^2 \omega t \, dt = \frac{I_0^2}{T} \int_0^T \frac{1 - \cos 2\omega t}{2} \, dt$$

The cosine averages to zero over a whole number of cycles, leaving

$$\overline{i^2} = \frac{I_0^2}{2} \quad \Rightarrow \quad I_{rms} = \sqrt{\overline{i^2}} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

The same argument for the voltage gives $V_{rms} = V_0 / \sqrt{2}$.

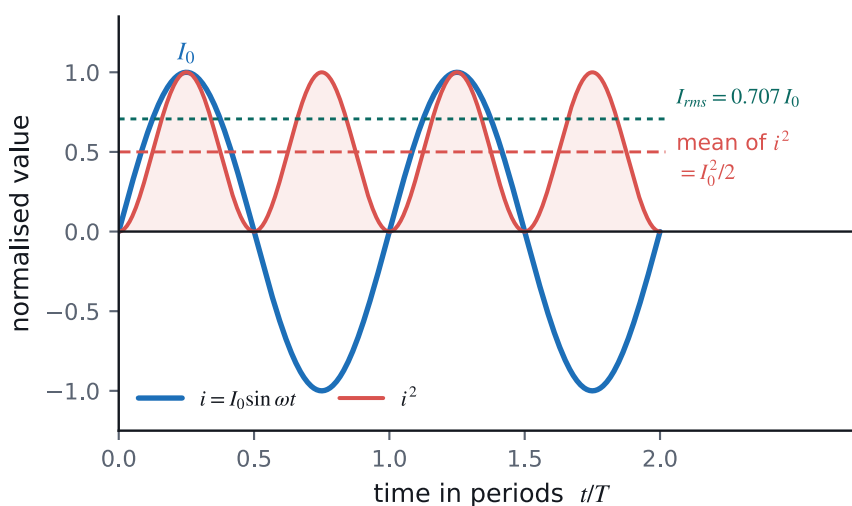


Figure 78. For $i = I_0 \sin \omega t$ the square i^2 is always positive and averages to $I_0^2/2$; hence $I_{rms} = I_0 / \sqrt{2} = 0.707 I_0$.

Quantity	Relation to peak value
r.m.s. value	$I_{rms} = I_0 / \sqrt{2} = 0.707 I_0$
Mean over half cycle	$I_{av} = 2I_0/\pi = 0.637 I_0$
Mean over full cycle	0
Form factor	$I_{rms}/I_{av} = 1.11$

230 V IS NOT THE PEAK

A "230 V" supply is 230 V **r.m.s.** Its peak is $V_0 = 230\sqrt{2} = 325$ V, and insulation must withstand that. Unless a question says "peak" or "maximum", every a.c. voltage and current quoted is an r.m.s. value.

WORKED EXAMPLE 19.1

Problem. An alternating current is given by $i = 14.14\sin(314 t)$ A. Find (a) the peak value, (b) the r.m.s. value, (c) the frequency and period, and (d) the mean value over half a cycle.

Solution. Comparing with $i = I_0\sin\omega t$: $I_0 = 14.14$ A and $\omega = 314$ rad s⁻¹.

(a) $I_0 = 14.14$ A.

(b)

$$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{14.14}{1.414} = 10.0 \text{ A}$$

(c)

$$f = \frac{\omega}{2\pi} = \frac{314}{6.283} = 50 \text{ Hz}, \quad T = \frac{1}{f} = 0.02 \text{ s}$$

(d)

$$I_{av} = \frac{2I_0}{\pi} = \frac{2 \times 14.14}{3.1416} = 9.0 \text{ A}$$

19.2 AC through a resistor, a capacitor and an inductor

Take the applied voltage as $v = V_0\sin\omega t$ in each case.

Pure resistor. By Ohm's law the current follows the voltage instant by instant:

$$i = \frac{v}{R} = \frac{V_0}{R}\sin\omega t = I_0\sin\omega t$$

Current and voltage are **in phase**; the phase difference is zero and the opposition is simply R .

Pure inductor. The only voltage in the circuit is the back e.m.f., so $v = L di/dt$:

$$di = \frac{V_0}{L}\sin\omega t dt \implies i = -\frac{V_0}{\omega L}\cos\omega t = \frac{V_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

So the current **lags** the voltage by $\pi/2$ (a quarter cycle), and the peak current is $I_0 = V_0/\omega L$. The quantity

$$X_L = \omega L = 2\pi fL$$

is the **inductive reactance**, measured in ohms. It is zero for d.c. ($f=0$) and grows with frequency — an inductor is a low-pass element, which is why a **choke** controls a.c. without wasting energy.

Pure capacitor. The charge is $q = Cv = CV_0\sin\omega t$, so

$$i = \frac{dq}{dt} = \omega CV_0 \cos \omega t = \omega CV_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

The current **leads** the voltage by $\pi/2$, with $I_0 = \omega CV_0$. The quantity

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

is the **capacitive reactance** in ohms. It is infinite for d.c. (a capacitor blocks direct current) and falls as frequency rises.

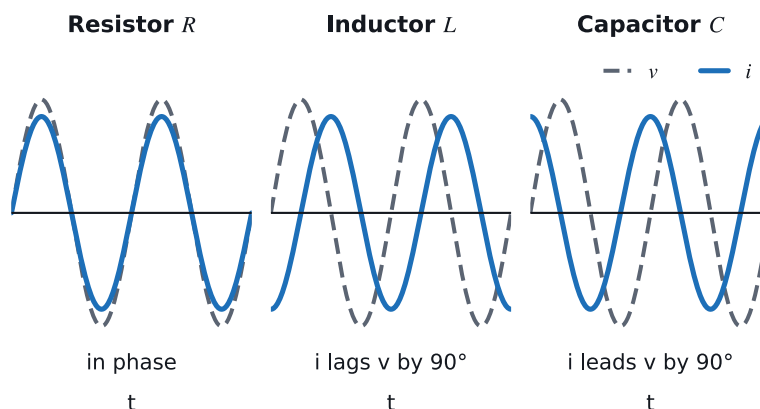


Figure 79. Voltage and current waveforms. In R they are in phase; in L the current lags by 90° ; in C the current leads by 90° .

Element	Opposition	Phase of i relative to v	Behaviour with f
Resistor R	R (ohm)	in phase	independent of f
Inductor L	$X_L = 2\pi f L$	lags by 90°	increases with f
Capacitor C	$X_C = 1/2\pi f C$	leads by 90°	decreases with f

19.3 Phasor diagram

Adding sine functions of different phase by trigonometry is slow. Instead each sinusoid is represented by a **phasor**: a rotating vector whose length is the peak (or r.m.s.) value and which rotates anticlockwise with angular velocity ω . The projection of the phasor on the vertical axis gives the instantaneous value.

Because every phasor in a circuit rotates at the same ω , the angles between them stay fixed, and the diagram can be drawn frozen at $t=0$. Sinusoids of the same frequency then add like vectors — by the parallelogram or triangle rule.

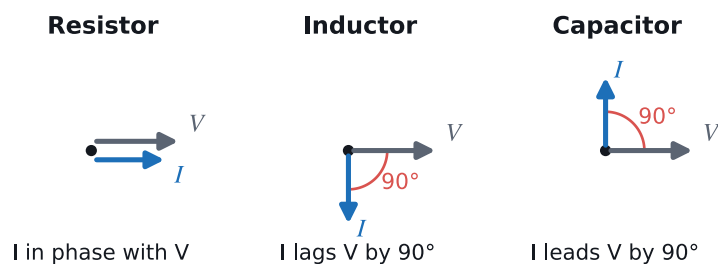


Figure 80. Phasor diagrams, frozen at one instant; all phasors rotate anticlockwise at ω . V and I are collinear for R (drawn apart for clarity), I is 90° behind V for L and 90° ahead for C .

19.4 Series circuits containing combination of resistance, capacitance and inductance

In a series circuit the **current is the same everywhere**, so the current phasor is drawn first and the three voltage phasors are set against it: V_R along the current, V_L ahead of it by 90° and V_C behind it by 90° . Since V_L and V_C are opposite, they subtract.

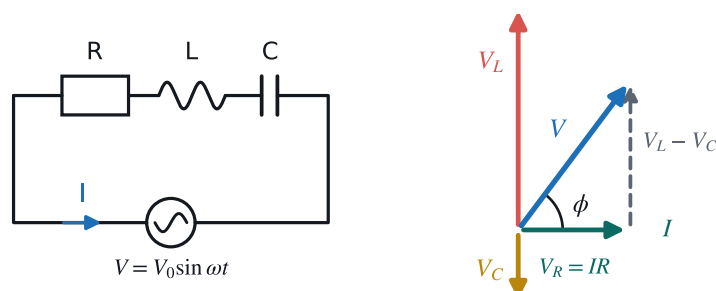


Figure 81. Series LCR circuit and its phasor diagram. V_L and V_C are antiparallel, so the resultant of V_R and $(V_L - V_C)$ gives the supply voltage V at phase angle ϕ .

IMPEDANCE OF A SERIES LCR CIRCUIT

Let the common current be I (r.m.s.). The three voltage magnitudes are

$$V_R = IR, \quad V_L = IX_L, \quad V_C = IX_C$$

V_L and V_C are in antiphase, so their resultant has magnitude $|V_L - V_C|$ and is perpendicular to V_R . By Pythagoras,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = I\sqrt{R^2 + (X_L - X_C)^2}$$

Hence the **impedance** $Z = V/I$ is

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

and the current lags the voltage by the **phase angle** ϕ where

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$$

- If $X_L > X_C$: $\phi > 0$, the circuit is **inductive** and the current lags.
- If $X_L < X_C$: $\phi < 0$, the circuit is **capacitive** and the current leads.
- If $X_L = X_C$: $\phi = 0$, the circuit behaves as pure resistance — resonance.

Dropping one element gives the special cases $Z = \sqrt{R^2 + X_L^2}$ for an LR circuit and $Z = \sqrt{R^2 + X_C^2}$ for a CR circuit.

NEVER ADD R , X_L AND X_C ARITHMETICALLY

$Z \neq R + X_L + X_C$. Resistance and reactance are at right angles on the phasor diagram, so they combine by Pythagoras. In the same way the voltmeter readings V_R , V_L and V_C do not add up to the supply voltage — V_L alone can be far larger than V .

WORKED EXAMPLE 19.2

Problem. A resistor of $30 \, \Omega$, an inductor of $0.191 \, \text{H}$ and a capacitor of $159 \, \mu\text{F}$ are connected in series across a $220 \, \text{V}$, $50 \, \text{Hz}$ supply. Find the reactances, the impedance, the current, the phase angle, and the voltage across each element.

Solution. $\omega = 2\pi f = 2\pi(50) = 314.2 \, \text{rad s}^{-1}$.

$$X_L = \omega L = 314.2 \times 0.191 = 60 \, \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{314.2 \times 159 \times 10^{-6}} = 20 \, \Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{30^2 + 40^2} = \sqrt{2500} = 50 \, \Omega$$

$$I = \frac{V}{Z} = \frac{220}{50} = 4.4 \, \text{A}$$

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{40}{30} = 1.333 \Rightarrow \phi = 53.1^\circ$$

Since $X_L > X_C$ the current **lags** the voltage by 53.1° .

$$V_R = IR = 4.4 \times 30 = 132 \, \text{V}, \quad V_L = 4.4 \times 60 = 264 \, \text{V}, \quad V_C = 4.4 \times 20 = 88 \, \text{V}$$

Check: $\sqrt{132^2 + (264 - 88)^2} = \sqrt{17424 + 30976} = \sqrt{48400} = 220 \, \text{V}$. Note that V_L alone exceeds the supply voltage.

19.5 Series resonance; quality factor

SERIES RESONANCE

A series LCR circuit is in resonance when the inductive and capacitive reactances are equal, so that the impedance is a minimum ($Z=R$), the current is a maximum ($I_{\max} = V/R$) and the current is exactly in phase with the applied voltage.

Setting $X_L = X_C$:

$$\omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0 = \frac{1}{\sqrt{LC}} \quad \text{and} \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At resonance the circuit behaves as a pure resistance and draws maximum power. Because a large current flows, the voltages across L and C can be very much larger than the supply — they are equal and opposite and cancel each other. For this reason the series resonant circuit is called an **acceptor** circuit, and it is used to tune a radio receiver to one station.

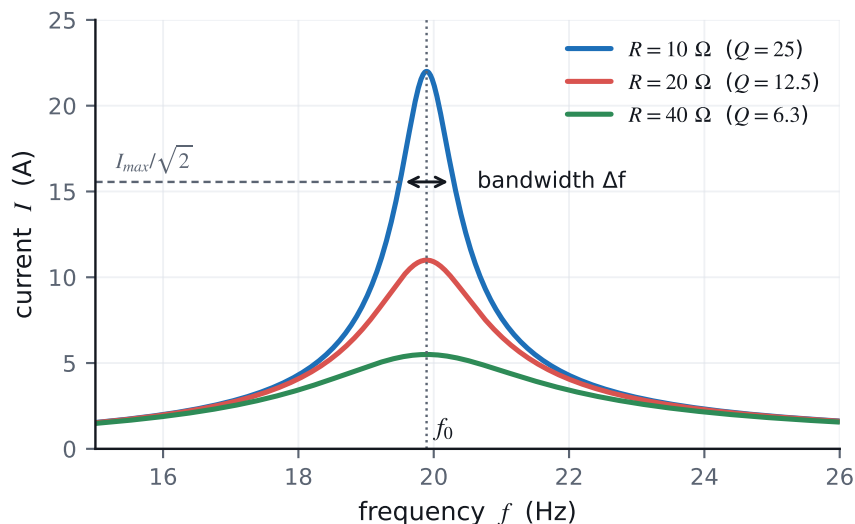


Figure 82. Resonance curves for a series LCR circuit. A smaller R gives a taller, sharper peak — a higher quality factor Q and a narrower bandwidth.

The **sharpness** of the peak is measured by the **quality factor**:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{f_0}{\Delta f}$$

where the **bandwidth** $\Delta f = f_2 - f_1$ is the range of frequencies over which the current stays above $I_{\max}/\sqrt{2}$. Equivalently, Q is the factor by which the voltage across L (or C) at resonance exceeds the supply voltage: $Q = V_L/V$. A large Q means a sharp, selective circuit.

WORKED EXAMPLE 19.3

Problem. A series circuit has $L = 2.0 \text{ H}$, $C = 32 \text{ } \mu\text{F}$ and $R = 10 \text{ } \Omega$, connected to a 220 V variable-frequency supply. Find (a) the resonant angular frequency and frequency, (b) the current at resonance, (c) the quality factor, (d) the voltage across the inductor at resonance, and (e) the bandwidth.

Solution.

(a)

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.0 \times 32 \times 10^{-6}}} = \frac{1}{8.0 \times 10^{-3}} = 125 \text{ rad s}^{-1}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{125}{6.283} = 19.9 \text{ Hz}$$

(b) At resonance $Z = R$, so

$$I = \frac{V}{R} = \frac{220}{10} = 22 \text{ A}$$

(c)

$$Q = \frac{\omega_0 L}{R} = \frac{125 \times 2.0}{10} = 25$$

(d)

$$V_L = IX_L = I\omega_0 L = 22 \times 125 \times 2.0 = 5500 \text{ V}$$

The capacitor voltage is $V_C = I/\omega_0 C = 22/(125 \times 32 \times 10^{-6}) = 5500 \text{ V}$ as well — equal and opposite, so they cancel. Note $V_L/V = 5500/220 = 25 = Q$, as expected.

(e)

$$\Delta f = \frac{f_0}{Q} = \frac{19.9}{25} = 0.80 \text{ Hz}$$

19.6 Power in AC circuits: power factor**AVERAGE POWER AND THE POWER FACTOR**

With $v = V_0 \sin \omega t$ and $i = I_0 \sin(\omega t - \phi)$, the instantaneous power is

$$p = vi = V_0 I_0 \sin \omega t \sin(\omega t - \phi)$$

Using $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$,

$$p = \frac{V_0 I_0}{2} [\cos \phi - \cos(2\omega t - \phi)]$$

Averaged over a complete cycle the second term vanishes, so

$$P_{av} = \frac{V_0 I_0}{2} \cos \phi = \frac{V_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cos \phi = V_{rms} I_{rms} \cos \phi$$

The factor $\cos \phi$ is the **power factor** of the circuit:

$$\cos \phi = \frac{R}{Z} = \frac{\text{true power}}{\text{apparent power}}$$

The product $V_{rms}I_{rms}$ is the **apparent power**, measured in volt-ampere (VA); P_{av} is the **true power** in watts.

Circuit	ϕ	Power factor	Average power
Pure resistance	0	1	$V_{rms}I_{rms}$
Pure inductance	-90°	0	zero
Pure capacitance	$+90^\circ$	0	zero
Series LCR	$\tan^{-1} \frac{X_L - X_C}{R}$	R/Z	$V_{rms}I_{rms}\cos \phi$
LCR at resonance	0	1	V^2/R

A current that flows with $\cos \phi = 0$ delivers no net energy at all: energy is stored in the field for a quarter cycle and returned in the next. This is called **wattless current**, and it is why a choke coil is preferred to a rheostat for controlling a.c. — the choke limits current without dissipating heat.

Industries are charged for apparent power, so a low power factor is expensive. A factory in Balaju with many induction motors (inductive load, $\cos \phi$ perhaps 0.6) connects large capacitors in parallel to cancel the lagging current and push the power factor towards 1.

WORKED EXAMPLE 19.4

Problem. A coil connected to a 200 V, 50 Hz supply draws a current of 5.0 A and consumes 600 W. Find its impedance, resistance, inductive reactance, inductance and power factor. What capacitance in series would make the power factor unity?

Solution.

$$Z = \frac{V}{I} = \frac{200}{5.0} = 40 \, \Omega$$

True power $P = I^2 R$, so

$$R = \frac{P}{I^2} = \frac{600}{25} = 24 \, \Omega$$

$$X_L = \sqrt{Z^2 - R^2} = \sqrt{1600 - 576} = \sqrt{1024} = 32 \, \Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{32}{314.2} = 0.102 \, \text{H}$$

$$\cos \phi = \frac{R}{Z} = \frac{24}{40} = 0.60 \quad (\text{lagging})$$

For unity power factor we need $X_C = X_L = 32 \, \Omega$:

$$C = \frac{1}{2\pi f X_C} = \frac{1}{314.2 \times 32} = 9.95 \times 10^{-5} \text{ F} \approx 100 \text{ } \mu\text{F}$$

Chapter summary

- For $i = I_0 \sin \omega t$: $I_{rms} = I_0 / \sqrt{2} = 0.707 I_0$, I_{av} (half cycle) $= 2I_0 / \pi = 0.637 I_0$, and the full-cycle mean is zero. Meters read r.m.s.
- Resistor: i in phase with v . Inductor: i lags by 90° , $X_L = 2\pi f L$. Capacitor: i leads by 90° , $X_C = 1/2\pi f C$.
- A phasor is a vector of length equal to the peak (or r.m.s.) value, rotating at ω ; sinusoids of the same frequency add like vectors.
- Series LCR: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\tan \phi = (X_L - X_C)/R$; $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.
- Resonance: $X_L = X_C$, so $f_0 = 1/2\pi \sqrt{LC}$, $Z_{min} = R$, $I_{max} = V/R$ and $\phi = 0$ (acceptor circuit).
- Quality factor $Q = \omega_0 L/R = (1/R)\sqrt{L/C} = f_0/\Delta f = V_L/V$; a large Q means a sharp resonance and a narrow bandwidth.
- Power: $P_{av} = V_{rms} I_{rms} \cos \phi$, with power factor $\cos \phi = R/Z$. Pure L or C consumes no power — wattless current.

Practice questions

Group A — Multiple choice (1 mark each)

1. The r.m.s. value of the current $i = I_0 \sin \omega t$ is [1] (a) I_0 (b) $I_0/2$ (c) $I_0/\sqrt{2}$ (d) $2I_0/\pi$
2. In a purely capacitive a.c. circuit, the current [1] (a) is in phase with the voltage (b) leads the voltage by 90° (c) lags the voltage by 90° (d) leads by 45°
3. At series resonance the impedance of an LCR circuit is [1] (a) maximum and equal to X_L (b) zero (c) minimum and equal to R (d) equal to $X_L + X_C$
4. The average power consumed in one cycle by a pure inductor is [1] (a) $V_{rms} I_{rms}$ (b) $V_{rms} I_{rms}/2$ (c) $V_0 I_0$ (d) zero
5. The quality factor of a series resonant circuit is [1] (a) $R/\omega_0 L$ (b) $\omega_0 L/R$ (c) $\omega_0 C R$ (d) $\Delta f/f_0$

ANSWERS TO GROUP A

1. (c) — the mean of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$. **2.** (b) — $i = \omega C V_0 \sin(\omega t + \pi/2)$. **3.** (c) — $X_L - X_C = 0$, so $Z = \sqrt{R^2 + 0} = R$. **4.** (d) — $\phi = 90^\circ$, so $\cos \phi = 0$ and energy is only stored and returned. **5.** (b) — equivalently $1/\omega_0 C R$ or $f_0/\Delta f$.

Group B — Short answer (5 marks each)

1. Define the r.m.s. value of an alternating current and show that $I_{rms} = I_0/\sqrt{2}$ for a sinusoidal current. [5]
2. Derive an expression for the current when an alternating voltage $V_0 \sin \omega t$ is applied to a pure inductor. Show that the current lags the voltage by $\pi/2$ and define inductive reactance. [5]
3. A $100 \text{ } \Omega$ resistor, a 0.50 H inductor and a $10 \text{ } \mu\text{F}$ capacitor are joined in series across a 200 V , 50 Hz supply. Calculate the impedance and the current. [5]

4. What is meant by the power factor of an a.c. circuit? Explain what is meant by a wattless current, and state why a choke coil is preferred to a resistor for controlling alternating current. [5]
5. A series circuit has $L = 0.50 \text{ H}$, $C = 20 \mu\text{F}$ and $R = 10 \Omega$. Find the resonant frequency and the quality factor. [5]

ANSWERS TO GROUP B

3. $\omega = 314.2 \text{ rad s}^{-1}$, $X_L = 314.2 \times 0.50 = 157.1 \Omega$; $X_C = 1/(314.2 \times 10 \times 10^{-6}) = 318.3 \Omega$. So $X_L - X_C = -161.2 \Omega$ and

$$Z = \sqrt{100^2 + 161.2^2} = \sqrt{10000 + 25995} = 189.7 \Omega$$

$$I = \frac{200}{189.7} = 1.05 \text{ A}$$

The circuit is capacitive, so the current leads the voltage.

4. Outline: the power factor is $\cos \phi = R/Z = \text{true power} / \text{apparent power}$. A wattless (idle) current is one flowing at $\cos \phi = 0$, as in a pure L or C : energy is stored for a quarter cycle and returned in the next, so the net power over a cycle is zero. A choke limits the current by its reactance X_L , which consumes no power, whereas a resistor would waste the same energy as heat.

5.

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.50 \times 20 \times 10^{-6}}} = \frac{1}{2\pi(3.162 \times 10^{-3})} = 50.3 \text{ Hz}$$

$$\omega_0 = 1/\sqrt{LC} = 1/(3.162 \times 10^{-3}) = 316.2 \text{ rad s}^{-1}, \text{ so}$$

$$Q = \frac{\omega_0 L}{R} = \frac{316.2 \times 0.50}{10} = 15.8$$

Group C — Long answer (8 marks each)

1. (a) Draw the phasor diagram for a series LCR circuit carrying an alternating current and hence derive expressions for its impedance and the phase angle between the current and the applied voltage. [4]
(b) What is series resonance? Obtain an expression for the resonant frequency, sketch the variation of current with frequency for two different resistances, and define the quality factor. [4]
2. An inductor of 0.20 H with negligible resistance, a 100Ω resistor and a $50 \mu\text{F}$ capacitor are connected in series across a 220 V , 50 Hz supply. Calculate (a) the impedance, (b) the current, (c) the phase angle, (d) the power factor and (e) the power dissipated. [8]

ANSWERS TO GROUP C

1. Outline: (a) draw I along the reference direction with V_R along I , V_L at $+90^\circ$ and V_C at -90° ; the resultant gives $V = I\sqrt{R^2 + (X_L - X_C)^2}$, so $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\tan \phi = (X_L - X_C)/R$. (b) Resonance is the condition $X_L = X_C$, giving $f_0 = 1/2\pi\sqrt{LC}$, $Z = R$ and $I_{\max} = V/R$; sketch the curves of §19.5 showing a sharper peak for smaller R ; $Q = \omega_0 L/R = f_0/\Delta f$.

2. $\omega = 2\pi(50) = 314.2 \text{ rad s}^{-1}$.

$$X_L = 314.2 \times 0.20 = 62.8 \, \Omega, \quad X_C = \frac{1}{314.2 \times 50 \times 10^{-6}} = 63.66 \, \Omega$$

(a) $X_L - X_C = 62.8 - 63.7 = -0.83 \, \Omega$, so

$$Z = \sqrt{100^2 + (0.83)^2} = 100.0 \, \Omega$$

(b) $I = 220/100.0 = 2.2 \, \text{A}$.

(c) $\tan \phi = -0.83/100 = -0.0083$, so $\phi = -0.48^\circ$ — the current leads very slightly, because the supply is almost exactly at the resonant frequency ($f_0 = 1/2\pi\sqrt{0.20 \times 50 \times 10^{-6}} = 50.3 \, \text{Hz}$).

(d) $\cos \phi = R/Z = 100/100.0 \approx 1.00$.

(e) $P = I^2 R = (2.2)^2 \times 100 = 484 \, \text{W}$.

UNIT 20

Electrons

Modern Physics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Millikan's oil drop experiment
- Motion of electron beam in electric and magnetic fields
- Thomson's experiment to determine specific charge of electrons

The electron was discovered twice over. In 1897 J. J. Thomson measured the **ratio** of its charge to its mass and showed that the same particle comes out of every cathode, whatever the gas or the metal. In 1909 R. A. Millikan measured the **charge itself** and found that it never comes in fractions. Put the two numbers together and the mass of the electron drops out. This unit is those two experiments, plus the piece of mechanics they both rest on: how a beam of electrons moves when an electric or a magnetic field pushes it sideways.

WHAT THE EXAMINER WANTS

Two set-piece experiments — Millikan and Thomson — each a full Group C answer: labelled diagram, theory, working formula, result. Between them sits a small formula bank, $v = \sqrt{2eV_a/m}$, $y = eEL^2/2mv^2$, $r = mv/eB$ and $v = E/B$, which carries almost every Group B numerical in this unit.

20.1 Millikan's oil drop experiment

Millikan's question was simple: is electric charge continuous, like the volume of water in a jug, or does it come in indivisible lumps? He answered it by weighing single charged oil drops against an electric field.

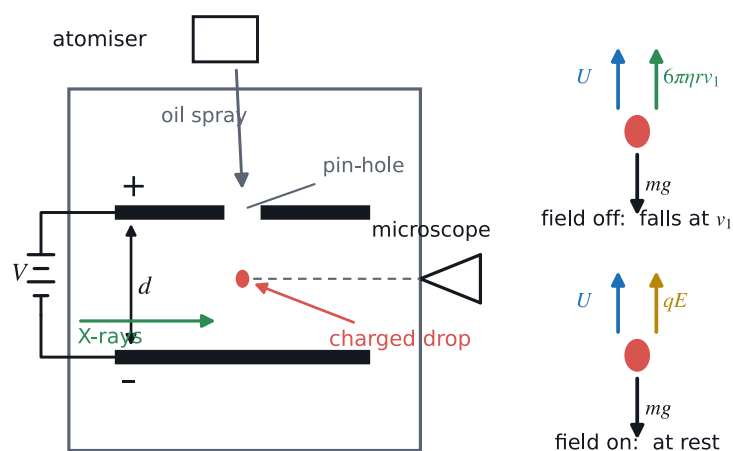


Figure 83. Millikan's oil-drop apparatus (left) and the force balance on one drop (right): with the field off the drop falls at terminal velocity; with the field on it can be held stationary.

Fine drops of low-volatility oil are sprayed from an **atomiser** into the space between two horizontal metal plates a distance d apart. Friction in the nozzle leaves each drop with a small charge; a beam of X-rays can add or remove further electrons. A drop is watched through a travelling microscope against an illuminated scale.

THE CHARGE ON A DROP

Stage 1 — field off. The drop falls and quickly reaches a constant (terminal) velocity v_1 , when weight is balanced by upthrust plus the viscous drag given by Stokes' law:

$$\frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi\eta r v_1$$

where ρ is the density of the oil, σ that of air and η the viscosity of air. Rearranging gives the radius of the drop:

$$r = \sqrt{\frac{9\eta v_1}{2(\rho - \sigma)g}}$$

Stage 2 — field on. The pd V is adjusted until the drop hangs **stationary**. Now the drag is zero, and the upward electric force balances the apparent weight:

$$qE = \frac{qV}{d} = \frac{4}{3}\pi r^3 (\rho - \sigma)g = 6\pi\eta r v_1$$

$$q = \frac{6\pi\eta r v_1 d}{V} = \frac{18\pi d}{V} \sqrt{\frac{\eta^3 v_1^3}{2(\rho - \sigma)g}}$$

Every quantity on the right is measurable, so q is measurable.

In practice a drop is rarely balanced exactly. Millikan more often let it rise with a steady velocity v_2 against the field. The drag then acts downwards and

$$qE = \frac{4}{3}\pi r^3 (\rho - \sigma)g + 6\pi\eta r v_2 = 6\pi\eta r (v_1 + v_2) \implies q = \frac{6\pi\eta r (v_1 + v_2) d}{V}$$

Timing the same drop up and down many times, and using X-rays to change its charge in between, he could watch q jump in steps — and the steps were always the same size.

Millikan repeated this for thousands of drops. The charges were never random: every one was a whole-number multiple of the same small quantity.

$$q = ne, \quad n = 1, 2, 3, \dots, \quad e = 1.60 \times 10^{-19} \text{ C}$$

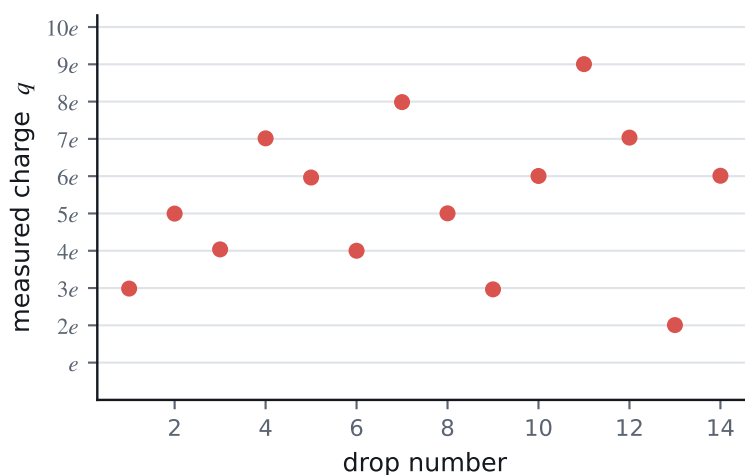


Figure 84. Charges measured on successive drops. Every value falls on a line $q = ne$ with $e = 1.6 \times 10^{-19} \text{ C}$; no drop ever carries a fraction of e .

QUANTISATION OF CHARGE

Electric charge is not continuous. It exists only in integral multiples of the elementary charge $e = 1.6 \times 10^{-19} \text{ C}$. Millikan's experiment is the direct experimental proof, and it also gives the value of e .

WEIGHT IS NOT mg HERE

The drop floats in air, so you must use the **apparent** weight $\frac{4}{3}\pi r^3(\rho - \sigma)g$, not $\frac{4}{3}\pi r^3\rho g$. Students also forget that the viscous force vanishes when the drop is stationary — Stokes' drag depends on speed, and the speed is zero.

WORKED EXAMPLE 20.1

Problem. In a Millikan apparatus the plates are 1.0 cm apart. An oil drop of mass $4.9 \times 10^{-15} \text{ kg}$ is held stationary when the pd across the plates is 1000 V . Find the charge on the drop and the number of electrons it carries. (Neglect upthrust; $g = 9.8 \text{ m s}^{-2}$.)

Solution. The field between the plates is

$$E = \frac{V}{d} = \frac{1000}{1.0 \times 10^{-2}} = 1.0 \times 10^5 \text{ V m}^{-1}$$

For equilibrium, $qE = mg$, so

$$q = \frac{mg}{E} = \frac{4.9 \times 10^{-15} \times 9.8}{1.0 \times 10^5} = \frac{4.802 \times 10^{-14}}{1.0 \times 10^5} = 4.8 \times 10^{-19} \text{ C}$$

Number of electronic charges:

$$n = \frac{q}{e} = \frac{4.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 3$$

The drop carries **3 excess electrons**.

20.2 Motion of electron beam in electric and magnetic fields

An electron released from rest at a cathode and accelerated through a pd V_a gains kinetic energy equal to the work done on it:

$$eV_a = \frac{1}{2}mv^2 \implies v = \sqrt{\frac{2eV_a}{m}}$$

The speed depends on $\sqrt{V_a}$, not on V_a — doubling the accelerating voltage multiplies the speed by only $\sqrt{2}$.

Transverse electric field

The beam now enters, at right angles, the field between two plates of length L and separation d held at pd V_d . Inside, $E = V_d/d$ and the electron has a constant sideways acceleration $a = eE/m$, while its forward speed v is unchanged. This is exactly projectile motion: uniform velocity along the beam, uniform acceleration across it, so the path is a **parabola**.

$$x = vt, \quad y = \frac{1}{2}\frac{eE}{m}t^2 = \frac{eE}{2mv^2}x^2$$

At the far edge of the plates $x = L$, so the deflection there is

$$y = \frac{eEL^2}{2mv^2} = \frac{eV_dL^2}{2mdv^2} = \frac{V_dL^2}{4V_ad}$$

using $v^2 = 2eV_a/m$. Notice that e and m have cancelled: the deflection of a cathode-ray spot depends only on the two voltages and the geometry. Beyond the plates there is no field, so the electron travels in a straight line along the tangent, making an angle θ with the axis where $\tan \theta = eEL/mv^2 = 2y/L$.

Transverse magnetic field

A magnetic field B perpendicular to the velocity exerts a force $F = evB$ that is always perpendicular to the velocity. It therefore changes the direction but never the speed: the path is a **circular arc** of radius r given by

$$evB = \frac{mv^2}{r} \implies r = \frac{mv}{eB}, \quad T = \frac{2\pi m}{eB}$$

The period T is independent of the speed — the principle of the cyclotron.

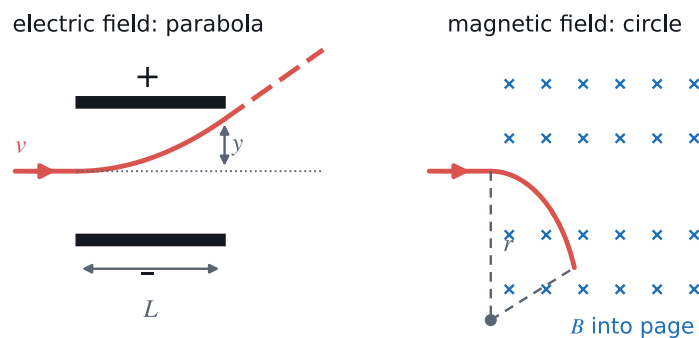


Figure 85. An electron beam in a transverse electric field follows a parabola (left); in a transverse magnetic field it follows a circular arc of radius $r = mv/eB$ (right).

Crossed fields — the velocity selector

Apply E and B at right angles to each other and to the beam, so that the electric force eE and the magnetic force evB act in opposite directions. Only electrons of one particular speed pass straight through:

$$eE = evB \implies v = \frac{E}{B}$$

Faster electrons are bent by the magnetic force, slower ones by the electric force. This "null method" is the heart of Thomson's experiment, and it is accurate because setting a deflection to zero is much easier than measuring it.

These three results are the working parts of every electron-beam instrument. The cathode-ray oscilloscope uses one pair of plates for the horizontal sweep and another for the vertical signal, so the spot traces the waveform directly. Television picture tubes and electron microscopes steer the beam magnetically, because a magnetic deflection needs no high voltage inside the tube. The mass spectrometer puts a velocity selector in front of a magnetic analyser, so that $r = mv/qB$ then sorts ions purely by mass.

Field	Force	Work done	Path	Speed
Electric E	eE , along the field line	non-zero	parabola	changes
Magnetic B	evB , perpendicular to v	zero	circle	constant

WORKED EXAMPLE 20.2

Problem. An electron is accelerated through 2000 V and then enters midway between two parallel plates 5.0 cm long and 2.0 cm apart, with 200 V across them. Find (a) the speed of the electron, (b) its deflection as it leaves the plates. ($e = 1.6 \times 10^{-19}$ C, $m = 9.1 \times 10^{-31}$ kg.)

Solution.

(a) From $eV_a = \frac{1}{2}mv^2$,

$$v = \sqrt{\frac{2eV_a}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 2000}{9.1 \times 10^{-31}}} = \sqrt{7.03 \times 10^{14}} = 2.65 \times 10^7 \text{ m s}^{-1}$$

(b) Using the deflection formula,

$$y = \frac{V_d L^2}{4V_a d} = \frac{200 \times (5.0 \times 10^{-2})^2}{4 \times 2000 \times 2.0 \times 10^{-2}} = \frac{200 \times 2.5 \times 10^{-3}}{160} = 3.1 \times 10^{-3} \text{ m}$$

so the beam is deflected about 3.1 mm — comfortably less than the 1.0 cm half-gap, so the electron does get out.

20.3 Thomson's experiment to determine specific charge of electrons

The **specific charge** of a particle is the ratio e/m — charge per unit mass. Thomson measured it for cathode rays and found a value about 1800 times larger than that for the lightest known ion, the hydrogen ion. Either the charge was enormous or the mass was tiny; the constancy of the result for every cathode material showed it was a new, very light particle common to all matter.

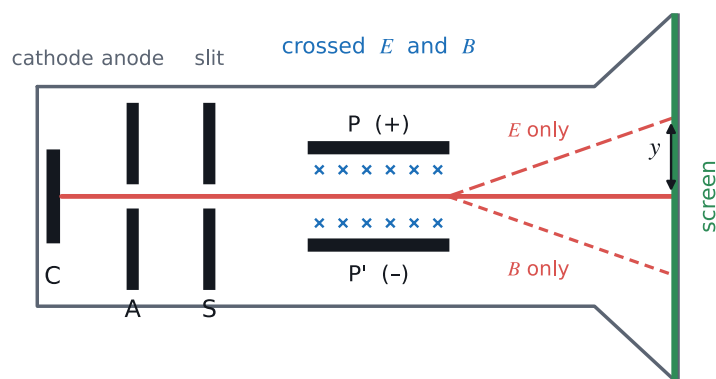


Figure 86. Thomson's apparatus. The beam is accelerated, collimated, then passed through crossed electric and magnetic fields; the spot position is read on the fluorescent screen.

Method. The tube is evacuated. Electrons boiled off the cathode C are accelerated by the anode A, collimated by the slit S, and strike a fluorescent screen, where they make a bright spot. Between the plates P and P' both an electric field E (from the plate pd) and a magnetic field B (from coils outside the tube, perpendicular to E) can be applied.

GETTING e/m FROM TWO MEASUREMENTS

Step 1 — find the speed. Switch on both fields and adjust B until the spot returns to its undeflected position. The two forces are then equal and opposite:

$$eE = evB \implies v = \frac{E}{B}$$

Step 2 — find the radius. Switch the electric field off, leaving B alone. The magnetic force now provides the centripetal force and the beam bends into a circle of radius r :

$$evB = \frac{mv^2}{r} \implies \frac{e}{m} = \frac{v}{Br}$$

Combining the two,

$$\frac{e}{m} = \frac{E}{B^2 r}$$

The radius is obtained from the measured deflection y of the spot and the geometry of the tube; for a short field region of length L , $r \approx L^2/2y$.

Alternative route. If instead the electric deflection y is measured with the magnetic field off, then $y = eEL^2/2mv^2$, and substituting $v = E/B$ from Step 1 gives $e/m = 2yE/B^2L^2$.

Quantity	Modern accepted value
Specific charge of electron, e/m	$1.76 \times 10^{11} \text{ C kg}^{-1}$
Electronic charge, e (Millikan)	$1.60 \times 10^{-19} \text{ C}$
Mass of electron, $m = e \div (e/m)$	$9.11 \times 10^{-31} \text{ kg}$
e/m for the hydrogen ion	$9.58 \times 10^7 \text{ C kg}^{-1}$

The last two rows are the punchline: the electron's specific charge is 1837 times that of the hydrogen ion, so the electron is about 1837 times lighter than the lightest atom.

WORKED EXAMPLE 20.3

Problem. In a Thomson-type tube the deflecting plates are 1.0 cm apart with 300 V across them. A magnetic field of $1.0 \times 10^{-3} \text{ T}$ leaves the beam undeflected. When the electric field is switched off the beam bends into a circle of radius 17.0 cm . Find the speed of the electrons, their specific charge, and hence the electron mass.

Solution. The electric field is

$$E = \frac{V}{d} = \frac{300}{1.0 \times 10^{-2}} = 3.0 \times 10^4 \text{ V m}^{-1}$$

From the null condition,

$$v = \frac{E}{B} = \frac{3.0 \times 10^4}{1.0 \times 10^{-3}} = 3.0 \times 10^7 \text{ m s}^{-1}$$

From the circular path,

$$\frac{e}{m} = \frac{v}{Br} = \frac{3.0 \times 10^7}{1.0 \times 10^{-3} \times 0.170} = 1.76 \times 10^{11} \text{ C kg}^{-1}$$

Taking $e = 1.6 \times 10^{-19} \text{ C}$ from Millikan,

$$m = \frac{e}{(e/m)} = \frac{1.6 \times 10^{-19}}{1.76 \times 10^{11}} = 9.1 \times 10^{-31} \text{ kg}$$

WHICH FORMULA, WHICH EXPERIMENT

Millikan gives e ; Thomson gives e/m . Neither gives m on its own. If a question asks for the mass of the electron, it is asking you to combine the two.

Chapter summary

- Millikan balanced the apparent weight of a charged oil drop against an electric force: $q = mgd/V$ for a stationary drop, or $q = 6\pi\eta rv_1 d/V$ with $r = \sqrt{9\eta v_1 / 2(\rho - \sigma)g}$.
- Every measured charge is an integral multiple of $e = 1.60 \times 10^{-19} \text{ C}$: charge is quantised.

- An electron accelerated through V_a has speed $v = \sqrt{2eV_a/m}$.
- In a transverse electric field the path is a parabola with deflection $y = eEL^2/2mv^2 = V_d L^2/4V_a d$ at the edge of the plates.
- In a transverse magnetic field the path is a circle, $r = mv/eB$, period $T = 2\pi m/eB$; the magnetic force does no work, so the speed is unchanged.
- Crossed fields select one speed: $v = E/B$.
- Thomson's tube gives $e/m = E/B^2 r = 2yE/B^2 L^2$; the accepted value is $1.76 \times 10^{11} \text{ C kg}^{-1}$, which with Millikan's e gives $m = 9.11 \times 10^{-31} \text{ kg}$.

Practice questions

Group A — Multiple choice (1 mark each)

1. Millikan's oil-drop experiment established that [1] (a) electrons have mass (b) charge is quantised (c) e/m is constant (d) oil is an insulator
2. An electron accelerated from rest through a pd V acquires a speed proportional to [1] (a) V (b) $\sqrt{V}$ (c) V^2 (d) $1/V$
3. A charged particle enters a uniform magnetic field at right angles to it. Its path is [1] (a) a straight line (b) a parabola (c) a circle (d) a helix
4. In crossed electric and magnetic fields, the particles that pass undeflected have speed [1] (a) EB (b) E/B (c) B/E (d) E^2/B
5. The specific charge of an electron is about [1] (a) $1.6 \times 10^{-19} \text{ C kg}^{-1}$ (b) $9.1 \times 10^{-31} \text{ C kg}^{-1}$ (c) $1.76 \times 10^{11} \text{ C kg}^{-1}$ (d) $9.58 \times 10^7 \text{ C kg}^{-1}$

ANSWERS TO GROUP A

- 1.** (b) — every drop charge came out as an integral multiple of e . **2.** (b) — $eV = \frac{1}{2}mv^2$ gives $v \propto \sqrt{V}$. **3.** (c) — the force evB stays perpendicular to v , so the speed is constant and the path closes into a circle. **4.** (b) — setting $eE = evB$ gives $v = E/B$. **5.** (c) — $1.76 \times 10^{11} \text{ C kg}^{-1}$; (d) is the value for the hydrogen ion.

Group B — Short answer (5 marks each)

1. Describe Millikan's oil-drop experiment with a labelled diagram and derive an expression for the charge on a stationary drop. [5]
2. An oil drop of mass $1.96 \times 10^{-15} \text{ kg}$ carrying two excess electrons is held stationary between horizontal plates 5.0 mm apart. Find the pd applied across the plates. [5]
3. Show that the path of an electron entering a uniform transverse electric field is a parabola, and obtain the deflection at the end of the plates. [5]
4. An electron is accelerated through 500 V and then enters a magnetic field of $2.0 \times 10^{-3} \text{ T}$ at right angles. Find its speed and the radius of its circular path. [5]
5. Distinguish between the effects of an electric field and a magnetic field on a moving electron. [5]

ANSWERS TO GROUP B

2. $q = 2e = 3.2 \times 10^{-19} \text{ C}$ and $mg = 1.96 \times 10^{-15} \times 9.8 = 1.92 \times 10^{-14} \text{ N}$. For equilibrium $qV/d = mg$, so $V = mgd/q = (1.92 \times 10^{-14} \times 5.0 \times 10^{-3}) / (3.2 \times 10^{-19}) = 300 \text{ V}$.

4. $v = \sqrt{2eV_a/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 500 / 9.1 \times 10^{-31}} = \sqrt{1.76 \times 10^{14}} = 1.33 \times 10^7 \text{ m s}^{-1}$. Then $r = mv/eB = (9.1 \times 10^{-31} \times 1.33 \times 10^7) / (1.6 \times 10^{-19} \times 2.0 \times 10^{-3}) = 1.21 \times 10^{-23} / 3.2 \times 10^{-22} = 3.8 \times 10^{-2} \text{ m} = 3.8 \text{ cm}$.

5. Outline: the electric force eE is along the field and does work, so the speed changes and the path is a parabola; the magnetic force evB is perpendicular to v , does no work, so the speed is constant and the path is a circular arc. A magnetic field acts only on a moving charge; an electric field acts on a charge at rest as well.

Group C — Long answer (8 marks each)

- (a) Draw a labelled diagram of Thomson's apparatus for measuring the specific charge of the electron. [3] (b) Explain the crossed-field (null) method and derive $e/m = E/B^2 r$. Show how the mass of the electron follows when Millikan's value of e is used. [5]
- An electron accelerated through 1000 V enters midway between two plates 4.0 cm long and 1.0 cm apart, with a pd of 100 V between them. Find (a) the speed of the electron, (b) its deflection on leaving the plates, (c) the angle its path then makes with the axis, and (d) the magnetic flux density that would bring the beam back to the axis. [8]

ANSWER TO GROUP C QUESTION 2

(a) $v = \sqrt{2eV_a/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 1000 / 9.1 \times 10^{-31}} = \sqrt{3.52 \times 10^{14}} = 1.88 \times 10^7 \text{ m s}^{-1}$.

(b) $y = \frac{V_{dL}^2}{4V_{ad}} = \frac{100 \times (4.0 \times 10^{-2})^2}{4 \times 1000 \times 1.0 \times 10^{-2}} = \frac{0.16}{40} = 4.0 \times 10^{-3} \text{ m} = 4.0 \text{ mm}$.

(c) $\tan \theta = 2y/L = (2 \times 4.0 \times 10^{-3}) / (4.0 \times 10^{-2}) = 0.20$, so $\theta = 11.3^\circ$.

(d) For no deflection, $eE = evB$ with $E = V_d/d = 100/0.010 = 1.0 \times 10^4 \text{ V m}^{-1}$: $B = E/v = 1.0 \times 10^4 / 1.88 \times 10^7 = 5.3 \times 10^{-4} \text{ T}$.

UNIT 21

Photons

Modern Physics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Quantum nature of radiation
- Einstein's photoelectric equation; Stopping potential
- Measurement of Planck's constant

Interference and diffraction proved that light is a wave. Then, in 1887, Hertz noticed that ultraviolet light makes a spark jump more easily — light was knocking electrons out of metal. Nothing about that emission fitted wave theory. Einstein's answer in 1905 was to say that light is delivered in indivisible packets of energy called **photons**, and that a single photon, not a whole wave, ejects a single electron. This unit is that argument and the experiment that settles it.

THE ONE EQUATION

$$hf = \phi_0 + K_{max}$$

Everything in this unit — threshold frequency, stopping potential, the straight line whose slope measures Planck's constant — is this equation read in a different way. Learn it first, then learn what each symbol is measured by.

21.1 Quantum nature of radiation

Classical physics treats a light beam as a continuous wave whose energy spreads smoothly over the wavefront. Planck (1900), trying to explain the spectrum of a hot body, was forced to assume the opposite: that an oscillator of frequency f can only gain or lose energy in whole lumps of size hf . Einstein went further and said the lumps are real, and travel through space as particles of light.

PHOTON

A photon is a quantum of electromagnetic radiation: a packet carrying energy $E = hf = hc/\lambda$ that is emitted, travels and is absorbed as a single indivisible unit. Here $h = 6.63 \times 10^{-34} \text{ J s}$ is Planck's constant.

Properties of a photon, as the NEB expects them listed:

- Energy $E = hf = hc/\lambda$; momentum $p = E/c = h/\lambda$.
- Rest mass **zero**; it travels at $c = 3 \times 10^8 \text{ m s}^{-1}$ in vacuum and cannot exist at rest. Its effective mass is $E/c^2 = hf/c^2$.
- Electrically neutral, so unaffected by electric and magnetic fields.

- Energy does not change with the brightness of the beam; **intensity** means the number of photons crossing unit area per second, not the size of each one.
- In a collision with an electron, total energy and total momentum are conserved.

A useful shortcut: since $hc = 1.99 \times 10^{-25} \text{ J m}$, the photon energy in electron-volts is

$$E \text{ (eV)} \approx \frac{1242}{\lambda \text{ (nm)}} \approx \frac{1240}{\lambda \text{ (nm)}}$$

so visible light (400–700 nm) carries roughly 1.8–3.1 eV per photon — the right size to move an outer electron, which is exactly why chemistry and vision work with visible light.

WORKED EXAMPLE 21.1

Problem. A sodium street lamp radiates 40 W of light of wavelength 589 nm. Find the energy of one photon in joules and in eV, and the number of photons emitted per second.

Solution. The photon energy is

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{589 \times 10^{-9}} = \frac{1.989 \times 10^{-25}}{5.89 \times 10^{-7}} = 3.38 \times 10^{-19} \text{ J}$$

$$E = \frac{3.38 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.11 \text{ eV}$$

The number emitted per second is the power divided by the energy per photon:

$$n = \frac{P}{E} = \frac{40}{3.38 \times 10^{-19}} = 1.18 \times 10^{20} \text{ s}^{-1}$$

More than 10^{20} photons a second — which is why a lamp looks continuous and the graininess of light never shows up in everyday life.

The photoelectric effect

When light of high enough frequency falls on a clean metal surface, electrons (called **photoelectrons**) are emitted at once. The apparatus is a photocell: an evacuated tube with a large emitting cathode C and a small collecting anode A, connected to a variable pd that can be reversed.

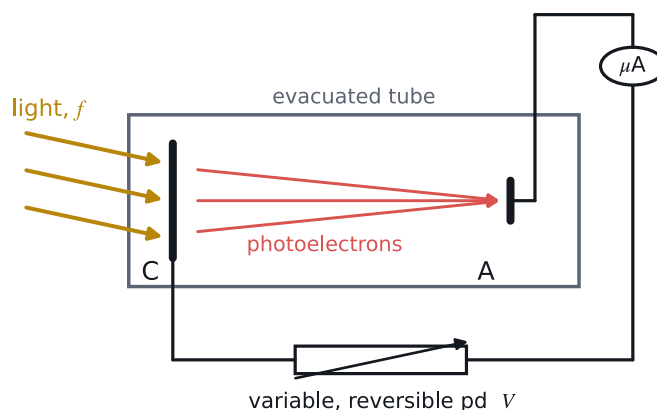


Figure 87. Apparatus for studying photoelectric emission. Monochromatic light of frequency f falls on the cathode C; the potential of the anode A can be made positive or negative and the current read on the microammeter.

Four experimental facts come out, and not one of them is what a wave theory predicts.

Observation	Wave theory predicts	What actually happens
Effect of frequency	any frequency works if you wait	below a threshold frequency f_0 there is no emission at all, however bright
Effect of intensity	brighter light gives faster electrons	brighter light gives more electrons, each with the same maximum energy
Maximum kinetic energy	depends on intensity	depends only on frequency, increasing linearly with f
Time lag	seconds to hours for a dim source	emission within 10^{-9} s, however dim

"BRIGHTER MEANS MORE ENERGETIC" IS WRONG

Doubling the intensity doubles the number of photons, so it doubles the saturation current — but each photon still carries hf , so the maximum kinetic energy and the stopping potential do not change at all.

21.2 Einstein's photoelectric equation; Stopping potential

Einstein's picture: one photon is absorbed by one electron, all or nothing. Part of the energy hf is spent freeing the electron from the metal; the rest appears as kinetic energy.

WORK FUNCTION

The **work function** ϕ_0 of a metal is the minimum energy needed to remove an electron from its surface. An electron emitted from the surface itself loses only ϕ_0 and so carries away the maximum kinetic energy; electrons from deeper down lose more and come out slower.

$$hf = \phi_0 + K_{max} \implies K_{max} = \frac{1}{2}mv_{max}^2 = hf - \phi_0$$

Writing $\phi_0 = hf_0$, where f_0 is the **threshold frequency**,

$$K_{max} = h(f - f_0), \quad f_0 = \frac{\phi_0}{h}, \quad \lambda_0 = \frac{hc}{\phi_0}$$

HOW THE EQUATION EXPLAINS ALL FOUR FACTS

1. If $f < f_0$ then $hf < \phi_0$: no electron can escape, whatever the intensity. **Threshold frequency explained.**
2. Intensity = number of photons per second, so it fixes the number of photoelectrons (the saturation current) and nothing else.
3. $K_{max} = hf - \phi_0$ is a straight line in f with slope h — independent of intensity.
4. Absorption is a single one-photon event, so there is no waiting time.

Stopping potential. To measure K_{max} , make the anode negative with respect to the cathode. Electrons are then decelerated, and slower ones turn back before arriving. Make the reverse pd large enough and even the fastest electron is stopped: the current falls to zero. That value V_0 is the **stopping potential**, and

$$eV_0 = K_{max} = hf - \phi_0 \implies V_0 = \frac{hf}{e} - \frac{\phi_0}{e}$$

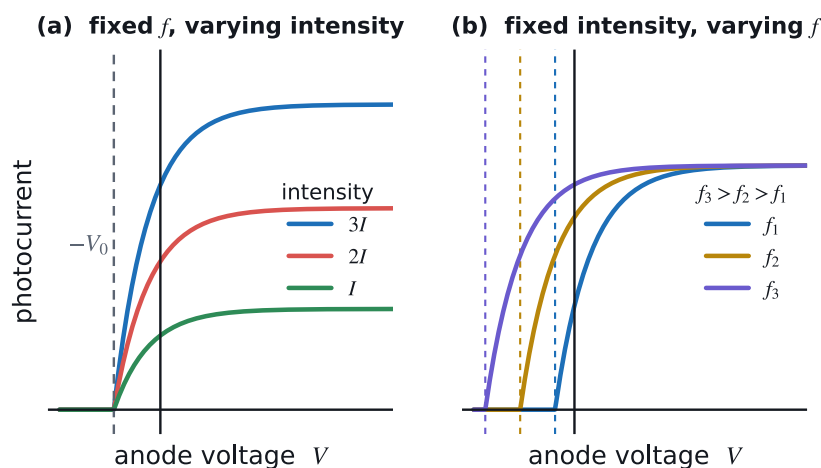


Figure 88. Photocurrent against anode voltage. (a) At fixed frequency, raising the intensity raises the saturation current but leaves V_0 unchanged. (b) At fixed intensity, raising the frequency increases V_0 but not the saturation current.

Two features of these graphs are examined constantly. The current does **not** fall to zero at $V=0$: some electrons leave the cathode with enough energy to reach the anode unaided. And the current levels off at a **saturation** value when the anode voltage is high enough to collect every emitted electron.

WORKED EXAMPLE 21.2

Problem. Light of wavelength 400 nm falls on a metal of work function 2.0 eV. Find (a) the photon energy, (b) the maximum kinetic energy of the photoelectrons, (c) the stopping potential, (d) the maximum speed of the electrons, and (e) the threshold wavelength of the metal.

Solution.

(a) $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{4.0 \times 10^{-7}} = 4.97 \times 10^{-19} \text{ J} = 3.11 \text{ eV}$.

(b) $K_{max} = hf - \phi_0 = 3.11 - 2.0 = 1.11 \text{ eV} = 1.11 \times 1.6 \times 10^{-19} = 1.78 \times 10^{-19} \text{ J}$.

(c) $V_0 = K_{max}/e = 1.11 \text{ V}$ (the number of eV read as volts).

(d) From $K_{max} = \frac{1}{2}mv_{max}^2$,

$$v_{max} = \sqrt{\frac{2K_{max}}{m}} = \sqrt{\frac{2 \times 1.78 \times 10^{-19}}{9.1 \times 10^{-31}}} = \sqrt{3.91 \times 10^{11}} = 6.25 \times 10^5 \text{ m s}^{-1}$$

(e) $\lambda_0 = \frac{hc}{\phi_0} = \frac{1.989 \times 10^{-25}}{2.0 \times 1.6 \times 10^{-19}} = 6.2 \times 10^{-7} \text{ m} = 620 \text{ nm}$ — in the orange, so red light would not work on this metal at all.

21.3 Measurement of Planck's constant

The relation $V_0 = (h/e)f - \phi_0/e$ is the equation of a **straight line** in f . Millikan measured it point by point in 1916 — ironically, trying to disprove Einstein — and got a value of h agreeing with Planck's to better than 1%.

Procedure. Monochromatic light of known frequency f is obtained from a mercury lamp with a set of filters or a monochromator. For each frequency the reverse pd is increased until the microammeter reads zero, giving V_0 . The points (f, V_0) are plotted.

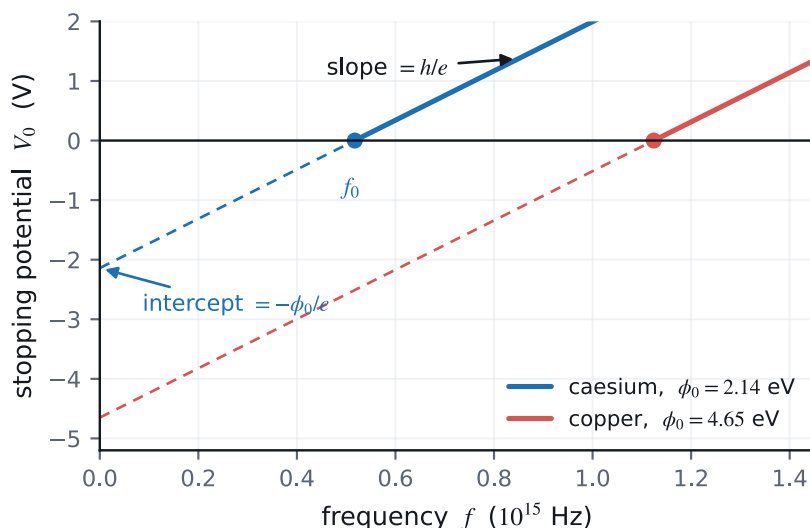


Figure 89. Stopping potential against frequency for two metals. Both lines have the same slope h/e ; the intercepts give the work functions and threshold frequencies.

Reading the graph:

Feature of the graph	What it gives
Slope	h/e , hence $h = e \times \text{slope}$
Intercept on the V_0 axis	$-\phi_0/e$, hence the work function
Intercept on the f axis	the threshold frequency f_0
Lines for different metals	parallel — the slope h/e is a universal constant

That last row is the deep result. The slope does not depend on the metal, on the intensity or on the apparatus. Planck's constant is a property of nature, not of the experiment.

If only two readings are available, the graph reduces to two equations:

$$eV_1 = hf_1 - \phi_0, \quad eV_2 = hf_2 - \phi_0 \implies h = \frac{e(V_1 - V_2)}{f_1 - f_2}$$

Metal	ϕ_0 (eV)	λ_0 (nm)
Caesium	2.14	580
Potassium	2.30	540
Sodium	2.75	452
Copper	4.65	267
Platinum	5.65	220

Caesium works with ordinary yellow light, which is why photocells and older TV camera tubes used caesium-coated cathodes; copper and platinum need ultraviolet.

WORKED EXAMPLE 21.3

Problem. In a photoelectric experiment the stopping potential is 1.85 V for light of wavelength 300 nm and 0.82 V for 400 nm. Find Planck's constant and the work function of the cathode.

Solution. The frequencies are

$$f_1 = \frac{c}{\lambda_1} = \frac{3.0 \times 10^8}{300 \times 10^{-9}} = 1.0 \times 10^{15} \text{ Hz}, \quad f_2 = \frac{3.0 \times 10^8}{400 \times 10^{-9}} = 7.5 \times 10^{14} \text{ Hz}$$

Subtracting the two photoelectric equations removes ϕ_0 :

$$h = \frac{e(V_1 - V_2)}{f_1 - f_2} = \frac{1.6 \times 10^{-19} (1.85 - 0.82)}{1.0 \times 10^{15} - 7.5 \times 10^{14}} = \frac{1.648 \times 10^{-19}}{2.5 \times 10^{14}} = 6.59 \times 10^{-34} \text{ J s}$$

within 1% of the accepted $6.63 \times 10^{-34} \text{ J s}$. The work function follows from either reading:

$$\phi_0 = hf_1 - eV_1 = 6.59 \times 10^{-19} - 2.96 \times 10^{-19} = 3.63 \times 10^{-19} \text{ J} = 2.27 \text{ eV}$$

KEEP THE UNITS IN ONE FAMILY

Work in electron-volts when the data is in eV and volts — then $K_{\max}$ in eV is numerically the stopping potential in volts, and you need no conversions at all. Convert to joules only at the last step, when a speed is asked for.

Chapter summary

- Radiation is emitted and absorbed in quanta: a photon has $E = hf = hc/\lambda$, $p = h/\lambda$, zero rest mass, and travels at c .
- Intensity fixes the number of photons per second; frequency fixes the energy of each one.
- Photoelectric emission occurs only above a threshold frequency f_0 , is instantaneous, gives a current proportional to intensity, and gives a maximum kinetic energy proportional to $(f - f_0)$.
- Einstein's equation: $hf = \phi_0 + K_{\max}$, with $\phi_0 = hf_0 = hc/\lambda_0$.
- Stopping potential: $eV_0 = K_{\max}$, so $V_0 = (h/e)f - \phi_0/e$; V_0 is independent of intensity.

- A graph of V_0 against f is a straight line of slope h/e , intercept $-\phi_0/e$ and f -intercept f_0 ; the slope is the same for every metal, giving $h = 6.63 \times 10^{-34} \text{ J s}$.

Practice questions

Group A — Multiple choice (1 mark each)

1. The momentum of a photon of wavelength λ is [1] (a) $h\lambda$ (b) h/λ (c) hc/λ (d) λ/h
2. The stopping potential in a photoelectric experiment depends on [1] (a) the intensity only (b) the frequency only (c) the distance of the source (d) the area of the cathode
3. The slope of a stopping potential versus frequency graph equals [1] (a) h (b) h/e (c) e/h (d) ϕ_0
4. If the intensity of the incident light is doubled at constant frequency, the saturation current [1] (a) halves (b) is unchanged (c) doubles (d) becomes zero
5. The rest mass of a photon is [1] (a) $9.1 \times 10^{-31} \text{ kg}$ (b) $1.67 \times 10^{-27} \text{ kg}$ (c) h/c^2 (d) zero

ANSWERS TO GROUP A

1. (b) — $p = E/c = hf/c = h/\lambda$.
2. (b) — $eV_0 = hf - \phi_0$ contains no intensity term.
3. (b) — $V_0 = (h/e)f - \phi_0/e$, so the slope is h/e .
4. (c) — twice as many photons eject twice as many electrons.
5. (d) — a photon has zero rest mass; h/c^2 is its effective mass.

Group B — Short answer (5 marks each)

1. State Einstein's photoelectric equation and use it to explain the existence of a threshold frequency and the failure of intensity to change the maximum kinetic energy. [5]
2. Light of wavelength 3000 Å falls on a metal whose threshold wavelength is 5000 Å . Find the work function and the stopping potential. [5]
3. Explain, with sketches, how the photocurrent varies with the anode voltage for (i) two different intensities at the same frequency and (ii) two different frequencies at the same intensity. [5]
4. Ultraviolet light of wavelength 300 nm falls on a sodium surface of work function 2.28 eV . Calculate the maximum kinetic energy of the photoelectrons in eV, the stopping potential, and the threshold frequency of sodium. [5]
5. Give three observations of the photoelectric effect that the wave theory of light cannot explain, and say what the photon theory says about each. [5]

ANSWERS TO GROUP B

2. $\phi_0 = hc/\lambda_0 = 1.989 \times 10^{-25} / (5.0 \times 10^{-7}) = 3.98 \times 10^{-19} \text{ J} = 2.49 \text{ eV}$. Photon energy $= 1.989 \times 10^{-25} / (3.0 \times 10^{-7}) = 6.63 \times 10^{-19} \text{ J} = 4.14 \text{ eV}$. So $K_{\max} = 4.14 - 2.49 = 1.65 \text{ eV}$ and $V_0 = 1.65 \text{ V}$.
4. Photon energy $= 1.989 \times 10^{-25} / (3.0 \times 10^{-7}) = 6.63 \times 10^{-19} \text{ J} = 4.14 \text{ eV}$. $K_{\max} = 4.14 - 2.28 = 1.86 \text{ eV}$, so $V_0 = 1.86 \text{ V}$. Threshold frequency $f_0 = \phi_0/h = (2.28 \times 1.6 \times 10^{-19}) / (6.63 \times 10^{-34}) = 5.50 \times 10^{14} \text{ Hz}$ (that is $\lambda_0 = 545 \text{ nm}$).

Group C — Long answer (8 marks each)

- (a) Describe, with a labelled diagram, an experiment to study the photoelectric effect, and define stopping potential. [4] (b) Derive Einstein's photoelectric equation and explain how a graph of stopping potential against frequency is used to measure Planck's constant and the work function. [4]
- A photocell has a caesium cathode of work function 2.14 eV and is illuminated with light of wavelength 400 nm . Find (a) the energy of each photon in eV, (b) the maximum kinetic energy of the photoelectrons, (c) the stopping potential, (d) the maximum speed of the electrons, and (e) the longest wavelength that would still cause emission. State what happens to answers (c) and (e) if the intensity is doubled. [8]

ANSWER TO GROUP C QUESTION 2

(a) $E = hc/\lambda = 1.989 \times 10^{-25} / (4.0 \times 10^{-7}) = 4.97 \times 10^{-19} \text{ J} = 3.11 \text{ eV}$.

(b) $K_{\text{max}} = 3.11 - 2.14 = 0.97 \text{ eV} = 0.97 \times 1.6 \times 10^{-19} = 1.55 \times 10^{-19} \text{ J}$.

(c) $V_0 = 0.97 \text{ V}$.

(d) $v_{\text{max}} = \sqrt{2K_{\text{max}}/m} = \sqrt{2 \times 1.55 \times 10^{-19} / 9.1 \times 10^{-31}} = \sqrt{3.41 \times 10^{11}} = 5.8 \times 10^5 \text{ m s}^{-1}$.

(e) $\lambda_0 = hc/\phi_0 = 1.989 \times 10^{-25} / (2.14 \times 1.6 \times 10^{-19}) = 5.8 \times 10^{-7} \text{ m} = 580 \text{ nm}$.

Doubling the intensity changes **neither** (c) nor (e): it only doubles the number of photoelectrons, and hence the saturation current.

UNIT 22

Semiconductor devices

Modern Physics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- P-N Junction
- Semiconductor diode: Characteristics in forward and reverse bias
- Full wave rectification
- Logic gates: NOT, OR, AND, NAND and NOR

A semiconductor on its own is a poor conductor. Add one foreign atom in ten million and it becomes useful; join a piece doped one way to a piece doped the other and it becomes a **device**. Every rectifier and logic gate in a phone, an inverter or a solar charger in a Nepali village rests on the thin layer where those two pieces meet. This unit follows that layer to the diode, the power supply and the logic gate.

WHAT THE EXAMINER WANTS

The p-n junction and its two bias conditions, the diode characteristic curve, the full-wave rectifier (circuit, waveform, and the numbers $V_{dc} = 2V_m/\pi$ and $\eta = 81.2\%$), and the five logic-gate truth tables. Those four items have appeared in almost every recent paper.

22.1 P–N Junction

Pure (intrinsic) silicon or germanium has four valence electrons per atom and few free carriers. Doping changes that:

Type	Dopant (valency)	Examples	Majority carriers	Minority carriers
n-type	pentavalent (donor)	P, As, Sb	electrons	holes
p-type	trivalent (acceptor)	B, Al, Ga, In	holes	electrons

Both types are electrically **neutral** — doping adds a carrier and its parent ion together.

A **p-n junction** is a single crystal, one half doped p and the other n. The moment it is formed, two things happen.

1. **Diffusion.** Holes diffuse from p to n and electrons from n to p, because each carrier is far more concentrated on its own side.
2. **Recombination.** Near the boundary they meet and cancel, leaving a thin layer with almost no mobile charge — the **depletion region** — but with the fixed, charged dopant ions still in place: negative acceptor ions on the p side, positive donor ions on the n side.

Those immobile ions set up an electric field pointing from n to p, which opposes further diffusion. Equilibrium is reached when the field is just strong enough to stop the net flow. The pd across the layer is

the **barrier** (or contact) potential V_B : about 0.3 V for germanium and 0.7 V for silicon at room temperature. The depletion width is typically $0.5\ \mu\text{m}$.

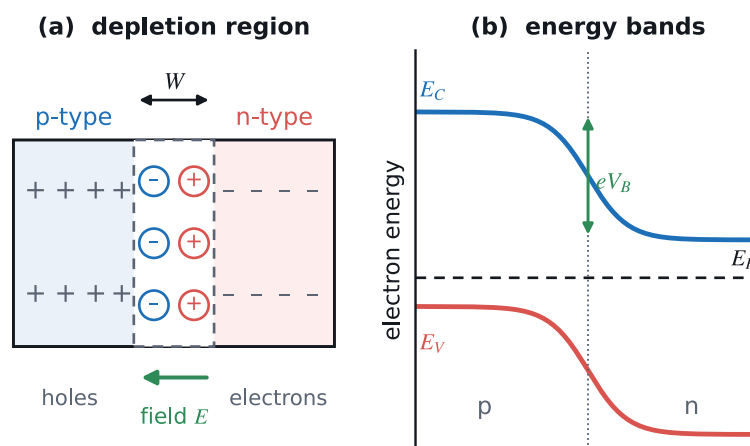


Figure 90. (a) The depletion region of an unbiased p-n junction: immobile ions left behind by diffusion set up a field from n to p. (b) The same junction in energy-band terms — the bands bend by eV_B and the Fermi level E_F is flat right across.

Biasing. Connect a battery across the junction and the barrier changes.

	Forward bias (p to +, n to −)	Reverse bias (p to −, n to +)
Applied field	opposes the barrier field	adds to the barrier field
Barrier height	reduced to $V_B - V$	raised to $V_B + V$
Depletion width	narrows	widens
Current	large, by majority carriers, mA	tiny, by minority carriers, μA
Resistance	low (a few ohms)	very high (megohms)

THE ONE-WAY STREET

A p-n junction conducts freely in one direction and hardly at all in the other. That asymmetry — not any exotic effect — is what makes the junction useful.

WORKED EXAMPLE 22.1

Problem. A silicon p-n junction has a barrier potential of 0.70 V across a depletion layer $0.50\ \mu\text{m}$ wide. Find the average electric field in the depletion layer. What happens to this field when the junction is reverse biased by 5.0 V, assuming the width stays the same?

Solution. The average field is the pd divided by the width:

$$E = \frac{V_B}{W} = \frac{0.70}{0.50 \times 10^{-6}} = 1.4 \times 10^6\ \text{V m}^{-1}$$

Under 5.0 V reverse bias the total pd across the layer is $0.70 + 5.0 = 5.7\ \text{V}$, so

$$E' = \frac{5.7}{0.50 \times 10^{-6}} = 1.14 \times 10^7 \text{ V m}^{-1}$$

an eightfold increase. (In reality W also widens, so the field grows a little less than this.) Fields of this size explain why a junction eventually breaks down.

22.2 Semiconductor diode: Characteristics in forward and reverse bias

A p-n junction with two leads is a **junction diode**. Its symbol is an arrowhead (the p side, or **anode**) touching a bar (the n side, or **cathode**); the arrow points the way conventional current can flow.

To obtain the characteristic, connect the diode in series with a resistor, a milliammeter and a variable pd; record current against voltage; then reverse the battery and repeat with a microammeter.

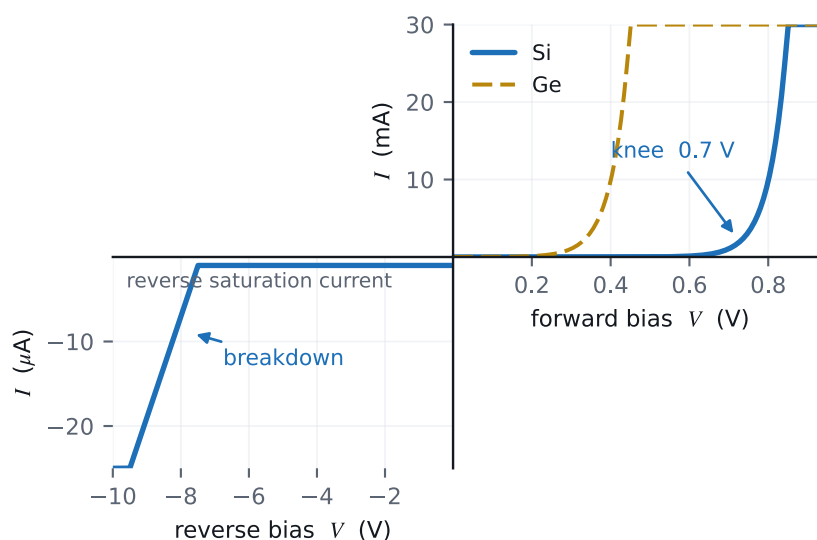


Figure 91. Static characteristic of a junction diode. Note the two different current scales: mA forward, μ A reverse.

Reading the curve:

- **Forward bias.** Almost no current until the applied voltage cancels most of the barrier. Beyond this **knee** (or cut-in) voltage — 0.7 V for Si, 0.3 V for Ge — the current rises very steeply and is limited only by the external resistance. The curve is **non-linear**: a diode does not obey Ohm's law.
- **Reverse bias.** A small, almost constant **reverse saturation current** of a few microamps, carried by thermally generated minority carriers. It doubles for roughly every 10°C rise but hardly changes with voltage.
- **Breakdown.** At a high enough reverse voltage the current suddenly becomes large (avalanche or Zener breakdown). An ordinary diode is destroyed; a Zener diode is designed to work there and is used as a voltage regulator.

The **dynamic** (a.c.) resistance in the forward region is the reciprocal of the slope at the working point:

$$r_d = \frac{\Delta V}{\Delta I}$$

DO NOT USE V/I

The static ratio V/I and the dynamic ratio $\Delta V/\Delta I$ are different numbers for a diode, because the graph is a curve and not a line through the origin. Examiners ask for the dynamic resistance, so always take a small change about the stated point.

WORKED EXAMPLE 22.2

Problem. (a) A silicon diode is joined in series with a $1.0\text{ k}\Omega$ resistor across a 5.0 V supply, connected in forward bias. Taking the diode drop as 0.7 V , find the current and the power dissipated in the diode. (b) On the forward characteristic, the current rises from 10 mA to 30 mA when the voltage goes from 0.70 V to 0.74 V . Find the dynamic resistance.

Solution.

(a) The resistor carries the whole of the remaining voltage:

$$I = \frac{V - V_D}{R} = \frac{5.0 - 0.7}{1.0 \times 10^3} = 4.3 \times 10^{-3}\text{ A} = 4.3\text{ mA}$$

$$P_D = V_D I = 0.7 \times 4.3 \times 10^{-3} = 3.0 \times 10^{-3}\text{ W} = 3.0\text{ mW}$$

(b) $r_d = \frac{\Delta V}{\Delta I} = \frac{0.74 - 0.70}{(30 - 10) \times 10^{-3}} = \frac{0.04}{0.02} = 2\text{ }\Omega$.

22.3 Full wave rectification

Rectification is the conversion of alternating current into direct current. A diode does it by conducting only on the half-cycles that forward bias it.

Half-wave rectifier. One diode in series with the load. The negative half of each cycle is simply thrown away, so the output is a series of gaps.

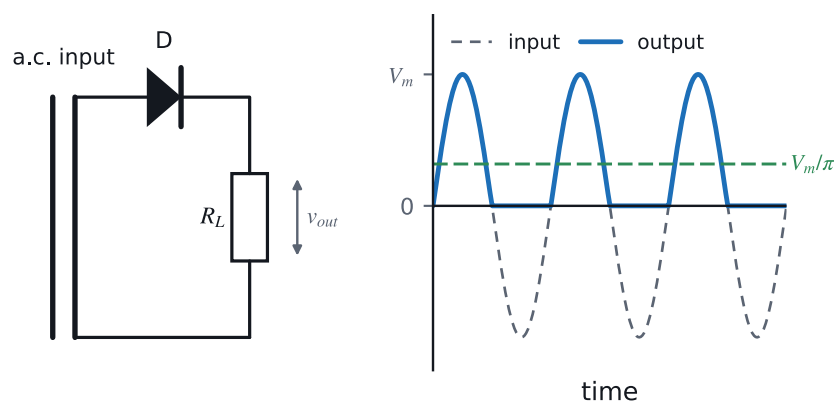


Figure 92. Half-wave rectifier: circuit (left) and output across the load (right). Only the positive half-cycles appear; the mean value is V_m/π .

Half-wave rectification wastes half the input. A **full-wave rectifier** uses both halves; there are two standard circuits.

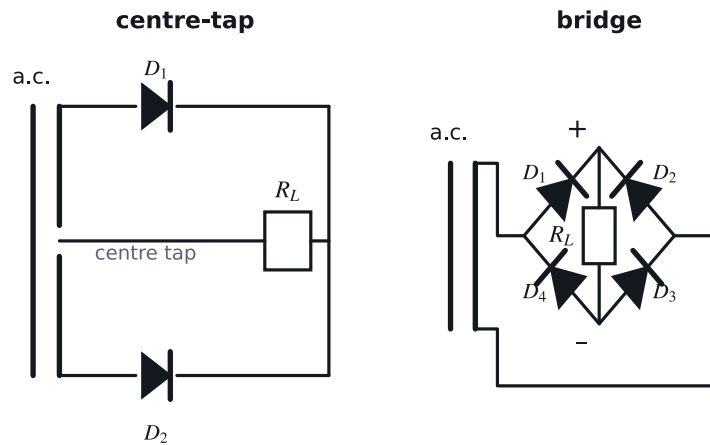


Figure 93. Full-wave rectifiers. Left: centre-tap circuit — D_1 conducts on one half-cycle, D_2 on the other. Right: bridge circuit — D_1D_3 conduct on one half-cycle, D_2D_4 on the other; the load current flows the same way both times.

In the **centre-tap** circuit the transformer secondary is split by an earthed centre tap, so the two halves always have opposite polarity. D_1 conducts while its half is positive, D_2 while the other half is positive; in both cases the current passes downward through R_L . In the **bridge** circuit no centre tap is needed: the diagonally opposite pairs D_1D_3 and D_2D_4 conduct alternately, again sending the current one way through the load.

MEAN CURRENT, RMS CURRENT AND EFFICIENCY

Write the load current on a conducting half-cycle as $i = I_m \sin \theta$. For a full-wave rectifier the load carries current for the whole cycle, so averaging over half a cycle is enough:

$$I_{dc} = \frac{1}{\pi} \int_0^\pi I_m \sin \theta \, d\theta = \frac{I_m}{\pi} [-\cos \theta]_0^\pi = \frac{2I_m}{\pi}$$

$$I_{rms} = \sqrt{\frac{1}{\pi} \int_0^\pi I_m^2 \sin^2 \theta \, d\theta} = \frac{I_m}{\sqrt{2}}$$

The rectification **efficiency** is the ratio of d.c. output power to a.c. input power, with r_f the forward resistance of the diode:

$$\eta = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + r_f)} = \frac{(2I_m/\pi)^2 R_L}{(I_m/\sqrt{2})^2 (R_L + r_f)} = \frac{8}{\pi^2} \cdot \frac{R_L}{R_L + r_f}$$

For $r_f \ll R_L$ this gives $\eta = 8/\pi^2 = 0.812$, i.e. **81.2%**. The same working for a half-wave rectifier, averaging over the whole cycle, gives $I_{dc} = I_m/\pi$, $I_{rms} = I_m/2$ and $\eta = 4/\pi^2 = 40.6\%$.

Quantity	Half-wave	Full-wave
I_{dc}	I_m/π	$2I_m/\pi$
V_{dc}	V_m/π	$2V_m/\pi$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$
Max efficiency	40.6%	81.2%
Ripple factor	1.21	0.48
Output frequency (50 Hz mains)	50 Hz	100 Hz
PIV per diode	V_m	$2V_m$ (centre-tap), V_m (bridge)

The output is still lumpy. A **filter** — most simply a large capacitor across the load — charges to the peak and discharges slowly between peaks, flattening the ripple.

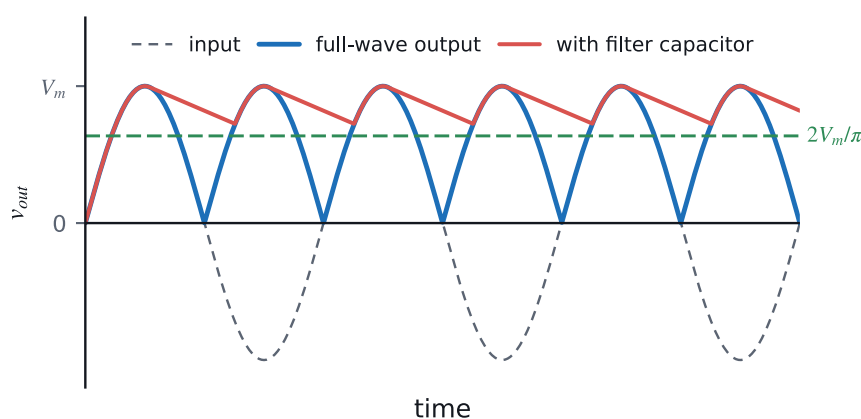


Figure 94. Full-wave output before and after a capacitor filter. Both halves of the input are used, so the output frequency is twice the input frequency and the mean value is $2V_m/\pi$.

WORKED EXAMPLE 22.3

Problem. A centre-tap full-wave rectifier feeds a load of $500\ \Omega$. Each half of the transformer secondary gives a peak voltage of 20 V . Taking the diodes as ideal, find (a) the peak load current, (b) the d.c. load current and d.c. output voltage, (c) the rms current, (d) the rectification efficiency, (e) the peak inverse voltage across each diode, and (f) the output ripple frequency for a 50 Hz supply.

Solution.

(a) $I_m = \frac{V_m}{R_L} = \frac{20}{500} = 0.040\text{ A} = 40\text{ mA}.$

(b) $I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 40}{3.142} = 25.5\text{ mA},$ and

$$V_{dc} = I_{dc}R_L = 25.5 \times 10^{-3} \times 500 = 12.7\text{ V} \quad \left(= \frac{2V_m}{\pi} = \frac{40}{3.142} \right)$$

(c) $I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{40}{1.414} = 28.3\text{ mA}.$

(d) $\eta = \frac{I_{dc}^2}{I_{rms}^2} = \left(\frac{25.5}{28.3} \right)^2 = 0.812 = 81.2\%$
(*diodes ideal, so $r_f = 0$*).

(e) When one diode conducts, the other has the whole secondary across it: $PIV = 2V_m = 40 \text{ V}$.

(f) Both half-cycles produce an output pulse, so the ripple frequency is $2 \times 50 = 100 \text{ Hz}$.

22.4 Logic gates: NOT, OR, AND, NAND and NOR

A **logic gate** is a circuit whose input and output voltages are restricted to two levels, written 1 (high, typically +5 V) and 0 (low, near 0 V). Diodes and transistors make the gates; Boolean algebra describes them.

Gate	Boolean expression	Read as
NOT	$Y = \bar{A}$	inverter
OR	$Y = A + B$	A or B
AND	$Y = A \cdot B$	A and B
NAND	$Y = \overline{A \cdot B}$	AND then NOT
NOR	$Y = \overline{A + B}$	OR then NOT

The truth tables, which is what the board actually asks you to write:

A	$Y = \bar{A}$
0	1
1	0

A	B	OR	AND	NAND	NOR
0	0	0	0	1	1
0	1	1	0	1	0
1	0	1	0	1	0
1	1	1	1	0	0

The NAND column is the AND column inverted, and NOR is OR inverted — that is all the bar means.

READING THE TABLES IN ONE LINE EACH

- **OR** — output 0 only when all inputs are 0.
- **AND** — output 1 only when all inputs are 1.
- **NAND** — output 0 only when all inputs are 1.
- **NOR** — output 1 only when all inputs are 0.

Universal gates. NAND and NOR are called universal because every other gate can be built from copies of just one of them. With NAND:

Required gate	How to build it from NAND gates
NOT	tie both inputs of one NAND together: $\overline{A \cdot A} = \bar{A}$
AND	a NAND followed by a NAND used as NOT
OR	invert both inputs with NANDs, then feed a third NAND: $\overline{\bar{A} \cdot \bar{B}} = A + B$

The last line is De Morgan's theorem. Check it: for $A = B = 0$, $\bar{A} \cdot \bar{B} = 1$ and the output is 0; for every other combination $\bar{A} \cdot \bar{B} = 0$ and the output is 1 — exactly the OR column.

NAND IS NOT "NOT THEN AND"

NAND means AND **then** NOT. Inverting first and then ANDing gives $\bar{A} \cdot \bar{B}$, which is NOR, not NAND. Order matters.

22.5 The junction transistor and its configurations

Two junctions back to back in one crystal make a **bipolar junction transistor**, n-p-n or p-n-p. The **emitter** is heavily doped and supplies the carriers, the **base** is very thin and lightly doped, the **collector** is the largest region. In normal operation the emitter-base junction is **forward** biased and the collector-base junction **reverse** biased, and because the base is thin over 95% of the injected carriers sweep straight through to the collector:

$$I_E = I_B + I_C, \quad \alpha = \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$$

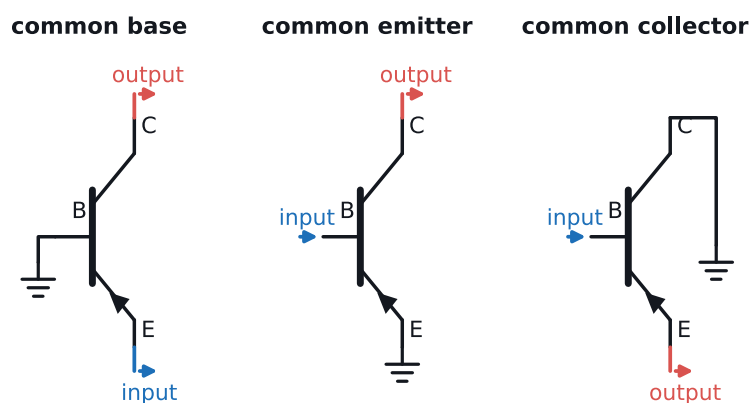


Figure 95. The three transistor configurations, drawn for an n-p-n transistor. The terminal common to the input and output loops gives each its name.

Configuration	Common terminal	Current gain	Voltage gain	Main use
Common base	base	$\alpha < 1$	high	high-frequency amplifier
Common emitter	emitter	β , typically 20–500	high	general-purpose amplifier, switch
Common collector	collector	$\approx \beta$	< 1	impedance matching (emitter follower)

WORKED EXAMPLE 22.4

Problem. A transistor has $\alpha = 0.98$ and an emitter current of 2.0 mA. Find the collector current, the base current and the current gain β in the common-emitter configuration.

Solution.

$$I_C = \alpha I_E = 0.98 \times 2.0 = 1.96 \text{ mA}$$

$$I_B = I_E - I_C = 2.0 - 1.96 = 0.04 \text{ mA} = 40 \text{ } \mu\text{A}$$

$$\beta = \frac{I_C}{I_B} = \frac{1.96}{0.04} = 49 \quad \left(\text{check: } \frac{\alpha}{1-\alpha} = \frac{0.98}{0.02} = 49 \right)$$

Chapter summary

- Diffusion across a p-n junction leaves a depletion region of immobile ions and a barrier potential V_B (0.3 V Ge, 0.7 V Si) that stops further diffusion.
- Forward bias lowers the barrier, narrows the depletion layer and gives a large majority-carrier current; reverse bias does the opposite and leaves only a microamp reverse saturation current until breakdown.
- The diode characteristic is non-linear: no current below the knee voltage, then a steep rise. Dynamic resistance $r_d = \Delta V / \Delta I$.
- Half-wave: $I_{dc} = I_m / \pi$, $\eta_{max} = 40.6\%$, ripple 1.21, output 50 Hz.
- Full-wave: $I_{dc} = 2I_m / \pi$, $V_{dc} = 2V_m / \pi$, $I_{rms} = I_m / \sqrt{2}$, $\eta = (8/\pi^2)R_L / (R_L + r_f)$, maximum 81.2%, ripple 0.48, output 100 Hz.
- PIV is V_m for half-wave and bridge, $2V_m$ for the centre-tap circuit.
- Gates: OR gives 0 only for all-zero inputs; AND gives 1 only for all-one inputs; NAND and NOR are their complements and are universal gates.
- Transistor: $I_E = I_B + I_C$, $\alpha = I_C / I_E$, $\beta = I_C / I_B = \alpha / (1 - \alpha)$.

Practice questions

Group A — Multiple choice (1 mark each)

1. In a p-type semiconductor the majority charge carriers are [1] (a) electrons (b) holes (c) protons (d) positive ions

2. The barrier potential of a silicon p-n junction at room temperature is about [1] (a) 0.1 V (b) 0.3 V (c) 0.7 V (d) 1.4 V
3. The maximum efficiency of a full-wave rectifier is [1] (a) 40.6% (b) 50% (c) 81.2% (d) 100%
4. The peak inverse voltage across each diode of a centre-tap full-wave rectifier is [1] (a) V_m (b) $2V_m$ (c) $V_m/2$ (d) V_m/π
5. A two-input NAND gate gives output 0 only when [1] (a) both inputs are 0 (b) both inputs are 1 (c) any one input is 1 (d) any one input is 0
6. For a transistor with $\alpha = 0.95$, the value of β is [1] (a) 9.5 (b) 19 (c) 95 (d) 0.95

ANSWERS TO GROUP A

1. (b) — trivalent dopants create holes. 2. (c) — 0.7 V for Si, 0.3 V for Ge. 3. (c) — $\eta_{max} = 8/\pi^2 = 81.2\%$.
 4. (b) — the non-conducting diode has the whole secondary across it. 5. (b) — NAND is AND inverted, and AND is 1 only for 1,1. 6. (b) — $\beta = \alpha/(1 - \alpha) = 0.95/0.05 = 19$.

Group B — Short answer (5 marks each)

1. Explain how the depletion region and barrier potential are formed at a p-n junction, and state what happens to each under forward and reverse bias. [5]
2. Draw the forward and reverse characteristic curves of a junction diode and explain the knee voltage, the reverse saturation current and breakdown. [5]
3. A silicon diode in series with a resistor R is connected across a 9.0 V battery in forward bias, and the current is 20 mA. Find R and the power dissipated in the diode. [5]
4. A half-wave rectifier delivers current to a $300\ \Omega$ load from a supply of peak voltage 30 V. Taking the diode as ideal, find the peak current, the d.c. current, the d.c. output voltage and the PIV. [5]
5. Draw the truth tables of the NAND and NOR gates, and show how a NAND gate can be used as a NOT gate. [5]

ANSWERS TO GROUP B

3. The diode drops 0.7 V, so the resistor has $9.0 - 0.7 = 8.3\text{ V}$ across it and $R = 8.3/(20 \times 10^{-3}) = 415\ \Omega$. Power in the diode $= 0.7 \times 20 \times 10^{-3} = 14\text{ mW}$.
 4. $I_m = V_m/R_L = 30/300 = 0.10\text{ A}$; $I_{dc} = I_m/\pi = 0.10/3.142 = 31.8\text{ mA}$; $V_{dc} = I_{dc}R_L = 31.8 \times 10^{-3} \times 300 = 9.55\text{ V}$ (which is V_m/π); $PIV = V_m = 30\text{ V}$.
 5. Outline: write both truth tables from the tables in §22.4. Joining the two inputs of a NAND gives $Y = \overline{A \cdot A} = \overline{A}$, which is NOT.

Group C — Long answer (8 marks each)

1. (a) Draw the circuit of a centre-tap full-wave rectifier and explain its action over one complete cycle, sketching the input and output waveforms. [4] (b) Derive expressions for the d.c. current and the rectification efficiency, and hence show that the maximum efficiency is 81.2%. [4]
2. A bridge rectifier is supplied from a transformer whose secondary voltage is 24 V rms. The load resistance is 1.0 k Ω and the diodes may be treated as ideal. Find (a) the peak voltage, (b) the peak current, (c) the d.c. output voltage and current, (d) the rms current, (e) the efficiency, and (f) the PIV of each diode. State one advantage of the bridge circuit over the centre-tap circuit. [8]

ANSWER TO GROUP C QUESTION 2

(a) $V_m = \sqrt{2} \times 24 = 33.9 \text{ V}.$

(b) $I_m = V_m/R_L = 33.9/1000 = 33.9 \text{ mA}.$

(c) $V_{dc} = 2V_m/\pi = (2 \times 33.9)/3.142 = 21.6 \text{ V}$ and $I_{dc} = V_{dc}/R_L = 21.6 \text{ mA}.$

(d) $I_{rms} = I_m/\sqrt{2} = 33.9/1.414 = 24.0 \text{ mA}.$

(e) $\eta = (I_{dc}/I_{rms})^2 = (21.6/24.0)^2 = 0.81 = 81\%.$

(f) For a bridge, $PIV = V_m = 33.9 \text{ V}.$

Advantage: no centre-tapped transformer is needed, and the PIV per diode is only V_m instead of $2V_m$, so cheaper diodes can be used for the same output.

UNIT 23

Quantization of energy

Modern Physics • 8 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Bohr's theory of hydrogen atom
- Spectral series; Excitation and ionization potentials
- Energy level; Emission and absorption spectra
- De Broglie Theory; Duality
- Uncertainty principle
- X-rays: Nature and production; uses
- X-rays diffraction; Bragg's law

Rutherford's nuclear atom had a fatal flaw: an orbiting electron is an accelerating charge, so classical physics says it must radiate and spiral into the nucleus in about 10^{-8} s. Atoms do not do this, and the light they emit is not a continuous rainbow but a set of sharp lines. Bohr's answer was that energy inside an atom is **quantized**. This unit follows that idea from the hydrogen atom to X-rays and to the matter waves behind Bohr's rule.

WHAT THE EXAM WANTS FROM THIS UNIT

The heaviest Modern Physics unit, and a reliable source of a Group C question. Master five things: derive r_n and E_n for hydrogen; use $1/\lambda = R(1/n_f^2 - 1/n_i^2)$; quote $\lambda = h/p = 12.27/\sqrt{V}$ Å; draw the X-ray spectrum; apply $2d\sin\theta = n\lambda$.

23.1 Bohr's theory of hydrogen atom

Bohr kept Rutherford's nucleus but added three postulates.

BOHR'S POSTULATES

1. **Stationary orbits.** An electron can revolve only in certain circular orbits in which it does **not** radiate energy. The centripetal force is supplied by the electrostatic attraction of the nucleus.
2. **Quantization of angular momentum.** The allowed orbits are those for which the angular momentum is an integral multiple of $h/2\pi$:

$$mvr = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

1. **Frequency condition.** Radiation is emitted or absorbed only when the electron jumps between two stationary orbits, and then

$$h\nu = E_i - E_f$$

n is the **principal quantum number**.

RADIUS, SPEED AND ENERGY OF THE n^{th} ORBIT

For a hydrogen atom (nuclear charge $+e$), the Coulomb attraction provides the centripetal force:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r} \quad \Rightarrow \quad mv^2 r = \frac{e^2}{4\pi\epsilon_0} \quad (i)$$

From the second postulate, $v = nh/(2\pi mr)$. Substituting into (i):

$$m\left(\frac{nh}{2\pi mr}\right)^2 r = \frac{e^2}{4\pi\epsilon_0} \quad \Rightarrow \quad \frac{n^2 h^2}{4\pi^2 m r} = \frac{e^2}{4\pi\epsilon_0}$$

$$r_n = \frac{\epsilon_0 h^2}{\pi m e^2} n^2$$

Putting in the constants, $r_1 = 5.29 \times 10^{-11} \text{ m} = 0.529 \text{ \AA}$ (the **Bohr radius**), and $r_n \propto n^2$.

Eliminating r instead gives the orbital speed

$$v_n = \frac{e^2}{2\epsilon_0 n h} = \frac{c}{137 n}$$

so $v_1 = 2.19 \times 10^6 \text{ m s}^{-1}$, about $1/137$ of the speed of light.

Energy. From (i) the kinetic energy is $K = \frac{1}{2}mv^2 = e^2/(8\pi\epsilon_0 r)$, and the electrostatic potential energy is $U = -e^2/(4\pi\epsilon_0 r)$. Hence

$$E = K + U = \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}$$

Substituting r_n :

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2} \text{ eV}$$

Note $U = -2K$ and $E = -K$: the energy is negative, so the electron is **bound**. $E = 0$ means a free, stationary electron.

BOHR'S MODEL IS NOT GENERAL

It works for hydrogen and hydrogen-like ions (He^+ , Li^{2+} , where $E_n = -13.6 Z^2/n^2$ eV and $r_n = 0.529 n^2/Z$ Å), but fails for any atom with two or more electrons. Do not apply it to helium or sodium.

23.2 Spectral series; Excitation and ionization potentials

Combining the frequency condition with E_n and $\nu = c/\lambda$:

$$\frac{1}{\lambda} = \frac{me^4}{8\epsilon_0^2 h^3 c} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where $R = 1.097 \times 10^7 \text{ m}^{-1}$ is the **Rydberg constant**. All transitions ending on the same n_f form one **spectral series**.

Series	n_f	n_i	Region	Longest λ	Series limit
Lyman	1	2, 3, ...	Ultraviolet	121.6 nm	91.2 nm
Balmer	2	3, 4, ...	Visible	656.3 nm	364.6 nm
Paschen	3	4, 5, ...	Infrared	1875 nm	820.4 nm
Brackett	4	5, 6, ...	Infrared	4051 nm	1458 nm
Pfund	5	6, 7, ...	Far infrared	7460 nm	2279 nm

The **series limit** is the $n_i \rightarrow \infty$ line, $\lambda_{\min} = n_f^2/R$.

ORDER OF THE SERIES

Lyman, Balmer, Paschen, Brackett, Pfund — "Little Boys Play Better Football" — ending on $n_f = 1, 2, 3, 4, 5$.

Excitation potential. The p.d. through which an electron must be accelerated so that, on colliding with an atom, it can lift that atom from the ground state to the level n :

$$V_{\text{exc}} = \frac{E_n - E_1}{e}$$

For hydrogen the first excitation potential is $(-3.4) - (-13.6) = 10.2$ V and the second is $(-1.51) - (-13.6) = 12.09$ V.

Ionization potential. The potential needed to take the electron from $n = 1$ to $n = \infty$: for hydrogen $V_{\text{ion}} = 13.6$ V, so the **ionization energy** is 13.6 eV.

WORKED EXAMPLE 23.1

Problem. Find (a) the radius and energy of the $n = 3$ orbit of hydrogen, and (b) the wavelength of the photon emitted in the transition $n = 3 \rightarrow 2$.

Solution.

(a) $r_3 = 0.529 \times 3^2 = 4.76 \text{ \AA} = 4.76 \times 10^{-10} \text{ m}$; $E_3 = -13.6/9 = -1.51 \text{ eV}$.

(b) $\Delta E = E_3 - E_2 = -1.51 - (-3.40) = 1.89 \text{ eV}$.

Using $\lambda = hc/\Delta E$ with $hc = 1240 \text{ eV nm}$,

$$\lambda = \frac{1240}{1.89} = 656 \text{ nm}$$

This is the red H_α line of the Balmer series.

23.3 Energy level; Emission and absorption spectra

Drawing the allowed energies as horizontal lines gives the **energy-level diagram**. Since $E_n \propto -1/n^2$, the levels crowd together as n grows and converge on the ionization limit $E = 0$. Above it lies a continuum: a free electron may have any energy.

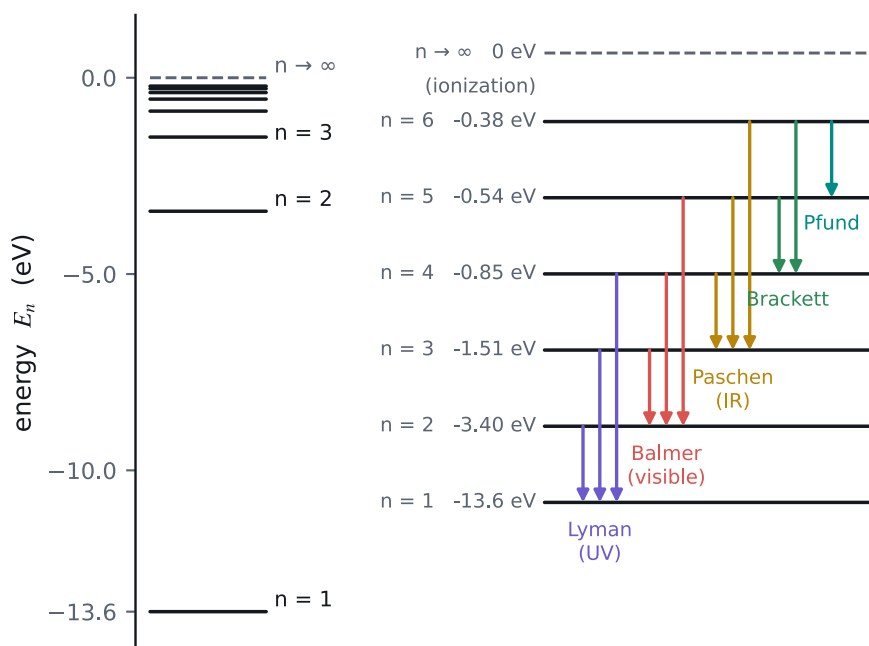


Figure 96. Left: hydrogen levels drawn to scale in energy — note how they crowd towards the ionization limit. Right: the same levels drawn equally spaced (not to scale) so that the five named series can be shown. Each arrow is one emitted photon.

Emission spectrum. An atom excited by heat, discharge or collision is unstable and drops back within about 10^{-8} s , emitting a photon of energy $E_i - E_f$. The result is a **bright-line spectrum**: coloured lines on a dark background.

Absorption spectrum. White light passed through cool hydrogen gas loses exactly those photons that can lift atoms to a higher level, giving a **dark-line spectrum** at the same wavelengths as the emission lines.

Feature	Emission spectrum	Absorption spectrum
Produced by	hot/excited gas	cool gas in front of a hot continuum source
Appearance	bright lines on dark ground	dark lines on bright ground
Transition	$n_i \rightarrow n_f$ (downward)	$n_f \rightarrow n_i$ (upward)
Lines observed	all series	mainly Lyman (from the ground state)

The dark **Fraunhofer lines** in sunlight are an absorption spectrum produced by the Sun's cooler outer gases, and they are how we know what stars are made of.

WORKED EXAMPLE 23.2

Problem. A hydrogen atom in its ground state is bombarded by an electron accelerated through 12.5 V. To which level can the atom be excited, and what wavelengths can it then emit?

Solution. The electron brings 12.5 eV. Energies needed from $n=1$ are $E_2 - E_1 = 10.2$ eV, $E_3 - E_1 = 12.09$ eV, $E_4 - E_1 = 12.75$ eV. Since $12.09 < 12.5 < 12.75$, the atom can reach $n=3$ but not $n=4$.

From $n=3$ three transitions are possible:

$3 \rightarrow 2$: $\Delta E = 1.89$ eV, $\lambda = 1240/1.89 = 656$ nm (Balmer) $3 \rightarrow 1$: $\Delta E = 12.09$ eV, $\lambda = 1240/12.09 = 103$ nm (Lyman) $2 \rightarrow 1$: $\Delta E = 10.2$ eV, $\lambda = 1240/10.2 = 122$ nm (Lyman)

23.4 De Broglie Theory; Duality

Light behaves as a wave in diffraction and as photons in the photoelectric effect. In 1924 de Broglie proposed that this **duality** is universal: every moving particle has an associated wave of wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

For a photon, $E = h\nu$ and $E = pc$ give $p = h/\lambda$; de Broglie read this backwards for matter. For an electron accelerated from rest through a p.d. V , $\frac{1}{2}mv^2 = eV$, so $p = \sqrt{2meV}$ and

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (V \text{ in volts})$$

WHY ORDINARY OBJECTS SHOW NO WAVE BEHAVIOUR

$h = 6.63 \times 10^{-34}$ J s is minute. A 60 kg student running at 5 m s^{-1} has $\lambda = 2.2 \times 10^{-36}$ m — smaller than any possible obstacle, so no diffraction is observable. Wave behaviour appears only when λ is comparable to the aperture, i.e. for electrons and atoms.

The **Davisson-Germer experiment** (1927) confirmed this: electrons accelerated through 54 V and scattered from a nickel crystal gave a strong beam at 50° , matching the predicted $12.27/\sqrt{54} = 1.67$ \AA.

De Broglie's hypothesis also **explains Bohr's second postulate**: a stable orbit is a standing wave that joins on to itself round the circumference,

$$2\pi r = n\lambda = \frac{nh}{mv} \Rightarrow mvr = \frac{nh}{2\pi}$$

which is exactly the quantization rule Bohr had to assume.

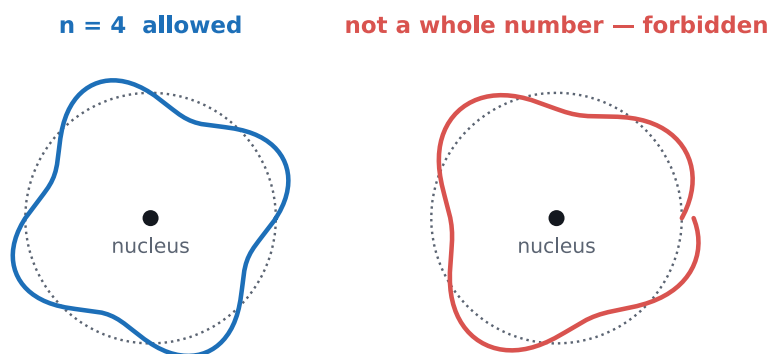


Figure 97. A Bohr orbit is a standing electron wave: the circumference must hold a whole number of de Broglie wavelengths, $2\pi r = n\lambda$. Left $n = 4$ (allowed), right a non-integral fit (the wave cancels itself, so the orbit is not allowed).

WORKED EXAMPLE 23.3

Problem. Calculate the de Broglie wavelength of an electron accelerated through 100 V. Take $h = 6.63 \times 10^{-34}$ J s, $m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C.

Solution.

$$p = \sqrt{2meV} = \sqrt{2(9.1 \times 10^{-31})(1.6 \times 10^{-19})(100)}$$

$$p = \sqrt{2.912 \times 10^{-47}} = 5.40 \times 10^{-24} \text{ kg m s}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{5.40 \times 10^{-24}} = 1.23 \times 10^{-10} \text{ m} = 1.23 \text{ \AA}$$

Check with the shortcut: $12.27/\sqrt{100} = 1.227 \text{ \AA}$. This is about the spacing of atoms in a crystal, which is why electron diffraction works.

23.5 Uncertainty principle

If an electron is a wave spread over a region, asking for its exact position is meaningless. Heisenberg made this quantitative in 1927.

HEISENBERG'S UNCERTAINTY PRINCIPLE

It is impossible to measure simultaneously both the position and the momentum of a particle with unlimited accuracy. If Δx and Δp are the uncertainties in position and in the momentum component along the same direction, then

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

An equivalent statement for energy and time is $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$.

This is not a limitation of our instruments; it is a property of nature. Two standard consequences:

- **Electrons cannot exist inside the nucleus.** Confining an electron to $\Delta x \approx 10^{-14}$ m needs $\Delta p \geq h/(4\pi\Delta x)$, an energy of about 20 MeV, whereas beta particles carry only a few MeV. The electron is created at the moment of decay, not stored.
- **Natural line width.** A state living for $\Delta t \approx 10^{-8}$ s has an energy blur $\Delta E \geq h/4\pi\Delta t$, so spectral "lines" are never infinitely sharp.

UNCERTAINTY IS NOT EXPERIMENTAL ERROR

Δx and Δp are not mistakes in reading a scale: even with perfect apparatus the product cannot fall below $h/4\pi$. The principle links x with p_x only — Δx and Δp_y are unrestricted.

WORKED EXAMPLE 23.4

Problem. The position of an electron is measured to an accuracy of 10^{-10} m. Find the minimum uncertainty in its velocity.

Solution.

$$\Delta v \geq \frac{h}{4\pi m \Delta x} = \frac{6.63 \times 10^{-34}}{4\pi(9.1 \times 10^{-31})(10^{-10})}$$

Denominator $= 4 \times 3.1416 \times 9.1 \times 10^{-31} \times 10^{-10} = 1.144 \times 10^{-39}$.

$$\Delta v \geq \frac{6.63 \times 10^{-34}}{1.144 \times 10^{-39}} = 5.8 \times 10^5 \text{ m s}^{-1}$$

The uncertainty is about a quarter of the electron's own orbital speed in hydrogen — so the idea of a definite "orbit" is not really tenable.

23.6 X-rays: Nature and production; uses

X-rays were discovered by Röntgen in 1895. They are **electromagnetic waves** of wavelength roughly 0.01 Å to 100 Å (0.001–10 nm), produced when fast electrons are suddenly stopped by a heavy target — the reverse of the photoelectric effect.

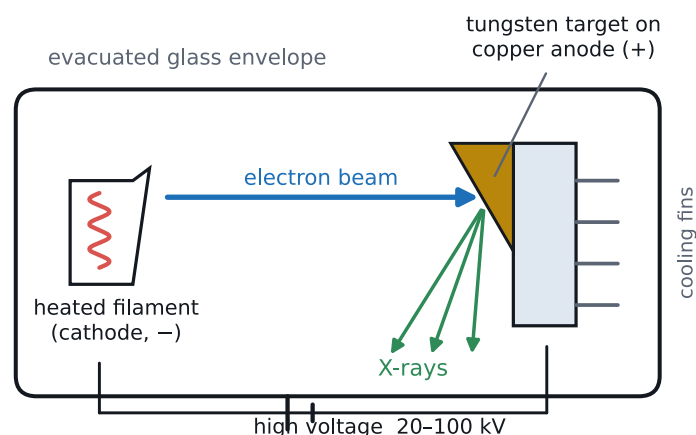


Figure 98. Coolidge X-ray tube. Electrons boiled off the hot filament are accelerated through 20–100 kV onto a tungsten target set in a copper anode; less than 1% of their energy becomes X-rays and the rest becomes heat.

Two processes go on at the target, giving two parts to the spectrum.

1. **Continuous (Bremsstrahlung) spectrum.** An electron decelerating in the field of a target nucleus radiates, and may lose any fraction of its energy, so all wavelengths above a limit appear. The limit comes from an electron that gives up all its energy in one collision:

$$eV = h\nu_{\max} = \frac{hc}{\lambda_{\min}} \quad \Rightarrow \quad \lambda_{\min} = \frac{hc}{eV} = \frac{1.24 \times 10^{-6}}{V} \text{ m}$$

This **Duane–Hunt limit** depends only on the accelerating voltage, not on the target material.

1. **Characteristic spectrum.** A fast electron knocks an electron out of the K shell of a target atom; an L-shell electron falling into the vacancy emits the sharp K_{α} line, an M-shell electron the K_{β} line. These positions depend only on the **target element** (Moseley's law).

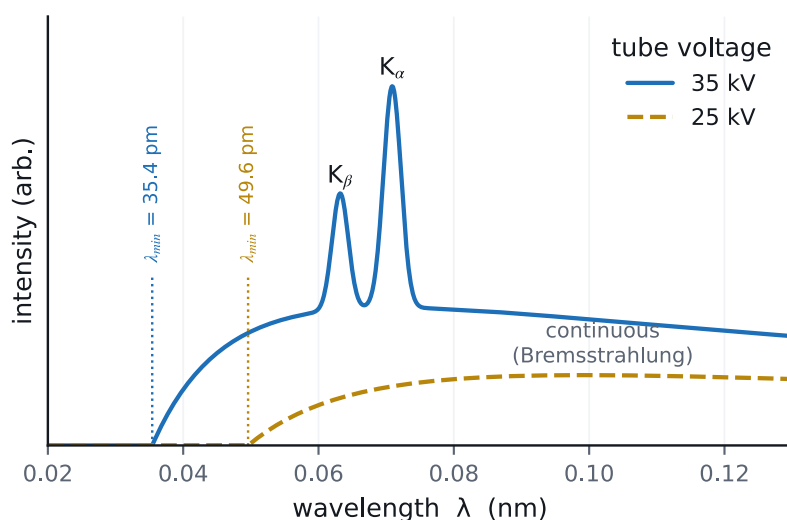


Figure 99. X-ray spectrum of a molybdenum target at two accelerating voltages. The continuous background cuts off sharply at $\lambda_{\min} = hc/eV$, which moves left as V rises, while the characteristic K_{α} and K_{β} lines stay fixed.

Properties. X-rays travel in straight lines at the speed of light, are undeflected by electric and magnetic fields, penetrate matter, ionize gases, affect photographic plates, cause fluorescence and are diffracted by crystals.

Use	Basis
Medical radiography, CT scan	bone (high Z) absorbs more than soft tissue
Radiotherapy	hard X-rays destroy malignant cells
Airport and customs scanning	differential absorption reveals hidden metal
Detecting cracks in castings	voids transmit more
X-ray crystallography	λ comparable to atomic spacing

Hard and soft X-rays. A high tube voltage gives short wavelength and high penetrating power ("hard" X-rays, used in radiotherapy); a low voltage gives "soft", less penetrating X-rays used for imaging soft tissue.

23.7 X-rays diffraction; Bragg's law

An optical grating has lines about 10^{-6} m apart — far too coarse to diffract X-rays of wavelength 10^{-10} m. Laue realised in 1912 that the atomic planes of a crystal, spaced about 10^{-10} m apart, form a natural three-dimensional grating.

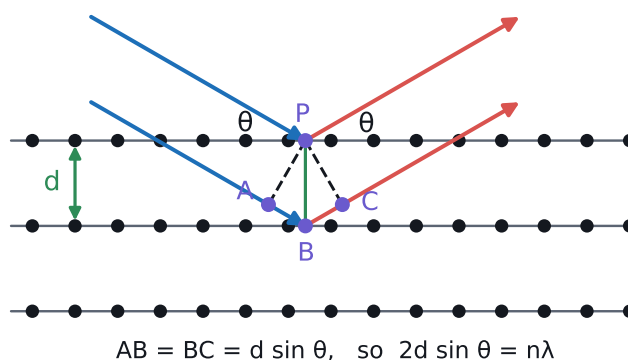


Figure 100. Bragg reflection. B lies directly below P, a distance d away. PA and PC are wavefronts, so the lower ray travels the extra path $AB + BC = 2d \sin \theta$.

BRAGG'S LAW

Take two atomic planes a distance d apart and a beam striking them at a **glancing angle** θ . With B vertically below P, the lower ray travels the extra distance $AB + BC$, where PA and PC are wavefronts. From the right-angled triangles PAB and PCB,

$$AB = BC = d \sin \theta \quad \Rightarrow \quad \text{path difference} = 2d \sin \theta$$

Constructive interference (a bright spot) requires the path difference to be a whole number of wavelengths:

$$2d \sin \theta = n\lambda, \quad n = 1, 2, 3, \dots$$

θ IS MEASURED FROM THE PLANE, NOT FROM THE NORMAL

In optics the angle of incidence is measured from the normal; in Bragg's law θ is the **glancing angle**, measured from the plane itself. Using the normal angle replaces $\sin \theta$ by $\cos \theta$ and is always wrong.

Since $\sin \theta \leq 1$, Bragg reflection needs $\lambda \leq 2d$. The law is used both ways: with λ known it measures d (X-ray crystallography, which gave us the structure of DNA); with d known it measures λ (the X-ray spectrometer).

WORKED EXAMPLE 23.5

Problem. A beam of X-rays of wavelength $1.54 \text{ \AA}$ gives a first-order Bragg reflection from a set of crystal planes at a glancing angle of 15.9° . Find the spacing of the planes. What is the largest order that can be observed?

Solution. For $n = 1$,

$$d = \frac{n\lambda}{2\sin \theta} = \frac{1 \times 1.54 \times 10^{-10}}{2\sin 15.9^\circ} = \frac{1.54 \times 10^{-10}}{2 \times 0.2740}$$

$$d = 2.81 \times 10^{-10} \text{ m} = 2.81 \text{ \AA}$$

Highest order: $\sin \theta \leq 1$ needs $n \leq 2d/\lambda = 2(2.81)/1.54 = 3.65$, so $n = 3$ is the highest order observable.

Chapter summary

- Bohr: $mvr = nh/2\pi$; $r_n = \epsilon_0 h^2 n^2 / \pi m e^2 = 0.529 n^2 \text{ \AA}$; $v_n = c/137n$; $E_n = -13.6/n^2 \text{ eV}$ (multiply by Z^2 , divide r_n by Z , for hydrogen-like ions).
- $h\nu = E_i - E_f$ gives $1/\lambda = R(1/n_f^2 - 1/n_i^2)$, $R = 1.097 \times 10^7 \text{ m}^{-1}$. Series: Lyman (UV, $n_f = 1$), Balmer (visible), Paschen, Brackett, Pfund (IR).
- Hydrogen: first excitation potential 10.2 V, ionization potential 13.6 V. Emission gives bright lines, absorption dark lines at the same wavelengths.
- De Broglie: $\lambda = h/p = 12.27/\sqrt{V} \text{ \AA}$ for an accelerated electron; $2\pi r = n\lambda$ reproduces Bohr's quantization.
- Uncertainty: $\Delta x \Delta p \geq h/4\pi$ and $\Delta E \Delta t \geq h/4\pi$.
- X-ray tube: continuous spectrum with $\lambda_{\min} = hc/eV$ (depends on V) plus characteristic K_α , K_β lines (depend on the target).
- Bragg: $2d\sin \theta = n\lambda$, θ measured from the atomic plane.

Practice questions**Group A — Multiple choice (1 mark each)**

1. The radius of the n^{th} Bohr orbit of hydrogen is proportional to [1] (a) n (b) n^2 (c) $1/n$ (d) $1/n^2$
2. The Balmer series of hydrogen lies in the [1] (a) ultraviolet (b) visible (c) infrared (d) X-ray region

- The minimum wavelength of the continuous X-ray spectrum depends on [1] (a) the target material (b) the filament current (c) the accelerating voltage (d) the gas pressure in the tube
- The de Broglie wavelength of an electron accelerated through 400 V is about [1] (a) 0.61 Å (b) 1.23 Å (c) 2.45 Å (d) 12.27 Å
- In Bragg's law $2d\sin\theta = n\lambda$, the angle θ is measured between the incident ray and the [1] (a) normal to the crystal plane (b) crystal plane (c) reflected ray (d) surface of the crystal only

ANSWERS TO GROUP A

1. (b) — $r_n = (\epsilon_0 h^2 / \pi m e^2) n^2$. 2. (b) — transitions to $n = 2$ give 656–410 nm, which is visible light. 3. (c) — $\lambda_{\min} = hc/eV$ contains only V ; the characteristic lines depend on the target. 4. (a) — $12.27/\sqrt{400} = 12.27/20 = 0.61$ Å. 5. (b) — θ is the glancing angle, measured from the plane.

Group B — Short answer (5 marks each)

- State Bohr's postulates and hence derive an expression for the radius of the n^{th} orbit of a hydrogen atom. [5]
- Distinguish between emission and absorption spectra. Why do the dark lines of the solar absorption spectrum occur at exactly the wavelengths of the bright lines of a laboratory hydrogen lamp? [5]
- Calculate the wavelength of the first line and of the series limit of the Lyman series. Take $R = 1.097 \times 10^7 \text{ m}^{-1}$. [5]
- An X-ray tube operates at 40 kV. Find the minimum wavelength of the emitted X-rays. Explain why the spectrum also contains sharp lines whose wavelengths do not change when the voltage is raised. [5]

ANSWERS TO GROUP B

1. Postulates as in §23.1. Equating Coulomb and centripetal force, $e^2/4\pi\epsilon_0 r^2 = mv^2/r$; putting $v = nh/2\pi mr$ gives $r_n = \epsilon_0 h^2 n^2 / \pi m e^2 = 0.529 n^2$ Å.

2. See the comparison table in §23.3. The same pair of levels is involved in both cases — upward in absorption, downward in emission — and the energy gap $E_i - E_f$ is a property of the atom, so $\lambda = hc/\Delta E$ is identical.

3. First line, $n_i = 2 \rightarrow n_f = 1$: $1/\lambda = 1.097 \times 10^7 (1 - 1/4) = 8.228 \times 10^6 \text{ m}^{-1}$, so $\lambda = 1.216 \times 10^{-7} \text{ m} = 121.6 \text{ nm}$. Series limit, $n_i = \infty$: $1/\lambda = 1.097 \times 10^7 (1)$, so $\lambda = 9.12 \times 10^{-8} \text{ m} = 91.2 \text{ nm}$. Both are ultraviolet.

4. $\lambda_{\min} = hc/eV = (6.63 \times 10^{-34})(3 \times 10^8) / [(1.6 \times 10^{-19})(4 \times 10^4)] = 3.11 \times 10^{-11} \text{ m} = 0.311 \text{ Å}$. The sharp lines are the characteristic spectrum, whose photon energies are fixed by the **target's** own energy levels, not by the tube voltage.

Group C — Long answer (8 marks each)

- (a) Using Bohr's postulates, derive an expression for the total energy of the electron in the n^{th} orbit of a hydrogen atom, and show that it equals $-13.6/n^2$ eV. [5] (b) Draw an energy-level diagram and mark the Lyman, Balmer and Paschen series. Calculate the ionization potential and the first excitation potential of hydrogen. [3]

2. (a) State de Broglie's hypothesis and show that it leads to Bohr's quantization of angular momentum. [4] (b) A beam of electrons is accelerated through 10 kV and falls on a crystal whose planes are $2.0 \text{ \AA}$ apart. Find the de Broglie wavelength and the glancing angle for first-order Bragg reflection. [4]

ANSWER TO GROUP C QUESTION 2(B)

$$\lambda = 12.27/\sqrt{10000} = 12.27/100 = 0.1227 \text{ \AA} = 1.227 \times 10^{-11} \text{ m}.$$

For $n = 1$: $\sin\theta = \frac{n\lambda}{2d} = \frac{1.227 \times 10^{-11}}{2 \times 2.0 \times 10^{-10}} = 0.03068$, so $\theta = 1.76^\circ$ — a very small glancing angle, because the electron wavelength is far shorter than the plane spacing.

UNIT 24

Radioactivity and nuclear reaction

Modern Physics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Alpha-particles, Beta-particles, Gamma rays
- Laws of radioactive disintegration
- Half-life, mean-life and decay constant
- Geiger-Muller Tube
- Carbon dating
- Medical use of nuclear radiation and possible health hazard

Becquerel found in 1896 that uranium salts fogged a wrapped photographic plate in a locked drawer. No chemistry, no heating and no light could switch the effect on or off — it came from inside the nucleus. **Radioactivity** is the spontaneous breakup of an unstable nucleus, and because it is spontaneous and random it obeys a statistical law rather than a mechanical one. This unit sets out that law, the instrument that counts the decays, and the two places Nepali students meet radioactivity in ordinary life: dating old things and treating cancer.

WHAT THE EXAM WANTS FROM THIS UNIT

The derivation of $N = N_0 e^{-\lambda t}$ from $dN/dt = -\lambda N$, the three relations $T_{1/2} = 0.693/\lambda$, $\tau = 1/\lambda = 1.44 T_{1/2}$ and $A = \lambda N$, the labelled GM tube, and a dating or activity numerical. Learn to spot "n half-lives" problems — they need no calculus at all.

24.1 Alpha-particles, Beta-particles, Gamma rays

A nucleus is held together by the short-range **strong nuclear force** against the electrostatic repulsion of its protons. Plotting the **binding energy per nucleon** against mass number shows where stability lies.

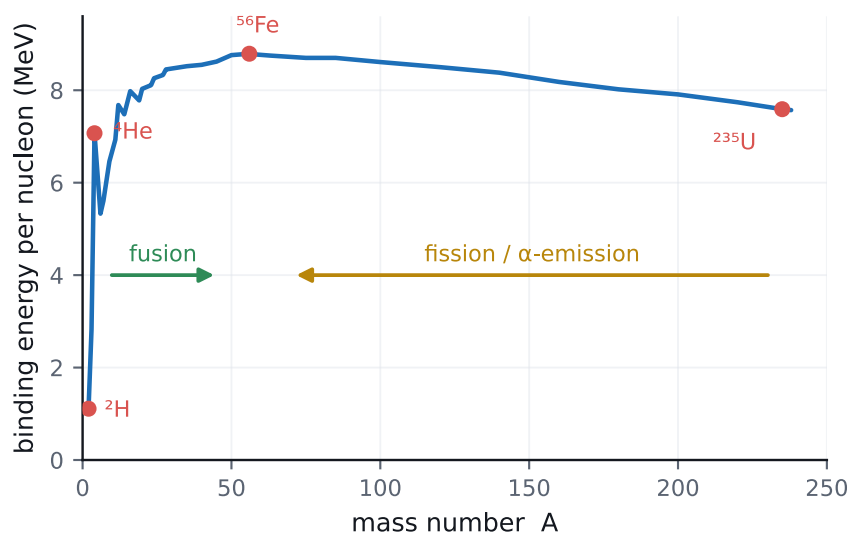


Figure 101. Binding energy per nucleon against mass number. The peak near $A \approx 56$ (iron and nickel) is the most tightly bound region; light nuclei gain energy by fusing and heavy nuclei by splitting or by emitting an *alpha*-particle.

Nuclei with $A > 209$ have no stable form: emitting particles moves them towards the peak and releases energy. Three kinds of radiation come out.

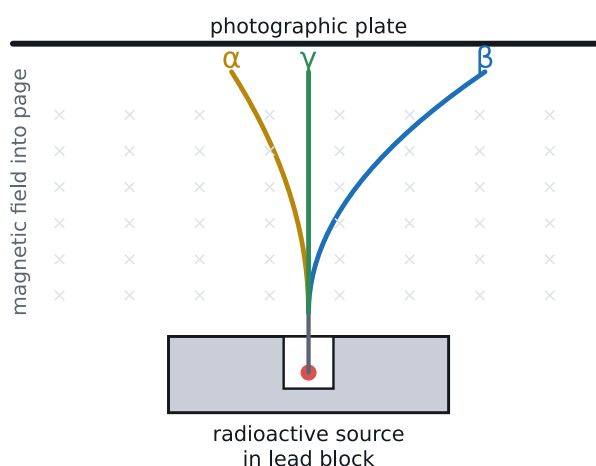


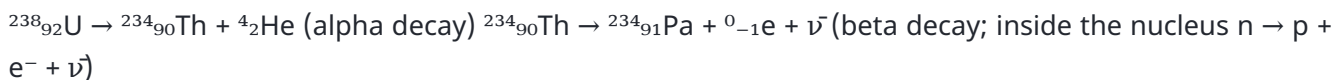
Figure 102. Separating the three radiations. With the magnetic field into the page, the positive *alpha*-particles bend one way, the negative *beta*-particles bend much further the other way (they are far lighter), and the uncharged *gamma*-rays go straight on.

Property	Alpha (α)	Beta (β^-)	Gamma (γ)
Nature	helium nucleus, ${}^4_2\text{He}$	fast electron	electromagnetic photon
Charge	$+2e$	$-e$	0
Mass	$4u = 6.64 \times 10^{-27} \text{ kg}$	$9.1 \times 10^{-31} \text{ kg}$	0 (rest mass)
Speed	$\approx 0.05c$	up to $0.99c$	c
Ionizing power	very high (10^4)	moderate (10^2)	low (1)
Penetrating power	low (1)	moderate (100)	very high (10^4)
Stopped by	a sheet of paper	3 mm aluminium	several cm of lead
Deflection in field	small	large, opposite way	none
Effect on nucleus	$A - 4, Z - 2$	A same, $Z + 1$	no change

IONIZING AND PENETRATING POWER RUN OPPOSITE WAYS

α is the best ionizer and the worst penetrator; γ is the reverse. The reason is the same for both: an α -particle is heavy and doubly charged, so it loses energy fast in a short distance.

The last row is the **Soddy–Fajans displacement law**:



Gamma emission follows either of these: the daughter nucleus is left excited and drops to its ground state by emitting a photon, typically 0.1–3 MeV.

THE BETA PARTICLE IS NOT AN ORBITAL ELECTRON

The nucleus contains no electrons (see the uncertainty-principle argument in Unit 23). The β -particle is **created** at the instant a neutron converts into a proton. An antineutrino is emitted with it, which is why β energies form a continuous spectrum while α energies are sharp.

The energy released in a decay is the **Q-value**, $Q = \Delta m \times 931.5 \text{ MeV}$, where Δm is the mass lost. A positive Q means the decay is energetically allowed.

WORKED EXAMPLE 24.1

Problem. Find the energy released when ${}^{238}\text{U}$ decays by alpha emission. Atomic masses: ${}^{238}\text{U} = 238.050788 \text{ u}$, ${}^{234}\text{Th} = 234.043601 \text{ u}$, ${}^4\text{He} = 4.002603 \text{ u}$.

Solution. Mass of products $= 234.043601 + 4.002603 = 238.046204 \text{ u}$.

$$\Delta m = 238.050788 - 238.046204 = 0.004584 \text{ u}$$

$$Q = 0.004584 \times 931.5 = 4.27 \text{ MeV}$$

Almost all of this appears as kinetic energy of the light α -particle, which is why α -particles from a given nuclide all have the same energy.

24.2 Laws of radioactive disintegration

THE LAWS OF RADIOACTIVE DISINTEGRATION

1. Radioactivity is a **spontaneous and random** nuclear process. It is not affected by temperature, pressure, chemical combination or any other physical condition. Which particular nucleus decays next cannot be predicted.
2. The **rate of disintegration** at any instant is directly proportional to the number of undecayed nuclei present at that instant.

THE EXPONENTIAL DECAY LAW

Let N be the number of undecayed nuclei at time t . By the second law,

$$-\frac{dN}{dt} \propto N \quad \Rightarrow \quad \frac{dN}{dt} = -\lambda N$$

where λ is the **decay constant**, a property of the nuclide. The minus sign shows N decreases. Separating the variables,

$$\frac{dN}{N} = -\lambda dt$$

and integrating from $N = N_0$ at $t = 0$ to N at time t :

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt \quad \Rightarrow \quad \ln \frac{N}{N_0} = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

The **activity** A is the number of disintegrations per second, $A = -dN/dt = \lambda N$. Multiplying the decay law by λ ,

$$A = A_0 e^{-\lambda t}$$

so activity — which is what a counter actually measures — falls off with the same exponential. The SI unit of activity is the **becquerel** (1 Bq = 1 disintegration per second); the older unit is the **curie**, 1 Ci = 3.7×10^{10} Bq.

DECAY CONSTANT

λ is the probability per unit time that a given nucleus will decay. Its unit is s^{-1} . From $N = N_0 e^{-\lambda t}$, putting $t = 1/\lambda$ gives $N = N_0/e$, so λ is the reciprocal of the time in which the number of undecayed nuclei falls to $1/e$ (about 37%) of its initial value.

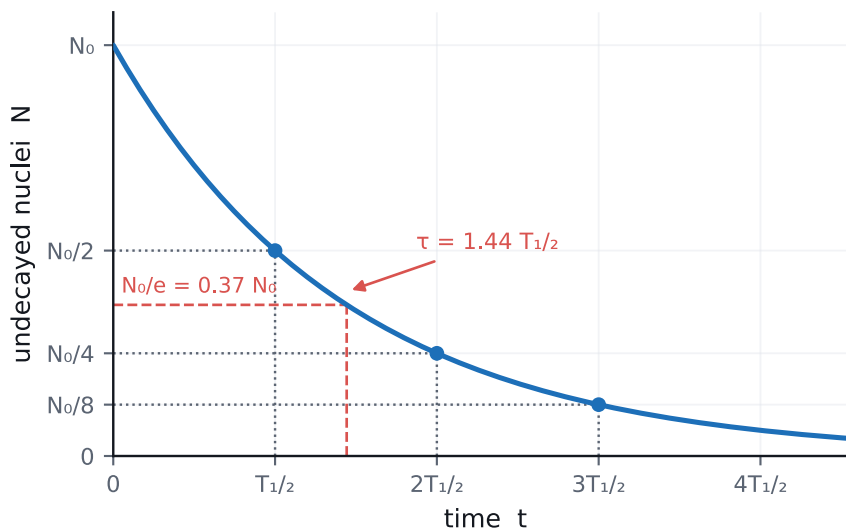
24.3 Half-life, mean-life and decay constant

Figure 103. Radioactive decay curve. In every half-life $T_{1/2}$ the number of undecayed nuclei is halved, whatever the starting point. The mean life $\tau = 1.44 T_{1/2}$ is where the curve has fallen to $1/e = 0.37$ of N_0 .

HALF-LIFE AND MEAN LIFE

Half-life $T_{1/2}$ is the time in which half the nuclei present decay. Putting $N = N_0/2$ at $t = T_{1/2}$ in $N = N_0 e^{-\lambda t}$:

$$\frac{1}{2} = e^{-\lambda T_{1/2}} \Rightarrow e^{\lambda T_{1/2}} = 2 \Rightarrow \lambda T_{1/2} = \ln 2$$

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Mean life τ is the average lifetime of all the nuclei. The number decaying between t and $t + dt$ is $\lambda N dt$, each having lived a time t , so

$$\tau = \frac{1}{N_0} \int_0^\infty t \lambda N_0 e^{-\lambda t} dt = \lambda \int_0^\infty t e^{-\lambda t} dt = \lambda \cdot \frac{1}{\lambda^2}$$

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693} = 1.44 T_{1/2}$$

After n half-lives ($n = t/T_{1/2}$), the surviving fraction is

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$$

which handles most exam numericals without any logarithms.

$T_{1/2}$	Nuclide	Typical use
110 minutes	fluorine-18	PET brain and cancer scans
6.0 hours	technetium-99m	bone and kidney scans
8.0 days	iodine-131	thyroid diagnosis and therapy
5.27 years	cobalt-60	teletherapy, sterilizing equipment
5730 years	carbon-14	archaeological dating
4.5×10^9 years	uranium-238	dating rocks and the Earth

WORKED EXAMPLE 24.2

Problem. The activity of a radioactive sample falls from 8000 counts per minute to 500 counts per minute in 24 hours. Find the half-life, the decay constant and the mean life.

Solution. $\frac{A_0}{A} = \frac{8000}{500} = 16 = 2^4$, so 4 half-lives have passed.

$$T_{1/2} = \frac{24 \text{ h}}{4} = 6 \text{ h}$$

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{6} = 0.1155 \text{ h}^{-1} = \frac{0.1155}{3600} = 3.21 \times 10^{-5} \text{ s}^{-1}$$

$$\tau = \frac{1}{\lambda} = \frac{1}{0.1155} = 8.66 \text{ h}$$

WORKED EXAMPLE 24.3

Problem. Calculate the activity of 1.0 g of radium-226, whose half-life is 1600 years. Take 1 year = 3.156×10^7 s and $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.

Solution. Number of nuclei in 1.0 g:

$$N = \frac{1.0}{226} \times 6.022 \times 10^{23} = 2.665 \times 10^{21}$$

Decay constant:

$$\lambda = \frac{0.693}{1600 \times 3.156 \times 10^7} = \frac{0.693}{5.050 \times 10^{10}} = 1.372 \times 10^{-11} \text{ s}^{-1}$$

$$A = \lambda N = (1.372 \times 10^{-11})(2.665 \times 10^{21}) = 3.66 \times 10^{10} \text{ Bq}$$

This is almost exactly 3.7×10^{10} Bq — and that is no accident: the curie was originally defined as the activity of one gram of radium.

HALF-LIFE IS NOT "HALF THE TOTAL LIFE"

A sample never disappears. After $2T_{1/2}$ one quarter remains, after $3T_{1/2}$ one eighth, and so on — the decay law is exponential, so the number never reaches zero. Also, $T_{1/2}$ is fixed for a nuclide: it cannot be changed by heating, cooling or chemical reaction.

24.4 Geiger-Müller Tube

The GM tube converts a single ionizing particle into an electrical pulse big enough to count.

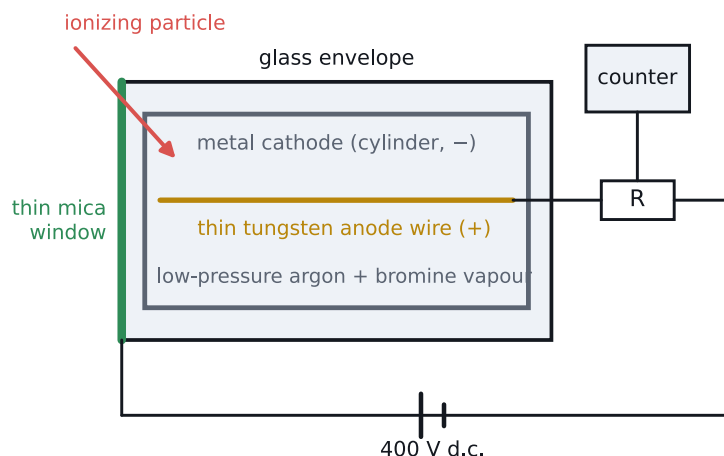


Figure 104. Geiger-Müller tube. A particle entering the mica window ionizes the argon; the freed electrons are accelerated to the thin central anode, where they trigger an avalanche that produces a pulse across R .

Construction. A cylindrical metal tube (the cathode) is sealed in glass with a fine tungsten wire along its axis (the anode). One end is closed by a thin **mica window** so that even α -particles can get in. The tube is filled with **argon** at about 0.1 atmosphere together with a little **bromine or ethyl alcohol vapour**, which acts as a quenching agent. A p.d. of about 400 V is applied through a high resistance R .

Working. An ionizing particle entering the window produces a few ion pairs. The electric field near the thin wire is intense, so the freed electrons are accelerated hard enough to ionize further atoms. The result is an **avalanche** of about 10^8 electrons reaching the anode within a microsecond. The current through R produces a voltage pulse, which is amplified and counted.

Quenching and dead time. The positive argon ions drift slowly to the cathode, where they would release fresh electrons and restart the discharge. The quenching vapour absorbs their energy and dissociates instead, stopping the pulse. During the roughly $200\ \mu\text{s}$ the ions take to clear — the **dead time** — the tube cannot register another particle, so true counts exceed recorded counts at high rates.

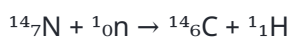
The plateau. Plotting count rate against tube voltage for a fixed source gives a flat **plateau** about 100 V wide. The tube is operated in the middle of the plateau, so that small supply fluctuations do not change the reading. Above the plateau a continuous discharge starts and destroys the tube.

WHAT A GM TUBE CAN AND CANNOT DO

It counts α , β and γ particles, but the pulse size is the same whatever the particle's energy. So a GM tube measures **how many**, never **how energetic** — energy measurement needs a scintillation counter instead.

24.5 Carbon dating

Cosmic-ray neutrons striking nitrogen in the upper atmosphere make radioactive carbon-14:



The ${}^{14}\text{C}$ mixes into atmospheric CO_2 , so every living plant and animal keeps a fixed ratio ${}^{14}\text{C} : {}^{12}\text{C}$ of about 1.3×10^{-12} , giving an activity of **15.3 disintegrations per minute per gram of carbon**. At death, intake stops while the ${}^{14}\text{C}$ keeps decaying back to nitrogen ($T_{1/2} = 5730$ years):



Measuring the present activity A therefore dates the specimen:

$$A = A_0 e^{-\lambda t} \quad \Rightarrow \quad t = \frac{1}{\lambda} \ln \frac{A_0}{A} = \frac{T_{1/2}}{0.693} \ln \frac{A_0}{A}$$

After about 10 half-lives the remaining activity is lost in the background, so the method is useful to roughly **50,000 years**. It works only on once-living material — wood, charcoal, bone, cloth. Radiocarbon dating of charcoal and timber beneath the Maya Devi temple at **Lumbini** placed a wooden structure there in about the sixth century BCE.

WORKED EXAMPLE 24.4

Problem. A piece of wood from an old temple beam gives 9.2 disintegrations per minute per gram of carbon. Living wood gives 15.3 per minute per gram. Take $T_{1/2}({}^{14}\text{C}) = 5730$ years. How old is the beam?

Solution.

$$\lambda = \frac{0.693}{5730} = 1.209 \times 10^{-4} \text{ year}^{-1}$$

$$t = \frac{1}{\lambda} \ln \frac{A_0}{A} = \frac{1}{1.209 \times 10^{-4}} \ln \frac{15.3}{9.2}$$

$$\ln(1.663) = 0.5087 \quad \Rightarrow \quad t = \frac{0.5087}{1.209 \times 10^{-4}} = 4.21 \times 10^3 \text{ years}$$

The beam is about **4200 years** old.

24.6 Medical use of nuclear radiation and possible health hazard

Diagnosis. A short-lived γ -emitting **tracer** is injected and a gamma camera images where it collects. Technetium-99m ($T_{1/2} = 6 \text{ h}$) is used for bone, kidney and heart scans; iodine-131 concentrates in the thyroid; fluorine-18 is the tracer in PET scans. Short half-lives are chosen so that the activity dies away within a day.

Therapy. Cancer cells divide fast and are more easily killed by radiation than normal cells. A **cobalt-60 teletherapy** unit aims 1.17 and 1.33 MeV γ -rays at a tumour from several directions, so the tumour receives a large dose while surrounding tissue receives little. In **brachytherapy** a sealed iridium-192 seed is placed inside or beside the tumour. Iodine-131 is drunk as a solution to destroy an over-active thyroid.

Sterilization. Syringes, dressings and surgical gloves are sealed in plastic and sterilized by γ -rays, which kill bacteria without heating.

Health hazards. Radiation ionizes the molecules of living cells and can break DNA strands. The effects are of two kinds:

- **Somatic** — affecting the exposed person: radiation burns, hair loss, sterility, cataract, anaemia, leukaemia and other cancers.
- **Genetic** — mutations in reproductive cells that are passed to children.

Quantity	Definition	SI unit
Absorbed dose D	energy absorbed per kg of tissue	gray (Gy), $1 \text{ Gy} = 1 \text{ J kg}^{-1}$
Equivalent dose H	$H = D \times Q$ (quality factor: 1 for β, γ ; 20 for α)	sievert (Sv)
Activity A	disintegrations per second	becquerel (Bq)

Natural background radiation in Nepal delivers roughly 2–3 mSv per year from cosmic rays, rocks and radon. Safety follows three simple rules: keep the **time** of exposure short, keep your **distance** (intensity falls as $1/r^2$), and use **shielding** — lead aprons, thick concrete walls, remote handling tongs. Workers wear film badges or thermoluminescent dosimeters, and radioactive waste is sealed in lead-lined containers.

Chapter summary

- Binding energy per nucleon peaks near $A \approx 56$ at about 8.8 MeV; nuclei move towards the peak by fusing, splitting or emitting α -particles.
- $\alpha = {}^4_2\text{He}$ ($A - 4, Z - 2$); β^- = electron created when $n \rightarrow p$ ($Z + 1$); γ = photon (no change). Ionizing power $\alpha > \beta > \gamma$; penetrating power is the reverse.
- $Q = \Delta m \times 931.5 \text{ MeV}$ is the energy released in a decay.
- Decay law: $dN/dt = -\lambda N$ gives $N = N_0 e^{-\lambda t}$ and $A = \lambda N = A_0 e^{-\lambda t}$. $1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$.
- $T_{1/2} = 0.693/\lambda$; $\tau = 1/\lambda = 1.44 T_{1/2}$; after n half-lives $N/N_0 = (1/2)^n$.
- A GM tube uses avalanche ionization in argon at 400 V, with a quenching vapour and a dead time of about 200 μs ; it counts particles but not their energy.
- Carbon dating: $t = (T_{1/2}/0.693) \ln(A_0/A)$ with $A_0 = 15.3 \text{ min}^{-1} \text{ g}^{-1}$ and $T_{1/2} = 5730 \text{ y}$; useful to about 50,000 years.
- Medical tracers are short-lived γ -emitters; therapy uses cobalt-60. Dose is measured in gray (absorbed) and sievert (equivalent).

Practice questions

Group A — Multiple choice (1 mark each)

- When a nucleus emits a β^- particle, its mass number and atomic number [1] (a) both decrease (b) A unchanged, Z increases by 1 (c) A decreases by 4, Z by 2 (d) both unchanged
- The decay constant of a nuclide of half-life 20 minutes is [1] (a) 0.0347 min^{-1} (b) 0.693 min^{-1} (c) 20 min^{-1} (d) 28.8 min^{-1}
- After 3 half-lives, the fraction of a radioactive sample left undecayed is [1] (a) $1/3$ (b) $1/6$ (c) $1/8$ (d) $1/9$
- The gas commonly used in a Geiger-Müller tube is [1] (a) hydrogen (b) argon (c) oxygen (d) nitrogen
- Carbon dating cannot be used to find the age of [1] (a) an old wooden beam (b) charcoal from a fireplace (c) a granite boulder (d) a bone

ANSWERS TO GROUP A

1. (b) — a neutron becomes a proton, so Z rises by 1 and A is unchanged. 2. (a) — $\lambda = 0.693/20 = 0.0347 \text{ min}^{-1}$. 3. (c) — $(1/2)^3 = 1/8$. 4. (b) — argon, an inert gas, with a bromine or alcohol quenching vapour. 5. (c) — granite was never alive, so it never took up ^{14}C .

Group B — Short answer (5 marks each)

- State the laws of radioactive disintegration and hence derive $N = N_0 e^{-\lambda t}$. [5]
- Define half-life and mean life, and show that $\tau = 1.44 T_{1/2}$. [5]
- Calculate the binding energy per nucleon of ^4_2He . Take mass of $^4\text{He} = 4.002603 \text{ u}$, of $^1_1\text{H} = 1.007825 \text{ u}$, of a neutron $= 1.008665 \text{ u}$ and $1 \text{ u} = 931.5 \text{ MeV}$. [5]
- A radioactive sample has a half-life of 10 days. Find its decay constant in s^{-1} and the fraction of the sample remaining after 30 days. [5]

ANSWERS TO GROUP B

2. Definitions as in §24.3. $T_{1/2} = 0.693/\lambda$ and $\tau = 1/\lambda$, so $\tau = T_{1/2}/0.693 = 1.44 T_{1/2}$.

3. ^4He has 2 protons and 2 neutrons. Sum of separate masses $= 2(1.007825) + 2(1.008665) = 2.015650 + 2.017330 = 4.032980 \text{ u}$. $\Delta m = 4.032980 - 4.002603 = 0.030377 \text{ u}$. Binding energy $= 0.030377 \times 931.5 = 28.30 \text{ MeV}$, so BE per nucleon $= 28.30/4 = 7.07 \text{ MeV}$.

4. $\lambda = 0.693/T_{1/2} = 0.693/10 = 0.0693 \text{ day}^{-1}$. In SI, $\lambda = 0.693/(10 \times 86400 \text{ s}) = 0.693/864000 = 8.02 \times 10^{-7} \text{ s}^{-1}$. After 30 days $= 3$ half-lives, $N/N_0 = (1/2)^3 = 1/8 = 0.125$, i.e. 12.5%.

Group C — Long answer (8 marks each)

- (a) Draw a labelled diagram of a Geiger-Müller tube and explain how it detects a single ionizing particle. [5] (b) What is meant by quenching and dead time? Why is the tube operated on the plateau of its characteristic curve? [3]

2. (a) Derive the relation between half-life and decay constant. [3] (b) A radioactive source contains 1.0×10^{20} nuclei of a nuclide whose half-life is 8 hours. Calculate the decay constant, the initial activity, and the activity after 24 hours. [5]

ANSWER TO GROUP C QUESTION 2(B)

$$\lambda = \frac{0.693}{8 \times 3600} = \frac{0.693}{28800} = 2.41 \times 10^{-5} \text{ s}^{-1}$$

$$A_0 = \lambda N_0 = (2.41 \times 10^{-5})(1.0 \times 10^{20}) = 2.41 \times 10^{15} \text{ Bq}$$

24 hours is exactly 3 half-lives, so $N = N_0/8 = 1.25 \times 10^{19}$ and

$$A = \lambda N = (2.41 \times 10^{-5})(1.25 \times 10^{19}) = 3.01 \times 10^{14} \text{ Bq}$$

(equivalently $A = A_0/8$).

UNIT 25

Recent trends in physics

Modern Physics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Seismology: Surface waves (Rayleigh and Love waves); Internal waves (S and P-waves); Wave patterns of Gorkha Earthquake 2015
- Gravitational Wave; Nanotechnology; Higgs Boson

The last unit looks outward. Gravitational waves, nanotechnology and the Higgs boson are results of the past two decades; seismology is the physics of the ground under Nepal. The mathematics is light — what is examined is whether you can explain the physics clearly and quote the key numbers.

WHAT THE EXAM WANTS FROM THIS UNIT

Descriptive answers with correct detail: the four wave types and their order of arrival, the facts of the Gorkha earthquake, what LIGO measured in 2015, what "nanoscale" means, and what the Higgs field does. Numbers earn marks: M_w 7.8, strain 10^{-21} , 1–100 nm, 125 GeV.

25.1 Seismology: Surface waves (Rayleigh and Love waves); Internal waves (S and P-waves); Wave patterns of Gorkha Earthquake 2015

Seismology is the study of elastic waves travelling through the Earth. Nepal sits where the Indian plate pushes north under Eurasia at about 20 mm per year along the **Main Himalayan Thrust**; strain stored on that locked fault for centuries is released in minutes as an earthquake. The point of rupture inside the Earth is the **focus**; the point on the surface above it is the **epicentre**.

Body waves: P and S

Waves travelling through the interior are **body waves** (internal waves).

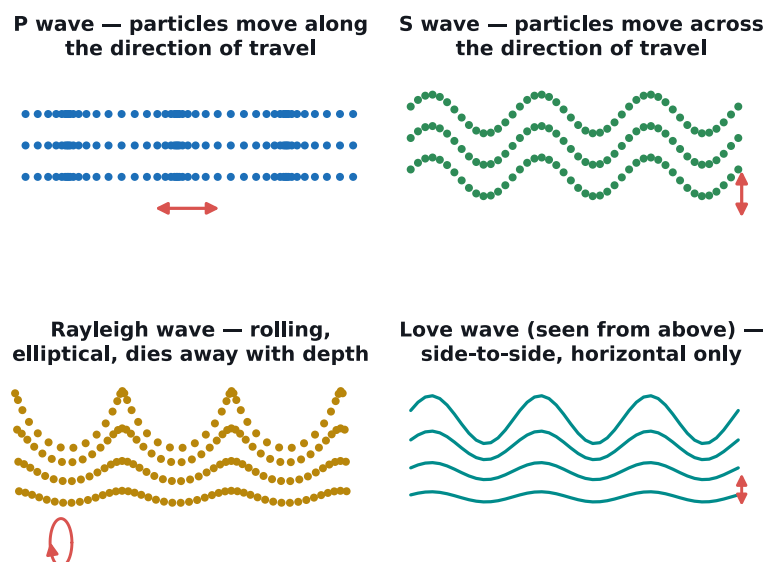


Figure 105. The four seismic wave types. P and S waves travel through the Earth's interior; Rayleigh and Love waves run along the surface and die away with depth.

P-waves (primary, longitudinal). The ground is alternately compressed and stretched along the direction of travel, like sound in air. At about 6 km s^{-1} in the crust they are the fastest waves and arrive first. A compression needs only a change of volume, so P-waves pass through **solids and liquids alike**.

S-waves (secondary, transverse). Particles move at right angles to the direction of travel, shearing the rock sideways, at about 3.5 km s^{-1} . A liquid has no rigidity, so **S-waves cannot pass through liquids** — the S-wave shadow zone beyond 103° from an epicentre is the direct evidence that the outer core is molten.

Surface waves: Rayleigh and Love

Surface waves are guided along the boundary between ground and air. They are slower than body waves, so they arrive last, but they carry the largest amplitude and do most of the damage, and they die away exponentially with depth.

Wave	Type	Particle motion	Typical speed	Note
P	body, longitudinal	along the ray	6 km s^{-1}	passes through liquids
S	body, transverse	perpendicular to the ray	3.5 km s^{-1}	blocked by liquids
Love	surface, transverse	horizontal, across the ray	3 km s^{-1}	needs a slow surface layer
Rayleigh	surface	retrograde vertical ellipse	2.7 km s^{-1}	"ground roll"; largest

ORDER OF ARRIVAL

P before **S** before **Love** before **Rayleigh** — "**P**lease **S**tart **L**eaving **R**ightaway".

Because P and S travel at different speeds, the gap between their arrival times measures the distance to the epicentre:

$$\Delta t = \frac{d}{v_S} - \frac{d}{v_P} \quad \Rightarrow \quad d = \Delta t \left(\frac{v_P v_S}{v_P - v_S} \right)$$

Three stations give three circles on a map, and the epicentre is where they meet.

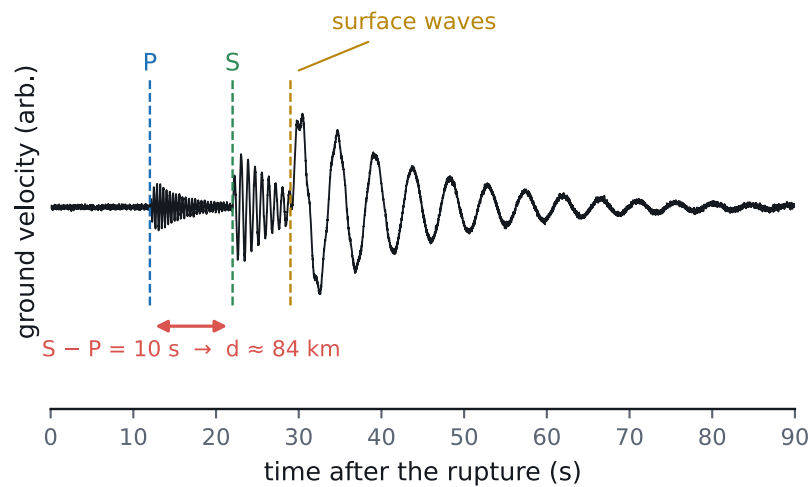


Figure 106. Shape of a seismogram close to the 2015 Gorkha epicentre. The small, fast P arrival is followed by the larger S waves, and then by long-period surface waves. Kathmandu's soft basin sediments amplified the 4–5 s surface-wave motion and kept it going for tens of seconds.

WORKED EXAMPLE 25.1

Problem. A Kathmandu station records the first P-wave at 11:56:22 and the first S-wave at 11:56:32. Taking $v_P = 6.0 \text{ km s}^{-1}$ and $v_S = 3.5 \text{ km s}^{-1}$, find the distance to the epicentre.

Solution. $\Delta t = 10 \text{ s}$.

$$d = \Delta t \left(\frac{v_P v_S}{v_P - v_S} \right) = 10 \times \frac{6.0 \times 3.5}{6.0 - 3.5} = 10 \times \frac{21}{2.5}$$

$$d = 10 \times 8.4 = 84 \text{ km}$$

Barpak, the 2015 epicentre, is about 80 km from Kathmandu.

Wave patterns of the Gorkha Earthquake 2015

Quantity	Value
Date and time	25 April 2015, 11:56 NST (06:11 UTC)
Moment magnitude	M_w 7.8 (Nepal's National Seismological Centre: M_L 7.6)
Epicentre	Barpak, Gorkha district (28.2° N, 84.7° E), ~80 km WNW of Kathmandu
Focal depth	about 15 km — shallow
Fault	Main Himalayan Thrust, dipping a few degrees
Rupture	~140 km eastwards at $\sim 3.3 \text{ km s}^{-1}$, maximum slip ~6 m
Largest aftershock	M_w 7.3, 12 May 2015, near Dolakha
Casualties	about 8,970 killed, 22,300 injured

Three features of the recorded waves explain the damage.

1. **Little high-frequency energy.** The fault slipped smoothly rather than jerkily, so peak ground acceleration in Kathmandu was only about $0.16g$ — far less than expected for M_w 7.8. Many ordinary houses therefore survived.
2. **Strong long-period shaking.** The soft lake sediments filling the Kathmandu valley resonated at a period of **4–5 s**, amplifying the surface waves and keeping the ground moving for tens of seconds. Tall flexible buildings, whose natural period is near 4–5 s, were shaken hardest; short stiff ones were not.
3. **Eastward directivity.** The rupture ran east, so the waves piled up in that direction. Sindhupalchok, east of the epicentre, lost the most lives.

MAGNITUDE AND INTENSITY ARE DIFFERENT QUANTITIES

Magnitude (M_w) is one number for the whole earthquake, fixed by the energy released at the source. **Intensity** (Modified Mercalli, I–XII) describes the shaking at one place and differs from village to village with distance and soil. One earthquake has one magnitude but many intensities; an increase of 1 in M_w means about 32 times more energy.

25.2 Gravitational Wave; Nanotechnology; Higgs Boson

Gravitational waves

Einstein's general theory of relativity (1915) describes gravity not as a force but as curvature of space-time. When a massive body accelerates non-symmetrically, that curvature ripples outwards at the speed of light: these ripples are **gravitational waves**.

A passing wave stretches space in one direction while squeezing it in the perpendicular direction. The quantity measured is the **strain**

$$h = \frac{\Delta L}{L}$$

and even for the strongest sources reaching Earth $h \approx 10^{-21}$ — which is why detection took a century.

LIGO. Each detector is a giant Michelson interferometer with two evacuated arms **4 km** long at right angles. A laser beam is split between the arms, reflected and recombined; a passing wave lengthens one arm and shortens the other, shifting the interference fringes. Two detectors 3,000 km apart (Hanford and Livingston, USA) must see the same signal within milliseconds, which rules out local disturbances.

WORKED EXAMPLE 25.2

Problem. A wave of strain $h = 1.0 \times 10^{-21}$ passes through a LIGO arm of length $L = 4.0$ km. Find the change in the arm's length and compare it with a proton radius, 0.84×10^{-15} m.

Solution.

$$\Delta L = hL = (1.0 \times 10^{-21})(4.0 \times 10^3) = 4.0 \times 10^{-18} \text{ m}$$

$$\frac{\Delta L}{r_{\text{proton}}} = \frac{4.0 \times 10^{-18}}{0.84 \times 10^{-15}} = 4.8 \times 10^{-3} \approx \frac{1}{210}$$

LIGO measures a length change about **one two-hundredth of a proton radius** — hence the multi-stage pendulums and ultra-high vacuum.

GW150914 — the first detection. On 14 September 2015 both LIGO detectors recorded a 0.2 s signal whose frequency swept up from 35 Hz to 250 Hz — a "chirp". Its source was two black holes of about 36 and 29 solar masses spiralling together 1.3 billion light-years away. They merged into one black hole of 62 solar masses, the missing **3 solar masses** being radiated as gravitational waves; the peak strain was 1.0×10^{-21} . The result was announced on 11 February 2016 and won the 2017 Nobel Prize in Physics for Weiss, Barish and Thorne.

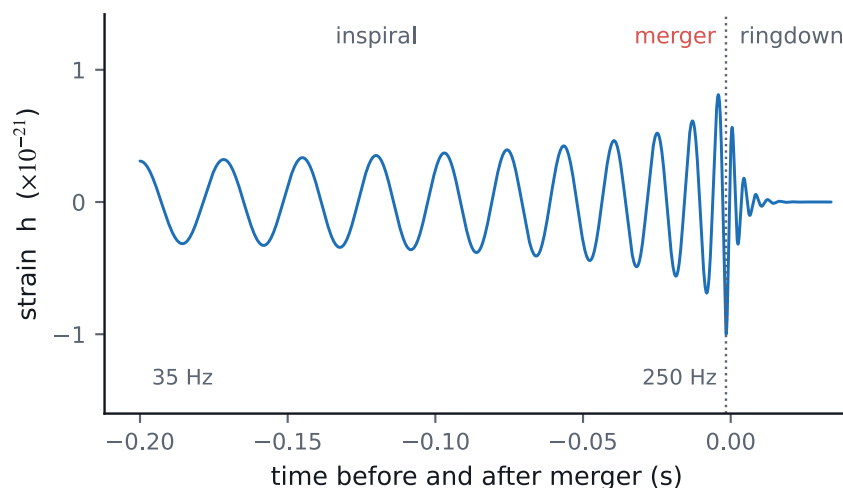


Figure 107. Model of the GW150914 chirp. As the two black holes spiral in, the frequency and amplitude rise together (inspiral), peak at the merger, and then decay as the new black hole settles down (ringdown).

Gravitational-wave astronomy is now routine. **GW170817** (2017), two merging neutron stars, was also seen by optical and gamma-ray telescopes — the start of "multi-messenger" astronomy. LIGO's fourth observing run ended in November 2025, and the catalogues now hold several hundred confirmed mergers.

GRAVITATIONAL WAVES ARE NOT ELECTROMAGNETIC WAVES

They are ripples in spacetime itself, not oscillating electric and magnetic fields. Nothing can shield you from them, and they come only from accelerating **mass** distributions that are not spherically symmetric — a perfectly spherical collapsing star radiates none.

Nanotechnology

NANOTECHNOLOGY

Nanotechnology is the design and use of structures with at least one dimension in the range **1 to 100 nanometres** ($1 \text{ nm} = 10^{-9} \text{ m}$), where the properties of matter differ from those of the bulk material.

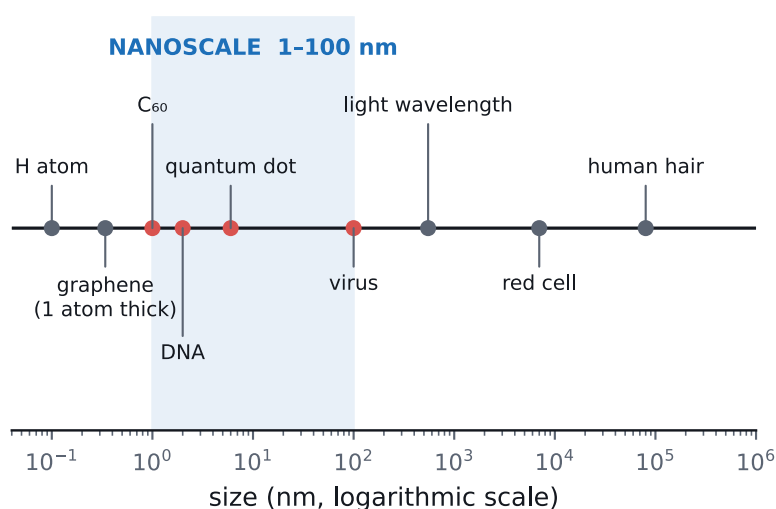


Figure 108. The nanoscale in context. The shaded band from 1 nm to 100 nm is the working range of nanotechnology — larger than single atoms, smaller than the wavelength of visible light.

Why the nanoscale is different. Two effects take over.

1. **Surface dominates.** For a cube of side L the surface-to-volume ratio is $6/L$, so shrinking a 1 cm grain to a 10 nm grain multiplies it by a million and almost every atom sits on the surface. Nanoparticles are therefore superb catalysts and absorbers.
2. **Quantum confinement.** In a particle only a few nanometres across the electron's de Broglie wave is squeezed and its energy levels spread apart (Unit 23), so the band gap — and hence the colour — depends on size. Semiconductor "quantum dots" of one material glow red at 6 nm and blue at 2 nm.

Nanomaterial	Structure	Property and use
Fullerene C ₆₀ (1985)	hollow ball of 60 carbon atoms, ~1 nm	drug carriers, lubricants
Carbon nanotube (1991)	rolled graphene cylinder, 1–2 nm	~100× the tensile strength of steel
Graphene (2004)	one carbon layer, 0.34 nm thick	strongest known material; Nobel Prize 2010
Quantum dot	semiconductor crystal 2–10 nm	size-tunable colour; QLED screens
TiO ₂ / ZnO nanoparticle	oxide crystal ~30 nm	absorbs UV, looks clear — sunscreen

Uses already in the field include targeted drug delivery, nanofiltration membranes that strip arsenic and iron from tube-well water, water-repellent coatings, chemical sensors and the transistors in every phone. The guiding idea goes back to Feynman's 1959 lecture *There's Plenty of Room at the Bottom*.

Higgs boson

The **Standard Model** lists the fundamental particles: six quarks, six leptons and the force-carrying bosons (photon, gluon, W, Z). Its equations, however, stay consistent only if all of them are **massless**, which they plainly are not.

The way out, proposed in 1964 by Peter Higgs and independently by Englert and Brout and by others, is the **Higgs field**, which fills all of space with a non-zero value. Particles moving through it are dragged, and that resistance is what we measure as mass: the top quark and the W and Z bosons couple strongly and are heavy, while the photon does not couple at all and stays massless.

The quantum of the Higgs field is the **Higgs boson**, and finding it was the last great test of the Standard Model.

Fact	Value
Announced	4 July 2012, CERN, by ATLAS and CMS
Machine	Large Hadron Collider — 27 km proton ring, 13 TeV
Mass	about 125 GeV/ c^2 (~133 proton masses)
Spin	0 — the only known fundamental scalar
Lifetime	$\sim 1.6 \times 10^{-22}$ s; seen only via its decay products
Nobel Prize	Physics 2013, Higgs and Englert

WORKED EXAMPLE 25.3

Problem. The Higgs boson has a rest energy of 125 GeV. Find its mass in kg and in proton masses ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$).

Solution.

$$E = 125 \times 10^9 \times 1.6 \times 10^{-19} = 2.0 \times 10^{-8} \text{ J}$$

$$m = \frac{E}{c^2} = \frac{2.0 \times 10^{-8}}{(3 \times 10^8)^2} = \frac{2.0 \times 10^{-8}}{9.0 \times 10^{16}} = 2.22 \times 10^{-25} \text{ kg}$$

$$\frac{m}{m_p} = \frac{2.22 \times 10^{-25}}{1.67 \times 10^{-27}} = 133$$

One Higgs boson weighs as much as 133 protons — about a caesium atom.

A SENTENCE EXAMINERS LIKE

"The Higgs field gives mass to the elementary particles; the Higgs boson is a ripple in that field, and detecting it in 2012 confirmed the mechanism." It does **not** explain everyday mass — over 98% of an atom's mass is the binding energy of the quarks inside its nucleons.

Chapter summary

- Body waves: **P** (longitudinal, $\sim 6 \text{ km s}^{-1}$, passes through liquids) and **S** (transverse, $\sim 3.5 \text{ km s}^{-1}$, blocked by liquids). Surface waves: **Love** (horizontal transverse) and **Rayleigh** (rolling ellipse, largest amplitude). Arrival order P, S, Love, Rayleigh.
- Epicentral distance from the S–P interval: $d = \Delta t \, v_P v_S / (v_P - v_S)$; three stations fix the epicentre.
- Gorkha: 25 April 2015, M_w 7.8, focus $\sim 15 \text{ km}$ deep near Barpak, rupture $\sim 140 \text{ km}$ eastwards on the Main Himalayan Thrust; PGA in Kathmandu only $\sim 0.16g$ but strong 4–5 s basin resonance; about 8,970 deaths.
- Gravitational waves are ripples of spacetime travelling at c , of strain $h = \Delta L/L \approx 10^{-21}$. LIGO's 4 km interferometers detected GW150914 on 14 September 2015 ($36 + 29 \rightarrow 62$ solar masses, 3 radiated).
- Nanotechnology works between **1 and 100 nm**, where surface-to-volume ratio ($6/L$) and quantum confinement change behaviour: fullerenes, nanotubes, graphene, quantum dots.
- The Higgs field gives elementary particles their mass; its quantum, the Higgs boson ($\approx 125 \text{ GeV}/c^2$, spin 0), was found at the LHC on 4 July 2012.

Practice questions

Group A — Multiple choice (1 mark each)

1. Which seismic wave cannot travel through a liquid? [1] (a) P-wave (b) S-wave (c) both (d) neither
2. The wave that usually does most damage to buildings is the [1] (a) P-wave (b) S-wave (c) Rayleigh wave (d) sound wave
3. The strain at LIGO from the first detected gravitational wave was about [1] (a) 10^{-9} (b) 10^{-15} (c) 10^{-21} (d) 10^{-31}
4. A structure is said to be nanoscale if one of its dimensions lies between [1] (a) 1 and 100 pm (b) 1 and 100 nm (c) 1 and 100 μm (d) 1 and 100 mm
5. The Higgs boson was discovered in 2012 at [1] (a) LIGO (b) Fermilab (c) CERN (d) the ISS

ANSWERS TO GROUP A

1. (b) — a liquid has no rigidity, so it cannot carry a shear wave. 2. (c) — surface waves have the largest amplitude and longest period. 3. (c) — $h \approx 1.0 \times 10^{-21}$ for GW150914. 4. (b) — $1 \text{ nm} = 10^{-9} \text{ m}$. 5. (c) — by ATLAS and CMS on CERN's Large Hadron Collider.

Group B — Short answer (5 marks each)

1. Distinguish between body waves and surface waves. Explain why P-waves arrive before S-waves, and why S-waves prove the outer core is liquid. [5]
2. A station notes an S–P interval of 25 s. Taking $v_P = 6.0 \text{ km s}^{-1}$ and $v_S = 3.5 \text{ km s}^{-1}$, find the distance to the epicentre, and explain why three stations are needed. [5]
3. What is a gravitational wave? Describe how LIGO detects one, and state two results from the 2015 detection. [5]
4. Define nanotechnology. Give two reasons why materials behave differently at the nanoscale, with an example of each. [5]

ANSWERS TO GROUP B

1. Body waves (P, S) travel through the interior; surface waves (Love, Rayleigh) run along the surface and decay with depth. P-waves are longitudinal and depend on the bulk modulus, which exceeds the shear modulus, so $v_P > v_S$ and P arrives first. S-waves need rigidity, and they vanish beyond 103° from the epicentre — the outer core must be liquid.
2. $d = \Delta t \cdot \frac{v_P v_S}{v_P - v_S} = 25 \times \frac{6.0 \times 3.5}{2.5} = 25 \times 8.4 = 210 \text{ km}$. One station gives only a circle of possible epicentres; two circles meet at two points; a third picks out the correct one.
3. A ripple in the curvature of spacetime travelling at c , produced by accelerating masses. LIGO uses a Michelson interferometer with 4 km arms: the wave stretches one arm and squeezes the other, shifting the fringes. GW150914 showed (i) that black-hole binaries exist and merge, and (ii) that the waves travel at the speed of light, as relativity predicts.
4. Definition as in §25.2. (i) The surface-to-volume ratio $6/L$ becomes huge, so nanoparticles are highly reactive — platinum in catalytic converters. (ii) Quantum confinement spaces out the energy levels, so the band gap depends on size — quantum dots change colour with diameter.

Group C — Long answer (8 marks each)

1. (a) Describe the four types of seismic wave, sketching the particle motion in each. [4] (b) Give an account of the wave pattern of the Gorkha Earthquake of 2015, and explain why damage in Kathmandu fell mainly on tall buildings. [4]
2. (a) What is the Higgs field, and what problem does it solve? [4] (b) The Higgs boson's rest energy is 125 GeV. Find its mass in kg, and the minimum uncertainty in that energy if it lives only $1.6 \times 10^{-22} \text{ s}$ ($h = 6.63 \times 10^{-34} \text{ J s}$). [4]

ANSWER TO GROUP C QUESTION 2(B)

$E = 125 \times 10^9 \times 1.6 \times 10^{-19} = 2.0 \times 10^{-8} \text{ J}$, so $m = E/c^2 = 2.0 \times 10^{-8} / 9.0 \times 10^{16} = 2.22 \times 10^{-25} \text{ kg}$. From $\Delta E \Delta t \geq h/4\pi$,

$$\Delta E \geq \frac{6.63 \times 10^{-34}}{4\pi \times 1.6 \times 10^{-22}} = \frac{6.63 \times 10^{-34}}{2.011 \times 10^{-21}} = 3.30 \times 10^{-13} \text{ J}$$

In electronvolts, $\Delta E = 3.30 \times 10^{-13} / 1.6 \times 10^{-19} \approx 2 \text{ MeV}$ — the measured "width" of the Higgs peak. The shorter the lifetime, the broader the peak.